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# What influences people's responses to public health messages for managing risks and preventing infectious diseases? A systematic review of the evidence and recommendations



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| Complete List of Authors: | Ghio, Daniela; University of Salford<br>Lawes- Wickwar, Sadie; University College London, Department of Primary Care and Population Health<br>Tang, Mei ; Newcastle University, Behavioural Science Policy Research Unit, Population Health Sciences<br>Epton, Tracy; University of Manchester, Manchester Centre for Health Psychology<br>Howlett, Neil; University of Hertfordshire, Psychology and Sport Sciences<br>Jenkinson, Elizabeth; University of the West of England, Department of Health and Social Sciences<br>Stanescu, Sabina; University of Southampton, School of Psychology<br>Westbrook, Juliette; University of Bath, Department of Psychology<br>Kassianos, Angelos; University College London, Department of Applied Health Research<br>Watson, Daniella; University of Southampton, Human Development and Health, Faculty of Medicine<br>Sutherland, Lisa; Behavioural Insight<br>Stanulewicz, Natalia; De Montfort University, Faculty of Health and Life Sciences, School of Applied Social Sciences<br>Guest, Ella; University of the West of England, Centre for Appearance Research, Department of Health and Social Sciences, Faculty of Health and Applied Sciences<br>Scanlan, Daniel; Education Support , Department of Communication, Policy, and Research<br>Carr, Natalie; Manchester Metropolitan University, Faculty of Health, Psychology, and Social Care<br>Chater, Angel; University of Bedfordshire, Psychology Department<br>Hotham, Sarah; University of Kent, Centre for Health Services Studies<br>Thorneloe, Rachael; Sheffield Hallam University, Centre for Behavioural Science and Applied Psychology<br>Armitage, Chris; University of Manchester, School of Psychological Sciences<br>Arden, Madelynne; Sheffield Hallam University, Centre for Behavioural Science and Applied Psychology<br>Hart, Jo; University of Manchester, Manchester Medical School<br>Byrne-Davis, Lucie; University of Manchester, Medical School<br>Keyworth, Christopher; University of Manchester, Division of Psychology & Mental Health |

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# What influences people's responses to public health messages for managing risks and preventing infectious diseases? A systematic review of the evidence and recommendations

Daniela Ghio,[1] Sadie Lawes- Wickwar,[2] Mei Yee Tang,[3] Tracy Epton, [4] Neil Howlett, [5] Elizabeth Jenkinson,[6] Sabina Stanescu, [7] Juliette Westbrook, [8] Angelos P. Kassianos [9] Daniella Watson, [10] Lisa Sutherland, [11] Natalia Stanulewicz, [12] Ella Guest, [13] Daniel Scanlan, [14] Natalie Carr, [15] Angel Chater, [16] Sarah Hotham, [17] Rachael Thorneloe, [18] Christopher J Armitage [4] Madelynne A Arden, [18] Jo Hart, [4] Lucie Byrne-Davis, [4] Chris Keyworth [4]

As part of the Health Psychology Exchange [healthpsychologyexchange@outlook.com]

Correspondence to [d.ghio@salford.ac.uk]

[1] Department of Psychology, Faculty of Health and Society, University of Salford, Manchester, United Kingdom

[2] Department of Primary Care and Population Health, University College London, London, United Kingdom

[3] Behavioural Science Policy Research Unit, Population Health Sciences, Newcastle University, United Kingdom

[4] Manchester Centre for Health Psychology, University of Manchester, Manchester and Manchester University NHS Foundation Trust, Manchester Academic Health Science Centre, Manchester, United Kingdom

[5] Department of Psychology, Sports, and Geography, School of Life and Medical Sciences, University of Hertfordshire, Hertfordshire, United Kingdom.

[6] Department of Health and Social Sciences, University of West England, Bristol, United Kingdom

[7] School of Psychology, University of Southampton, Southampton, United Kingdom

[8] Department of Psychology, University of Bath, Bath, United Kingdom

[9] Department of Applied Health Research, University College London, London, United Kingdom

[10] Human Development and Health, Faculty of Medicine, University of Southampton, United Kingdom

[11] Behavioural Insight, United Kingdom

[12] Faculty of Health and Life Sciences, School of Applied Social Sciences, De Montfort University, Leicester, United Kingdom

[13] Centre for Appearance Research, Department of Health and Social Sciences, Faculty of Health and Applied Sciences, University of the West of England, Bristol, United Kingdom

[14] Department of Communication, Policy, and Research, Education Support, London, UK

[15] Faculty of Health, Psychology, and Social Care, Manchester Metropolitan University, Manchester, United Kingdom

[16] Centre for Health, Wellbeing and Behaviour Change, University of Bedfordshire, Bedfordshire, United Kingdom

[17] Centre for Health Services Studies, University of Kent, Canterbury, United Kingdom

[18] Centre for Behavioural Science and Applied Psychology, Sheffield Hallam University, Sheffield, United Kingdom

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## Abstract

### *Background*

Population level behaviour change, requiring individual behaviour change such as hand hygiene and physical distancing, are central to reducing transmission of infectious diseases such as COVID-19. However, little is known about effective methods of communicating risk reducing information, and how populations might respond.

### *Objective*

To identify and synthesise evidence relating to: a) what characterises effective public-health messages for managing risk and preventing infectious disease, and b) what influences people's responses to public-health messages.

### *Design*

A systematic review was conducted, and data were analysed using narrative synthesis. The protocol is published on Prospero CRD42020188704.

### *Data sources*

Electronic databases were searched, including Ovid Medline, Ovid PsycINFO and Healthevidence.org , alongside PsyarXiv and OSF Preprints up to May 2020.

### *Study selection*

All study designs were included, including grey literature that: (a) evaluated a public-health messaging intervention targeted at adults, and (b) concerned a communicable disease spread via primary route of transmission of respiratory and/or touch (human to human contact). Non-English language papers were excluded.

### *Results*

Seventy eligible papers were identified (3 systematic reviews, 54 individual papers and 14 pre-prints). To influence behaviour effectively at the population level, public-health messages need to be acceptable, credible and trustworthy, to increase the public's understanding and perceptions of the health threat.

### *Discussion*

There are five key recommendations: (1) engage communities in the development of public-health messaging, (2) use credible and legitimate sources, (3) address uncertainty immediately and with transparency, (4) focus on unifying messages from all sources, and (5) develop messages aimed at increasing understanding, induce social responsibility and empower personal control. Embedding these principles of behavioural science into public health messaging is an important step towards more effective health-risk communication during epidemics/pandemics.


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**What is already known about this subject?**

- Population-level behaviour change is central to reducing transmission via primary route of respiratory and/or touch (human to human contact) of infectious diseases, such as COVID-19.
- These behaviour changes require rapid responses in the context of uncertainties and often rapidly evolving new knowledge, with reliance on public health messaging for effective public health strategies.
- We can learn from public health messaging from past pandemics/epidemics that required similar population-level behaviour change to reduce human to human transmission to determine recommendations for effective public health messaging.

**Strengths and Limitations of this study**

- Although searched systematically research may have been excluded through exclusion of non-English language papers.
- Papers included that were unpublished may be subjected to change once peer-reviewed.
- This study provides a synthesis of different evaluations of studies at different time points of an epidemic/pandemic for different populations to provide recommendations to address what influences people's responses to messages.
- Our results are relevant for guidance and preparations for current pandemic messaging and to inform preparedness for future infectious pandemics/ epidemics.

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**Background**

The outbreak of novel infectious diseases, including COVID-19, require rapid changes to existing (e.g. related to physical distance) and new (e.g. use of face-coverings) behaviours in the context of uncertainties and often rapidly-evolving new knowledge. Public-health messaging is one component of effective risk communication strategies to ensure sustained population level behaviour change. However, little is known about what characterises effective public-health messages for dealing with infectious diseases and what factors influence the public's response.

Drawing from the Behaviour Change Wheel (BCW) [1] system for intervention development, the "COM-B" model of behaviour change proposes public-health messages should be designed as multi-component strategies to support people's capability (the knowledge/skills), opportunity (societal norms/physical resources) and motivation (the desire/habit) to act; factors at the heart of the behaviour change wheel.[1] This theory has been used in current guidance to highlight core considerations for reducing the spread of COVID-19 [2], namely, increasing knowledge of personal protective behaviours (capability), signposting and making available resources (opportunity), and explaining why behaviour change is important, while mitigating emotional reactions (motivation).

When building public health campaigns or interventions, it is important to consider past evidence to build a behavioural diagnosis using COM-B; accepted modes of delivery; and an evaluation of outcomes. It is crucial to understand public-health messaging in rapidly changing epidemics/pandemics and for this, a review of the evidence base is needed.

***Review aims***

To conduct a rapid systematic review[3] to identify and synthesise evidence in order to provide evidence-based recommendations for designing and delivering public-health messages for health authorities and social care organisations dealing with infectious disease outbreaks. This review aims to identify:

a) What influences people's responses to public-health messages about health-risk communication

b) What characterises effective public-health messages for managing risk and preventing disease during epidemics/pandemics

## Methods

Systematic review methods[3] the searches started in May 2020 and analysis was completed in July 2020. Public-health messages were characterised as: messages delivered by mobile-technology, news broadcasts/newspapers, posters, leaflets and press conferences.

***Information sources***

The following databases were searched: Ovid PsycINFO, Ovid MEDLINE, PsyArXiv, OSF Preprints, and healthevidence.org (search strategies are presented in Appendix 1). The protocol for this systematic review is published on Prospero CRD42020188704. The SPICE



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criteria (Setting, Perspective, Phenomena of Interest, Comparison, Evaluation, Time Scope), was used to guide data extraction.[4]

### *Patient and Public Involvement*

The Patient Public Involvement and Engagement group within the Health Psychology Exchange group were consulted when developing the protocol.

### *Eligibility criteria*

All study designs were considered for inclusion (e.g. systematic reviews, empirical studies) and grey literature (e.g. guidelines, frameworks, and policy documents) with no date restrictions until 20 May 2020.

Papers were included if they:

- evaluated a public-health messaging intervention targeted at adults,
- concerned a communicable disease spread via primary route of transmission of respiratory and/or touch (human to human contact)
- were written in English.

Papers on HIV were excluded as they involved different preventative behaviours and therefore deemed to be out of scope of the review. Papers that focused exclusively on public-health messaging for vaccination uptake (intentions and uptake) during epidemic/pandemics were noted and the findings synthesised in a separate review.[5]

### *Study selection*

Titles/abstracts (80% double screened) and full texts were screened by 15 authors (Figure 1; for further breakdown of the included studies see Appendix 2 in supplementary materials). Conflicts over inclusion were resolved through discussions with 4 authors (Appendix 3 provides detail on each author's roles in screening, extraction and synthesis).

### *Data extraction*

Characteristics of the papers (e.g. type of message, quality of study), the type of health risk and results were extracted (Appendix 4). Two authors screened and completed a data quality check using Mixed Methods Appraisal Tool[6] for the 54 individual papers and AMSTAR[7] for the systematic reviews (Appendix 5).

### *Synthesis of results*

A narrative synthesis was conducted[8] to identify key themes with respect to: (a) what influenced people's responses to public-health messages in general and for subpopulations in particular during salient time points (further details presented in Appendix 6) and (b) interpreted recommendations for effective public-health messaging for managing risk and preventing disease during epidemics/pandemics, which are presented below. The synthesis involved combining the results of reviews and individual studies reporting (a) qualitative studies, (b) quantitative studies, and (c) both qualitative and quantitative studies, in order to describe the recommendations for effective delivery of public health messages. To establish trustworthiness in data analysis, discussions among several members of the study team were held at fortnightly intervals to develop the coding framework, and to discuss, refine, and group the emerging codes into overall



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explanatory themes. All study authors were involved in establishing the conceptual framework.

[insert figure 1 here]

## Results

A total of 70 papers rated as high to moderate-quality (about 50% of them scoring as high-quality and 32% of moderate-quality) were included: 3 systematic reviews, 56 individual peer-reviewed papers and 13 preprints. The papers focused mainly on Influenza A virus subtype H1N1 (n=20), Covid-19 (n=15) and Ebola (n=11), which have emerged at different points in time in the last 50 years timelines from initial outbreaks are highlighted in Figure 2. The included studies were conducted at various time points (e.g., beginning, during or post) during these pandemics and eleven did not report the timing (Appendix 6). Key variables included: (a) behaviours (e.g. hand-washing, quarantine, using tissues, physical-distancing), (b) cognitive factors (e.g. increase in awareness, perceived risk), and (c) emotions (worry, anxiety). (See Appendix 5 for full characteristics of papers).

[insert figure 2 here]

### *Recommendations*

Four recommendations were derived from the evidence (See Appendix 6 for a comprehensive report of the evidence on influences on effective public-health messaging, messages for sub-populations and communication at salient points in an epidemic/pandemic). Results were synthesised in Figure 3, including the recommendations and influences on behaviours. The recommendations are (1) engaging with different communities, (2) addressing uncertainty immediately, with transparency, (3) unified messaging, and (4) message framing. (See Appendix 7 for summary of mapping of evidence onto recommendations).

[insert figure 3 here]



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## 1. *Engaging with key stakeholders and communities*

**1a. Involve community leaders and others perceived as credible sources within the community**

One high-quality study indicated that messages designed without input from the target population may lead to low levels of public adherence to behaviour change messages. [9] A moderate-quality review found that this is especially important when dealing with inconsistencies and changing information. [10]

A high-quality systematic review found that the public pay more attention to messages if the community is engaged in its development [11] as this will ensure that the information is relatable,[9] and addresses the concerns, values, interests and priorities of the community.[10] Consequently, this may lead to heightened perceptions of personal risk. Two studies of low to moderate-quality, suggested that one way to do so is to include community leaders, to find people that are trusted[12] and allow faith-based organisations to help.[13] One moderate-quality study found that culturally and linguistically appropriate messages (e.g. in videos) can also help retain long-term knowledge of preventative behaviours.[14]

A preprint study stated that over time preferred expert sources are displaced by unofficial sources for information regarding epidemic/pandemics;[15] therefore, credible sources within the community can be perceived as valuable. A high-quality study found that students tend to perceive information from their university (from their own communities) as more credible than the media. [16] One low-quality study found that community engagement is also important for quickly disseminating messages which are translated into different languages.[17, 18]

**1b. Tailoring help make the key messages applicable to an individual's situation**

A high-quality systematic review found that those who are less likely to accept they are at personal risk of the threat (e.g. the young, least educated and hard to engage communities) are also less likely to adhere to the recommended behaviours.[19] This has important implications for reducing inequalities through tailored public health messaging. Social networks and having close ties to the community are drivers of better knowledge and behaviour change. Partnership with community leaders and/or community organizations should be used to reach out to the most vulnerable (this may include those who have a disability e.g. hearing/vision) and those who are least literate as well as non-Native speakers.[19]

Individuals desire information that fits with their experiences.[20-23] Studies of moderate-quality have shown that adaptable and personalised information, that is context-driven, is more effective in changing determinants of behaviour, especially in vulnerable



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groups.[24, 25] One moderate-quality study and two high-quality studies showed that messages should be tailored to take into account: (a) different levels of perceived susceptibility (e.g., younger adults see themselves as less vulnerable than older adults[26]), (b) likelihood of misunderstanding instructions (e.g., older people thought a campaign was referring to handkerchiefs rather than disposable tissues)[27] and (c) skills needed to enact the behaviour (e.g., migrant workers didn't know how to wear a face-covering).[28] One low-quality study found that there may even be differences in message preferences (e.g., older adults and mothers preferred messages that emphasised the protection of others).[27]

**1c. Consider any difficulties accessing information and levels of literacy**

Messages are typically delivered at a high literacy level.[19] When individuals do not understand the message, they may engage in behaviours that reflect their understanding.[29] Some target groups, such as those with low literacy levels, could particularly benefit from a simplified, clear, and consistent message. [30] Pilot-testing messages can help identify phrasing that can be confusing or unfamiliar.[31]

It is also important to ensure public-health messages reach non-native speakers by using translated materials as shown in a high-quality study.[32] Older people in non-native groups in particular may not understand English. Thus, messages should be communicated in the original languages and through generations. Translated information should be disseminated through culturally appropriate channels such as community visits, town hall meetings, and health and education and communication channels to complement mass media messages. This reinforces the need for community leaders (recommendation 1a) in aiding with translations and identifying the appropriate channels.

Some people have limited experiences of engaging in recommended behaviours (e.g. using face-coverings or a thermometer),[33] highlighting a need for training/skill development as part of a public-health campaign to improve health literacy, especially when it refers to 'new' behaviours.

**1d. Use different media for delivery and match delivery to the population's needs and perceptions**

Message delivery should be appropriate for the targeted population. For example, two studies of moderate to high-quality suggested that social media can reach younger people.[34, 35] A study of moderate-quality showed that social media can be effective in communicating messages fostering trust and providing opportunities for dialogue.[36] However, as highlighted in two studies of high-quality, there are disadvantages as (a) social media is also associated with misconceptions, such as what causes the disease[37] and (b) those without internet access (or a device) would be excluded.[38] A high-quality



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study found that those of lower socioeconomic status SES were less likely to use a website, and more likely to find public-health messages from TV and Radio to be confusing and contradictory than those of higher SES.[30] Whilst another high-quality study found that prompts alone do not increase handwashing.[39]

## *2. Addressing uncertainty immediately and with transparency*

### 2a. Address uncertainty and changing information that may exist during an ongoing public-health crisis

Public-health messages with emerging epidemics/pandemics are likely to involve much uncertainty about the virus and the appropriate preventative behaviours. A high-quality systematic review found that honest reporting about the threat, through a presentation of known and unknown factors, increases people's knowledge and makes attitudes and beliefs more positive; it also increases trust in the way the government is handling the emergency.[19]

As epidemics/pandemics develop the information and recommended behaviours may change. Changing information decreases trust in the government.[40] However, three studies of moderate to high-quality reported that prompt (i.e. giving information as it happens), stage appropriate (i.e., not seen as too lenient or too extreme for the risk level), and transparent messaging (i.e. that includes recommended preventative behaviours) reduces anxiety around the reported health risk.[41]

### 2b. Consistency and co-ordination between different sources of information

Different sources of information may give information and behaviour change instructions that are inconsistent with each other, especially as knowledge rapidly changes over time. Differences in the information and behaviour change instructions between countries should be explained. Information should be transparent, not be hidden and predictions should not be too optimistic – this will increase overall trust towards authorities.[42, 43]

### 2c. Be transparent: admit errors and unknowns whenever appropriate

A high-quality study showed that if uncertainties are not acknowledged or transparency is not provided, the consequences of the errors are addressed and this can reduce trust in agencies, thus reducing information acceptance and compliance with recommendations.[44]

Two studies of moderate to high-quality found that information needs to be released as early as possible at the start of the outbreak, whenever there is conflicting information



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and even if there are unknowns.[45, 46] Honest, open[25] and explicit information, transparency and assurances of personal data safety are important[47] especially when promoting preventative behaviours.[48] A low-quality study found that if official sources report the outbreak before unofficial/informal sources, then they become the leading indicator that people use for their information.[17] Delay can impact the public's trust in official sources that leads to beliefs that the threat is exaggerated by government or news media[22] and increases the chance of apathy and communication fatigue.[16]

**2d. Be transparent: identify sources of information**

Two high-quality studies reported that attempting to increase knowledge in the context of low levels of public trust in the source can make a message ineffective; and mistrust can increase if the perception is that the information is exaggerated[22, 49] or if the outbreak is perceived as uncontrolled.[50] Three preprint studies reported that trust is key to the acceptance of messaging that can lead to behaviour change (e.g. using masks, physical distance), thus there is greater adherence when the messages come from trusted sources. [15, 51, 52] Trustworthy sources can include public-health experts, organisations (e.g. Centers for Disease Control and Prevention) and state and local governments.[53, 54] A study found that believability of messages was reduced by scepticism towards the media and governments; in these instances, the threat was perceived as exaggerated and able to cause unnecessary panic which could influence low adoption of recommended behaviours.[22] Therefore, it is important to ensure transparency highlighting that information sources are credible and legitimate.[55]

## *3. Unified messages*

**3a. Deliver consistent, clear, core messages about risk and preventative behaviour across sources within the same time points**

Where possible, messages should be unified across sources; as shown by two high-quality studies, this is especially important for those groups who have lower literacy (e.g., lower SES) as they are more likely to perceive messages as confusing and inconsistent.[30, 38] A moderate-quality review found that unifying messages result in a greater understanding of the health message, greater perceived risk and clear guidelines for behaviour;[10, 30] these all contribute to increased trust. A high-quality study found that a repeated measures survey during the H1N1 pandemic, showed trust levels in the government decreased over time and this decrease was due to conflicting messages at concurrent time-points.[40] In two moderate-quality studies it was found that if messages from different sources provide clear, consistent instructions these are more likely to be recalled and adhered to.[56] When messages provided the public with clear consistent



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information, whilst admitting that information is evolving, reported the risks, and focused on risk-reducing actions, people were more likely to perform protective behaviours (e.g., clean objects, wash hands and use tissues when sneezing).[57]

**3b. Identify inconsistencies in messages from uncontrolled sources, especially when addressing key preventative behaviours**

Whilst official sources of information (such as governmental sources, or public-health bodies) are perceived as legitimate, individuals do not always find them useful as highlighted in one high-quality study and one moderate-quality study,[35, 36] and consequently seek other sources (e.g. unofficial sources such as family or social media). In a moderate-quality study, it was found that individuals seek information from multiple sources to meet different information needs.[45] Different unofficial sources of information may give information and behaviour change instructions that are inconsistent with each other and official sources. Additionally, as found in a high-quality study, unofficial sources may detract from understanding of preventative behaviours.[58] It is important to acknowledge and address the inconsistencies in unofficial sources.

When addressing inconsistency, a high-quality study found that it is also important to recognise that some messages may appear inconsistent due to varying levels of personal risk of different groups (conditional messages). To improve clarity these conditional messages should be identified and where possible explained or dispelled.[32]

**3c. Increase the public's awareness of the risks of the virus to their own health and the health of others**

Public-health messaging that included information about the threat can be effective in identifying symptoms and changing behaviours. From six high-quality studies, it was found that the key aim of public-health messages early in an epidemic/pandemic is typically to increase knowledge and awareness of the health-risk (see Appendix 6 for papers and priorities mapped onto timepoints).[16, 20, 22, 55, 59, 60] However, knowledge about the virus alone is not sufficient to change behaviour;[59] acknowledgement of unknown factors, how to identify symptoms, and how to prevent contracting and spreading the virus is also necessary as highlighted by five high-quality studies and a high-quality systematic review, two moderate-quality and one low-quality study. [19, 22, 32, 33, 41, 55, 61, 62] A low-quality study found that when perceived risk is low, ignoring recommended behaviours is rationalised.[63]



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## 4. Message framing

**4a. Increase factual knowledge of all aspects of a virus (e.g. symptoms) and benefits of preventative behaviour using an appropriate message frame**

Six high-quality studies found that framing and choice of language are influential in how individuals understand the threat and in turn behave. [22, 64-68] Positively framed messages (emphasising the benefits of prevention behaviours) may be effective.[64] For example, gain-framed signs (e.g., "stay healthy this season. Sanitize your hands") are more effective in influencing the use of hand sanitizer than signs that emphasised people's susceptibility to contamination.[65]

Non-narrative messages (i.e., factual) are more effective than narrative messages (e.g. storytelling such as a movie *Contagion*) in changing knowledge and perceived response efficacy related to prevention of influenza.[68] Factual and scientific knowledge in messages in the media can positively influence risk perceptions.[66] Whilst formal information (presented from credible sources) increased understanding.[69]

**4b. Consider framing messages around social responsibility and norms**

Cohesive social networks and having close ties to the community are drivers of better knowledge and compliance with preventive measures. There is some moderate-quality evidence to suggest that framing messages about others' risk, in addition to your own risk, is effective in increasing information seeking.[45] In a high-quality systematic review, it was found that being worried (about self or family members at risk) was an important predictor of compliance with recommended preventative behaviours, such as using tissues, hand gel and washing hands.[19] Especially when this tapped into being socially responsible through framing messages with positive social responsibility as highlighted in a high-quality study.[64]

The grey literature highlighted that developing prosocial messages and promoting positive emotional appeals, increased willingness to self-isolate especially when producing a strong, positive emotional response (such as fear).[70] Prosocial framing is effective in changing behaviour; this may be due to inducing compassion, activating social norms, altruism or moral duties.[54, 71]

**4c. Language choice improves trust, transparency, and risk perception**

A high-quality systematic review and a moderate-quality study, found that messages should sufficiently increase worry and perceived severity of risk to self and others[19, 45] so as highlighted in a high-quality study, recommendations are seen as proportionate[22] and



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behaviour change is more likely to occur.[19] However, in two high-quality studies, it was found that inducing too much fear has mixed results on behaviour: it can be counterproductive if this leads to panic.[49, 72]

There can also be confusion about the use of technical terms. A high-quality study found that it is difficult to differentiate between pandemic flu and seasonal flu especially when symptoms are framed as "flu-like symptoms." [22]

**4d. Frame the message to emphasise positive beliefs about one's own health and that preventative behaviour is within their control**

A study of moderate-quality found that when messages provided the public with clear consistent information (whilst admitting that information and evidence is evolving), focused on the practical actions that people can take to reduce their risk and emphasized the efficacy of those actions, people were more likely to perform preventative behaviours.[57] This may also further empower them and help them become socially responsible as shown in a low-quality study.[73] People are more likely to follow guidelines when there are fewer perceived barriers to perform recommended behaviours, when benefits are emphasised, and when contextual factors (e.g., anxiety about missed work) are addressed.[22]

People want messages about specific actions that they could take to protect themselves and their families during the epidemic/pandemic as shown in a high-quality study.[32] Furthermore, another high-quality study showed that messages that emphasise the benefits of prevention behaviours (positively framed) may be effective in increasing uptake and compliance.[64]. Empowering the individual to stop the spread of the disease is particularly important in the early stages (see Appendix 6 for breakdown of timepoints). In a moderate-quality study, it was suggested that including non-health benefits may improve adherence to recommendations. [63] Whilst in a preprint study, messages that are framed in relation to identity (e.g. "don't be a spreader" vs "don't spread") were thought to be more likely to predict physical distancing but no significant differences were found between the groups.[74]

## Discussion

The aim of this systematic review was to synthesis evidence-based recommendations for designing and delivering public-health messaging. We found that inconsistent messages across different sources can be detrimental to any public-health messaging campaigns,[75]- which is particularly true in times of public-health-crisis and uncertainty.[76] Miscommunication, or contradictions can result in lower compliance of preventative behaviours. Literacy is fundamental to the understanding and acceptance of any public-health messaging campaign. Therefore, public-health messaging should focus on clarity, simplicity, transparency and unified messaging even if tailoring to different groups



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regardless of level of literacy. Even when the message is understood, different populations have different barriers to address in order to change behaviour. A COM-B behavioural diagnosis[1] can help to inform public health strategies (e.g., lower SES groups find costs of face-coverings a barrier)[30] as suggested in recent British Psychological Society guidance (see Appendix 8).[2] Our review suggests that unified messaging can increase trust[45] build community resilience[32], and increase perceived risk and knowledge of threat[49]. Furthermore, consistent messaging through different time points during an evolving pandemic are important to consider for context-specific recommendations; these were rarely considered or reported in the studies included in the review.

We found evidence of mode of delivery as an important consideration of public-health messaging. Specifically, there should be careful consideration of how to communicate public-health messages, which should target specific populations. For example, making use of social media platforms to target younger populations,[16, 34] or the radio as a delivery method for older populations.[20] Translations for different cultures, and the inclusion of community leaders as part of public-health message campaigns, should also be prioritised. This would ensure that all populations are reached (and not stigmatized), and the knowledge, concerns, cultural values, interests, and priorities of the targeted populations are all considered.[77] However, as noted in one of the studies this can be very difficult when an in-house translation service does not exist, and the rapidly evolving scientific evidence will challenge the turnaround time for developing, translating, and disseminating information.[78]

***Limitations***

We acknowledge the present systematic review has limitations. Although we searched multiple databases systematically it is possible that relevant research was excluded from this review since we did not have the resources to translate non-English language papers in such a short space of time. The inclusion of unpublished literature in the review means some findings may change once these papers have been published in peer-reviewed journals. However, our inclusive approach means a range of types of messages and a variety of factors related to what influences public perceptions of messages have been identified and used to inform recommendations for messaging during current and future epidemic/pandemics.

Much of our evidence is consistent with relevant health behaviour models, (e.g., Protection Motivation Theory,[79] Health Action Process Approach,[80] and Health Belief Model[81]) with some studies reporting the use of the models in their design. These models state that people perform protective behaviours when they perceive (a) the health threat as sufficiently threatening (i.e., they are vulnerable to a severe risk), (b) the recommended behaviours to be effective, and (c) they have self-efficacy to perform the behaviour. We found evidence that such cognitive appraisals were important considerations for developing public-health messages, but further research is needed to examine effects on behaviour change.



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***Conclusions***

Increasing knowledge and awareness of health risks alone is unlikely to be sufficient to increase understanding of risk and subsequent risk reduction behaviours, and adherence to recommendations. Rather, taking a multifaceted approach to public-health messaging which considers all relevant drivers of behaviour (social, psychological and environmental factors), is recommended. Our four key recommendations should be considered when designing and delivering public-health messages: engage communities in the development of public-health messaging, using credible and legitimate sources, address uncertainty immediately and with transparency, focus on unified messages from all sources, and develop messages aimed at increasing understanding, inducing social responsibility and empowering personal control. These are being translated into practical guidelines for agencies. Embedding these principles of behavioural science into public-health messaging is an important step towards more effective health-risk communication.

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**Role of the funding source**

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Figure 1: Prisma Diagram

Figure 2: Timelines of pandemics/epidemics included in review



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Figure 3: Diagram of synthesis of recommendations and influences on behaviour

## References


1. Michie S, van Stralen MM, West R. The behaviour change wheel: A new method for characterising and designing behaviour change interventions. Implement Sci. 2011;6(1):42.
2. Chater AA, M.A.; Armitage C.J., Byre-Davis, L.; Chadwick, P.; Drury, J.; Hart, J.; Lewis, L.; McBride, E.; Perriard-Abdoh, S.; Thompson, S.; Whittaker, E.; O'Connor, D.;. Behavioural science and disease prevention: psychological guidance: British Psychological Society; 2020 [
3. Dobbins M. Rapid review guidebook. National Collaborating Centre for Methods and Tools, Hamilton, ON. 2017.
4. Booth A. Clear and present questions: formulating questions for evidence based practice. Library Hi Tech. 2006;24(3):355-68.
5. Lawes-Wickwar SG, Daniela; Tang, Mei Yee; Keyworth, Chris; Stanescu, Sabina; Westbrook, Juliette; Jenkinson Elizabeth; Kassianos, Angelos, P.; Scanlan, Daniel; Garnett Natalie; Laidlaw, Lynn; Byrne-Davis, Lucie; Chater, Angel; Epton Tracy; and members of the Health Psychology Exchange (HPX) Public Health MEssaging Review Group A rapid review of responses to public health messages encouraging vaccination to prevent the spread of a pandemic or epidemic virus. in submission. 2020.
6. Hong QN, Gonzalez-Reyes A, Pluye P. Improving the usefulness of a tool for appraising the quality of qualitative, quantitative and mixed methods studies, the Mixed Methods Appraisal Tool (MMAT). Journal of evaluation in clinical practice. 2018;24(3):459-67.
7. Shea BJ, Grimshaw JM, Wells GA, Boers M, Andersson N, Hamel C, et al. Development of AMSTAR: a measurement tool to assess the methodological quality of systematic reviews. BMC Med Res Methodol. 2007;7(1):10.
8. Popay J, Roberts H, Sowden A, Petticrew M, Arai L, Rodgers M, et al. Guidance on the conduct of narrative synthesis in systematic reviews. A product from the ESRC methods programme Version. 2006;1:b92.
9. Crosier A, McVey D, French J. 'By failing to prepare you are preparing to fail': Lessons from the 2009 H1N1 'swine flu' pandemic. : European Journal of Public Health. Vol.25(1), 2015, pp. 135-139.; 2015.
10. Barrelet C, Bourrier M, Burton-Jeangros C, Schindler M. Unresolved issues in risk communication research: the case of the H1N1 pandemic (2009-2011). Influenza other respi. 2013;7 Suppl 2:114-9.
11. Schiavo R, May Leung M, Brown M. Communicating risk and promoting disease mitigation measures in epidemics and emerging disease settings. Pathog Glob Health. 2014;108(2):76-94.
12. Lyu SY, Chen RY, Wang SF, Weng YL, Peng EY, Lee MB. Perception of spokespersons' performance and characteristics in crisis communication: experience of the 2003 severe acute respiratory syndrome outbreak in Taiwan. J Formos Med Assoc. 2013;112(10):600-7.
13. Santibanez S, Siegel V, O'Sullivan M, Lacson R, Jorstad C. Health communications and community mobilization during an Ebola response: partnerships with community and faith-based organizations. Public Health Rep. 2015;130(2):128-33.
14. Roess AA, Di Peppi R, Kinzoni EA, Molouania M, Kennedy E, Ibata SR, et al. Knowledge Gained and Retained from a Video-Centered, Community-Based Intervention for Ebola Prevention, Congo. J Health Commun. 2017;22(11):913-22.
15. Merkley E, Loewen PJ, Owen T, Ruths D. Anti-intellectualism and Information Preferences during the COVID-19 Pandemic. 2020.




For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

16. Mitchell T, Massoudi M, Swerdlow DL, Dee DL, Gould LH, Kutty PK, et al. Swine flu in college: early campus response to outbreak control measures. Am J Health Behav. 2014;38(3):448-64.
17. Menon KU. SARS revisited: managing "outbreaks" with "communications". Ann Acad Med Singapore. 2006;35(5):361-7.
18. Vaughan E, Tinker T. Effective health risk communication about pandemic influenza for vulnerable populations. Am J Public Health. 2009;99 Suppl 2:S324-32.
19. Lin L, Savoia E, Agboola F, Viswanath K. What have we learned about communication inequalities during the H1N1 pandemic: a systematic review of the literature. BMC Public Health. 2014;14:484.
20. Freiman AJ, Montgomery JP, Green JJ, Thomas DL, Kleiner AM, Boulton ML. Did H1N1 influenza prevention messages reach the vulnerable population along the Mississippi Gulf Coast? J Public Health Manag Pract. 2011;17(1):52-8.
21. Qian X, Ren R, Wang Y, Guo Y, Fang J, Wu ZD, et al. Fighting against the common enemy of COVID-19: a practice of building a community with a shared future for mankind. Infect. 2020;9(1):34.
22. Teasdale E, Yardley L. Understanding responses to government health recommendations: public perceptions of government advice for managing the H1N1 (swine flu) influenza pandemic. Patient Educ Couns. 2011;85(3):413-8.
23. Davis MD, Stephenson N, Lohm D, Waller E, Flowers P. Beyond resistance: social factors in the general public response to pandemic influenza. BMC Public Health. 2015;15:436.
24. Daellenbach K, Parkinson J, Krisjanous J. Just how prepared are you? An application of marketing segmentation and theory of planned behavior for disaster preparation. : Journal of Nonprofit & Public Sector Marketing. Vol.30(4), 2018, pp. 413-443.; 2018.
25. Qiu W, Chu C, Hou X, Rutherford S, Zhu B, Tong Z, et al. A Comparison of China's Risk Communication in Response to SARS and H7N9 Using Principles Drawn From International Practice. Disaster med. 2018;12(5):587-98.
26. Idoiaga N, De Montes LG, Valencia J-F. Communication and representation of risk in health crises: The influence of framing and group identity. : Revista de Psicologia Social. Vol.31(1), 2016, pp. 59-74.; 2016.
27. Jones SC, Waters L, Holland O, Bevins J, Iverson D. Developing pandemic communication strategies: Preparation without panic. : Journal of Business Research. Vol.63(2), 2010, pp. 126-132.; 2010.
28. Hickey JE, Gagnon AJ, Jitthai N. Knowledge about pandemic influenza preparedness among vulnerable migrants in Thailand. Health Promot Internation. 2016;31(1):124-32.
29. Bonwitt J, Dawson M, Kandeh M, Ansumana R, Sahr F, Brown H, et al. Unintended consequences of the 'bushmeat ban' in West Africa during the 2013-2016 Ebola virus disease epidemic. Soc Sci Med. 2018;200:166-73.
30. Aburto NJ, Pevzner E, Lopez-Ridaura R, Rojas R, Lopez-Gatell H, Lazcano E, et al. Knowledge and adoption of community mitigation efforts in Mexico during the 2009 H1N1 pandemic. Am J Prev Med. 2010;39(5):395-402.
31. Lapka C, Jupka K, Wray RJ, Jacobsen H. Applying cognitive response testing in message development and pre-testing.: Health Education Research. Vol.23(3), 2008, pp. 467-476.; 2008.
32. Gray L, MacDonald C, Mackie B, Paton D, Johnston D, Baker MG. Community responses to communication campaigns for influenza A (H1N1): a focus group study. BMC Public Health. 2012;12:205.
33. Hickey J, Gagnon AJ, Jitthai N. Pandemic preparedness: perceptions of vulnerable migrants in Thailand towards WHO-recommended non-pharmaceutical interventions: a cross-sectional study. BMC Public Health. 2014;14:665.



For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

34. Seltzer EK, Jean NS, Kramer-Golinkoff E, Asch DA, Merchant RM. The content of social media's shared images about Ebola: A retrospective study. : Public Health. Vol.129(9), 2015, pp. 1273-1277.; 2015.
35. Toppenberg-Pejcic D, Noyes J, Allen T, Alexander N, Vanderford M, Gamhewage G. Emergency risk communication: Lessons learned from a rapid review of recent gray literature on Ebola, Zika, and yellow fever. : Health Communication. Vol.34(4), 2019, pp. 437-455.; 2019.
36. Tully M, Dalrymple KE, Young R. Contextualizing nonprofits' use of links on Twitter during the West African Ebola virus epidemic. : Communication Studies. Vol.70(3), 2019, pp. 313-331.; 2019.
37. Winters M, Jalloh MF, Sengeh P, Jalloh MB, Conteh L, Bunnell R, et al. Risk Communication and Ebola-Specific Knowledge and Behavior during 2014-2015 Outbreak, Sierra Leone. Emerg Infect Dis. 2018;24(2):336-44.
38. Avery EJ, Kim S. Anticipating or precipitating crisis? Health agencies may not be heeding best practice advice in avian flu press releases. : Journal of Public Relations Research. Vol.21(2), 2009, pp. 187-197.; 2009.
39. Davis OL, Fante RM, Jacobi LL. The effectiveness of sign prompts to increase hand washing behaviors in restrooms. : North American Journal of Psychology. Vol.15(3), 2013, pp. 565-576.; 2013.
40. van der Weerd W, Timmermans DR, Beaujean DJ, Oudhoff J, van Steenbergen JE. Monitoring the level of government trust, risk perception and intention of the general public to adopt protective measures during the influenza A (H1N1) pandemic in The Netherlands. BMC Public Health. 2011;11:575.
41. Li R, Xie R, Yang C, Frost M. Perceptions on the risk communication strategy during the 2013 avian influenza A/H7N9 outbreak in humans in China: a focus group study. Western pac. 2016;7(3):21-8.
42. Lohiniva AL, Sane J, Sibenberg K, Puumalainen T, Salminen M. Understanding coronavirus disease (COVID-19) risk perceptions among the public to enhance risk communication efforts: a practical approach for outbreaks, Finland, February 2020. Euro Surveill. 2020;25(13):04.
43. Rogers WA, Street JM, Braunack-Mayer AJ, Hiller JE, FluViews T. Pandemic influenza communication: views from a deliberative forum. Health Expect. 2009;12(3):331-42.
44. Holmes BJ, Henrich N, Hancock S, Lestou V. Communicating with the public during health crises: Experts' experiences and opinions. : Journal of Risk Research. Vol.12(6), 2009, pp. 793-807.; 2009.
45. Cole J, Watkins C. International employees' concerns during serious disease outbreaks and the potential impact on business continuity: Lessons identified from the 2014-15 West African Ebola outbreak. J Bus Contin Emer Plan. 2015;9(2):149-62.
46. Johnson BB, Slovic P. Fearing or fearsome Ebola communication? Keeping the public in the dark about possible post-21-day symptoms and infectiousness could backfire. : Health, Risk & Society. Vol.17(5-6), 2015, pp. 458-471.; 2015.
47. Bradshaw EL, Ryan R, Noetel M, Saeri AK, Slattery P, Grundy E, et al. Information safety assurances affect intentions to use COVID-19 contact tracing applications, regardless of autonomy-supportive or controlling message framing. 2020.
48. Dai B, Fu D, Meng G, Liu B, Li Q, Liu X. The effects of governmental and individual predictors on COVID-19 protective behaviors in China: a path analysis model. Public Administration Review. 2020.
49. Jhummon-Mahadnac ND, Knott J, Marshall C. A cross-sectional study of pandemic influenza health literacy and the effect of a public health campaign. BMC Res Notes. 2012;5:377.
50. Sumo J, George G, Weah V, Skrip L, Rude JM, Clement P, et al. Risk communication during disease outbreak response in post-Ebola Liberia: experiences in Sinoe and Grand Kru counties. Pan Afr Med J. 2019;33(Suppl 2):4.



For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

51. Abu-Akel A, Spitz A, West R. Who is listening? Spokesperson Effect on Communicating Social and Physical Distancing Measures During the COVID-19 Pandemic. 2020.
52. Goldberg M, Gustafson A, Maibach E, Ballew MT, Bergquist P, Kotcher J, et al. Mask-wearing increases after a government recommendation: A natural experiment in the US during the COVID-19 pandemic. 2020.
53. Berry TR, Wharf-Higgins J, Naylor PJ. SARS wars: an examination of the quantity and construction of health information in the news media. Health Commun. 2007;21(1):35-44.
54. Wirz CD, Schwakopf JM, Brossard D, Brown LD, Brauer M. Self-reported compliance and attitudes about social distancing during the COVID-19 outbreak Preprint.
55. Yardley L, Miller S, Schlotz W, Little P. Evaluation of a Web-based intervention to promote hand hygiene: exploratory randomized controlled trial. J Med Internet Res. 2011;13(4):e107.
56. Miczo N, Danhour E, Lester KE, Bryant J. Memorable messages and the H1N1 flu virus. : Western Journal of Communication. Vol.77(5), 2013, pp. 625-644.; 2013.
57. Prati G, Pietrantoni L, Zani B. Compliance with recommendations for pandemic influenza H1N1 2009: the role of trust and personal beliefs. Health Educ Res. 2011;26(5):761-9.
58. McVernon J, Mason K, Petrony S, Nathan P, LaMontagne AD, Bentley R, et al. Recommendations for and compliance with social restrictions during implementation of school closures in the early phase of the influenza A (H1N1) 2009 outbreak in Melbourne, Australia. BMC Infect Dis. 2011;11:257.
59. Kavanagh AM, Bentley RJ, Mason KE, McVernon J, Petrony S, Fielding J, et al. Sources, perceived usefulness and understanding of information disseminated to families who entered home quarantine during the H1N1 pandemic in Victoria, Australia: a cross-sectional study. BMC Infect Dis. 2011;11:2.
60. Chang C. News coverage of health-related issues and its impacts on perceptions: Taiwan as an example. : Health Communication. Vol.27(2), 2012, pp. 111-123.; 2012.
61. Hoda J. Identification of information types and sources by the public for promoting awareness of Middle East respiratory syndrome coronavirus in Saudi Arabia. Health Educ Res. 2016;31(1):12-23.
62. Rim H, Ha JH, Kiousis S. The evidence of compelling arguments in agenda building: Relationships among public information subsidies, media coverage, and risk perceptions during a pandemic outbreak. : Journal of Communication Management. Vol.18(1), 2014, pp. 101-116.; 2014.
63. Rimi NA, Sultana R, Ishtiak-Ahmed K, Rahman MZ, Hasin M, Islam MS, et al. Understanding the failure of a behavior change intervention to reduce risk behaviors for avian influenza transmission among backyard poultry raisers in rural Bangladesh: a focused ethnography. BMC Public Health. 2016;16(1):858.
64. Basnyat I, Lee ST. Framing of Influenza A (H1N1) pandemic in a Singaporean newspaper. Health Promot Internation. 2015;30(4):942-53.
65. Updegraff JA, Emanuel AS, Gallagher KM, Steinman CT. Framing flu prevention-An experimental field test of signs promoting hand hygiene during the 2009-2010 H1N1 pandemic. : Health Psychology. Vol.30(3), 2011, pp. 295-299.; 2011.
66. Sell TK, Boddie C, McGinty EE, Pollack K, Smith KC, Burke TA, et al. Media Messages and Perception of Risk for Ebola Virus Infection, United States. Emerg Infect Dis. 2017;23(1):108-11.
67. Liao Q, Cowling B, Lam WT, Ng MW, Fielding R. Situational awareness and health protective responses to pandemic influenza A (H1N1) in Hong Kong: a cross-sectional study. PLoS ONE. 2010;5(10):e13350.
68. Bekalu MA, Bigman CA, McCloud RF, Lin LK, Viswanath K. The relative persuasiveness of narrative versus non-narrative health messages in public health emergency communication: Evidence from a field experiment. Prev Med. 2018;111:284-90.



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69. Liao Q, Cowling B, Lam WT, Ng MW, Fielding R. Situational awareness and health protective responses to pandemic influenza a (H1N1) in Hong Kong: A cross-sectional study. : PLoS ONE. Vol.5 2010, ArtID e13350.; 2010.
70. Heffner J, Vives ML, FeldmanHall O. Emotional responses to prosocial messages increase willingness to self-isolate during the COVID-19 pandemic. 2020.
71. Jordan J, Yoeli E, Rand D. Don't get it or don't spread it? Comparing self-interested versus prosocially framed COVID-19 prevention messaging. 2020.
72. Zikmund-Fisher BJ, Scherer AM, Knaus M, Das E, Fagerlin A. Discussion of Average versus Extreme Case Severity in Pandemic Risk Communications. Emerg Infect Dis. 2017;23(4):706-8.
73. Menon KU, Goh KT. Transparency and trust: Risk communications and the Singapore experience in managing SARS. : Journal of Communication Management. Vol.9(4), 2005, pp. 375-383.; 2005.
74. Yonemitsu F, Ikeda A, Yoshimura N, Sasaki K, Takashima K, Qian K, et al. Warning "Don't spread" vs."Don't be a spreader" to prevent the COVID-19 pandemic. 2020.
75. Hernandez-Garcia I, Gimenez-Julvez T. Assessment of Health Information About COVID-19 Prevention on the Internet: Infodemiological Study. JMIR Public Health Surveill. 2020;6(2):e18717.
76. Hooker C, Capon A, Leask J. Communicating about risk: strategies for situations where public concern is high but the risk is low. Public health res. 2017;27(1):15.
77. Lin L, Jung M, McCloud RF, Viswanath K. Media use and communication inequalities in a public health emergency: a case study of 2009-2010 pandemic influenza A virus subtype H1N1. Public Health Rep. 2014;129 Suppl 4:49-60.
78. Person B, Sy F, Holton K, Govert B, Liang A, National Center for Inectious Diseases SCOT. Fear and stigma: the epidemic within the SARS outbreak. Emerg Infect Dis. 2004;10(2):358-63.
79. Rogers RW. A protection motivation theory of fear appeals and attitude change. The journal of psychology. 1975;91(1):93-114.
80. Schwartzer R. Self-efficacy in the adoption and maintenance of health behaviours: Theoretical approaches and a new model. Self-efficacy: Thought control of action Washington, DC: Hemisphere. 1992:217-43.
81. Becker MH. The health belief model and personal health behavior. Health education monographs. 1974;2:324-473.

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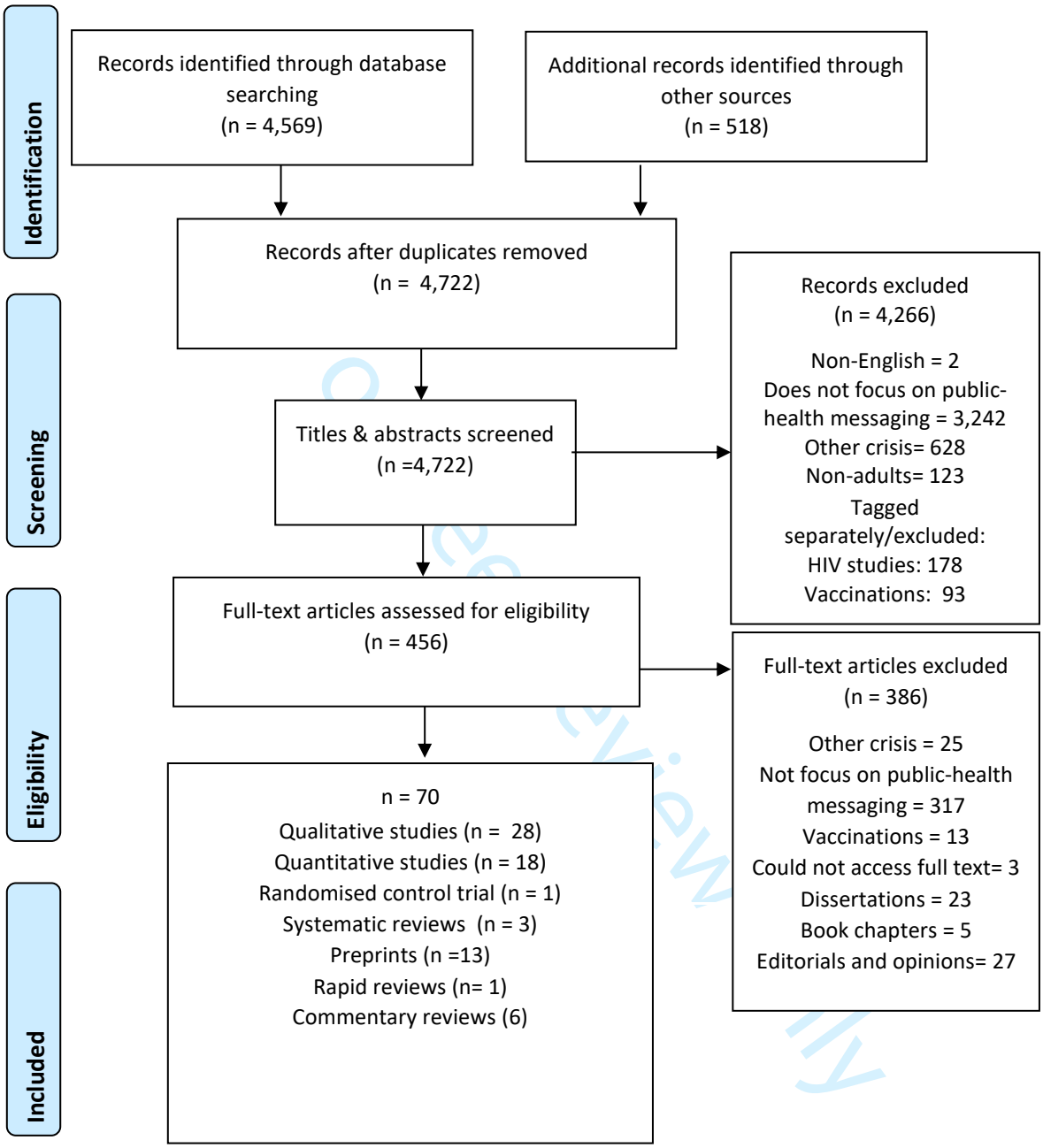



Figure 1: Prisma Diagram

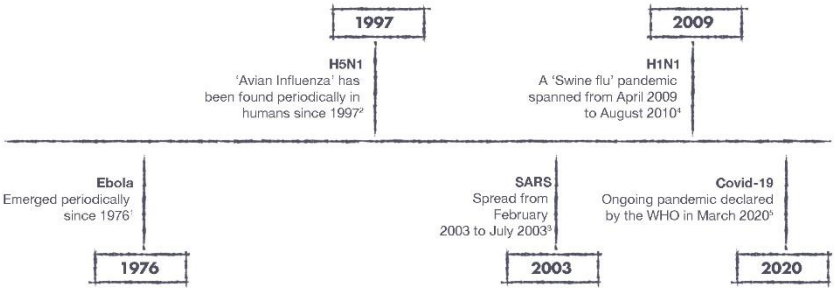



Figure 1: Timelines of pandemics/epidemics included in review

Note:[1]Centre for Disease Control & Prevention (CDC). Years of Ebola Virus Disease Outbreak. 2019. Available from: https://www.cdc.gov/vhf/ebola/history/chronology.html; [2]WHO. H5N1 avian influenza: Timeline of major events. 2012. Available from: https://www.who.int/influenza/human_animal_interface/H5N1_avian_influenza_update.pdf; [3]CDC. CDC SARS Response Timeline. 2013. Available from: https://www.cdc.gov/about/history/sars/timeline.htm; [4]CDC. 2009 H1N1 Pandemic Timeline. Available from: https://www.cdc.gov/flu/pandemic-resources/2009-pandemic-timeline.html; [5]WHO. Rolling updates on coronavirus disease (COVID-19). 2020. Available from: https://www.who.int/emergencies/diseases/novel-coronavirus-2019/events-as-they-happen

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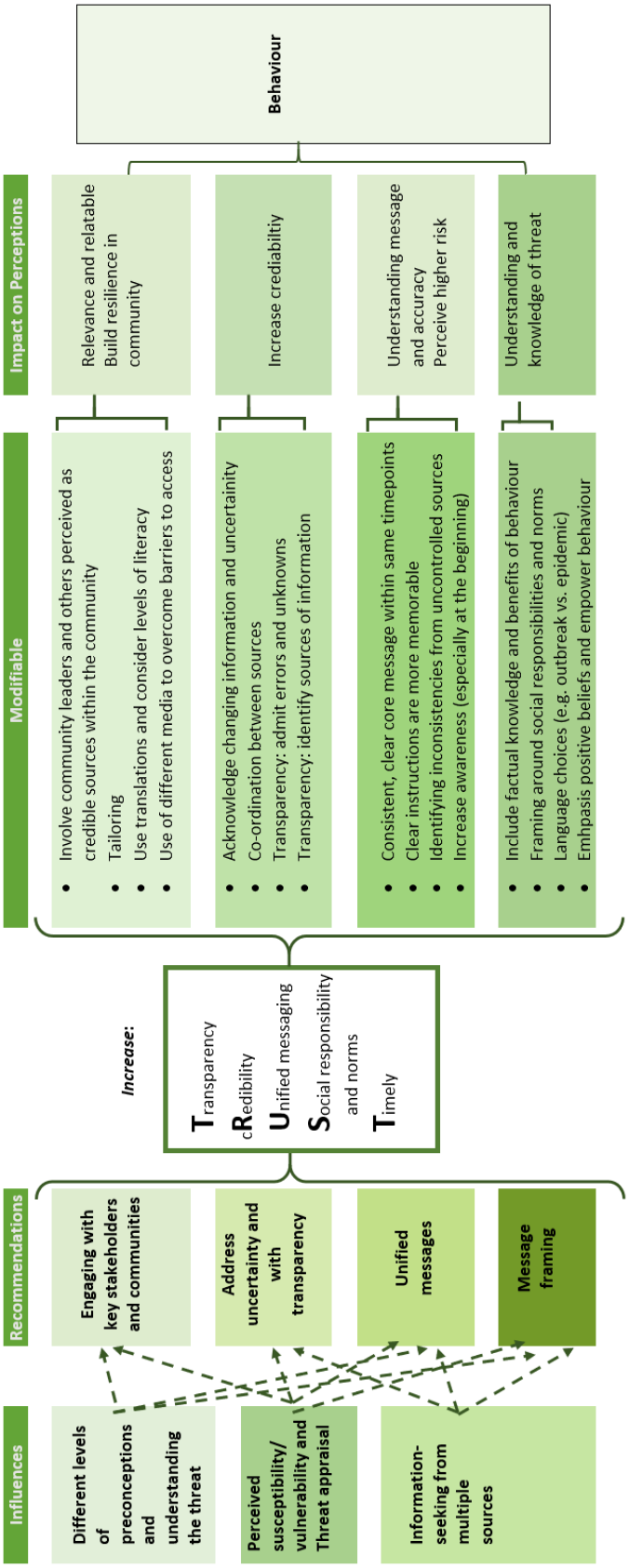



Figure 1: Diagram of synthesis of recommendations and influences on behaviour

# Supplementary Appendix

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# Appendix 1: Search Terms

## *Ovid PsycINFO & Ovid MEDLINE*

| # | Search Terms | Results | Notes (e.g. where search terms were adapted from) |
|---|---|---|---|
| ***Health and risk communication terms*** | | | |
| 1 | public service announcements/ | 370 | |
| 2 | communication/ or messages/ or persuasive communication/ or scientific communication/ or social communication/ | 65306 | Imploded and selected a few narrower terms as the others were irrelevant |
| 3 | ((public health or health or risk or emergenc* or crisis* or catastroph* or disaster* or outbreak*) adj3 (communication* or campaign* or information* or plan* or message* or uncertaint* or alert* or awareness or recommendation* or guideline* or guidance or measure*)).tw. | 54841 | |
| 4 | 1 or 2 or 3 | 116708 | |
| ***Pandemic and epidemic terms*** | | | |
| 5 | exp Pandemics/ | 473 | |
| 6 | exp Epidemics/ | 3399 | |
| 7 | exp Influenza/ | 1396 | Includes Swine Flu, no MeSH terms for SARs, MERs, and Ebola so these searched as keywords only |
| 8 | exp Emergency Preparedness/ | 1192 | |
| 9 | ("SARS" or coronavirus or severe acute respiratory syndrome).tw. | 587 | |
| 10 | (ebola or ebolavirus or ebola virus).tw. | 411 | |
| 11 | ("MERS-CoV" or "MERS or Middle East Respiratory Syndrome coronavirus").tw. | 11 | |
| 12 | (influenza or swine flu or H1N1 or pH1N1* or pdmH1N1* or nH1N1*).tw. | 2490 | |
| 13 | (pandemic* or epidemic*).tw. | 14247 | |
| 14 | 5 or 6 or 7 or 8 or 9 or 10 or 11 or 12 or 13 | 17926 | |
| 15 | 4 and 14 | 1525 | |

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*Health Evidence (Reviews)*

[(Public health messag*) OR ("Risk communication") AND Pandemic OR Epidemic] (224 results)

*Grey Literature Searches*

OSF Preprints + PsyArXiv Preprints

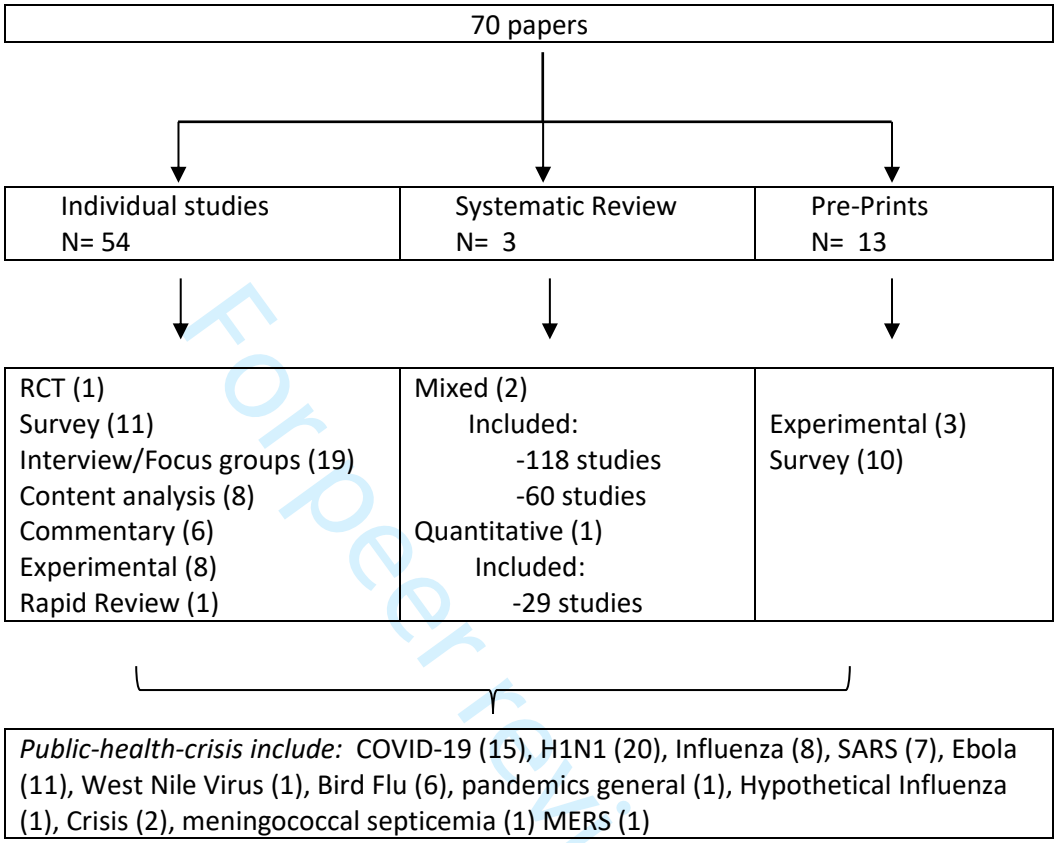
| **Search category** | **Search terms** |
|---|---|
| Intervention | (Public health messag* OR communication OR plan OR alert OR health campaign OR health recommendation* OR public information OR medical information OR public awareness OR community engagement OR risk reduction OR health promotion) |
| Phenomena of interest | (Pandemic OR catastrophe OR cris?s OR outbreak OR emergency OR Coronavirus OR Covid* OR SARS* OR ebola* OR MERS OR epidemic) |
| Combination | (Public health messag* OR communication OR plan OR alert OR health campaign OR health recommendation* OR public information OR medical information OR public awareness OR community engagement OR risk reduction OR health promotion) AND (Pandemic OR catastrophe OR cris?s OR outbreak OR emergency OR Coronavirus OR Covid* OR SARS* OR ebola* OR MERS OR epidemic) |
| Limitations | Arts & Humanities; Life Sciences; Medicine and Health Sciences; Physical Sciences and Mathematics; Social and Behavioral Sciences |
| Results | 518 |

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## Appendix 2: Breakdown of studies




## Appendix 3: screening, data extraction and synthesis details

| *Task* | *Authors* |
|---|---|
| *Screening of titles and abstracts (1st and 2nd)* | NH, EJ, SS, JW, APK, DW, LS, NS, EG, DS, NC, DG, MYT, SLW, CK |
| *Conflicts resolved* | DG, MYT, SLW, CK |
| *Development of guidance document* | DG, MYT, SLW, CK, TE, AC, SH, LBD |
| *Pilot data extraction* | AG, SLW |
| *Data extraction* | NH, EJ, SS, JW, APK, DW, LS, NS, EG, DS, NC, DG, MYT, SLW, CK |
| *Quality check* | JW, SS, DG, MYT |
| *Analysis* | DG, MYT, SLW, CK |
| *Iterations and discussions* | DG, MYT, SLW, CK, TE, NH, EJ, AC, SH, RT, CJA, MAA, JH, LBD |
| *Agreed on final categories* | DG, MYT, SLW, CK, TE, NH, EJ, AC, SH, RT, CJA, MAA, JH, LBD |



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Appendix 4: Data and study characteristics table

| | Country | Study Type | Health Threat | Intervention type | Target behaviour | Main findings |
|---|---|---|---|---|---|---|
| **Yardley (2011)** | UK | parallel-group pragmatic exploratory trial design | H1N1 pandemic | Website | hand washing | Increased hand washing, targeted/changed attitudes |
| **Kavanagh (2011)** | Australia | cross sectional survey | H1N1 pandemic 2009 | official sources (health dept, schools) and unofficial sources (media, family and friends, health care providers) | quarantine | Compliance was higher when they understood the messages. make messages easy to understand for everyone. Official sources are trusted/legitimised more than unofficial sources |
| **Freiman (2011)** | USA | face to face interviews | H1N1 pandemic 2009 | TV, newspaper, internet, family/friends, workplace, radio, other | Reporting on a range of preventative behaviours | Variable - did not always have the desired effect, or an effect at all. relied on the internet - however a lot of the target population did not have access to this - and thus were missed. Even those with internet access were unlikely to visit the CDC's website to learn about H1N1 |
| **Hickey (2014)** | Quatar | Cross-sectional survey | H1N1 pandemic 2009 | Non-pharmaceutical interventions (e.g. using a thermometer, wearing a face-covering, hand washing, and household | using a thermometer, wearing a face-covering, hand washing, and household disinfection practices | Attitudes towards recommended NPIs were generally negative or uncertain, limited experience applying these interventions (e.g., using a thermometer, wearing a face-covering) and inadequate hand washing and household disinfection practices |



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| | Country | Study Type | Health Threat | Intervention type | Target behaviour | Main findings |
|---|---|---|---|---|---|---|
| | | | | disinfection practices) | | |
| **Jhummon-Mahadnac (2012)** | Australia | Cross-sectional survey | H1N1 pandemic 2009 | Public Education Campaigns | Working from home, postponing social gatherings, wearing facemasks, oseltamivir treatment if sick | 68% (175) reported behaviour change because of the pandemic. Gaps in knowledge included failure to identify certain high risk groups. Recall of government campaigns was significantly associated with a higher knowledge score. 60% (151) thought that authorities and media had exaggerated the threat; only 40% (101) would comply with recommended measures in a future pandemic. |
| **Gray (2012)** | New Zealand | Focus groups | H1N1 pandemic 2009 | Ministry of Health campaigns | General protective behaviours | Four major themes: personal and community risk, building community strategies, responsibility and information sources. People wanted messages about specific actions that they could take to protect themselves and their families and to mitigate any consequences. They wanted transparent and factual communication where both good and bad news is conveyed by people who they could trust. |
| **Davis (2015)** | Australia & Scotland | Qualitative interviews and focus groups | H1N1 pandemic 2009 | Broadcast and electronic media | Hygiene (e.g. covering mouth when coughing or sneezing, washing hands, cleaning | Participants wanted information but to be able to interpret advice based on own situation. Information on hygiene and social |

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| | Country | Study Type | Health Threat | Intervention type | Target behaviour | Main findings |
|---|---|---|---|---|---|---|
| | | | | | surfaces), social isolation, vaccination | isolation endorsed but seen as short-term solution as being social is a central part of life. Participants had an individualistic approach to risk, decide whether to adopt PH behaviours based on risk to themselves, gender norms play a part in understanding and engagement with hygiene and social isolation. |
| **Person, (2004)** | USA | Focus groups and rapid situational assessment | SARS | Community outreach strategies | N/A Community resilience | SARS-related stigmatization was occurring more frequently within the Asian community than from outsiders directed toward the Asian community. The team also found that those persons with SARS-like symptoms who used traditional herbal physicians and pharmacies were less likely to be referred to, or seek out, public health officials, suggesting that further research into strategies to reach this population is needed. Conducting community visits also showed that CDC was responding to the needs of the community at risk for SARS-related fear, stigmatization, and discrimination and was modelling positive behaviours to the public. |

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| | Country | Study Type | Health Threat | Intervention type | Target behaviour | Main findings |
|---|---|---|---|---|---|---|
| **Liao, (2010)** | Hong Kong | telephone interviews | H1N1 pandemic 2009 | | hand hygiene and social distancing | There was a relationship between trust in formal information and understanding (positive), trust in information info and susceptibility (negative) |
| **Aburto (2009)** | Mexico | cross sectional representative household survey | H1N1 pandemic 2009 | | N/A - reporting on a range of preventative behaviours | Risk mediated behaviours in some cases e.g. if they perceived the threat of the virus to be high, they were in turn more likely to avoid crowds. People who did comply, were more likely to if they had knowledge about the virus. |
| **Qian (2020)** | China | Content analysis | COVID-19 | Newspapers | N/A | The Chinese mass media news lagged when reporting the major developments of the viral spread. Prevention and control procedures, medical treatment, and research are major themes of the press but mainly focus on the whole society, while instructions on personal and individual prevention, clinic and medicine choices, and detection need to be further enhanced. |
| **Santibañez (2015)** | USA | Commentary | Ebola | | N/A | 10-step approach can be adapted for infectious disease response communications (in this case Ebola) allowing community and faith-based organisations to help spread effective health messages |
| **Basnyat & Lee (2014)** | Singapore | Qualitative | H1N1 pandemic | Singapore Ministry of Health press | N/A | The media did convey useful information for managing the pandemic from the Ministry of |



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| | Country | Study Type | Health Threat | Intervention type | Target behaviour | Main findings |
|---|---|---|---|---|---|---|
| | | | | release in newspapers | | Health press release (i.e. social responsibility, benefits of lockdown) and these were often in the tone/ semantics of colonial-era ideologies that are present within the culture in Singapore and reflect that of the government. Beneficial for the government to provide more than basic information, such as about coping with anxiety, and consider how other types of media (e.g. social media, blogs) can be used too. |
| **Davis, (2013)** | USA | Experimental design with 3 conditions | Influenza A; Influenza B; HINT | Posters | Hand washing | Prompts alone were not effective at increasing hand washing.<br><br>In the education building female bathroom soap usage significantly increased during the first poster prompt (following baseline), then fell below baseline during the second prompt (i.e., warning of influenza transmission). In the business building bathroom women's soap usage did not change significantly from baseline to first poster condition but decreased slightly on the second condition. Women in the education building used significantly more soap overall |

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| | Country | Study Type | Health Threat | Intervention type | Target behaviour | Main findings |
|---|---|---|---|---|---|---|
| | | | | | | than women in the business building; there were no significant differences between men by building type. |
| **Jones (2010)** | Australia | Focus groups | Bird Flu | Campaigns prepared by advertising agency | Hand washing and Hygiene and self-protection measures (e.g. | In the early stages of a pandemic (i.e., prior to the identification of cases within a given country), communications should focus on increasing awareness of the disease and communicating important, but simple, protective behaviours to reduce the risk of transmission. Social marketing campaigns will also need to have an associated strategy for their social marketing campaigns' targeting intermediaries—such as general practitioners and other medical personnel, schools, business owners, and commercial and public organisations which could be utilized to disseminate information and resources. |
| **Miczo (2013)** | USA | Cross-sectional survey | H1N1 pandemic | Campaigns on campus (flyers, posters, emails). Mass media (various, including television & radio) other sources (e.g. medical) | Various - Self-isolation, hand hygiene, respiratory etiquette, sanitizing frequently touched surfaces, not attending public events if sick, vaccination, early intervention for high-risk students and staff | No significant results found for the impact of the health messages on health behaviour performance. The most frequently mentioned messages students remembered were: to wash hands (56.9%), avoidance (23.5%) (e.g. staying away from others when feeling ill) and |



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| | Country | Study Type | Health Threat | Intervention type | Target behaviour | Main findings |
|---|---|---|---|---|---|---|
| | | | | | members, and cancelling travel plans | getting a flu shot/ vaccination (22.1%). Positive health self-concept was positively related to the argument message dimension and flu-prevention behaviours; negative health self-concept was negatively related to the argument message dimension and hand washing. Campus most frequently cited source and useful from campus health viewpoint. There was limited success in attempting to assess memorable messages using a dimensional approach. |
| **Lohiniva (2020)** | Finland | Qualitative | COVID-19 | e-mail and social media | Risk perceptions | Risk perception domain: catastrophic - potential strong wording describing the epidemic, catastrophe, worldwide threat. Concept - Emotional response. Recommendation - Avoid downplaying strong feelings. Concept - Localised epidemic; stigmatising attitude towards foreign nationals. Recommendation - Humanise infected people by telling stories. Location of transmission: public transportation linked with airports and foreign passengers. Concept - Crowded places; stigmatising attitudes towards |



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| | Country | Study Type | Health Threat | Intervention type | Target behaviour | Main findings |
|---|---|---|---|---|---|---|
| | | | | | | foreign nationals. Recommendation - Emphasise handwashing and cough etiquette as effective ways to prevent COVID-19. Concept - Unreliable information Recommendation - Repeat information and provide an explanation (reason). |
| **Winters (2018)** | Sierra Leone | Cross-sectional | Ebola | A variety of sources: mostly electronic media, followed by community sources, print and new media. About half received government communications and most were exposed to more than 2 types of info. | Not reported | increased protective behaviours. Strong dose-response between info exposure and protective behaviours, however exposure to all sources (except electronic and print media) was associated with misconceptions |
| **Sumo (2019)** | Liberia | Interpersonal communication' and dialogue | meningococcal septicemia | Community engagement (religious leaders, elders etc), radio talk shows, house to house visits, radio ads (jingles) | hand-washing, avoiding gatherings, risk perceptions | Increased trust and adherence. Contact tracing successful and epidemic contained, trust increased and adherence to messages increased (e.g. reporting sick people to health facility). |
| **Chang (2012)** | Taiwan | Experiment (2 by 3 factorial design) | H1N1 pandemic | Newspapers | Risk perceptions | The amount of exposure to news stories increases its impact on the public. When a health issue is |



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| | Country | Study Type | Health Threat | Intervention type | Target behaviour | Main findings |
|---|---|---|---|---|---|---|
| | | | | | | ambiguous, greater exposures to news coverage with high alarm frames, as opposed to low alarm frames, evoke more fear and increase participants' perceptions of the severity of the issue, as well as their vulnerability to it. Exposures to news do not increase prevention or treatment efficacy. Repetition exacerbates the effects of media exposure on perceived severity and vulnerability but not fear. The amount of exposure increases its impact. |
| **Daellenbach (2018)** | Australia, New Zealand | Quantitative online survey | Crisis | | Risk perceptions | Adaptable and personalised to be more effective. Segmentation based on Theory of Planned Behaviour variables. The study highlighting that vulnerability may be identified based on such factors as critical awareness, perceived barriers, and ultimately preparation undertaken, including community-related factors. |
| **Tully (2019)** | West Africa | Content analysis | Ebola | Hyperlinks - embedded content in social media, e.g.Twitter used for | Risk perceptions | Increased credibility and awareness of messages. Organizations produce proportionally fewer tweets than news outlets - maintaining an active Twitter presence allows |



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| | Country | Study Type | Health Threat | Intervention type | Target behaviour | Main findings |
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| | | | | information dissemination. | | response organisations to promote their work and position themselves centrally to the unfolding crisis and conversation. Results suggested that 'owned' and 'earned' media served as informational resources and promotional pieces for the organizations. Repeatedly linking to sites controlled by the organization and/or produced by news media and other reliable sources served as a low-cost way for organizations to keep pace with the rapid flow of information on social media. |
| **Toppenberg (2018)** | Various | Rapid Review of Grey Literature | Ebola | | Awareness only | Literature strongly underlines the central importance of local communities. A one-size-fits-all approach does not work. For maximum effectiveness, local communities need to be involved with and own emergency risk communication processes, preferably well before an emergency occurs. Social media can open new avenues for communication but is not a general panacea and should not be viewed as a replacement for traditional modes of communication. In general, the gray literature indicates movement toward greater |

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| | Country | Study Type | Health Threat | Intervention type | Target behaviour | Main findings |
|---|---|---|---|---|---|---|
| | | | | | | recognition of emergency risk communication as a vitally important element of public health. |
| **Teasdale (2011)** | UK | Focus groups | H1N1 pandemic | Government + author-developed messages | Government advice was stay at home if you have flu-like symptoms and continue to go to work if you do not (behaviour not tested in this study though) | Participants were sceptical about the feasibility and appropriateness of self-diagnosis of pandemic flu. Participants actively evaluated recommended actions according to their own beliefs and reasoning about flu and their perceptions of the costs of carrying out the recommended actions. For most people the experience of the H1N1 pandemic was relatively mild compared to previous influenza pandemics of the 20th century. Consequently, the recommendation to stay at home was perceived as inappropriate. Written government communications aiming to improve and maintain population health may also need to address the recipients' perspective. Attempts should be made to elicit and address common doubts and concerns, to reduce perceived barriers to recommended behaviours, emphasize benefits, to find ways to support people to |

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| | Country | Study Type | Health Threat | Intervention type | Target behaviour | Main findings |
|---|---|---|---|---|---|---|
| | | | | | | adopt them and consider the likely contextual factors that may affect perceptions of the advice. |
| **Zikmund-Fischer (2017)** | Netherlands | experiment | hypothetical influenza | mock news article | cover mouth, wash hands | Participants are more influenced by average than extreme case information, presenting both is counterproductive. |
| **Lapka, (2008)** | USA | Focus groups and interviews | Influenza | | Various | Cognitive Response Testing (CRT) is easily applied to various subject matters and effectively highlights problematic terminology and phrases, allowing for beneficial and efficient message revisions. CRT is a straightforward, appropriate and useful method in health promotion message development and pre-testing. |
| **Sell (2017)** | USA | Content analysis of news coverage | Ebola | News coverage | risk perception | more scientific knowledge to be used in media coverage. Certain risk messages about Ebola were used more frequently than others by US news media, which may have affected risk perception during the outbreak. Some messages increase, some decrease risk perception. |
| **Li (2016)** | China | focus group | H7N9 | News from various sources | Risk perception | Most people learnt about virus from tv, info was felt to be released in timely and transparent manner, most useful was info about preventative behaviours, and evolving |



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| | Country | Study Type | Health Threat | Intervention type | Target behaviour | Main findings |
|---|---|---|---|---|---|---|
| | | | | | | outbreak trends. Gov info perceived as trustworthy, but young people trusted social media more and wanted gov to publish news in such way. Health recommendations seen as easy to follow and clear. Anxiety affected by disease severity, distance from pandemic, high level of news coverage, trust in gov ability to fight the issue. Health communication needs to be specific and practical |
| **Rimi (2016)** | Bangladesh | ethnography | avian influenza | Posters | Prevention (handling poultry) | Incorporating non-health benefits might improve acting on recommendations.<br>More people heard about the bird flu after the intervention, and had more awareness around the disease, as well as performed preventative behaviours. Not all behaviours changed in the desired direction though (e.g., selling sick poultry, separating it). Reported behaviours were not consistent with the observations done by the researchers though. People rationalised ignoring recommended behaviours - low perception of risk - the flu not present among their birds, also finance issues and not worrying |



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| | Country | Study Type | Health Threat | Intervention type | Target behaviour | Main findings |
|---|---|---|---|---|---|---|
| | | | | | | about getting ill. Also inconvenience was mentioned, as well as social pressure not to follow the recommendations, and scepticism about the necessity of the intervention - some conspiracy theories. |
| **Rogers (2009)** | Australia | Deliberative forum | Influenza | hypothetical | Risk awareness | The forum wanted full and frank information about the potential risk and international developments including numbers of cases and fatalities. Forum members understood that predictions about the pandemic influenza were uncertain, this did not lead to loss of confidence in the experts or the information they imparted. Third, the forum recommended releasing geographically localizing information about initial cases. The forum recommended increasing use of television and websites, including those targeting youth and rural and remote groups as the pandemic developed. The forum recommended immediate activities to educate and build awareness and swift action if and when Australia has its first cases of PI. |

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| | Country | Study Type | Health Threat | Intervention type | Target behaviour | Main findings |
|---|---|---|---|---|---|---|
| **Prati (2011)** | Italy | Cross-sectional survey | H1N1 | Educational campaign | Various – all preventative behaviours | Recommended behaviours such as cleaning objects, social distancing and washing hands were related to all the psychosocial factors except for trust in the institutional response to the outbreak and trust in medical science. Trust in medical science was significantly associated with cleaning objects. To use tissues when sneezing was predicted by all the psychosocial factors, except for trust in medical science and control. These results showed that media trust, trust in the Ministry of Health (regardless of what was done to reduce this risk), worry and perceived severity of illness predicted all the recommended behaviours. Exposure to the Topo Gigio campaign did have a limited influence on complying with health related recommendations but only with regard the use of tissues when sneezing. |
| **Updegraff (2011)** | USA | Experimental design with 3 conditions | H1N1 | Posters/ signs placed besides hand sanitizer units around university campus | Use of hand sanitizer dispensers | All signs resulted in significantly greater usage than no sign, but they were not equally effective. Dispensers with the gain-framed signs had the greatest usage of all, with 66.4% more use than |



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| | Country | Study Type | Health Threat | Intervention type | Target behaviour | Main findings |
|---|---|---|---|---|---|---|
| | | | | | | dispensers with no signs (p <.001). Loss-framed signs were associated a 58.4% increase in use over no sign (p <.001). The social norms signs (44.3% increase) and the perceived susceptibility signs (40.6% increase) were associated with somewhat lower increases in usage compared to the gain-framed and loss-framed signs, but both led to significantly more usage than no sign at all (both p's < .01). Gain-framed signs received 12.5% more usage than dispensers in the other sign conditions combined, a significant difference (p = .029). Usage of sanitizer dropped consistently over time, closely mirroring temporal trends in public interest in H1N1. This study showed that the relatively simple strategy of placing theoretically grounded cue-to-action reminder signs at the point-of-use significantly promoted usage. The worst-performing sign emphasized people's susceptibility to contamination. |
| **Lyu (2012)** | Taiwan | Qualitative Interviews | SARS | News reports/ public announcements | Perception of spokesperson | The essential/favourable characteristics for spokespersons |

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| | Country | Study Type | Health Threat | Intervention type | Target behaviour | Main findings |
|---|---|---|---|---|---|---|
| | | | | | | should include: having professional capability, having better media interactions, involvement in policy making, having trustworthiness, and having favourable (or positive) personality and traits. |
| **Holmes (2009)** | Canada | Semi-structured interviews | crisis | Hypothetical | Risk perceptions | The notion of uncertainty – and the difficulties associated with communicating in uncertain situations – arose frequently. Recommendations: is to set and get agreement on a goal or goals (there may be different goals set for different audiences or specific situations). View all other aspects of communications planning and implementation through this ethical lens. Media representatives should be engaged immediately in discussions about a potential emerging infectious disease outbreak, including the role media should play and how the public health community can help them fulfil that role. Building trust with the public. |
| **Johnson & Slovic (2015)** | USA | Experimental - 2 studies | Ebola | Mock story | Risk perceptions | Informing Americans about the small likelihood of post-21-days Ebola symptoms would not increase perceived risk and |

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| | Country | Study Type | Health Threat | Intervention type | Target behaviour | Main findings |
|---|---|---|---|---|---|---|
| | | | | | | distrust and might diminish negative reactions to the media reporting a case who developed symptoms of Ebola after 21 days. Thus, public health officials wanting appropriate public responses to potential or actual epidemics may benefit from early communication of unpleasant infectious-disease facts before events reveal them, and signal officials' lack of preparedness for public reactions. |
| **Seltzer (2015)** | USA | Content analysis | Ebola | Instagram/ Flickr pictures relating to Ebola | Risk awareness | Images of health care workers and professionals [308 (25%)], West Africa [75 (6%)], the Ebola virus [59 (5%)], and artistic renderings of Ebola [64 (5%)]. Also identified were images with accompanying embedded text related to Ebola and associated: facts [68 (6%)], fears [40 (3%)], politics [46 (4%)], and jokes [284 (23%). Image sharing platforms are being used for information exchange about public health crises, like Ebola. Use differs by platform and discerning these differences can help inform future uses for health care professionals and researchers seeking to assess public fears and |



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| | Country | Study Type | Health Threat | Intervention type | Target behaviour | Main findings |
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| | | | | | | misinformation or provide targeted education/awareness interventions. |
| **Crosier (2015)** | UK. Italy, Hungary | Qualitative interviews | H1N1 | Government poster | Public compliance and engagement | The study found a lack of planning and a low value attached to the skills required to produce effective communications. There was a dearth of good quality audience research to inform the development of communications. Little thought had been given to the tone, targeting or channelling of messages. Instead, communications were characterized by a 'one size fits all' and a 'top down', expert-led response. There was also little effort to evaluate the impact of communications, but where this was done, very low levels of public compliance and engagement with key behavioural messages were found. |
| **Goldberg (2015)** | USA | Report | Ebola and others | Checklist | Risk awareness and perceptions | 5 stages: agree on a common goal, coordinate the leadership, develop a coordination strategy, launch a communication operation, maximize communication effectiveness |



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| | Country | Study Type | Health Threat | Intervention type | Target behaviour | Main findings |
|---|---|---|---|---|---|---|
| **Rim (2014)** | USA | content analysis + correlating with survey results | H1N1 | News media | Various | The more info from health authorities, the more news coverage. The same correlation for severity of content. Increased information subsidies with salience of the severity attribute was linked with increased H1N1 salience in media coverage. But no relationship between the severity attribute use in media coverage, and the public perception of risk. Framing that effectively empowers the public to engage in desired behaviour should be further studied for the success of a public health campaign. |
| **Mitchell (2014)** | USA | Focus groups | Influenza A H1N1 | University mobile message alerts | Various preventative behaviours | Reported concern and commitment to recommendations decreased rapidly. Initial university messaging and response was critical in shaping participants' later perceptions. |
| **Bonwitt (2018)** | Sierra Leone & Guinea | Longitudinal qualitative/ ethnographic study | Ebola | posters | Various preventative behaviours | All study participants, irrespective of age or gender, were aware of wild mammals acting as a source of transmission for Ebola virus disease (EVD); respondents gave a variety of answers when asked what species of animal could transmit Ebola - this confusion |



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| | Country | Study Type | Health Threat | Intervention type | Target behaviour | Main findings |
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| | | | | | | may have rested in the content of public health messages, which were inconsistent in the species shown to be potentially hazardous; People simply refused to believe that wild meat could pose any health risk; Those that believed or partially believed in the link between EVD and wild meat adopted various strategies to mitigate the perceived risk of infection. Most admitted to only refraining from eating those animals that they understood as posing a risk for EVD; The widespread suspicion over the risks posed by wild animals prompted discussions during village meetings to discuss the veracity of public health messages.<br>Rather than outright rejection of these messages, people elaborated situated hypotheses to make sense of the conflicting and incomplete information they had received. This process of contextualisation helped to bridge the disjuncture between fear and the highly routine nature of eating meat. |



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| | Country | Study Type | Health Threat | Intervention type | Target behaviour | Main findings |
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| **Bekalu (2017)** | USA | Experimental | Pandemic influenza | Video clips | washing hands frequently, staying home when sick and avoiding unnecessary travel | A non-narrative message format may be more effective than its narrative counterpart to communicate basic prevention information during public health emergencies. Compared with the narrative and/or fictional version, the more didactic and factual format was found to be more effective in changing knowledge and perceived response efficacy related to prevention of pandemic influenza. |
| **Roess (2017)** | DR Congo | Pre- post interventional study | Ebola | Four films | Risk perceptions and awareness | a culturally and linguistically appropriate video-centred intervention was effective in improving knowledge, attitudes and behaviours related to EVD in Congo. The results also demonstrate retention of the knowledge one year after exposure to the intervention. |
| **Avery (2009)** | USA | Content Analysis of avian flu press releases | Avian flu | Press releases issued by federal and global health agencies | Not reported | The threat was not localized; outlets for publics to contact for more information were not always present, and there was inconsistency in how avian flu was referenced. There was a steady increase in the volume of releases since 2004, and public health information officers |



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| | Country | Study Type | Health Threat | Intervention type | Target behaviour | Main findings |
|---|---|---|---|---|---|---|
| | | | | | | generally seemed to recognize the importance of partnerships and timeliness in pandemic preparedness. Public health communicators must establish a strong sense of efficacy in an informed public to insure that audiences will adhere to protocol rather than succumb to hysteria in the face of a wide-scale crisis such as pandemic flu. They must also provide information accessible to all publics, including those who do not have Internet access. |
| **Crouse Quinn (2008)** | USA | Commentary/proposal of a model for crisis and emergency and risk communication in a pandemic | Pandemics (general) | | Not reported | Overall proposed model: pre-disaster phase (ongoing risk education, partnership formation, community engagement, opportunity for deliberation on difficult policies and procedures); during an emergency (more one way, emergency risk communication, more need for immediate action, community partners' engagement in response); increased capacity and resilience of communities at times of disasters and pandemics |
| **Berry (2007)** | Canada | Content analysis of the construction of | SARS/other infectious | Media reports - newspaper, radio, television, and | Not reported | The results of the content analysis showed that in 2003, there was far more information |



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| | Country | Study Type | Health Threat | Intervention type | Target behaviour | Main findings |
|---|---|---|---|---|---|---|
| | | news reports on health topics | diseases/general health | internet news on health-related topics | | available in the news media on SARS and West Nile Virus than on other health topics with greater population prevalence, such as obesity or heart disease. The number of articles about SARS in 1 year was greater than for any other individual topic across all 5 years, with the exception of smoking (which in 2003 had only 36 articles, compared to SARS, which had 164) and "other" topics. Expert sources were cited far more often than nonexperts. |
| **Menon (2005)** | Singapore | Commentary – narrative review | SARS | | Risk perceptions | 1. Providing more information is more effective than less information, 2. it always better to over-react than to under-react. 3. fear and ignorance of any disease is worse than the disease itself. It reduces all to impotence and defeat. Information is an all-powerful tool to fight fear. It empowers people and allows them to become socially responsible. |
| **Idoiaga (2016)** | Spain | Experimental | Flu epidemic | Mass media | Risk perceptions | The human interest framing increased the perception of risk, especially when the proximity of the epidemic was high; this effect was explained by people's emotional response. |



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| | Country | Study Type | Health Threat | Intervention type | Target behaviour | Main findings |
|---|---|---|---|---|---|---|
| | | | | | | Furthermore, youth projected the risk towards 'the other' in order to protect their invulnerability identity. |
| **Cole & Watkins (2015)** | Liberia, Sierra Leone or Guinea | interviews | Ebola | WHO, CDC, professional media (international, local) informal information e.g. rumours | Information-seeking | Their recall can be grouped into three broad source categories (official, international and local media, and informal) Throughout later information-seeking behaviour, interviewees maintained a distinction between the three categories. This affected how they trusted and processed information from each, although generally they tended to use all three simultaneously. Multiple source information gathering, especially following up from informal information (personal experiences) because they felt that was more rumour and wanted confirmation from official sources but felt official sources were slow with the information. Consistency provided reassurance and helped to be trusted. perception of risk, triggered by hearing that someone they knew personally, or felt an affinity to, had contracted the virus. know who employees see as their |



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| | Country | Study Type | Health Threat | Intervention type | Target behaviour | Main findings |
|---|---|---|---|---|---|---|
| | | | | | | community and 'people like me'. Behavioural changes seemed to kick in at the border between the near and real at-risk rather than at the border between far at-risk and near at-risk. Those in the far at-risk group generally showed more concern for people they knew in higher risk categories than for themselves. |
| **Qiu (2018)** | China | qualitative comparative case study | SARS, H7N9 | Media | Risk perceptions and awareness | 8 key principles of risk communication<br>Trust is the basis by maintaining an open and honest attitude and engage key stakeholders.<br>Delaying in reporting cases caused distrust, need accurate, timely, honest reporting, transparency helped subdue rumours, maintain social stability and helped maintain trust, and ensured more cooperation to reduce the spread of the disease.<br>Being honest and open, planning well, being empathetic and caring, Accepting and involving the public as a partner regular feedback from the public and interaction between government and community. |
| **Vaughan (2009)** | Not reported | Commentary – narrative review | Pandemic Influenza | | Message acceptance | Barriers to accepting messages include, environmental factors, |

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| | Country | Study Type | Health Threat | Intervention type | Target behaviour | Main findings |
|---|---|---|---|---|---|---|
| | | | | | | social and cultural characteristics, language preferences (translation, culturally sensitive) difficulty of and attitudes towards public health interventions |
| **Van der Weerd (2011)** | Netherlands | repeated measure cross-sectional telephone survey | H1N1 | | information-seeking | During the course of the pandemic the majority of respondents wanted to receive information from health services/health care providers and media. Wanted information on how to prevent infection, what to do in the event of illness, symptom, risks , consequences and number of infected cases. Trust in governmental information depressed by the time of period 3. At first, mistrust was because information was incomplete, kept secret or withheld. Later, because they thought the situation was exaggerated, information unclear and information was contradicted itself. Fear/worry was sig associated with intention to adopt protective measures during all time periods. |
| **Menon (2006)** | Singapore | Narrative review/ commentary | SARS | Mixed - TV, technology | Not reported | Technology (The Infrared Fever Screening System), a SARS-dedicated TV channel, transparency, leadership (highest |



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| | Country | Study Type | Health Threat | Intervention type | Target behaviour | Main findings |
|---|---|---|---|---|---|---|
| | | | | | | levels of government came into the picture very early in the crisis and took the lead), communication tools (moderating public fear through dialogue with grassroots leaders and community, briefings for foreign business groups, diplomats, religious groups, trade associations, cartoons, hotlines, songs on TV), earning the trust and confidence of Singaporeans, rebuttal of negative reports in the foreign media claiming that foreign visitors had been infected with SARS whilst transiting Singapore, outbreak communications across cultures. |
| **Hoda (2016)** | Saudi Arabia | Survey | MERS | Promoting public awareness of MERS in the Saudi population | Risk awareness | The internet was the most commonly used source of information (39.5%) and the most endorsed channel for a MERS awareness campaign. Physicians were the preferred source of information (45.6%), followed by other health care providers (31.3%). In univariate multinomial logistic regression models, males and individuals aged ≤27 years were more likely to seek information from the internet than from physicians. Residents |



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| | Country | Study Type | Health Threat | Intervention type | Target behaviour | Main findings |
|---|---|---|---|---|---|---|
| | | | | | | of southern and western Riyadh preferred physicians as a credible source of information over the Ministry of Health. |
| | | | **Systematic Review** | | | |
| **Barrelet (2013)** | | Systematic Review | H1N1, H5N1 | | Risk perception, vaccine perception, rumours | Risk perceptions vary across time, social groups, cultures, and countries. Social resistance to vaccination is an enduring phenomenon that should not be downplayed by public health officials. Competing narratives (official statements and presentations, rumours, conspiracy theories, alternative worldviews and explanations, urban legends, hoaxes, etc.) about public health crises always coexist in the public sphere. Trust building is a key aspect of risk perceptions. |
| **Schiavo (2014)** | | Systematic review | Influenza, SARS | Media | Handwashing, immunizations, infection control precautions | most studies found behaviour changed |
| **Lin (2014)** | | Systematic review | H1N1 | Non-specific variety of interventions, including 'Websites', ' Commercial television', 'Health department' | Preventive behaviours such as hygiene and social distancing practices in 70% of the studies (N = 64), | To reduce communication inequalities during a large scale emergency, such as a pandemic, public health officials should focus their communication efforts on the young, the less |



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| | Country | Study Type | Health Threat | Intervention type | Target behaviour | Main findings |
|---|---|---|---|---|---|---|
| | | | | | risk perceptions (70%, N = 64), levels of knowledge and awareness about the pandemic (53%, N = 49), and emotional responses such as fear and worry (47%, N = 43). Factors influencing the H1N1 vaccination acceptance rate (26%, N = 24) were also frequently investigated. | educated and the indigent because there is evidence that these are the people at risk of not knowing about the threat, perceiving the threat to be of low risk and ultimately being less likely to follow recommended behaviours. An honest reporting of what the threat looks like, through a presentation of known and unknown factors, seems to have a better impact on people's knowledge, attitudes and beliefs, including trust in the way the government is handling the emergency. Consequently, there is some evidence that better knowledge and trust are likely to be associated with the adoption of recommended behaviours (i.e. immunization practices). Social networks and ties to the community are also drivers of better knowledge and compliance with preventive measures; these results suggest that non-traditional channels of communication (i.e. partnership with community leaders or organizations) should be used to reach out to the most vulnerable |



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| | Country | Study Type | Health Threat | Intervention type | Target behaviour | Main findings |
|---|---|---|---|---|---|---|
| | | | | | | and those in need of a better understanding of the risks and actions needed to be able to protect themselves. Public health communication messages are still delivered at a literacy level that does not meet the needs of the less educated. |
| | | | **Pre-print** | | | |
| **Abu-Akel, Spitz & West (2020)** | Switzerland | Pilot experimental survey study | COVID-19 | Social distancing | Social distancing | The government official was more effective at garnering support for social distancing than the celebrity, particular in older respondents. Support and compliance relating to social distancing was higher in older people, despite them having lower percieved risk of threat. Females were overall more in support of social distancing. Higher support and compliance was related to greater concern about the situation, concern for others health, greater belief in social distancing measures and feeling more constrained by them. Inverse relationship between city size and practise of social distancing. |
| **Bilancini, (2020)** | Italy | Four-condition, between-subjects experiment. | COVID-19 | Official coronavirus-related messages | Behaviours relating to the COVID-19 emergency | No statistically significant difference between the 4 conditions. Authors conclude that stronger 'nudging' interventions |

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| | Country | Study Type | Health Threat | Intervention type | Target behaviour | Main findings |
|---|---|---|---|---|---|---|
| | | | | | | may be needed to influence behaviour. |
| **Blagov (2020)** | USA | Online survey | COVID-19 | Public health messages worded to target different personality traits (self-centered, responsible, compassionate, avoidant & sociable). Looks like she were developed by the researchers. | Staying at home, physically distancing and practising handwashing | Agreeableness and conscientiousness predicted endorsement of social distancing and hygiene. Meanness and disinhibition (and overall psychopathy) as well as Machiavellianism (less so) predicted lower intent for social distancing. Together with boldness, the psychopathy traits predicted endorsement of risky, venturesome behavior under the pretend scenario that one is a disease carrier. Meanness and disinhibition predicted endorsement of knowing and possibly deliberate behaviour that puts others at risk of infection. |
| **Dai (2020)** | China | Cross-sectional online survey | COVID-19 | Information on government emergency management measures relating to COVID-19 | Protective behaviours, including preventive (i.e. wearing masks, disinfectants), avoidant (i.e. stringent quarantine, avoiding public places), and management of disease behaviors (i.e. seeking professional protection or treatment information, paying for preventive and therapeutic drugs) | Governmental information transparency, positive propaganda, rumour refutation, and supplies positively predict the protective behaviours. Individual factors such as perceived control, positive emotions, and risk perception mediate role in predicting protective behaviours. |



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| | Country | Study Type | Health Threat | Intervention type | Target behaviour | Main findings |
|---|---|---|---|---|---|---|
| **Merkley, (2020)** | Canada | 2-part online experimental survey study | COVID-19 | News stories relating to the pandemic - signalling information from experts or non-experts. | Information on hygiene (i.e. cleaning surfaces), social gatherings, testing and scientific discoveries. | Study 1: We find that our respondents were on average 3 points more likely to select news stories with headlines featuring experts (p~0.007). News stories with headlines featuring experts scored 2 points higher in credibility (p<0.002). In support of H1, we find that the effect of receiving headlines featuring experts is heterogeneous across levels of anti-intellectualism. The marginal effects on story selection are shown in the left panel of Figure 2. Respondents with the lowest levels of anti-intellectualism are six points more likely to select stories featuring experts (p<0.001), but this effect disappears once the mid-point of the scale is reached.<br>Study 2: Our respondents were 19 points more likely to select COVID-19 news (p<0.001). They also viewed such news as 6 points more credible (p<0.001) and 19 points more important (p<0.001). Once again, anti-intellectualism strongly conditions news preferences. Respondents with the highest levels of trust in experts were 25 points more likely to select COVID-19 news (p<0.001), and viewed it as 8 points more credible (p<0.001) and 24 points more important (p<0.001). In all three cases this effect weakens as anti- |



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| | Country | Study Type | Health Threat | Intervention type | Target behaviour | Main findings |
|---|---|---|---|---|---|---|
| | | | | | | intellectualism rises, though it does not entirely vanish for the most part. |
| **Everett, (2020)** | USA | online experimental survey study | COVID-19 | Social media (Facebook post) either from a citizen or leader. | Behavioural intentions for: Hand washing, avoiding gatherings, isolation, cancelling holidays | Stronger intentions if messages were shared by 'leader' than citizen, modest effects of using deontological and virtue-based messaging (compared to utilitarian) |
| **Jordan (2020)** | USA | Three-part online experimental study | COVID-19 | PH messages/ fliers about COVID-19 emphasising personal, public or personal & public protective messages. | Hand washing, social distancing, staying at home, wearing a mask. | In the early stages of the pandemic, prosocial (i.e. public) messaging was more effective. However, as the pandemic progressed, all three types of framing (i.e. prosocial/ public, self-interested/ personal and both) were equally as effective. Self-interested framing was never more effective than prosocial framing and did not improve efficacy of prosocial framing. |
| **Goldberg, (2020)** | USA | Online Survey | COVID-19 | N/A - measured behaviours as a result of seeing national guidance from the CDC | mask buying and mask wearing | Once the CDC recommendation had been disseminated for at least one full day, there were large increases in reported mask wearing and mask buying. |
| **Yonemitsu, (2020)** | Japan | Two wave online survey | COVID-19 | Online messages based on 2 frameworks: identity-related or behaviour-related | restricting going out, wearing masks, avoiding conversations where people are close, follow 'coughing etiquette' | Prediction is that messages focusing on identity will be more powerful than the ones focused on behaviour. |
| **Wirz, (2020)** | USA | Online Survey | COVID-19 | N/A - survey asking whether people were practising guidelines (e.g. social distancing) and questions about | Social distancing | Perceived effectiveness of social distancing for decreasing the risk of infection and perceived social norms related to social distancing were strongly correlated with reported social distancing, |



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| | Country | Study Type | Health Threat | Intervention type | Target behaviour | Main findings |
|---|---|---|---|---|---|---|
| | | | | barriers and communication. | | suggesting that individuals engaged in social distancing if they thought it worked and if they thought their peers were doing it. If they learned that their social distancing behaviour would help others (e.g., family, older members of the community) and that there are serious health consequences for people like them, they would engage in more social distancing. Respondents in this group reported often getting news from national news networks and social media, followed by newspapers/news magazines. Respondents reported having the highest levels of trust in information from the CDC, public health experts, and the WHO. These groups were followed by university scientists and friends and family. The White House, state and local governments, and the news media were among the least trusted sources for information. |
| **Bradshaw, (2020)** | Australia | Randomized experimental online study | COVID-19 | Online vignettes about the contact tracing app | Downloading and using a contact tracing app | Autonomy-supportive and controlling message framing did not differentially affect intended uptake. However, there was a main effect of information safety. Those in high information safety conditions reported higher intentions to use the application |



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| | Country | Study Type | Health Threat | Intervention type | Target behaviour | Main findings |
|---|---|---|---|---|---|---|
| | | | | | | and to recommend it to others than those in low information safety conditions, regardless of message framing. Australians appeared more willing to assent to authority regarding contact tracing insofar as their data safety can be assured. |
| **Dorison, (2020)** | Global | Experimental online survey | COVID-19 | Main gov recommendations, randomised to either have a gain or a loss frame | intentions to follow recommendations, support for anti-infection policies and information seeking | N/A |
| **Heffner (2020)** | USA | Within subject design, online survey | COVID-19 | 2 messages about staying at home, one focused on 'fear', the other on 'altruism' | Intentions to self-isolate, emotional response | Both messages were associated with increased willingness to self-isolate. The prosocial message was more effective at boosting willingness to self-isolate if it produced a strong, positive, and arousing emotional response. Fear messages created a stronger emotional response. |

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# Appendix 5- Data quality table using Mixed Methods appraisal tool and AMSTAR

| Studies | | | | Criteria from the mixed methods appraisal tool | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | S1 | S2 | 1.1 | 1.2 | 1.3 | 1.4 | 1.5 | 2.1 | 2.2 | 2.3 | 2.4 | 2.5 | 3.1 | 3.2 | 3.3 | 3.4 | 3.5 | 4.1 | 4.2 | 4.3 | 4.4 | 4.5 | 5.1 | 5.2 | 5.3 | 5.4 | 5.5 |
| Yardley (2011) | parallel-group pragmatic exploratory trial design (RCT) | 4 | **** | 1 | 1 | | | | | | 1 | 1 | 1 | 0 | 1 | | | | | | | | | | | | | | | |
| Kavanagh (2011) | cross sectional survey | 4 | **** | 1 | 1 | | | | | | | | | | | | | | | | 1 | 1 | 1 | 0 | 1 | | | | | |
| Freiman (2011) | face to face interviews | 5 | ***** | 1 | 1 | | | | | | | | | | | | | | | | 1 | 1 | 1 | 1 | 1 | | | | | |
| Hickey (2014) | Cross-sectional survey | 5 | ***** | 1 | 1 | | | | | | | | | | | | | | | | 1 | 1 | 1 | 1 | 1 | | | | | |
| Jhummon-Mahadnac (2012) | Cross-sectional survey | 5 | ***** | 1 | 1 | | | | | | | | | | | | | | | | 1 | 1 | 1 | 1 | 1 | | | | | |
| Gray (2012) | Focus groups | 5 | ***** | 1 | 1 | 1 | 1 | 1 | 1 | 1 | | | | | | | | | | | | | | | | | | | | |
| Davis (2015) | Qualitative interviews and focus groups | 3 | *** | 1 | 1 | 1 | 1 | 0 | 1 | 0 | | | | | | | | | | | | | | | | | | | | |
| Person, (2004) | Focus groups and rapid situational assessment | 2 | ** | 1 | 1 | 1 | 1 | 0 | 0 | 0 | | | | | | | | | | | | | | | | | | | | |
| Liao, (2010) | telephone interviews (surveys) | 4 | **** | | | | | | | | | | | | | | | | | | 1 | 1 | 1 | 0 | 1 | | | | | |
| Aburto (2009) | cross sectional representative household survey | 4 | **** | | | | | | | | | | | | | | | | | | 1 | 1 | 1 | 0 | 1 | | | | | |
| Qian (2020) | Content analysis | 2 | ** | 1 | 1 | 0 | 0 | 1 | 1 | 0 | | | | | | | | | | | | | | | | | | | | |
| Santibañez (2015) | Commentary | | n/a | | | | | | | | | | | | | | | | | | | | | | | | | | | |

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|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Basnyat & Lee (2014) | Qualitative | 4 | **** | 1 | 1 | 1 | 0 | 1 | 1 | 1 | | | | | | | | | | | | | | | | | | | | | |
| Davis, (2013) | Experimental design with 3 conditions | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| Jones (2010) | Focus groups | 2 | ** | 1 | 1 | 0 | 0 | 1 | 0 | 1 | | | | | | | | | | | | | | | | | | | | | |
| Miczo (2013) | Cross-sectional survey | 2 | ** | 1 | 1 | | | | | | | | | | | | | | | | 0 | 0 | 1 | 0 | 1 | | | | | | |
| Lohiniva (2020) | Qualitative | 4 | **** | 1 | 1 | 1 | 1 | 1 | 0 | 1 | | | | | | | | | | | | | | | | | | | | | |
| Winters (2018) | Cross-sectional | 5 | ***** | 1 | 1 | | | | | | | | | | | | | | | | 1 | 1 | 1 | 1 | 1 | | | | | | |
| Sumo (2019) | Interpersonal communication' and dialogue | | n/a | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| Chang (2012) | Experiment (2 by 3 factorial design) | 5 | ***** | 1 | 1 | | | | | | | | | | | | | | | | | | | | | 1 | 1 | 1 | 1 | 1 | |
| Daellenbach (2018) | Quantitative online survey | 3 | *** | 1 | 1 | | | | | | | | | | | | | | | | 1 | 0 | 1 | 0 | 1 | | | | | | |
| Tully (2019) | Content analysis | 3 | *** | 1 | 1 | 1 | 1 | 0 | 1 | 0 | | | | | | | | | | | | | | | | | | | | | |
| Toppenberg (2018) | Rapid Review of Grey Literature | 5 | ***** | 1 | 1 | 1 | 1 | 1 | 1 | 1 | | | | | | | | | | | | | | | | | | | | | |
| Qiu (2018) | Focus groups and interviews | 5 | ***** | 1 | 1 | 1 | 1 | 1 | 1 | 1 | | | | | | | | | | | | | | | | | | | | | |
| Teasdale (2011) | Focus groups | 4 | **** | 1 | 1 | 1 | 1 | 1 | 1 | 0 | | | | | | | | | | | | | | | | | | | | | |
| Zikmund-Fischer (2017) | experiment | 2 | ** | 1 | 1 | | | | | | 0 | 0 | 0 | 1 | 1 | | | | | | | | | | | | | | | | |
| Lapka (2008) | Focus Group and Interviews | 5 | **** | 1 | 1 | 1 | 1 | 1 | 1 | 1 | | | | | | | | | | | | | | | | | | | | | |
| Sell (2017) | Content analysis of news coverage | 5 | **** | 1 | 1 | 1 | 1 | 1 | 1 | 1 | | | | | | | | | | | | | | | | | | | | | |
| Li (2016) | Focus group | 5 | **** | 1 | 1 | 1 | 1 | 1 | 1 | 1 | | | | | | | | | | | | | | | | | | | | | |
| Rimi (2016) | ethnography | 3 | *** | 1 | 1 | 1 | 0 | 1 | 1 | 0 | | | | | | | | | | | | | | | | | | | | | |
| Rogers (2009) | Deliberative forum | 3 | *** | 1 | 1 | 1 | 1 | 1 | 0 | 1 | | | | | | | | | | | | | | | | | | | | | |

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|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Prati (2011) | Cross-sectional survey | 3 | *** | 1 | 1 | | | | | | | | | | | | | | | | | 1 | 0 | 1 | 0 | 1 | | | | | |
| Updegraff (2011) | Experimental design with 3 conditions | 4 | **** | 1 | 1 | | | | | | 1 | 1 | 1 | 0 | 1 | | | | | | | | | | | | | | | | |
| Lyu (2012) | Qualitative Interviews | 3 | *** | 1 | 1 | 1 | 0 | 1 | 1 | 0 | | | | | | | | | | | | | | | | | | | | | |
| Holmes (2009) | Semi-structured interviews | 5 | **** | 1 | 1 | 1 | 1 | 1 | 1 | 1 | | | | | | | | | | | | | | | | | | | | | |
| Johnson (2015) | Experimental – 2 studies (both studies) | 3 | *** | 1 | 1 | | | | | | 1 | 1 | 1 | 0 | 0 | | | | | | | | | | | | | | | | |
| Seltzer (2015) | Content analysis | 3 | *** | 1 | 1 | 1 | 0 | 1 | 0 | 0 | | | | | | | | | | | | | | | | | | | | | |
| Crosier (2015) | Qualitative Interviews | 4 | **** | 1 | 1 | 1 | 1 | 1 | 1 | 0 | | | | | | | | | | | | | | | | | | | | | |
| Goldberg (2015) | Report | | n/a | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| Rim (2014) | Content analysis +correlation with survey results | 1 | * | 1 | 1 | 0 | 0 | 1 | 0 | 0 | | | | | | | | | | | | | | | | | | | | | |
| Mitchell (2014) | Focus groups and interviews | 5 | **** | 1 | 1 | 1 | 1 | 1 | 1 | 1 | | | | | | | | | | | | | | | | | | | | | |
| Bonwitt (2018) | Experimental | 4 | **** | 1 | 1 | 1 | 0 | 1 | 1 | 1 | | | | | | | | | | | | | | | | | | | | | |
| Bekalu (2017) | Pre/post intervention study | 4 | **** | 1 | 1 | | | | | | 1 | 1 | 1 | 0 | 1 | | | | | | | | | | | | | | | | |
| Roess (2017) | Pre-post interventional study | 3 | *** | 1 | 1 | 0 | 0 | 1 | 1 | 1 | | | | | | | | | | | | | | | | | | | | | |
| Avery (2009) | Content analysis | 5 | **** | 1 | 1 | 1 | 1 | 1 | 1 | 1 | | | | | | | | | | | | | | | | | | | | | |
| Crouse Quinn (2008) | Commentary | | n/a | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| Berry (2007) | Content analysis | 4 | **** | 1 | 1 | 1 | 1 | 1 | 1 | 0 | | | | | | | | | | | | | | | | | | | | | |
| Menon (2005) | Narrative review | | n/a | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| Idoiaga (2016) | experimental | 2 | ** | 1 | 1 | | | | | | 0 | 0 | 1 | 0 | 1 | | | | | | | | | | | | | | | | |

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|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Cole & Watkins (2015) | interviews | 2 | ** | 1 | 1 | 1 | 1 | 0 | 0 | 0 | | | | | | | | | | | | | | | | | | | |
| Vaughan (2009) | Narrative review | | n/a | | | | | | | | | | | | | | | | | | | | | | | | | | |
| Van der Weerd (2011) | Cross-sectional telephone survey | 5 | **** | 1 | 1 | | | | | | | | | | | | | | | 1 | 1 | 1 | 1 | 1 | | | | | |
| Menon (2006) | Narrative review/commentary | | n/a | | | | | | | | | | | | | | | | | | | | | | | | | | |
| Hoda (2016) | Survey | 2 | ** | 1 | 1 | | | | | | | | | | | | | | | 0 | 0 | 1 | 0 | 1 | | | | | |

Data Quality check of Systematic Reviews with AMSTAR

| **Citation** | **1** | **2** | **3** | **4** | **5** | **6** | **7** | **8** | **9** | **10** | **11** |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Schiavo et al | 1 | 1 | 1 | 1 | 1 | 0 | 1 | 1 | 1 | 0 | 1 |
| Lin et al | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 0 | 1 |
| Barrelet et al | 1 | 1 | 0 | 1 | 1 | 0 | 0 | 0 | 1 | 0 | 1 |

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## Appendix 6: Influences on effective public-health messaging

Most of the studies were qualitative work and reported on determinants of intent to adhering to guidelines by reporting what influences people's responses to public health messaging. These are: preconceptions and understanding of the threat, perceived susceptibility and perceived risk severity (threat appraisal(Rogers, 1975)).

Table: Recommendation cross-referenced to relevant evidence in synthesis

| Recommendation | Cross-reference to narrative synthesis |
|---|---|
| **1. Engaging with key stakeholders and communities** | |
| 1.a. involve community leaders and others perceived as credible sources within the community | Community engagement<br><br>messages for sub-populations<br><br>increasing trust |
| 1.b. Tailoring help make the key messages applicable to an individual's situation | |
| 1.c. Consider any difficulties accessing information and levels of literacy | |
| 1.d. Use different media for delivery and match delivery to the population's needs and perceptions | |
| **2. Addressing uncertainty immediately with transparency** | |
| 2.a. Address uncertainty and changing information that may exist during an ongoing public-health crisis | Increase trust<br><br>Preconceptions and understanding threat<br><br>Timing – beginning of health-crisis |
| 2.b. Consistency and co-ordination between different sources of information | |
| 2.c. Be transparent: admit errors and unknowns whenever appropriate | |
| 2.d. Be transparent: identify sources of information | |
| **3. Unified messages** | |
| 3.a. make core messages consistent | Increase trust<br><br>Threat appraisal<br><br>Preconceptions and understanding threat |
| 3.b. identify inconsistencies across sources | |
| 3.c. increase awareness of the risks of the virus to their own health and the health of others | |
| **4. Message framing** | |
| 4.a. increase understanding of health threat | Preconceptions and understanding threat<br><br>Threat appraisal<br><br>Community engagement |
| 4.b. to consider social responsibility | |
| 4.c. language choice to explain severity | |
| 4.d. promote sense of personal control | |

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*Preconceptions and understanding of the threat*

Of the 70 studies included, 26 specified that public health messaging included information about the threat which can be effective in identifying symptoms and changing behaviours. Some of these, including the only randomised control trial, reported changes in handing washing (Yardley, Miller, Schlotz, & Little, 2011), changes in knowledge and higher compliance to quarantine (Kavanagh et al., 2011). However, this particular study found the messages were not effective in encouraging quarantine behaviour suggesting that increasing knowledge alone did not change behaviour (Kavanagh et al., 2011). More comprehensive reporting on what the threat is and looks like, including presentation of known and unknown factors has a positive impact on people's knowledge, attitudes and beliefs. Self-reports, in the qualitative studies and in surveys, suggest changes in hygiene and unspecified protective behaviours following messages of how a virus (e.g. H1N1, SARS, Ebola) is transmitted, and the ability to idenitfy symptoms, and how to prevent transmision (Chidgzey, Davis, Williams, & Reeve, 2015; Gray et al., 2012; Hickey, Gagnon, & Jitthai, 2014; Hoda, 2016; R. Li, Xie, Yang, & Frost, 2016; Rim, Ha, & Kiousis, 2014; Teasdale & Yardley, 2011; Yardley et al., 2011). In the systematic reivew, public health messaging campaigns using posters (e.g. "*Catch it, Bin it, Kill it"*) increased knowledge about tramsission, and successfully increased self-reported prevention behaviours (hygiene when using tissues and sneezing). However, compliance to these behaviours was also predicted by worrry (Lin, Savoia, Agboola, & Viswanath, 2014). The systmatic review argued that individuals with higher education were better informed about H1N1. The review concluded that messages and communication are still delivered at a high literacy level (Lin, Savoia, et al., 2014). Whilst most of these papers highlighted that people will have different preconceptions and concerns to address in the public health messages, three papers also highlighted that people may have limited experiences of engaging in the preventative behaviour e.g. using face masks or using a thermometer (Einarsdottir, Passa, & Gunnlaugsson, 2001; Hickey et al., 2014; Swaddiwudhipong et al., 1992). Such cases highlighted a need for training/skill development as part of a public health campaign.

People that were able to identify symptoms of H1N1 were more likely to avoid crowds and washed hands more (Aburto et al., 2010). Framing and choice of language were found to be influential in behaviour and how individuals understand the threat (Teasdale & Yardley, 2011). Positively framed messages (emphasising the benefits of prevention behaviours) were considered 'robust contingency measures' (Basnyat & Lee, 2015). Gain-framed signs were significantly more effective in influencing the use of hand sanitizer as cue-to-action reminder signs at the point-of use during the H1N1 pandemic (Updegraff, Emanuel, Gallagher, & Steinman, 2011). The worst-performing signs emphasised people's susceptibility to contamination (Updegraff et al., 2011). Communication strategies were characterized in three case studies from the H1N1 pandemic from Italy, Hungary and England as "top-down" expert-led responses by public health communications specialists because there was no audience research. With no audience research, there was a perceived lack of planning and consideration to the tone, targeting or channelling of public health



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messages. Lack of consideration of these aspects led to low levels of public compliance and engagement with behaviour (Crosier, McVey, & French, 2015). A non-narrative message format (i.e., factual) may be more effective than a narrative counterpart to communicate basic prevention information during public health emergencies. Compared with the narrative and/or fictional version, the more didactic and factual format was found to be more effective in changing knowledge and perceived response efficacy related to prevention of pandemic influenza however, behaviour or intent were not measured in the study (Bekalu, Bigman, McCloud, Lin, & Viswanath, 2018). When messages provided the public with clear consistent information whilst admitting that information is evolving, reported the risks, and focused on the practical actions that people can do to reduce their risk and emphasized the efficacy of those actions, people were more likely to clean objects, wash hands and use tissues when sneezing (Prati, Pietrantoni, & Zani, 2011). A pre-print manuscript found that for one to perceive and understand the extent of a threat, messages that are framed in relation to identity (e.g. “don’t be a spreader” vs “don’t spread”) were more likely to predict preventative behaviour following the Japanese Ministry of Health guidelines (Yonemitsu, 2020).

When individuals do not understand the message or the preventative behaviour being communicated, participants did not outright reject public health messages. Instead, people adjusted the key underlining message to what they thought made sense, and engaged in behaviour consistent with the adjusted message (Bonwitt et al., 2018). As there is general uncertainty around a pandemic flu it is hard for people to differentiate this from seasonal flu especially when symptoms are framed as “flu-like symptoms” (Teasdale & Yardley, 2011). In a qualitative study, people reported being more likely to follow guidelines when there were fewer perceived barriers to perform the recommended behaviours, when there are emphasised benefits and when contextual factors are considered such as feeling guilty and anxious about missing work and not wanting to let people down (Teasdale & Yardley, 2011). High levels of knowledge was associated with exposure to messages and was a mediator between different sources and complying with recommended protective behaviour. However, engaging with social media was associated with misconceptions of public health information (Winters et al., 2018).

*Threat appraisal: perceived susceptibility/vulnerability*

Increase in exposure to any type of information was associated with increased knowledge, and as level of outbreak increased (that increased the level of threat appraisal), protective behaviours increased, suggesting a strong dose-response (Johnson & Slovic, 2015; Winters et al., 2018). As perceptions of higher risk increase so did self-reported behaviours such as handing washing, wearing a facemask and using hand gels (Aburto et al., 2010). However, salience (i.e. prominence) in media coverage did not have a direct relationship with increasing perceived risk. One study found that there was no relationship between salience



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in media coverage of H1N1, or the severity of content and the public perception of risk (Rim et al., 2014).

Being worried about H1N1 was associated with media attention and number of H1N1 cases, and overall an important predictor of compliance with recommended preventative behaviours, such as using tissues, hand gel and washing hands (Lin, Savoia, et al., 2014). Worry led to effective compliance and increased adaptive behaviour in one included study (Gutteling, Terpstra, & Kerstholt, 2018). Evoking fear increased participants' perceptions of the severity of the issue and their vulnerability to the risk but did not increase prevention behaviours or treatment efficacy (Chang, 2012). Moreover, panic could be counterproductive (Zikmund-Fisher, Scherer, Knaus, Das, & Fagerlin, 2017); providing a "worse-case" scenario (that there is a high risk and there is a pandemic) and increasing knowledge of the health threat simultaneously, has been found to induce panic in those receiving the message (Jhummon-Mahadnac, Knott, & Marshall, 2012). Believability of messages was reduced by scepticism towards the media and governments; in these instances the threat was perceived as exaggerated and able to cause unnecessary panic which could influence low adoption of recommended action (Teasdale & Yardley, 2011). In a preprint manuscript, messages arousing an emotional response (such as fear) was more effective in boosting willingness to self-isolate alongside prosocial messages that are producing strong and positive emotional response (Heffner, 2020).

Perceiving a threat would mean that population warning systems (method where local, regional or national authorities contact the public en masse) can work but social pressure in addition to increasing perceived risk is also required (Gutteling et al., 2018). Worry about self or family members led to increased perceptions of disease severity and a stronger belief of being susceptible to the infection, leading to a greater compliance with recommended behaviours (Lin, Savoia, et al., 2014). Perceived susceptibility in the messaging (e.g. "this can happen to someone like me") was more likely to lead to engagement in physical distancing than the geographical closeness of the disease (e.g. cases in the local area) thereby appraising the threat as high (Cole & Watkins, 2015). If risk is perceived to be low (e.g. H1N1 pandemic likened to influenza pandemics) recommendations such as "stay at home" were perceived as extreme and inappropriate (Teasdale & Yardley, 2011). This then has implications for increasing distrust in the messages and sources. Behaviours such as usage of a sanitizer dropped over time mapping the temporal trends in public interest in H1N1 (Updegraff et al., 2011).

Individuals were more likely to be concerned about the risk to themselves rather than the risk they would pose to others (M. D. Davis, Stephenson, Lohm, Waller, & Flowers, 2015). A recent systematic review concluded that beliefs about personal susceptibility to H1N1, and perceiving it as potentially severe (infection rate), is linked with high levels of worry about self or family members at risk (Lin, Savoia, et al., 2014). Efficient public health messaging (where individuals were more likely to comply with behaviour (e.g. avoiding crowds; (Aburto et al., 2010) tapped into worry about self or family members by framing messages with positive social responsibility (Basnyat & Lee, 2015). Concern for family led to reporting more



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information-seeking and appraising the threat as a higher risk to self and their family (Cole & Watkins, 2015). Prosocial framing and focus on activating social norms were found in the grey literature to be effective in increased engagement in physical distancing (Jordan, 2020; Wirz 2020). Tailoring public health messages with compassion may increase acceptance of the messages (Blagov, 2020). Such prosocial framing in the public health messaging personalised the risk and increasing the perceived susceptibility and vulnerability.

### Increasing trust

An important finding that 60% of the included studies identified the need to increase the public's trust in the messages during pandemics and epidemics (Aburto et al., 2010; Cole & Watkins, 2015; Gray et al., 2012; Hickey et al., 2014; Holmes, Henrich, Hancock, & Lestou, 2009; Jhummon-Mahadnac et al., 2012; Johnson & Slovic, 2015; Kavanagh et al., 2011; Liao, Cowling, Lam, Ng, & Fielding, 2010; Lohiniva, Sane, Sibenberg, Puumalainen, & Salminen, 2020; Lyu et al., 2013; Person et al., 2004; Prati et al., 2011; Qiu et al., 2018; Sumo et al., 2019). Participants cited trust as the main reason they complied, and observational studies found trust to be essential in complying with preventative behaviours (R. Li et al., 2016; Menon & Goh, 2005), both when communicating public health advice from public health agencies (Li et al., 2016) or from the government (Menon & Goh, 2005). Arguably, the need to increase trust highlights that acceptability of the message is reliant on the right message given in the right way (framing cross-reference), by an appropriate messenger (cross-reference credible source) at the right time.

One of the systematic reviews, concluded that competing narratives should be addressed to support trust-building, and this can be achieved through involving the public with two-way communications in the design of the messages (Barrelet, Bourrier, Burton-Jeangros, & Schindler, 2013). Whilst the other systematic review, found that an honest reporting of what the threat looks like, through a presentation of known and unknown factors, seems to have a better impact on people's knowledge, attitudes and beliefs, including trust in the way the government is handling the emergency (Lin, Savoia, et al., 2014).

In a repeated measures survey during H1N1 pandemic, trust levels in the government decreased over time, higher levels of government trust led to higher intention to adhere to the guidelines and change in trust was due to conflicting messages at the same time points and as they change over time (van der Weerd, Timmermans, Beaujean, Oudhoff, & van Steenbergen, 2011). Anxiety was mitigated by messages that were prompt and delivered at the right time for the point of the pandemic/epidemic, transparent, recommended preventative behaviours, and provided evolving outbreak trends (i.e. giving information as it happens) (R. Li et al., 2016). People are more likely to engage in behaviour inconsistent with the message if messages lack clarity or there is inconsistent messaging both over time and between sources. This eventually results in individuals distrusting the message. Therefore, one paper argues besides information and excessive exposure of such information there is also a need for behavioural modelling (to see the behaviour by others) (O. L. Davis, Fante, &



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Jacobi, 2013). Information in messages needs to be relevant and trustworthy to influence behaviour change and can include: catastrophic potential, probability of dying, and reasons for exposure. Providing explanations for perceived differences from other countries, not being too optimistic and not hide information can increase overall trust towards authorities (Lohiniva et al., 2020). Attempting to increase knowledge in the context of low levels of public trust in the source can make a message ineffective, and mistrust can increase if the perception is that the information is exaggerated (Jhummon-Mahadnac et al., 2012; Teasdale & Yardley, 2011). When an epidemic (e.g. Ebola) seemed contained, and contact tracing was successful, trust increased and, in response, so did the adherence to messages (e.g. reporting to health facility) (Sumo et al., 2019).

*Multiple and credible sources*

Individuals seek information from multiple sources to meet different information needs (Cole & Watkins, 2015). Whilst official sources of information (such as governmental sources, or public health bodies) are seen as legitimate, individuals do not always find them useful (Toppenberg-Pejcic et al., 2019; Tully, Dalrymple, & Young, 2019) thereby seek other sources. When official sources take a long time to provide information, individuals rely on informal sources such as rumours, word of mouth, social media (Cole & Watkins, 2015; van der Weerd et al., 2011); the value of information is at its highest when the outbreak is emerging. If mass media do not report the outbreak in time (before unofficial/informal sources start reporting the outbreak) it fails to become the leading indicator that people use for their information (J. Li, Xu, Cuomo, Purushothaman, & Mackey, 2020). Delay can impact the public's trust in official sources (e.g. government) and has led to beliefs that the threat is exaggerated by government or news and media (J. Li et al., 2020; Teasdale & Yardley, 2011) especially when information is conflicting. Furthermore, a delay in addressing rumours circulating among the public increases the chances of apathy and communication fatigue (Mitchell et al., 2014).

The grey literature reported that individuals seek information from sources that they consider trustworthy these include public health experts, WHO and Centers for Disease Control and Prevention (CDC) and state and local governments; news media were the least trusted sources (Wirz 2020). Another preprint found that there was greater compliance and acceptance of messages when the messages came from government officials and credible sources (Abu-Akel, 2020). The effect on behaviour change (self-reported mask wearing and mask buying) from messaging was greater amongst individuals who trusted the Centers for Disease Control and Prevention (CDC), government agencies and scientists (M. H. Goldberg, Gustafson, A., Maibach, E.W., Ballew, M.T., Bergquist, P., Kotcher, J.E., Marlon, J.R., Rosenthal, S.A., Leiserowitz, A.,, 2020) and these messages were better received (Merkley, 2020). Explicit information, data safety and transparency were important (Bradshaw 2020) especially when describing protective behaviours (Dai, 2020). Whilst individuals generally preferred expert sources for the initial information-seeking, they eventually relied on other



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sources (and perceptions such as “anti-intellectualism” increased); eventually leading to discontinuing use of official sources for information during a pandemic (Merkley, 2020).

### Community engagement and social networks

Social networks and close ties to the community are drivers of better knowledge and compliance with preventive measures. A systematic review suggests that non-traditional channels of communication (i.e. partnership with community leaders or organizations) should be used to reach out to the most vulnerable (can include those who have a disability e.g. hearing/vision/intellectual) and those who are least literate as well as non-Native speakers. Public health communication messages are still delivered at a literacy level that does not meet the needs of the less educated. To reduce communication inequalities, those who are less likely to identify risk associated with the threat and so less likely to adhere to the behaviours should be targeted. These include the young, least educated and hard to reach communities (Lin, Savoia, et al., 2014). The public pay more attention if the community is engaged in the intervention (Schiavo, May Leung, & Brown, 2014) as communication is a dynamic process and to address the concerns, values, interests and priorities the public need to be partners in communication (Barrelet et al., 2013).

The inclusion of various communities increases the chances of developing materials with which all individuals can identify (Crosier et al., 2015). This in turn increases the chances of perceiving the message is relevant to their self and important people in their lives thus heightening risk perception, and changing behaviours (Crouse Quinn, 2008; Gray et al., 2012; Person et al., 2004). Individuals want information that fits with their experiences (Freiman et al., 2011; Teasdale & Yardley, 2011). Adaptable and personalised information, that is context-driven, is more effective in changing determinants of behaviour (e.g. especially in vulnerable groups) (Daellenbach, Parkinson, & Krisjanous, 2018; Qiu et al., 2018). Core messages should be consistent to ensure trust but also considerations need to be made to ensure the involvement of communities to address their needs (e.g. including community leaders, translations, access to resources). Message delivery should be appropriate for the targeted population; for example, social media can open new avenues but should not to other types of delivery (Toppenberg-Pejcic et al., 2019). There is a use for social media, as it can be effective in communicating messages (Lwin, Lu, Sheldenkar, & Schulz, 2018), fostering trust and providing opportunities for dialogue (Tully et al., 2019). Social media can be especially useful for reaching young people who trust social media channels (R. Li et al., 2016). Social-economic status (SES) is related to information access and understanding; those of lower SES were less likely to use a website, and more likely to find public health messages from TV and Radio to be confusing and contradictory than those of higher SES (Aburto et al., 2010).

The grey literature highlighted that developing prosocial messages and identity, increased willingness to self-isolate especially when producing a strong, positive emotional response (Heffner, 2020). Another preprint found that messages should not just target those who are at risk but those who do not perceive themselves at risk from COVID-19 (e.g. young people



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and hard to reach groups). Messages must also address barriers to physical distancing and outline benefits of carrying them out, emphasising altruism through messaging from credible sources using social media (Wirz 2020). Another preprint found that prosocial messages were the most effective type of framing, tapping into people's sense of morality especially at the beginning of a pandemic. This changed over time and self-interest messaging were just as effective (Jordan, 2020).

## Messages for sub-populations

### Young (Student populations) and older adult populations

Three studies examined perceptions and experiences of H1N1 messages with student populations (Idoiaga, De Montes, & Valencia, 2016; Miczo, Danhour, Lester, & Bryant, 2013; Mitchell et al., 2014). An experimental study on framing found that younger students (mean age = 21.05) perceived themselves as less vulnerable than older students (over the age of 60) in falling ill with flu when reading a news report that described a flu epidemic (Idoiaga et al., 2016). The authors suggest this may be due to the 'social invulnerability identity' of younger people. This is consistent with social cognitive approaches to health risk perception, which suggests people may be "unrealistically optimistic" about a particular health threat (Weinstein, 1987).

Campus and mass media were reported, by students, as the most common places for getting information about the H1N1. Students tend to perceive information from the university as more credible than the media (Mitchell et al., 2014).

Focus groups exploring reactions to two different campaign ideas (for a potential avian influenza epidemic in Australia) found that there was some confusion amongst older adults whereby they presumed the 'WIPE' part of the WASH, WIPE, and WEAR' message was referring to handkerchiefs, rather than disposable tissues. Older adults (and separately mothers) predominantly preferred messages which emphasised protecting others (e.g. that they should wear a mask if they were sick in order to protect others) (Jones, Waters, Holland, Bevins, & Iverson, 2010). On the other hand, regional and frequent travellers' groups thought it was equally important to protect themselves from other people's germs, and also protect others from their germs (Jones et al., 2010).

### Vulnerable groups

The importance of both protecting themselves and others was evident in findings from focus groups conducted with vulnerable groups in New Zealand during the influenza H1N1 2009 pandemic (Gray et al., 2012). People wanted messages about specific actions that they could take to protect themselves and their families during the pandemic.

During the influenza H1N1 2009 pandemic, a cross-sectional survey was conducted with migrant workers (i.e., anyone who did not have Thai citizenship) to explore their perceptions



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of and ability to implement non-pharmaceutical interventions (NPIs; e.g., using face masks when sick) proposed by the WHO (Hickey et al., 2014). Certain groups of migrant workers were considered as a vulnerable group due to traditions in raising poultry and swine, poor personal hygiene and sanitation, low health literacy, limited access to healthcare, and high frequency of cross-border communication. Attitudes towards recommended NPIs were generally negative or ambivalent. Barriers included never having worn a face mask and not knowing the correct way to wear one. This highlights the need to educate vulnerable groups about performing NPIs during pandemics as migrants are at risk of propagating the spread of a pandemic virus. It is also important that public education campaigns can reach migrants living in remote and difficult to access areas, and to accommodate for the diverse cultural and linguistic needs.

The importance of ensuring that public health messages can reach different linguistic groups was also echoed by the Pacific Peoples' participants in the focus groups conducted by Gray et al. (Gray et al., 2012). Older people in the group are unlikely to understand English so messages should be communicated in the original languages. However, as noted by Person et al., (Person et al., 2004) this can be very difficult when an in-house translation service does not exist, and the rapidly evolving scientific evidence will challenge the turnaround time for developing, translating, and disseminating information. To help address this, the team prioritised key SARS documents for translation and back-translations conducted by professional translation services ensured accuracy. The translated information should be disseminated through culturally appropriate channels such as community visits, town hall meetings, and health and education and communication channels to complement mass media messages. This is similar to reflections of Singapore's experience in dealing with the SARS outbreak (Menon, 2006) where there was ongoing transparent dialogue with grassroots leaders and communities, and religious groups.

Social Economic Status (SES)

A cross-sectional survey with 1010 adult Italians during the influenza H1N1 2009 pandemic found that respondents who were women, and were facing economic hardship, were more likely to clean objects, wash their hands, and use tissues when sneezing (Prati et al., 2011).

A Mexican study, which explored the public's ability and willingness to adopt community mitigation efforts during the influenza H1N1 2009 pandemic, found that respondents from lower SES groups were more likely to report costs of masks being a barrier to their use (Aburto et al., 2010). Respondents from lower SES tertials were also more likely to report that mitigation recommendations were contradictory or confusing. The authors highlight that it is important for future campaigns to tailor the messaging to people of lower SES who are more likely to have lower literacy levels and reduced ability to interpret messages. Messages should be coordinated, consistent, and simplified for these groups.



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*Evaluation of messages at salient points in an epidemic/pandemic*

This presents the papers that had specific aims for the time of the study (either beginning, during, or post a pandemic/epidemic) and these are mapped to recommendations developed in this review. Evaluations collecting data at different time points over the course of a pandemic can tell us about any changes in patterns of public behaviour or perceptions as the situation evolves.

Table 1: Papers organised according to timepoints of epidemic/pandemic

| | Study | Aim of messages | Mapped to Recommendations | |
|---|---|---|---|---|
| **Beginning of epidemic/pandemic** | Yardley et al 2011°<br>Chang et al, 2012°<br>Teasdale et al, 2011°<br>Mitchell et al, 2014°<br>Kavanagh et al, 2011°<br>Freiman et al, 2001° | Increase knowledge and awareness | • Early announcements<br>• Consistency of messages<br>• Acknowledge uncertainties<br>• Timing of recommendations / announcements / alerts | **Addressing uncertainty and increasing trust** |
| | Aburto et al, 2010°<br>Hickey et al, 2014†<br>Roess et al, 2017†<br>Bonwitt et al, 2018†<br>Winters et al, 2018†<br>Lohiniva et al, 2020 ¥ | Empower use of behaviours to prevent spread of virus within the first 6 months | • Integrate community /community leaders in risk comms and into planning<br>• Social responsibility<br>• Tailor to increase accessibility | **Community engagement** |
| | | | • Different sources of information – ensure consistency<br>• Clear instructions<br>• Language to explain severity | **Message framing and unified messaging** |
| **During epidemic/pandemic/epidemic** | Jhummon-Mahadnac et al, 2010°<br>Gray et al, 2012°<br>Prati et al, 2011°<br>Updegraff et al, 2011° | Assess efficacy of messages/ campaigns | • Responding to concerns questions<br>• Skilled / credible spokespeople<br>• Admitting mistakes<br>• Acknowledge uncertainties<br>• Acknowledge unknowns and knowns<br>• Labelling recommendations as interim | **Addressing uncertainty and increasing trust** |
| | Rim et al, 2014°<br>Johnson & Slovic, 2015†<br>Toppenberg et al, 2018† | Update and possibly introduce new behaviours based on new information | | |
| | Tully et al, 2019†<br>Davis et al, 2013‡ | Continue with raising awareness and maintain behaviours | • Tailor information to users needs (accessibility)<br>• Use of local spokespeople and influencers<br>• Translate and use different sources with consistency in messaging | **Community engagement** |
| | Bekalu et al, 2017‡<br>Rimi et al 2016+<br>Avery et al, 2009+ | Manage misinformation | • Increase understanding of health threat<br>• Promote sense of personal control | **Message framing and unified messaging** |

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| | Study | Aim of messages | Mapped to Recommendations | |
|---|---|---|---|---|
| | | | • Social responsibility | |
| **Post epidemic/pandemic** | Davis et al, 2015°<br>Basnyat & lee, 2014°<br>Miczo et al, 2013°<br>Crosier et al 2015°<br>Barrelet et al 2013°<br>Person et al, 2004•<br>Qui et al, 2017⁺•<br>Lyu et al, 2012•<br>Li et al 2016⁺<br>Sell et al 2017†<br>Sumo et al 2019†<br>Cole & Watkins, 2015† | Assess efficacy of messages/ campaigns<br>Manage misinformation | • Responding to concerns questions<br>• Skilled / credible spokespeople<br>• Admitting mistakes<br>• Acknowledge uncertainties<br>• Acknowledge unknowns and knowns<br>• Public release of evaluations and reviews | **Addressing uncertainty and increasing trust** |
| | | Evaluation for lessons learnt and Preparation training | • Translate and use different sources with consistency in messaging<br>• Equity considerations | **Community engagement** |
| | | | • identify inconsistencies<br>• Core messages consistent<br>• understanding of health threat | **Message framing and unified messaging** |

Note a: °Study at beginning, during or post H1N1; ¥ Study during COVID-19; †Study at the beginning, during or post Ebola; ‡Study at beginning, during or post influenza season; ⁺Study at beginning, during or post avian flu; • Study at beginning, during or post SARS

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Appendix 7: Rapid review findings mapped to the BPS COVID-19 Behavioural Science and Disease Prevention Psychological Guidance

| **Recommendation in review** | **BPS Behavioural Science guidance** |
|---|---|
| Address uncertainty | 2. Deliver messages from a credible source in relatable terms to the target audience |
| Engaging communities | 1. Minimise the 'I' and emphasise the 'we'<br><br>4. Identify what influences each preventative behaviour and ensure policies, messaging and interventions target all relevant drivers<br><br>6. Avoid unintended negative consequences |
| Unified messaging | 5. Clearly specify behaviours and their effectiveness<br><br>7. Create clear channels of access for health literacy<br><br>8. Use behavioural scientists and the psychological evidence base to support the COVID-19 response<br><br>9. Make a pledge to work together, through a multi-disciplinary approach. #CombatCovid19Together |
| Message framing | 3. Create worry but not fear |

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## AUTHORS

Professor Madelynne A. Arden is an HCPC registered Health Psychologist and a Professor of Health Psychology at Sheffield Hallam University, specialising in the development and evaluation of theory and evidence based behaviour change interventions.

Professor Christopher J. Armitage is an HCPC registered Health Psychologist, a Fellow of the Academy of Social Sciences and Professor of Health Psychology at the University of Manchester. He specialises in using psychological theory to develop tools to facilitate behaviour change among diverse populations.

Professor Lucie Byrne-Davis is an HCPC registered Health Psychologist, Principal Fellow of the Higher Education Academy, Fellow of the European Health Psychology Society and a Professor of Health Psychology at the University of Manchester, specialising in health professional behaviour change and global health.

Natalie Carr is a health psychology MSc student at Manchester Metropolitan University.

Professor Angel Chater is an HCPC registered Health Psychologist, a Professor in Health Psychology and Behaviour Change at the University of Bedfordshire and Leads the British Psychological Society's Division of Health Psychology and COVID-19 Behavioural Science and Disease Prevention Taskforce.

Dr Tracy Epton is a lecturer and researcher, at the University of Manchester, specialising in behaviour change.

Dr Daniela Ghio is a lecturer at the University of Salford, specialising in intervention development and health behaviour change.

Ella Guest is a Research Associate at the Centre for Appearance Research in Bristol and Stage 2 Trainee in final year at UWE, Bristol.

Professor Jo Hart is an HCPC registered Health Psychologist and a professor of health professional education at University of Manchester, specialising in health professional education and behaviour change.

Dr Sarah Hotham is a Senior Research Fellow and BPS Chartered Psychologist at the Centre for Health Services Studies, University of Kent specialising in applied heath research and training of healthcare professionals.



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Dr Neil Howlett is a Senior Research Fellow in the Department of Psychology and Sport Sciences, University of Hertfordshire, specialising in physical activity and wider behaviour change methodology.

Dr Elizabeth Jenkinson is a Senior Lecturer in Health Psychology and an HCPC registered Health Psychologist at the University of the West of England, Bristol.

Dr Angelos P. Kassianos is a Senior Research Fellow, a Fellow of Higher Education Academy and an HCPC registered Health Psychologist, at UCL (Department of Applied Health Research) specialising in disease prevention, medical education as well as developing and evaluating evidence-based behaviour change digital interventions.

Dr Chris Keyworth is a Research Fellow, at The University of Manchester, specialising in health communication and behaviour change.

Dr Sadie Lawes- Wickwar is a HCPC registered Health Psychologist. Associate Fellow of BPS & Member of DHP. Visiting Lecturer in Health Psychology at City, University of London.

Daniel Scanlan has a MSc in Health Psychology and works as a Research Assistant at Education Support.

Sabina Stanescu is a Health Psychology Trainee and a PhD Candidate within Psychology at the University of Southampton.

Dr Natalia Stanulewicz is a Lecturer in Psychology at De Montfort University. Her research focuses on health behaviours, wellbeing as well as emotional and individual characteristics predicting prosocial behaviours.

Lisa Sutherland - Behavioural Insight Consultant. MSc in Health Psychology 20 years working in digital, social marketing and behavioural change intervention design.

Dr Rachael Thorneloe is a Senior Research Fellow in the Centre for Behavioural Science and Applied Psychology, Sheffield Hallam University, specialising in behaviour change.

Dr Mei Yee Tang is a Research Associate at the Behavioural Science Policy Research Unit at Newcastle University.

Daniella Watson is a Trainee Health Psychologist & Global Health PhD student (University of Southampton).

Juliette Westbrook is a Research Assistant at the University of Reading and studying MSc Health Psychology at the University of Bath.

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Contribution Author List

Conceptualisation: DG, CK, TE, MYT, SLW, AC, SH, RT, CJA, MAA, JH, LBD

Study Design: TE, CK, MYT, SLW, DG

Screening of titles and abstracts (1st and 2nd) NH, EJ, SS, JW, APK, DW, LS, NS, EG, DS, NC, DG, MYT, SLW, CK

Conflicts resolved DG, MYT, SLW, CK

Development of guidance document DG, MYT, SLW, CK, TE, AC, SH, LBD

Pilot data extraction AG, SLW

Data extraction NH, EJ, SS, JW, APK, DW, LS, NS, EG, DS, NC, DG, MYT, SLW, CK

Quality check JW, SS, DG, MYT

Analysis DG, MYT, SLW, CK

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# PRISMA 2009 Checklist

| Section/topic | # | Checklist item | Reported on page # |
|---|---|---|---|
| **TITLE** | | | |
| Title | 1 | Identify the report as a systematic review, meta-analysis, or both. | 1 |
| **ABSTRACT** | | | |
| Structured summary | 2 | Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number. | 2 |
| **INTRODUCTION** | | | |
| Rationale | 3 | Describe the rationale for the review in the context of what is already known. | 3 |
| Objectives | 4 | Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS). | 3 |
| **METHODS** | | | |
| Protocol and registration | 5 | Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number. | 3 |
| Eligibility criteria | 6 | Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale. | 4 |
| Information sources | 7 | Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched. | 4 |
| Search | 8 | Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated. | S2-S3 |
| Study selection | 9 | State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis). | 4 |
| Data collection process | 10 | Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators. | 4 |
| Data items | 11 | List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made. | 4, S5 |
| Risk of bias in individual studies | 12 | Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis. | 4 |
| Summary measures | 13 | State the principal summary measures (e.g., risk ratio, difference in means). | N/A |
| Synthesis of results | 14 | Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., $I^2$) for each meta-analysis. | N/A |

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# PRISMA 2009 Checklist



| Section/topic | # | Checklist item | Reported on page # |
|---|---|---|---|
| Risk of bias across studies | 15 | Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies). | N/A |
| Additional analyses | 16 | Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified. | N/A |
| **RESULTS** | | | |
| Study selection | 17 | Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram. | 5 Figure 1 |
| Study characteristics | 18 | For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations. | S5 |
| Risk of bias within studies | 19 | Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12). | S41 |
| Results of individual studies | 20 | For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot. | N/A |
| Synthesis of results | 21 | Present results of each meta-analysis done, including confidence intervals and measures of consistency. | 5 |
| Risk of bias across studies | 22 | Present results of any assessment of risk of bias across studies (see Item 15). | 5 |
| Additional analysis | 23 | Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]). | 5 |
| **DISCUSSION** | | | |
| Summary of evidence | 24 | Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers). | 12 |
| Limitations | 25 | Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias). | 13 |
| Conclusions | 26 | Provide a general interpretation of the results in the context of other evidence, and implications for future research. | 13 |
| **FUNDING** | | | |
| Funding | 27 | Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review. | 4 |

*From:* Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(7): e1000097. doi:10.1371/journal.pmed1000097

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BMJ Open

# What influences people's responses to public health messages for managing risks and preventing infectious diseases? A rapid systematic review of the evidence and recommendations



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# What influences people's responses to public health messages for managing risks and preventing infectious diseases? A rapid systematic review of the evidence and recommendations

Daniela Ghio,[1] Sadie Lawes- Wickwar,[2] Mei Yee Tang,[3] Tracy Epton, [4] Neil Howlett, [5] Elizabeth Jenkinson,[6] Sabina Stanescu, [7] Juliette Westbrook, [8] Angelos P. Kassianos [9] Daniella Watson, [10] Lisa Sutherland, [11] Natalia Stanulewicz, [12] Ella Guest, [13] Daniel Scanlan, [14] Natalie Carr, [15] Angel Chater, [16] Sarah Hotham, [17] Rachael Thorneloe, [18] Christopher J Armitage [4,19] Madelynne A Arden, [18] Jo Hart, [4] Lucie Byrne-Davis, [4] Chris Keyworth [4]

As part of the Health Psychology Exchange [healthpsychologyexchange@outlook.com]

Correspondence to [tracy.epton@manchester.ac.uk]

[1] Department of Psychology, Faculty of Health and Society, University of Salford, Manchester, United Kingdom

[2] Department of Primary Care and Population Health, University College London, London, United Kingdom

[3] Behavioural Science Policy Research Unit, Population Health Sciences, Newcastle University, United Kingdom

[4] Manchester Centre for Health Psychology, University of Manchester, Manchester

[5] Department of Psychology, Sports, and Geography, School of Life and Medical Sciences, University of Hertfordshire, Hertfordshire, United Kingdom.

[6] Department of Health and Social Sciences, University of West England, Bristol, United Kingdom

[7] School of Psychology, University of Southampton, Southampton, United Kingdom

[8] Department of Psychology, University of Bath, Bath, United Kingdom

[9] Department of Applied Health Research, University College London, London, United Kingdom

[10] Human Development and Health, Faculty of Medicine, University of Southampton, United Kingdom

[11] Behavioural Insight, United Kingdom

[12] Faculty of Health and Life Sciences, School of Applied Social Sciences, De Montfort University, Leicester, United Kingdom

[13] Centre for Appearance Research, Department of Health and Social Sciences, Faculty of Health and Applied Sciences, University of the West of England, Bristol, United Kingdom

[14] Department of Communication, Policy, and Research, Education Support, London, UK

[15] Faculty of Health, Psychology, and Social Care, Manchester Metropolitan University, Manchester, United Kingdom

[16] Centre for Health, Wellbeing and Behaviour Change, University of Bedfordshire, Bedfordshire, United Kingdom

[17] Centre for Health Services Studies, University of Kent, Canterbury, United Kingdom

[18] Centre for Behavioural Science and Applied Psychology, Sheffield Hallam University, Sheffield, United Kingdom

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[19]Manchester University NHS Foundation Trust, Manchester Academic Health Science Centre, Manchester; and NIHR Greater Manchester Patient Safety Translational Research Centre, University of Manchester, Manchester, United Kingdom

**Abstract**

***Background***

Individual behaviour changes, such as hand hygiene and physical distancing, are required on a population scale to reduce transmission of infectious diseases such as COVID-19. However, little is known about effective methods of communicating risk reducing information, and how populations might respond.

***Objective***

To synthesise evidence relating to what: a) characterises effective public health messages for managing risk and preventing infectious disease, b) influences people's responses to messages.

***Design***

A rapid systematic review was conducted. Protocol is published on Prospero CRD42020188704.

***Data sources***

Electronic databases were searched: Ovid Medline, Ovid PsycINFO and Healthevidence.org, and grey literature (PsyarXiv, OSF Preprints) up to May 2020.

***Study selection***

All study designs were included that: (a) evaluated public health messaging interventions targeted at adults, (b) concerned a communicable disease spread via primary route of transmission of respiratory and/or touch. Outcomes included preventative behaviours, perceptions/awareness and intentions. Non-English language papers were excluded.

***Synthesis***

Due to high heterogeneity studies were synthesised narratively focusing on determinants of intentions in the absence of measured adherence/preventative behaviours. Themes were developed independently by two researchers and discussed within team to reach consensus. Recommendations were translated from narrative synthesis to provide evidence-based methods in providing effective messaging.

***Results***

Sixty-eight eligible papers were identified. Characteristics of effective messaging include delivery by credible sources, community engagement, increasing awareness/knowledge, mapping to stage of epidemic/pandemic. To influence intent effectively, public health



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messages need to be acceptable, increase understanding/perceptions of health threat and perceived susceptibility.

***Discussion***

There are four key recommendations: (1) engage communities in development of messaging, (2) address uncertainty immediately and with transparency, (3) focus on unifying messages from sources, and (4) frame messages aimed at increasing understanding, social responsibility and personal control. Embedding principles of behavioural science into public health messaging is an important step towards more effective health-risk communication during epidemics/pandemics.

**Strengths and Limitations**

- Whilst we conducted a rapid review, we ensured that we completed it in a systematic manner with a broad initial search (e.g. no restriction on study design) to develop recommendations from lessons in risk communication that we can translate to the current pandemic.
- The rapid review included all study designs with high heterogeneity, some of which were low quality, so findings should be interpreted tentatively.
- The focus of most the studies included in the review was on determinants of intention and not behaviour, therefore we acknowledge that the recommendations may not lead to successful enactment of target behaviours (e.g. handwashing) even though they may be helpful in increasing intentions.
- We were unable to conduct backward and forward citation searching on the included studies, this may have resulted in relevant literature not being captured.
- We had strong stakeholder engagement as part of the team with different expertise of behaviour science and public health that provided feedback from initial design through to development of recommendations to be used by public health practitioners.

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**Background**

The outbreak of novel infectious diseases, including COVID-19, require rapid changes to existing (e.g. related to physical distance) and new (e.g. use of face coverings) behaviours in the context of uncertainties and often rapidly-evolving new knowledge. Public health messaging is one component of effective risk communication strategies to ensure sustained population level behaviour change. However, little is known about what characterises effective public health messages for dealing with infectious diseases and what factors influence the public's response.

Drawing from the Behaviour Change Wheel (BCW) [1] system for intervention development, the "COM-B" model of behaviour change proposes that public health messages should be designed as multi-component strategies to support people's capability (the knowledge/skills), opportunity (societal norms/physical resources) and motivation (the desire/habit) to act; factors at the heart of the behaviour change wheel.[1] This theory has been used in current guidance to highlight core considerations for reducing the spread of COVID-19 [2], namely, increasing knowledge of personal protective behaviours (capability), signposting and making available resources (opportunity), and explaining why behaviour change is important, while mitigating emotional reactions (motivation).

When building public health campaigns or interventions, it is important to consider past evidence to build a behavioural diagnosis using COM-B; accepted modes of delivery; and an evaluation of outcomes. It is crucial to understand public health messaging in rapidly changing epidemics/pandemics and for this, a review of the evidence base is needed.

***Review aims***

To conduct a rapid systematic review[3] to identify and synthesise evidence in order to provide evidence-based recommendations for designing and delivering public health messages for health authorities and social care organisations dealing with infectious disease outbreaks. This review aims to identify:

a) What influences people's responses to public health messages about health-risk communication

b) What characterises effective public health messages for managing risk and preventing disease during epidemics/pandemics

## Methods

Rapid systematic review methods[3] the searches started in May 2020 and analysis was completed in July 2020. Public health messages were characterised as: messages delivered by mobile-technology, news broadcasts/newspapers, posters, leaflets and press conferences.

***Information sources***

The following databases were searched: Ovid PsycINFO, Ovid MEDLINE, PsyArXiv, OSF Preprints, and healthevidence.org (search strategies are presented in Appendix 1). The



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protocol for this rapid systematic review is published on Prospero CRD42020188704. The SPICE criteria (Setting, Perspective, Phenomena of Interest, Comparison, Evaluation, Time Scope), was used to guide data extraction.[4] Our search strategy was piloted with a scoping review to ensure that the terms were capturing all relevant literature and to also choose which databases to search. These terms were then shared within the team and with public health practitioners and behaviour science experts for feedback using an iterative process to finalise our search terms.

***Patient and Public Involvement***

This work was a rapid response to a request by Public Health Practitioners to the Health Psychology Exchange consortium. The Patient Public Involvement and Engagement group within the Health Psychology Exchange group were consulted when developing the protocol.

***Ethical Approval***

No ethical approval was sought to undertake this rapid systematic review.

***Eligibility criteria***

All study designs were considered for inclusion (e.g. systematic reviews, empirical studies) and grey literature (e.g. guidelines, frameworks, and policy documents) with no date restrictions until 20 May 2020.

Papers were included if they:

- evaluated a public health messaging intervention targeted at adults aged 18 years and above (no limitations on population or region),
- concerned a communicable disease spread via primary route of transmission of respiratory and/or touch (human to human contact)
- were written in English.

Papers on HIV were excluded as they involved different preventative behaviours and therefore deemed to be out of scope of the review. Papers that focused exclusively on public health messaging for vaccination uptake (intentions and uptake) during epidemic/pandemics were noted and the findings synthesised in a separate review.[5]

To ensure a broad range of literature relating to epidemics/pandemics/ health crisis communication could be captured studies were not excluded based on outcome. However, outcomes of interest included preventative behaviours (e.g. handwashing, quarantining), perceptions (e.g. risk), intent, and awareness.

***Study selection***

Titles/abstracts (80% double screened) and full texts were screened by 15 authors (Figure 1; for further breakdown of the included studies see Appendix 2 in supplementary materials). Conflicts over inclusion (2.3% had disagreements) were resolved through discussions with 4 authors (Appendix 3 provides detail on each author's roles in screening, extraction and synthesis).

***Data extraction***



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Characteristics of the papers (e.g. type of message, quality of study), the type of health risk and results were extracted (Appendix 4). Four authors (JW, SS, NC, DS) screened and completed a data quality check using Mixed Methods Appraisal Tool[6] for the 54 individual papers, the 11 pre-prints and AMSTAR[7] for the systematic reviews (Appendix 5). Overall, there was a moderate agreement level between the reviewers with 61% level of agreement. Disagreements were resolved through discussion with moderators.

***Synthesis of results***

A narrative synthesis was conducted[8] to identify key themes with respect to: (a) what influenced people's responses to public health messages in general and for subpopulations in particular during salient time points (further details presented in Appendix 6) and (b) interpreted recommendations for effective public health messaging for managing risk and preventing disease during epidemics/pandemics, which are presented below. The synthesis involved combining the results of reviews and individual studies reporting (a) qualitative studies, (b) quantitative studies, and (c) both qualitative and quantitative studies, in order to describe the recommendations for effective delivery of public health messages. These were exported into NVivo (Version 12) to data manage the combined results of different papers. To establish trustworthiness in data analysis, discussions among several members of the study team were held at fortnightly intervals to develop the coding framework, and to discuss, refine, and group the emerging codes into overall explanatory themes. All study authors were involved in establishing the conceptual framework.

[insert figure 1 PRISMA Diagram]

## Results

A total of 68 papers rated as high to moderate-quality (about 50% of them scoring as high-quality and 32% of moderate-quality) were included: 3 systematic reviews, 54 individual peer-reviewed papers and 11 preprints. The papers focused mainly on Influenza A virus subtype H1N1 (n=20), Covid-19 (n=15) and Ebola (n=11), other diseases (n= 12) which have emerged at different timepoints in the last 50 years. The timelines from initial outbreaks are highlighted in Figure 2. The included studies were conducted at various time points (e.g., beginning, during or post) during these pandemics and eleven did not report the timing (Appendix 6). Key variables included: (a) behaviours (e.g. hand-washing, quarantine, using tissues, physical-distancing), (b) cognitive factors (e.g. increase in awareness, perceived risk), and (c) emotions (worry, anxiety). (See Appendix 5 for full characteristics of papers).

[insert figure 2 here Timelines of pandemics/epidemics]

A narrative analysis of the papers was conducted on what was mostly qualitative work that reported on determinants of intent to adhere to guidelines. These were organised according to preconceptions and understanding of the threat, perceived susceptibility, and perceived risk severity (threat appraisal). This narrative analysis is presented in



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Appendix 6. Across the different themes and sub-themes developed about community engagement, messages for sub-populations, increasing trust, perceptions and understanding of threat and threat appraisal we developed four areas of recommendations to provide evidence-based steps to be taken to provide effective public health messaging during pandemics/epidemics. These recommendations are cross-referenced to the narrative synthesis in Table 1 and the recommendations are reported below with evidence summarised.

| Recommendation | Cross-reference to narrative synthesis in appendix 6 |
|---|---|
| **1. Engaging with key stakeholders and communities** | |
| 1.a. Involve community leaders and others perceived as credible sources within the community | Community engagement<br><br>Messages for sub-populations<br><br>Increase trust |
| 1.b. Tailoring helps to make the key messages applicable to an individual's situation | |
| 1.c. Consider any difficulties accessing information and levels of literacy | |
| 1.d. Use different media for delivery and match delivery to the population's needs and perceptions | |
| **2. Addressing uncertainty immediately with transparency** | |
| 2.a. Address uncertainty and changing information that may exist during an ongoing public health crisis | Increase trust<br><br>Preconceptions and understanding threat<br><br>Timing – beginning of health-crisis |
| 2.b. Consistency and co-ordination between different sources of information | |
| 2.c. Be transparent: admit errors and unknowns whenever appropriate | |
| 2.d. Be transparent: identify sources of information | |
| **3. Unified messages** | |
| 3.a. Make core messages consistent | Increase trust<br><br>Threat appraisal<br><br>Preconceptions and understanding threat |
| 3.b. Identify inconsistencies across sources | |
| 3.c. Increase awareness of the risks of the virus to their own health and the health of others | |
| **4. Message framing** | |
| 4.a. Increase understanding of health threat | Preconceptions and understanding threat<br><br>Threat appraisal |
| 4.b. Consider social responsibility | |
| 4.c. Language choice to explain severity | |



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| | |
|---|---|
| 4.d. Promote sense of personal control | Community engagement |

Table 1: Recommendations mapped and cross-referenced to Narrative Synthesis

## *Recommendations*

Four recommendations were derived from the evidence (See Appendix 6 for a comprehensive report of the evidence on influences on effective public health messaging, messages for sub-populations and communication at salient points in an epidemic/pandemic). Results were synthesised in Figure 3, including the recommendations and influences on behaviours. The recommendations are (1) engaging with different communities, (2) addressing uncertainty immediately, with transparency, (3) unified messaging, and (4) message framing.

[insert figure 3 Diagram of synthesis of recommendations]

### *1. Engaging with key stakeholders and communities*

**1a. Involve community leaders and others perceived as credible sources within the community**

One high-quality study indicated that messages designed without input from the target population may lead to low levels of public adherence to behaviour change messages. [9] A moderate-quality review found that this is especially important when dealing with inconsistencies and changing information. [10]

A high-quality systematic review found that the public pay more attention to messages if the community is engaged in its development [11] as this will ensure that the information is relatable,[9] and addresses the concerns, values, interests and priorities of the community.[10] Consequently, this may lead to heightened perceptions of personal risk. Two studies of low to moderate-quality, suggested that one way to do so is to include community leaders, to find people that are trusted[12] and allow faith-based organisations to help.[13] One moderate-quality study found that culturally and linguistically appropriate messages (e.g. in videos) can also help retain long-term knowledge of preventative behaviours.[14]

A preprint study stated that over time preferred expert sources (e.g. government websites) are displaced by unofficial sources (e.g. social media) for information regarding epidemic/pandemics;[15] therefore, developing ties within the community (e.g. trusted spokesperson) can be helpful to provide accurate information. A high-quality study found that students tend to perceive information from their university (from their own communities) as more credible than the media. [16] One low-quality study found that



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community engagement is also important for quickly disseminating messages which are translated into different languages.[17, 18]

**1b. Tailoring helps to make the key messages applicable to an individual's situation**

A high-quality systematic review found that those who are less likely to accept they are at personal risk of the threat (e.g. the young, least educated and hard to engage communities) are also less likely to adhere to the recommended behaviours.[19] This has important implications for reducing inequalities through tailored public health messaging. Social networks and having close ties to the community are drivers of better knowledge and behaviour change. Partnership with community leaders and/or community organizations should be used to reach out to the most vulnerable (this may include those who have a disability e.g. hearing/vision) and those who are least literate as well as non-Native speakers.[19]

Individuals desire information that fits with their experiences.[20-23] Studies of moderate-quality have shown that adaptable and personalised information, that is context-driven, is more effective in changing determinants of behaviour, especially in vulnerable groups.[24, 25] One moderate-quality study and two high-quality studies showed that messages should be tailored to take into account: (a) different levels of perceived susceptibility (e.g., younger adults see themselves as less vulnerable than older adults[26]), (b) likelihood of misunderstanding instructions (e.g., older people thought a campaign was referring to handkerchiefs rather than disposable tissues)[27] and (c) skills needed to enact the behaviour (e.g., migrant workers didn't know how to wear a face-covering).[28] One low-quality study found that there may be differences in message preferences (e.g., older adults and mothers preferred messages that emphasised the protection of others).[27]

**1c. Consider any difficulties accessing information and levels of literacy**

Messages are typically delivered at a high literacy level.[19] When individuals do not understand the message, they may engage in behaviours that reflect their understanding.[29] Some target groups, such as those with low literacy levels, could particularly benefit from a simplified, clear, and consistent message. [30] Pilot-testing messages can help identify phrasing that can be confusing or unfamiliar.[31]

It is also important to ensure public health messages reach non-native speakers by using translated materials as shown in a high-quality study.[32] Older people in non-native groups in particular may not understand English. Thus, messages should be communicated in the original languages and through generations. Translated information should be disseminated through culturally appropriate channels such as community visits, town hall meetings, and health and education and communication channels to complement mass



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media messages. This reinforces the need for community leaders (recommendation 1a) in aiding with translations and identifying the appropriate channels.

Some people have limited experiences of engaging in recommended behaviours (e.g. using face-coverings or a thermometer).[33] These limited experiences highlight a need for training/skill development to be included as part of a public health campaign. This will improve health literacy and self-efficacy, especially when it refers to 'new' behaviours. Including training/skill development fits in with taking a COM-B model approach in developing public health messaging as it increases an individual's physical/psychological capabilities. Benefits to taking this approach could be enhanced with equally improving motivation (reflective and automatic) by considering other recommendations (e.g. recommendation 4: Message framing) as well as considering opportunity for behaviour (social and physical) which can identify potential barriers (e.g. social norms).

**1d. Use different media for delivery and match delivery to the population's needs and perceptions**

Message delivery should be appropriate for the targeted population. For example, two studies of moderate to high-quality suggested that social media can reach younger people.[34, 35] A study of moderate-quality showed that social media can be effective in communicating messages fostering trust and providing opportunities for dialogue.[36] However, as highlighted in two studies of high-quality, there are disadvantages as (a) social media is also associated with misconceptions, such as what causes the disease[37] and (b) those without internet access (or a device) would be excluded.[38] A high-quality study found that those of lower socioeconomic status SES were less likely to use a website, and more likely to find public health messages from TV and Radio to be confusing and contradictory than those of higher SES.[30] In specific situations for example, messages through the use of posters in bathrooms to increase handwashing need to not just have prompts for the behaviour but also messages about transmission as a high-quality study found that prompts alone do not increase handwashing.[39] This is consistent with the Health Belief Model where cues of action can trigger behaviour but requires cognitive representations of perceived susceptibility and perceived barriers/costs to action.

## *2. Addressing uncertainty immediately and with transparency*

**2a. Address uncertainty and changing information that may exist during an ongoing public health crisis**



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Public health messages with emerging epidemics/pandemics are likely to involve much uncertainty about the virus and the appropriate preventative behaviours. A high-quality systematic review found that honest reporting about the threat, through a presentation of known and unknown factors, increases people's knowledge and makes attitudes and beliefs more positive; it also increases trust in the way the government is handling the emergency.[19]

As epidemics/pandemics develop the information and recommended behaviours may change. Changing information decreases trust in the government.[40] However, three studies of moderate to high-quality reported that prompt (i.e. giving information as it happens), stage appropriate (i.e., not seen as too lenient or too extreme for the risk level), and transparent messaging (i.e. that includes recommended preventative behaviours) reduces anxiety around the reported health risk.[41]

**2b. Consistency and co-ordination between different sources of information**

Different sources of information may give information and behaviour change instructions that are inconsistent with each other, especially as knowledge rapidly changes over time. Differences in the information and behaviour change instructions between countries should be explained. Information should be transparent, not be hidden and predictions should not be too optimistic – this will increase overall trust towards authorities.[42, 43]

**2c. Be transparent: admit errors and unknowns whenever appropriate**

A high-quality study showed that if uncertainties are not acknowledged or transparency is not provided, the consequences of the errors are addressed and this can reduce trust in agencies, thus reducing information acceptance and compliance with recommendations.[44]

Two studies of moderate to high-quality found that information needs to be released as early as possible at the start of the outbreak, whenever there is conflicting information and even if there are unknowns.[45, 46] Honest, open[25] and explicit information, transparency and assurances of personal data safety are important[47] especially when promoting preventative behaviours.[48] A low-quality study found that if official sources report the outbreak before unofficial/informal sources, then they become the leading indicator that people use for their information.[17] Delay can impact the public's trust in official sources that leads to beliefs that the threat is exaggerated by government or news media[22] and increases the chance of apathy and communication fatigue.[16]

**2d. Be transparent: identify sources of information**



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Two high-quality studies reported that attempting to increase knowledge in the context of low levels of public trust in the source can make a message ineffective; and mistrust can increase if the perception is that the information is exaggerated[22, 49] or if the outbreak is perceived as uncontrolled.[50] Three preprint studies reported that trust is key to the acceptance of messaging that can lead to behaviour change (e.g. using masks, physical distance), thus there is greater adherence when the messages come from trusted sources. [15, 51, 52] Sources that potentially can be perceived as credible by the general population can include public health experts, organisations (e.g. Centers for Disease Control and Prevention) and state and local governments.[53, 54] A study found that believability of messages was reduced by scepticism towards the media and governments; in these instances, the threat was perceived as exaggerated and able to cause unnecessary panic which could influence low adoption of recommended behaviours.[22] Therefore, it is important to ensure transparency highlighting that information sources are credible and legitimate.[55]

### *3. Unified messages*

**3a. Deliver consistent, clear, core messages about risk and preventative behaviour across sources within the same time points**

Where possible, messages should be unified across sources; as shown by two high-quality studies, this is especially important for those groups who have lower literacy (e.g., lower SES) as they are more likely to perceive messages as confusing and inconsistent.[30, 38] A moderate-quality review found that unifying messages result in a greater understanding of the health message, greater perceived risk and clear guidelines for behaviour;[10, 30] these all contribute to increased trust. A high-quality study found that a repeated measures survey during the H1N1 pandemic, showed trust levels in the government decreased over time and this decrease was due to conflicting messages at concurrent time-points.[40] In two moderate-quality studies it was found that if messages from different sources provide clear, consistent instructions these are more likely to be recalled and adhered to.[56] When messages provided the public with clear consistent information, whilst admitting that information is evolving, reported the risks, and focused on risk-reducing actions, people were more likely to perform protective behaviours (e.g., clean objects, wash hands and use tissues when sneezing).[57]

**3b. Identify inconsistencies in messages from uncontrolled sources, especially when addressing key preventative behaviours**

Whilst official sources of information (such as governmental sources, or public health bodies) are perceived as legitimate, individuals do not always find them useful as



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highlighted in one high-quality study and one moderate-quality study,[35, 36] and consequently seek other sources (e.g. unofficial sources such as family or social media). In a moderate-quality study, it was found that individuals seek information from multiple sources to meet different information needs.[45] Different unofficial sources of information may give information and behaviour change instructions that are inconsistent with each other and official sources. Additionally, as found in a high-quality study, unofficial sources may detract from understanding of preventative behaviours.[58] It is important to acknowledge and address the inconsistencies in unofficial sources.

When addressing inconsistency, a high-quality study found that it is also important to recognise that some messages may appear inconsistent due to varying levels of personal risk of different groups (conditional messages). To improve clarity these conditional messages should be identified and where possible explained or dispelled.[32]

**3c. Increase the public's awareness of the risks of the virus to their own health and the health of others**

Public health messaging that included information about the threat can be effective in identifying symptoms and changing behaviours. From six high-quality studies, it was found that the key aim of public health messages early in an epidemic/pandemic is typically to increase knowledge and awareness of the health-risk (see Appendix 6 for papers and priorities mapped onto timepoints).[16, 20, 22, 55, 59, 60] However, knowledge about the virus alone is not sufficient to change behaviour;[59] acknowledgement of unknown factors, how to identify symptoms, and how to prevent contracting and spreading the virus is also necessary as highlighted by five high-quality studies and a high-quality systematic review, two moderate-quality and one low-quality study. [19, 22, 32, 33, 41, 55, 61, 62] A low-quality study found that when perceived risk is low, ignoring recommended behaviours is rationalised.[63]

## *4. Message framing*

**4a. Increase factual knowledge of all aspects of a virus (e.g. symptoms) and benefits of preventative behaviour using an appropriate message frame**

Six high-quality studies found that framing and choice of language are influential in how individuals understand the threat and in turn behave. [22, 64-68] Positively framed messages (emphasising the benefits of prevention behaviours) may be effective.[64] For example, gain-framed signs (e.g., "stay healthy this season. Sanitize your hands") are more effective in influencing the use of hand sanitizer than signs that emphasised people's susceptibility to contamination.[65]



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Non-narrative messages (i.e., factual) are more effective than narrative messages (e.g. storytelling such as a movie *Contagion*) in changing knowledge and perceived response efficacy related to prevention of influenza.[68] Factual and scientific knowledge in messages in the media can positively influence risk perceptions.[66] Whilst formal information (presented from credible sources) increased understanding.[69]

**4b. Consider framing messages around social responsibility and norms**

Cohesive social networks and having close ties to the community are drivers of better knowledge and compliance with preventive measures. There is some moderate-quality evidence to suggest that framing messages about others' risk, in addition to your own risk, is effective in increasing information seeking.[45] In a high-quality systematic review, it was found that being worried (about self or family members at risk) was an important predictor of compliance with recommended preventative behaviours, such as using tissues, hand gel and washing hands.[19] The effects of worry about others at risk on compliance with preventative behaviours can potentially be amplified when combined with messages about being socially responsible. A high-quality study found framing messages with positive social responsibility to be useful for the public.[64]

The grey literature highlighted that developing prosocial messages and promoting positive emotional appeals, increased willingness to self-isolate especially when producing a strong, positive emotional response (such as fear).[70] Prosocial framing is effective in changing behaviour; this may be due to inducing compassion, activating social norms, altruism or moral duties.[54, 71]

**4c. Choice of language needs to be clear and appropriate to understanding the magnitude of risk**

A high-quality systematic review and a moderate-quality study, found that messages should sufficiently increase worry and perceived severity of risk to self and others[19, 45] so as highlighted in a high-quality study, recommendations are seen as proportionate[22] and behaviour change is more likely to occur.[19] However, in two high-quality studies, it was found that inducing too much fear has mixed results on behaviour: it can be counterproductive if this leads to panic.[49, 72]

There can also be confusion about the use of technical terms. A high-quality study found that it is difficult to differentiate between pandemic flu and seasonal flu especially when symptoms are framed as "flu-like symptoms." [22]

**4d. Frame the message to emphasise positive beliefs about one's own health and that preventative behaviour is within their control**



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A study of moderate-quality found that when messages provided the public with clear consistent information (whilst admitting that information and evidence is evolving), focused on the practical actions that people can take to reduce their risk and emphasized the efficacy of those actions, people were more likely to perform preventative behaviours.[57] This may also further empower them and help them become socially responsible as shown in a low-quality study.[73] People are more likely to follow guidelines when there are fewer perceived barriers to perform recommended behaviours, when benefits are emphasised, and when contextual factors (e.g., anxiety about missed work) are addressed.[22]

Recommendation of increasing self-efficacy through messaging is well supported in theoretical frameworks about behaviour change and risk communication for example, the Theory of Planned Behaviour[74], Health Belief Model[75], Protection Motivation Theory[76], the Health Action Process Approach[77], COM-B model[1] and Social Cognitive Theory[78, 79]. Increasing self-efficacy has positive implications on planning, intention possibly then behaviour. This review includes limited but high quality evidence that people want messages about specific actions that they could take to protect themselves and their families during the epidemic/pandemic[32] with messages that emphasise the benefits of prevention behaviours (positively framed) potentially effective in increasing uptake and compliance.[64]. informing the individual regarding preventative actions to stop the spread of the disease is particularly important in the early stages (see Appendix 6 for breakdown of timepoints). In a moderate-quality study, it was suggested that including non-health benefits may improve adherence to recommendations. [63]

## Discussion

The aim of this systematic review was to synthesis evidence-based recommendations for designing and delivering public health messaging. We found that inconsistent messages across different sources can be detrimental to any public health messaging campaigns,[80] which is particularly true in times of public health-crisis and uncertainty.[81] Miscommunication, or contradictions can result in lower compliance of preventative behaviours. Literacy is fundamental to the understanding and acceptance of any public health messaging campaign. Therefore, public health messaging should focus on clarity, simplicity, transparency and unified messaging even if tailoring to different groups regardless of level of literacy. Even when the message is understood, different populations have different barriers to address in order to change behaviour. A COM-B behavioural diagnosis[1] can help to inform public health strategies (e.g., lower SES groups find costs of face-coverings a barrier)[30] as suggested in recent British Psychological Society guidance (see appendix 7 to see how current recommendations map onto the guidance).[2] Our review suggests that unified messaging can increase trust[45] build community resilience[32], and increase perceived risk and knowledge of threat[49]. Furthermore, consistent messaging through different time points during an evolving



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pandemic are important to consider for context-specific recommendations; these were rarely considered or reported in the studies included in the review.

We found evidence of mode of delivery as an important consideration of public health messaging. Specifically, there should be careful consideration of how to communicate public health messages, which should target specific populations. For example, making use of social media platforms to target younger populations,[16, 34] or the radio as a delivery method for older populations.[20] Translations for different cultures, and the inclusion of community leaders as part of public health message campaigns, should also be prioritised. This would ensure that all populations are reached (and not stigmatized), and the knowledge, concerns, cultural values, interests, and priorities of the targeted populations are all considered.[82] However, as noted in one of the studies this can be very difficult when an in-house translation service does not exist, and the rapidly evolving scientific evidence will challenge the turnaround time for developing, translating, and disseminating information.[83]

***Limitations***

We acknowledge the present rapid systematic review has limitations. Although we searched multiple databases systematically it is possible that relevant research was excluded from this review since we did not have the resources to translate non-English language papers in such a short space of time or conduct backward and forward citation searching. The inclusion of unpublished literature in the review means some findings may change once these papers have been published in peer-reviewed journals. However, our inclusive approach means a range of types of messages and a variety of factors related to what influences public perceptions of messages have been identified and used to inform recommendations for messaging during current and future epidemic/pandemics.

The aim of this rapid review was to synthesise lessons learnt from previous epidemics/pandemics to provide evidence-based recommendations about what characteristics create effective messaging. The focus of most studies was on determinants of intent and not behaviour, which may have implications on successful enactment of target behaviours. As highlighted in theories (such as health action process approach[77]) intention formation is part of the process and key to planning and more work is needed to understand the translation into action. Inclusion of different infectious disease (although the messaging would be of similar behaviours) may have included different contextual influences that we could not account for when synthesising the data (e.g. different countries and different social norms or political influences). Furthermore, more work is needed to understand the moderating effects of individual differences on message acceptance.

Much of our evidence is consistent with components of relevant health behaviour models, (e.g., Protection Motivation Theory,[76] Health Action Process Approach,[77] and Health Belief Model[75]) with some studies reporting the use of the models in their design. These models state that people perform protective behaviours when they perceive (a)



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the health threat as sufficiently threatening (i.e., they are vulnerable to a severe risk), (b) the recommended behaviours to be effective, and (c) they have self-efficacy to perform the behaviour. We found evidence that such cognitive appraisals were important considerations for developing public health messages, but further research is needed to examine effects on behaviour change.

***Conclusions***

Increasing knowledge and awareness of health risks alone is unlikely to be sufficient to increase understanding of risk and subsequent risk reduction behaviours, and adherence to recommendations. Rather, taking a multifaceted approach to public health messaging which considers all relevant drivers of behaviour (social, psychological and environmental factors), is recommended. Our four key recommendations should be considered when designing and delivering public health messages: engage communities in the development of public health messaging, using credible and legitimate sources, address uncertainty immediately and with transparency, focus on unified messages from all sources, and develop messages aimed at increasing understanding, inducing social responsibility and empowering personal control. These are being translated into practical guidelines for agencies. Embedding these principles of behavioural science into public health messaging is an important step towards more effective health-risk communication.

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**Figure Captions:**

Figure 1: PRISMA Diagram

Figure 2: Timelines of pandemics/epidemics included in review

Note:1Centre for Disease Control & Prevention (CDC). Years of Ebola Virus Disease Outbreak. 2019. Available from: https://www.cdc.gov/vhf/ebola/history/chronology.html; 2.WHO. H5N1 avian influenza: Timeline of major events. 2012. Available from: https://www.who.int/influenza/human_animal_interface/H5N1_avian_influenza_update.pdf; 3CDC. CDC SARS Response Timeline. 2013. Available from: https://www.cdc.gov/about/history/sars/timeline.htm; 4CDC. 2009 H1N1 Pandemic Timeline. Available from: https://www.cdc.gov/flu/pandemic-resources/2009-pandemic-timeline.html; 5WHO. Rolling updates on coronavirus disease (COVID-19). 2020. Available from: https://www.who.int/emergencies/diseases/novel-coronavirus-2019/events-as-they-happen

Figure 3: Diagram of synthesis of recommendations and influences on behaviour



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## References


1. Michie S, van Stralen MM, West R. The behaviour change wheel: A new method for characterising and designing behaviour change interventions. Implement Sci. 2011;6(1):42.
2. Chater AA, M.A.; Armitage C.J., Byre-Davis, L.; Chadwick, P.; Drury, J.; Hart, J.; Lewis, L.; McBride, E.; Perriard-Abdoh, S.; Thompson, S.; Whittaker, E.; O'Connor, D.;. Behavioural science and disease prevention: psychological guidance: British Psychological Society; 2020 [
3. Dobbins M. Rapid review guidebook. National Collaborating Centre for Methods and Tools, Hamilton, ON. 2017.
4. Booth A. Clear and present questions: formulating questions for evidence based practice. Library Hi Tech. 2006;24(3):355-68.
5. Lawes-Wickwar SG, Daniela; Tang, Mei Yee; Keyworth, Chris; Stanescu, Sabina; Westbrook, Juliette; Jenkinson Elizabeth; Kassianos, Angelos, P.; Scanlan, Daniel; Garnett Natalie; Laidlaw, Lynn; Byrne-Davis, Lucie; Chater, Angel; Epton Tracy; and members of the Health Psychology Exchange (HPX) Public Health MEssaging Review Group A rapid review of responses to public health messages encouraging vaccination to prevent the spread of a pandemic or epidemic virus. in submission. 2020.
6. Hong QN, Gonzalez-Reyes A, Pluye P. Improving the usefulness of a tool for appraising the quality of qualitative, quantitative and mixed methods studies, the Mixed Methods Appraisal Tool (MMAT). Journal of evaluation in clinical practice. 2018;24(3):459-67.
7. Shea BJ, Grimshaw JM, Wells GA, Boers M, Andersson N, Hamel C, et al. Development of AMSTAR: a measurement tool to assess the methodological quality of systematic reviews. BMC Med Res Methodol. 2007;7(1):10.
8. Popay J, Roberts H, Sowden A, Petticrew M, Arai L, Rodgers M, et al. Guidance on the conduct of narrative synthesis in systematic reviews. A product from the ESRC methods programme Version. 2006;1:b92.
9. Crosier A, McVey D, French J. 'By failing to prepare you are preparing to fail': Lessons from the 2009 H1N1 'swine flu' pandemic. [References]: European Journal of Public Health. Vol.25(1), 2015, pp. 135-139.; 2015.
10. Barrelet C, Bourrier M, Burton-Jeangros C, Schindler M. Unresolved issues in risk communication research: the case of the H1N1 pandemic (2009-2011). Influenza other respi. 2013;7 Suppl 2:114-9.
11. Schiavo R, May Leung M, Brown M. Communicating risk and promoting disease mitigation measures in epidemics and emerging disease settings. Pathog Glob Health. 2014;108(2):76-94.
12. Lyu SY, Chen RY, Wang SF, Weng YL, Peng EY, Lee MB. Perception of spokespersons' performance and characteristics in crisis communication: experience of the 2003 severe acute respiratory syndrome outbreak in Taiwan. J Formos Med Assoc. 2013;112(10):600-7.
13. Santibanez S, Siegel V, O'Sullivan M, Lacson R, Jorstad C. Health communications and community mobilization during an Ebola response: partnerships with community and faith-based organizations. Public Health Rep. 2015;130(2):128-33.
14. Roess AA, Di Peppi R, Kinzoni EA, Molouania M, Kennedy E, Ibata SR, et al. Knowledge Gained and Retained from a Video-Centered, Community-Based Intervention for Ebola Prevention, Congo. J Health Commun. 2017;22(11):913-22.
15. Merkley E, Loewen PJ, Owen T, Ruths D. Anti-intellectualism and Information Preferences during the COVID-19 Pandemic. 2020.




For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

16. Mitchell T, Massoudi M, Swerdlow DL, Dee DL, Gould LH, Kutty PK, et al. Swine flu in college: early campus response to outbreak control measures. Am J Health Behav. 2014;38(3):448-64.
17. Menon KU. SARS revisited: managing "outbreaks" with "communications". Ann Acad Med Singapore. 2006;35(5):361-7.
18. Vaughan E, Tinker T. Effective health risk communication about pandemic influenza for vulnerable populations. Am J Public Health. 2009;99 Suppl 2:S324-32.
19. Lin L, Savoia E, Agboola F, Viswanath K. What have we learned about communication inequalities during the H1N1 pandemic: a systematic review of the literature. BMC Public Health. 2014;14:484.
20. Freiman AJ, Montgomery JP, Green JJ, Thomas DL, Kleiner AM, Boulton ML. Did H1N1 influenza prevention messages reach the vulnerable population along the Mississippi Gulf Coast? J Public Health Manag Pract. 2011;17(1):52-8.
21. Qian X, Ren R, Wang Y, Guo Y, Fang J, Wu ZD, et al. Fighting against the common enemy of COVID-19: a practice of building a community with a shared future for mankind. Infect. 2020;9(1):34.
22. Teasdale E, Yardley L. Understanding responses to government health recommendations: public perceptions of government advice for managing the H1N1 (swine flu) influenza pandemic. Patient Educ Couns. 2011;85(3):413-8.
23. Davis MD, Stephenson N, Lohm D, Waller E, Flowers P. Beyond resistance: social factors in the general public response to pandemic influenza. BMC Public Health. 2015;15:436.
24. Daellenbach K, Parkinson J, Krisjanous J. Just how prepared are you? An application of marketing segmentation and theory of planned behavior for disaster preparation. [References]: Journal of Nonprofit & Public Sector Marketing. Vol.30(4), 2018, pp. 413-443.; 2018.
25. Qiu W, Chu C, Hou X, Rutherford S, Zhu B, Tong Z, et al. A Comparison of China's Risk Communication in Response to SARS and H7N9 Using Principles Drawn From International Practice. Disaster med. 2018;12(5):587-98.
26. Idoiaga N, De Montes LG, Valencia J-F. Communication and representation of risk in health crises: The influence of framing and group identity. [References]: Revista de Psicologia Social. Vol.31(1), 2016, pp. 59-74.; 2016.
27. Jones SC, Waters L, Holland O, Bevins J, Iverson D. Developing pandemic communication strategies: Preparation without panic. [References]: Journal of Business Research. Vol.63(2), 2010, pp. 126-132.; 2010.
28. Hickey JE, Gagnon AJ, Jitthai N. Knowledge about pandemic influenza preparedness among vulnerable migrants in Thailand. Health Promot Internation. 2016;31(1):124-32.
29. Bonwitt J, Dawson M, Kandeh M, Ansumana R, Sahr F, Brown H, et al. Unintended consequences of the 'bushmeat ban' in West Africa during the 2013-2016 Ebola virus disease epidemic. Soc Sci Med. 2018;200:166-73.
30. Aburto NJ, Pevzner E, Lopez-Ridaura R, Rojas R, Lopez-Gatell H, Lazcano E, et al. Knowledge and adoption of community mitigation efforts in Mexico during the 2009 H1N1 pandemic. Am J Prev Med. 2010;39(5):395-402.
31. Lapka C, Jupka K, Wray RJ, Jacobsen H. Applying cognitive response testing in message development and pre-testing. [References]: Health Education Research. Vol.23(3), 2008, pp. 467-476.; 2008.
32. Gray L, MacDonald C, Mackie B, Paton D, Johnston D, Baker MG. Community responses to communication campaigns for influenza A (H1N1): a focus group study. BMC Public Health. 2012;12:205.
33. Hickey J, Gagnon AJ, Jitthai N. Pandemic preparedness: perceptions of vulnerable migrants in Thailand towards WHO-recommended non-pharmaceutical interventions: a cross-sectional study. BMC Public Health. 2014;14:665.



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34. Seltzer EK, Jean NS, Kramer-Golinkoff E, Asch DA, Merchant RM. The content of social media's shared images about Ebola: A retrospective study. [References]: Public Health. Vol.129(9), 2015, pp. 1273-1277.; 2015.
35. Toppenberg-Pejcic D, Noyes J, Allen T, Alexander N, Vanderford M, Gamhewage G. Emergency risk communication: Lessons learned from a rapid review of recent gray literature on Ebola, Zika, and yellow fever. [References]: Health Communication. Vol.34(4), 2019, pp. 437-455.; 2019.
36. Tully M, Dalrymple KE, Young R. Contextualizing nonprofits' use of links on Twitter during the West African Ebola virus epidemic. [References]: Communication Studies. Vol.70(3), 2019, pp. 313-331.; 2019.
37. Winters M, Jalloh MF, Sengeh P, Jalloh MB, Conteh L, Bunnell R, et al. Risk Communication and Ebola-Specific Knowledge and Behavior during 2014-2015 Outbreak, Sierra Leone. Emerg Infect Dis. 2018;24(2):336-44.
38. Avery EJ, Kim S. Anticipating or precipitating crisis? Health agencies may not be heeding best practice advice in avian flu press releases. [References]: Journal of Public Relations Research. Vol.21(2), 2009, pp. 187-197.; 2009.
39. Davis OL, Fante RM, Jacobi LL. The effectiveness of sign prompts to increase hand washing behaviors in restrooms. [References]: North American Journal of Psychology. Vol.15(3), 2013, pp. 565-576.; 2013.
40. van der Weerd W, Timmermans DR, Beaujean DJ, Oudhoff J, van Steenbergen JE. Monitoring the level of government trust, risk perception and intention of the general public to adopt protective measures during the influenza A (H1N1) pandemic in The Netherlands. BMC Public Health. 2011;11:575.
41. Li R, Xie R, Yang C, Frost M. Perceptions on the risk communication strategy during the 2013 avian influenza A/H7N9 outbreak in humans in China: a focus group study. Western pac. 2016;7(3):21-8.
42. Lohiniva AL, Sane J, Sibenberg K, Puumalainen T, Salminen M. Understanding coronavirus disease (COVID-19) risk perceptions among the public to enhance risk communication efforts: a practical approach for outbreaks, Finland, February 2020. Euro Surveill. 2020;25(13):04.
43. Rogers WA, Street JM, Braunack-Mayer AJ, Hiller JE, FluViews T. Pandemic influenza communication: views from a deliberative forum. Health Expect. 2009;12(3):331-42.
44. Holmes BJ, Henrich N, Hancock S, Lestou V. Communicating with the public during health crises: Experts' experiences and opinions. [References]: Journal of Risk Research. Vol.12(6), 2009, pp. 793-807.; 2009.
45. Cole J, Watkins C. International employees' concerns during serious disease outbreaks and the potential impact on business continuity: Lessons identified from the 2014-15 West African Ebola outbreak. J Bus Contin Emer Plan. 2015;9(2):149-62.
46. Johnson BB, Slovic P. Fearing or fearsome Ebola communication? Keeping the public in the dark about possible post-21-day symptoms and infectiousness could backfire. [References]: Health, Risk & Society. Vol.17(5-6), 2015, pp. 458-471.; 2015.
47. Bradshaw EL, Ryan R, Noetel M, Saeri AK, Slattery P, Grundy E, et al. Information safety assurances affect intentions to use COVID-19 contact tracing applications, regardless of autonomy-supportive or controlling message framing. 2020.
48. Dai B, Fu D, Meng G, Liu B, Li Q, Liu X. The effects of governmental and individual predictors on COVID-19 protective behaviors in China: a path analysis model. Public Administration Review. 2020.
49. Jhummon-Mahadnac ND, Knott J, Marshall C. A cross-sectional study of pandemic influenza health literacy and the effect of a public health campaign. BMC Res Notes. 2012;5:377.
50. Sumo J, George G, Weah V, Skrip L, Rude JM, Clement P, et al. Risk communication during disease outbreak response in post-Ebola Liberia: experiences in Sinoe and Grand Kru counties. Pan Afr Med J. 2019;33(Suppl 2):4.



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51. Abu-Akel A, Spitz A, West R. Who is listening? Spokesperson Effect on Communicating Social and Physical Distancing Measures During the COVID-19 Pandemic. 2020.
52. Goldberg M, Gustafson A, Maibach E, Ballew MT, Bergquist P, Kotcher J, et al. Mask-wearing increases after a government recommendation: A natural experiment in the US during the COVID-19 pandemic. 2020.
53. Berry TR, Wharf-Higgins J, Naylor PJ. SARS wars: an examination of the quantity and construction of health information in the news media. Health Commun. 2007;21(1):35-44.
54. Wirz CD, Schwakopf JM, Brossard D, Brown LD, Brauer M. Self-reported compliance and attitudes about social distancing during the COVID-19 outbreak Preprint.
55. Yardley L, Miller S, Schlotz W, Little P. Evaluation of a Web-based intervention to promote hand hygiene: exploratory randomized controlled trial. J Med Internet Res. 2011;13(4):e107.
56. Miczo N, Danhour E, Lester KE, Bryant J. Memorable messages and the H1N1 flu virus. [References]: Western Journal of Communication. Vol.77(5), 2013, pp. 625-644.; 2013.
57. Prati G, Pietrantoni L, Zani B. Compliance with recommendations for pandemic influenza H1N1 2009: the role of trust and personal beliefs. Health Educ Res. 2011;26(5):761-9.
58. McVernon J, Mason K, Petrony S, Nathan P, LaMontagne AD, Bentley R, et al. Recommendations for and compliance with social restrictions during implementation of school closures in the early phase of the influenza A (H1N1) 2009 outbreak in Melbourne, Australia. BMC Infect Dis. 2011;11:257.
59. Kavanagh AM, Bentley RJ, Mason KE, McVernon J, Petrony S, Fielding J, et al. Sources, perceived usefulness and understanding of information disseminated to families who entered home quarantine during the H1N1 pandemic in Victoria, Australia: a cross-sectional study. BMC Infect Dis. 2011;11:2.
60. Chang C. News coverage of health-related issues and its impacts on perceptions: Taiwan as an example. [References]: Health Communication. Vol.27(2), 2012, pp. 111-123.; 2012.
61. Hoda J. Identification of information types and sources by the public for promoting awareness of Middle East respiratory syndrome coronavirus in Saudi Arabia. Health Educ Res. 2016;31(1):12-23.
62. Rim H, Ha JH, Kiousis S. The evidence of compelling arguments in agenda building: Relationships among public information subsidies, media coverage, and risk perceptions during a pandemic outbreak. [References]: Journal of Communication Management. Vol.18(1), 2014, pp. 101-116.; 2014.
63. Rimi NA, Sultana R, Ishtiak-Ahmed K, Rahman MZ, Hasin M, Islam MS, et al. Understanding the failure of a behavior change intervention to reduce risk behaviors for avian influenza transmission among backyard poultry raisers in rural Bangladesh: a focused ethnography. BMC Public Health. 2016;16(1):858.
64. Basnyat I, Lee ST. Framing of Influenza A (H1N1) pandemic in a Singaporean newspaper. Health Promot Internation. 2015;30(4):942-53.
65. Updegraff JA, Emanuel AS, Gallagher KM, Steinman CT. Framing flu prevention-An experimental field test of signs promoting hand hygiene during the 2009-2010 H1N1 pandemic. [References]: Health Psychology. Vol.30(3), 2011, pp. 295-299.; 2011.
66. Sell TK, Boddie C, McGinty EE, Pollack K, Smith KC, Burke TA, et al. Media Messages and Perception of Risk for Ebola Virus Infection, United States. Emerg Infect Dis. 2017;23(1):108-11.
67. Liao Q, Cowling B, Lam WT, Ng MW, Fielding R. Situational awareness and health protective responses to pandemic influenza A (H1N1) in Hong Kong: a cross-sectional study. PLoS ONE. 2010;5(10):e13350.
68. Bekalu MA, Bigman CA, McCloud RF, Lin LK, Viswanath K. The relative persuasiveness of narrative versus non-narrative health messages in public health emergency communication: Evidence from a field experiment. Prev Med. 2018;111:284-90.



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69. Liao Q, Cowling B, Lam WT, Ng MW, Fielding R. Situational awareness and health protective responses to pandemic influenza a (H1N1) in Hong Kong: A cross-sectional study. [References]: PLoS ONE. Vol.5 2010, ArtID e13350.; 2010.
70. Heffner J, Vives ML, FeldmanHall O. Emotional responses to prosocial messages increase willingness to self-isolate during the COVID-19 pandemic. 2020.
71. Jordan J, Yoeli E, Rand D. Don't get it or don't spread it? Comparing self-interested versus prosocially framed COVID-19 prevention messaging. 2020.
72. Zikmund-Fisher BJ, Scherer AM, Knaus M, Das E, Fagerlin A. Discussion of Average versus Extreme Case Severity in Pandemic Risk Communications. Emerg Infect Dis. 2017;23(4):706-8.
73. Menon KU, Goh KT. Transparency and trust: Risk communications and the Singapore experience in managing SARS. [References]: Journal of Communication Management. Vol.9(4), 2005, pp. 375-383.; 2005.
74. Ajzen I. The theory of planned behavior. Organizational behavior and human decision processes. 1991;50(2):179-211.
75. Becker MH. The health belief model and personal health behavior. Health education monographs. 1974;2:324-473.
76. Rogers RW. A protection motivation theory of fear appeals and attitude change. The journal of psychology. 1975;91(1):93-114.
77. Schwartzer R. Self-efficacy in the adoption and maintenance of health behaviours: Theoretical approaches and a new model. Self-efficacy: Thought control of action Washington, DC: Hemisphere. 1992:217-43.
78. Bandura A. Social Cognitive Theory: An Agentic Perspective. Annual Review of Psychology. 2001;52(1):1-26.
79. Bandura A. Social cognitive theory of mass communication. Media psychology. 2001;3(3):265-99.
80. Hernandez-Garcia I, Gimenez-Julvez T. Assessment of Health Information About COVID-19 Prevention on the Internet: Infodemiological Study. JMIR Public Health Surveill. 2020;6(2):e18717.
81. Hooker C, Capon A, Leask J. Communicating about risk: strategies for situations where public concern is high but the risk is low. Public health res. 2017;27(1):15.
82. Lin L, Jung M, McCloud RF, Viswanath K. Media use and communication inequalities in a public health emergency: a case study of 2009-2010 pandemic influenza A virus subtype H1N1. Public Health Rep. 2014;129 Suppl 4:49-60.
83. Person B, Sy F, Holton K, Govert B, Liang A, National Center for Inectious Diseases SCOT. Fear and stigma: the epidemic within the SARS outbreak. Emerg Infect Dis. 2004;10(2):358-63.

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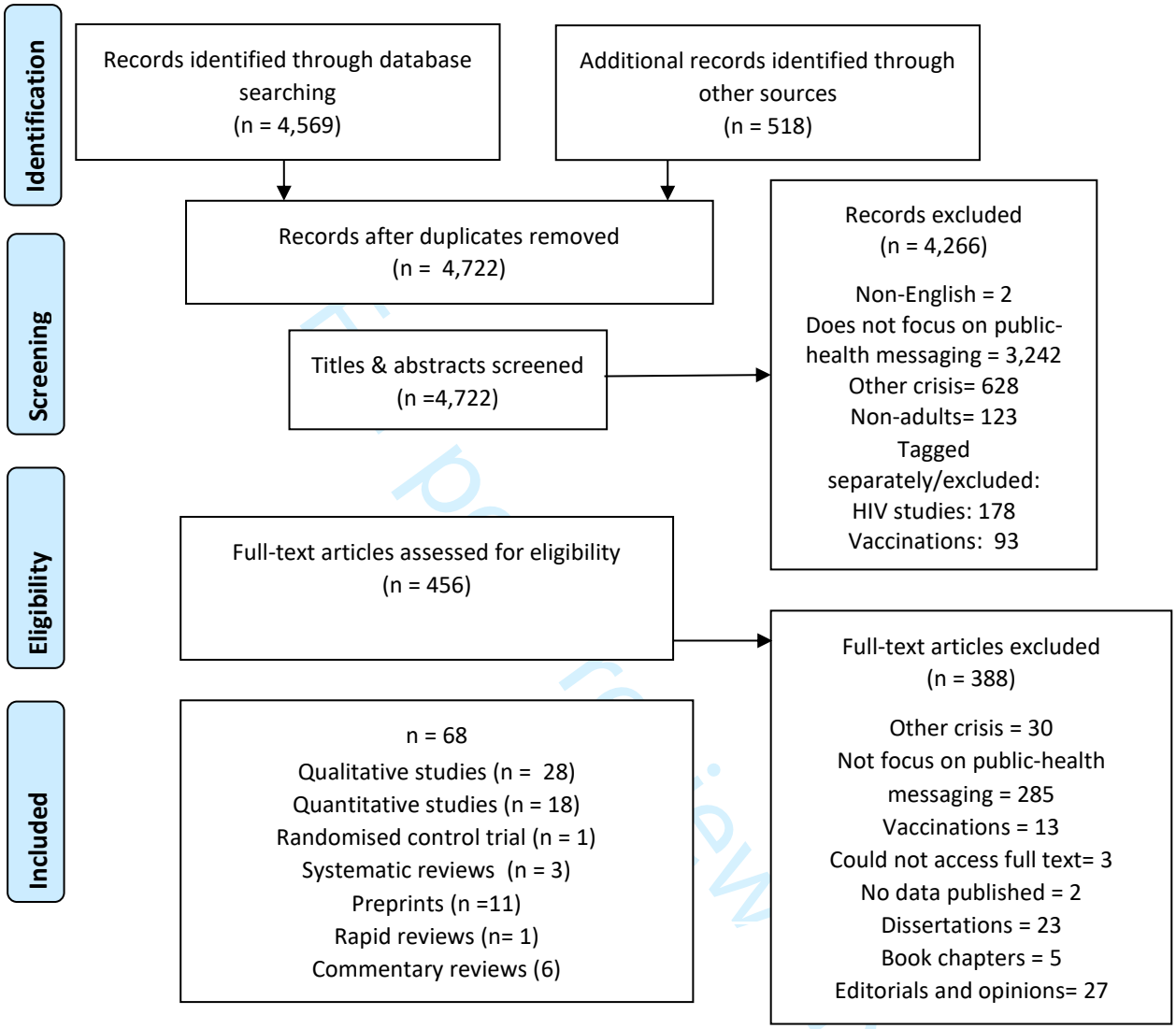



Figure 1: Prisma Diagram

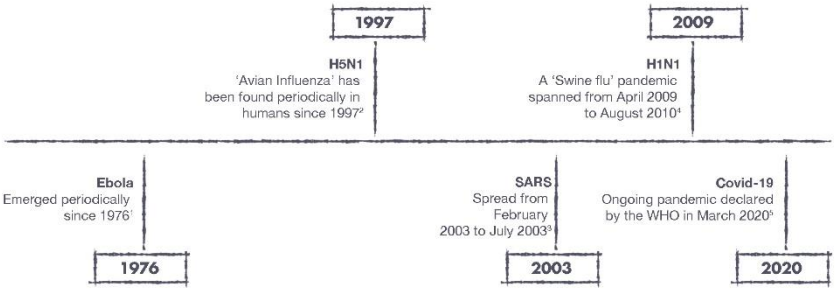



Figure 1: Timelines of pandemics/epidemics included in review

Note:[1]Centre for Disease Control & Prevention (CDC). Years of Ebola Virus Disease Outbreak. 2019. Available from: https://www.cdc.gov/vhf/ebola/history/chronology.html; [2]WHO. H5N1 avian influenza: Timeline of major events. 2012. Available from: https://www.who.int/influenza/human_animal_interface/H5N1_avian_influenza_update.pdf; [3]CDC. CDC SARS Response Timeline. 2013. Available from: https://www.cdc.gov/about/history/sars/timeline.htm; [4]CDC. 2009 H1N1 Pandemic Timeline. Available from: https://www.cdc.gov/flu/pandemic-resources/2009-pandemic-timeline.html; [5]WHO. Rolling updates on coronavirus disease (COVID-19). 2020. Available from: https://www.who.int/emergencies/diseases/novel-coronavirus-2019/events-as-they-happen

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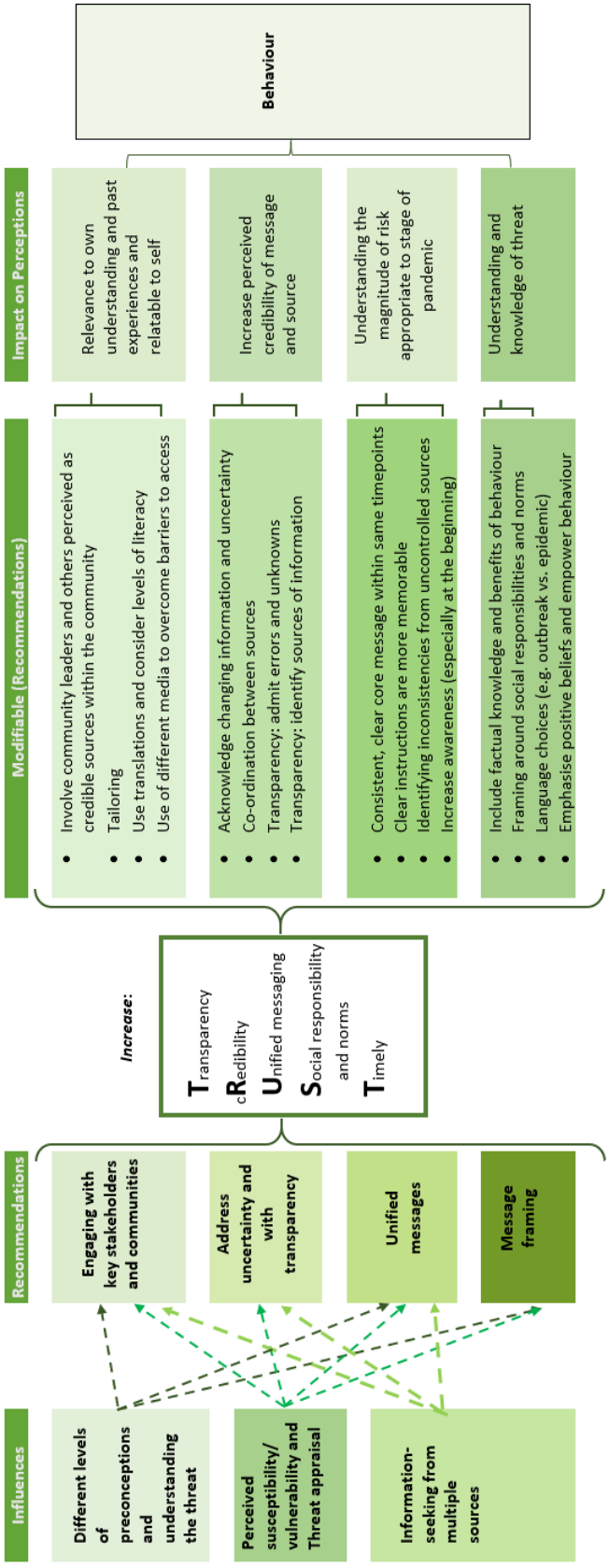



Figure 1: Diagram of synthesis of recommendations and influences on behaviour

# Supplementary Appendix

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# Appendix 1: Search Terms

*Ovid PsycINFO & Ovid MEDLINE*

| # | Search Terms | Results | Notes (e.g. where search terms were adapted from) |
|---|---|---|---|
| ***Health and risk communication terms*** | | | |
| 1 | public service announcements/ | 370 | |
| 2 | communication/ or messages/ or persuasive communication/ or scientific communication/ or social communication/ | 65306 | Imploded and selected a few narrower terms as the others were irrelevant |
| 3 | ((public health or health or risk or emergenc* or crisis* or catastroph* or disaster* or outbreak*) adj3 (communication* or campaign* or information* or plan* or message* or uncertaint* or alert* or awareness or recommendation* or guideline* or guidance or measure*)).tw. | 54841 | |
| 4 | 1 or 2 or 3 | 116708 | |
| ***Pandemic and epidemic terms*** | | | |
| 5 | exp Pandemics/ | 473 | |
| 6 | exp Epidemics/ | 3399 | |
| 7 | exp Influenza/ | 1396 | Includes Swine Flu, no MeSH terms for SARs, MERs, and Ebola so these searched as keywords only |
| 8 | exp Emergency Preparedness/ | 1192 | |
| 9 | ("SARS" or coronavirus or severe acute respiratory syndrome).tw. | 587 | |
| 10 | (ebola or ebolavirus or ebola virus).tw. | 411 | |
| 11 | ("MERS-CoV" or "MERS or Middle East Respiratory Syndrome coronavirus").tw. | 11 | |
| 12 | (influenza or swine flu or H1N1 or pH1N1* or pdmH1N1* or nH1N1*).tw. | 2490 | |
| 13 | (pandemic* or epidemic*).tw. | 14247 | |
| 14 | 5 or 6 or 7 or 8 or 9 or 10 or 11 or 12 or 13 | 17926 | |
| 15 | 4 and 14 | 1525 | |

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*Health Evidence (Reviews)*

[(Public health messag*) OR ("Risk communication") AND Pandemic OR Epidemic] (224 results)

*Grey Literature Searches*

OSF Preprints + PsyArXiv Preprints

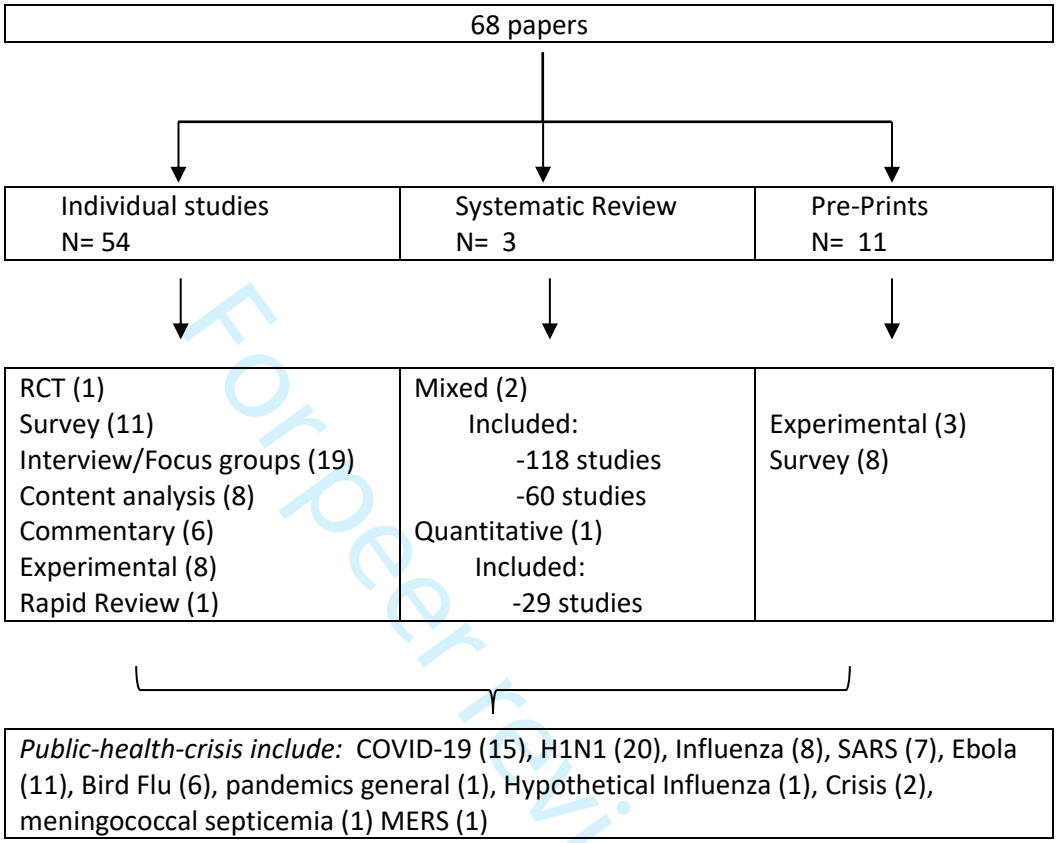
| **Search category** | **Search terms** |
|---|---|
| Intervention | (Public health messag* OR communication OR plan OR alert OR health campaign OR health recommendation* OR public information OR medical information OR public awareness OR community engagement OR risk reduction OR health promotion) |
| Phenomena of interest | (Pandemic OR catastrophe OR cris?s OR outbreak OR emergency OR Coronavirus OR Covid* OR SARS* OR ebola* OR MERS OR epidemic) |
| Combination | (Public health messag* OR communication OR plan OR alert OR health campaign OR health recommendation* OR public information OR medical information OR public awareness OR community engagement OR risk reduction OR health promotion) AND (Pandemic OR catastrophe OR cris?s OR outbreak OR emergency OR Coronavirus OR Covid* OR SARS* OR ebola* OR MERS OR epidemic) |
| Limitations | Arts & Humanities; Life Sciences; Medicine and Health Sciences; Physical Sciences and Mathematics; Social and Behavioral Sciences |
| Results | 518 |

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## Appendix 2: Breakdown of studies




## Appendix 3: screening, data extraction and synthesis details

| *Task* | *Authors* |
|---|---|
| *Screening of titles and abstracts (1st and 2nd)* | NH, EJ, SS, JW, APK, DW, LS, NS, EG, DS, NC, DG, MYT, SLW, CK |
| *Conflicts resolved* | DG, MYT, SLW, CK |
| *Development of guidance document* | DG, MYT, SLW, CK, TE, AC, SH, LBD |
| *Pilot data extraction* | AG, SLW |
| *Data extraction* | NH, EJ, SS, JW, APK, DW, LS, NS, EG, DS, NC, DG, MYT, SLW, CK |
| *Quality check* | JW, SS, DG, MYT, NC, DS |
| *Analysis* | DG, MYT, SLW, CK |
| *Iterations and discussions* | DG, MYT, SLW, CK, TE, NH, EJ, AC, SH, RT, CJA, MAA, JH, LBD |
| *Agreed on final categories* | DG, MYT, SLW, CK, TE, NH, EJ, AC, SH, RT, CJA, MAA, JH, LBD |



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Appendix 4: Data and study characteristics tables 1-4

**Table 1: Characteristics of Qualitative Studies**

| | Country | Study Type | Health Threat | Intervention type | Target behaviour | Main findings |
|---|---|---|---|---|---|---|
| **Avery (2009)** | USA | Content Analysis of avian flu press releases | Avian flu | Press releases issued by federal and global health agencies | Not reported | The threat was not localized; outlets for publics to contact for more information were not always present, and there was inconsistency in how avian flu was referenced. There was a steady increase in the volume of releases since 2004, and public health information officers generally seemed to recognize the importance of partnerships and timeliness in pandemic preparedness. Public health communicators must establish a strong sense of efficacy in an informed public to insure that audiences will adhere to protocol rather than succumb to hysteria in the face of a wide-scale crisis such as pandemic flu. They must also provide information accessible to all publics, including those who do not have Internet access. |
| **Basnyat & Lee (2014)** | Singapore | Qualitative | H1N1 pandemic | Singapore Ministry of Health press release in newspapers | N/A | The media did convey useful information for managing the pandemic from the Ministry of Health press release (i.e. social responsibility, benefits of lockdown) and these were often in the tone/ semantics of colonial-era ideologies that are present within the culture in Singapore and reflect that of the government. Beneficial for the government to provide more than basic information, such as about coping with anxiety, and consider how other types of media (e.g. social media, blogs) can be used too. |
| **Berry (2007)** | Canada | Content analysis of the construction of news reports on health topics | SARS/other infectious diseases/general health | Media reports - newspaper, radio, television, and internet news on health-related topics | Not reported | The results of the content analysis showed that in 2003, there was far more information available in the news media on SARS and West Nile Virus than on other health topics with greater population prevalence, such as obesity or heart disease. The number of articles about SARS in 1 year was greater than for any other individual topic across all 5 years, with the exception of smoking (which in 2003 had only 36 articles, compared to SARS, which had 164) and “other” topics. Expert sources were cited far more often than nonexperts. |
| **Bonwitt (2018)** | Sierra Leone & Guinea | Longitudinal qualitative/ | Ebola | posters | Various preventative behaviours | All study participants, irrespective of age or gender, were aware of wild mammals acting as a source of transmission for Ebola virus disease (EVD); respondents gave a variety of answers when asked |



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| | | | | | | |
|---|---|---|---|---|---|---|
| | | ethnographic study | | | | what species of animal could transmit Ebola - this confusion may have rested in the content of public health messages, which were inconsistent in the species shown to be potentially hazardous; People simply refused to believe that wild meat could pose any health risk; Those that believed or partially believed in the link between EVD and wild meat adopted various strategies to mitigate the perceived risk of infection. Most admitted to only refraining from eating those animals that they understood as posing a risk for EVD; The widespread suspicion over the risks posed by wild animals prompted discussions during village meetings to discuss the veracity of public health messages.<br>Rather than outright rejection of these messages, people elaborated situated hypotheses to make sense of the conflicting and incomplete information they had received. This process of contextualisation helped to bridge the disjuncture between fear and the highly routine nature of eating meat. |
| **Cole & Watkins (2015)** | Liberia, Sierra Leone or Guinea | interviews | Ebola | WHO, CDC, professional media (international, local) informal information e.g. rumours | Information-seeking | Their recall can be grouped into three broad source categories (official, international and local media, and informal) Throughout later information-seeking behaviour, interviewees maintained a distinction between the three categories. This affected how they trusted and processed information from each, although generally they tended to use all three simultaneously. Multiple source information gathering, especially following up from informal information (personal experiences) because they felt that was more rumour and wanted confirmation from official sources but felt official sources were slow with the information. Consistency provided reassurance and helped to be trusted. perception of risk, triggered by hearing that someone they knew personally, or felt an affinity to, had contracted the virus. know who employees see as their community and 'people like me'. Behavioural changes seemed to kick in at the border between the near and real at-risk rather than at the border between far at-risk and near at-risk. Those in the far at-risk group generally showed more concern for people they knew in higher risk categories than for themselves. |
| **Crosier (2015)** | UK. Italy, Hungary | Qualitative interviews | H1N1 | Government poster | Public compliance and engagement | The study found a lack of planning and a low value attached to the skills required to produce effective communications. There was a dearth of good quality audience research to inform the development of communications. Little thought had been given to |



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| | | | | | | |
|---|---|---|---|---|---|---|
| | | | | | | the tone, targeting or channelling of messages. Instead, communications were characterized by a 'one size fits all' and a 'top down', expert-led response. There was also little effort to evaluate the impact of communications, but where this was done, very low levels of public compliance and engagement with key behavioural messages were found. |
| **Davis (2015)** | Australia & Scotland | Qualitative interviews and focus groups | H1N1 pandemic 2009 | Broadcast and electronic media | Hygiene (e.g. covering mouth when coughing or sneezing, washing hands, cleaning surfaces), social isolation, vaccination | Participants wanted information but to be able to interpret advice based on own situation. Information on hygiene and social isolation endorsed but seen as short-term solution as being social is a central part of life. Participants had an individualistic approach to risk, decide whether to adopt PH behaviours based on risk to themselves, gender norms play a part in understanding and engagement with hygiene and social isolation. |
| **Freiman (2011)** | USA | face to face interviews | H1N1 pandemic 2009 | TV, newspaper, internet, family/friends, workplace, radio, other | Reporting on a range of preventative behaviours | Variable - did not always have the desired effect, or an effect at all. relied on the internet - however a lot of the target population did not have access to this - and thus were missed. Even those with internet access were unlikely to visit the CDC's website to learn about H1N1 |
| **Gray (2012)** | New Zealand | Focus groups | H1N1 pandemic 2009 | Ministry of Health campaigns | General protective behaviours | Four major themes: personal and community risk, building community strategies, responsibility and information sources. People wanted messages about specific actions that they could take to protect themselves and their families and to mitigate any consequences. They wanted transparent and factual communication where both good and bad news is conveyed by people who they could trust. |
| **Holmes (2009)** | Canada | Semi-structured interviews | crisis | Hypothetical | Risk perceptions | The notion of uncertainty – and the difficulties associated with communicating in uncertain situations – arose frequently. Recommendations: is to set and get agreement on a goal or goals (there may be different goals set for different audiences or specific situations). View all other aspects of communications planning and implementation through this ethical lens. Media representatives should be engaged immediately in discussions about a potential emerging infectious disease outbreak, including the role media should play and how the public health community can help them fulfil that role. Building trust with the public. |



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| | | | | | | |
|---|---|---|---|---|---|---|
| **Jones (2010)** | Australia | Focus groups | Bird Flu | Campaigns prepared by advertising agency | Hand washing and Hygiene and self-protection measures (e.g. | In the early stages of a pandemic (i.e., prior to the identification of cases within a given country), communications should focus on increasing awareness of the disease and communicating important, but simple, protective behaviours to reduce the risk of transmission. Social marketing campaigns will also need to have an associated strategy for their social marketing campaigns' targeting intermediaries—such as general practitioners and other medical personnel, schools, business owners, and commercial and public organisations which could be utilized to disseminate information and resources. |
| **Lapka, (2008)** | USA | Focus groups and interviews | Influenza | | Various | Cognitive Response Testing (CRT) is easily applied to various subject matters and effectively highlights problematic terminology and phrases, allowing for beneficial and efficient message revisions. CRT is a straightforward, appropriate and useful method in health promotion message development and pre-testing. |
| **Li (2016)** | China | focus group | H7N9 | News from various sources | Risk perception | Most people learnt about virus from tv, info was felt to be released in timely and transparent manner, most useful was info about preventative behaviours, and evolving outbreak trends. Gov info perceived as trustworthy, but young people trusted social media more and wanted gov to publish news in such way. Health recommendations seen as easy to follow and clear. Anxiety affected by disease severity, distance from pandemic, high level of news coverage, trust in gov ability to fight the issue. Health communication needs to be specific and practical |
| **Liao, (2010)** | Hong Kong | telephone interviews | H1N1 pandemic 2009 | | hand hygiene and social distancing | There was a relationship between trust in formal information and understanding (positive), trust in information info and susceptibility (negative) |
| **Lohiniva (2020)** | Finland | Qualitative | COVID-19 | e-mail and social media | Risk perceptions | Risk perception domain: catastrophic - potential strong wording describing the epidemic, catastrophe, worldwide threat. Concept - Emotional response. Recommendation - Avoid downplaying strong feelings. Concept - Localised epidemic; stigmatising attitude towards foreign nationals. Recommendation - Humanise infected people by telling stories. Location of transmission: public transportation linked with airports and foreign passengers. Concept - Crowded places; stigmatising attitudes towards foreign nationals. Recommendation - Emphasise handwashing and cough etiquette as effective ways to prevent COVID-19. Concept - Unreliable |



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| | | | | | | |
|---|---|---|---|---|---|---|
| | | | | | | information Recommendation - Repeat information and provide an explanation (reason). |
| **Lyu (2012)** | Taiwan | Qualitative Interviews | SARS | News reports/ public announcements | Perception of spokesperson | The essential/favourable characteristics for spokespersons should include: having professional capability, having better media interactions, involvement in policy making, having trustworthiness, and having favourable (or positive) personality and traits. |
| **Mitchell (2014)** | USA | Focus groups | Influenza A H1N1 | University mobile message alerts | Various preventative behaviours | Reported concern and commitment to recommendations decreased rapidly. Initial university messaging and response was critical in shaping participants' later perceptions. |
| **Person, (2004)** | USA | Focus groups and rapid situational assessment | SARS | Community outreach strategies | N/A Community resilience | SARS-related stigmatization was occurring more frequently within the Asian community than from outsiders directed toward the Asian community. The team also found that those persons with SARS-like symptoms who used traditional herbal physicians and pharmacies were less likely to be referred to, or seek out, public health officials, suggesting that further research into strategies to reach this population is needed. Conducting community visits also showed that CDC was responding to the needs of the community at risk for SARS-related fear, stigmatization, and discrimination and was modelling positive behaviours to the public. |
| **Qian (2020)** | China | Content analysis | COVID-19 | Newspapers | N/A | The Chinese mass media news lagged when reporting the major developments of the viral spread. Prevention and control procedures, medical treatment, and research are major themes of the press but mainly focus on the whole society, while instructions on personal and individual prevention, clinic and medicine choices, and detection need to be further enhanced. |
| **Qiu (2018)** | China | qualitative comparative case study | SARS, H7N9 | Media | Risk perceptions and awareness | 8 key principles of risk communication<br>Trust is the basis by maintaining an open and honest attitude and engage key stakeholders.<br>Delaying in reporting cases caused distrust, need accurate, timely, honest reporting, transparency helped subdue rumours, maintain social stability and helped maintain trust, and ensured more cooperation to reduce the spread of the disease. Being honest and open, planning well, being empathetic and caring, Accepting and involving the public as a partner regular feedback from the public and interaction between government and community. |



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| | | | | | | |
|---|---|---|---|---|---|---|
| **Rim (2014)** | USA | content analysis + correlating with survey results | H1N1 | News media | Various | The more info from health authorities, the more news coverage. The same correlation for severity of content. Increased information subsidies with salience of the severity attribute was linked with increased H1N1 salience in media coverage. But no relationship between the severity attribute use in media coverage, and the public perception of risk. Framing that effectively empowers the public to engage in desired behaviour should be further studied for the success of a public health campaign. |
| **Rimi (2016)** | Bangladesh | ethnography | avian influenza | Posters | Prevention (handling poultry) | Incorporating non-health benefits might improve acting on recommendations.<br>More people heard about the bird flu after the intervention, and had more awareness around the disease, as well as performed preventative behaviours. Not all behaviours changed in the desired direction though (e.g., selling sick poultry, separating it). Reported behaviours were not consistent with the observations done by the researchers though. People rationalised ignoring recommended behaviours - low perception of risk - the flu not present among their birds, also finance issues and not worrying about getting ill. Also inconvenience was mentioned, as well as social pressure not to follow the recommendations, and scepticism about the necessity of the intervention - some conspiracy theories. |
| **Sell (2017)** | USA | Content analysis of news coverage | Ebola | News coverage | risk perception | more scientific knowledge to be used in media coverage. Certain risk messages about Ebola were used more frequently than others by US news media, which may have affected risk perception during the outbreak. Some messages increase, some decrease risk perception. |
| **Seltzer (2015)** | USA | Content analysis | Ebola | Instagram/ Flickr pictures relating to Ebola | Risk awareness | Images of health care workers and professionals [308 (25%)], West Africa [75 (6%)], the Ebola virus [59 (5%)], and artistic renderings of Ebola [64 (5%)]. Also identified were images with accompanying embedded text related to Ebola and associated: facts [68 (6%)], fears [40 (3%)], politics [46 (4%)], and jokes [284 (23%). Image sharing platforms are being used for information exchange about public health crises, like Ebola. Use differs by platform and discerning these differences can help inform future uses for health care professionals and researchers seeking to assess public fears and misinformation or provide targeted education/awareness interventions. |



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| | | | | | | |
|---|---|---|---|---|---|---|
| **Sumo (2019)** | Liberia | Interpersonal communication' and dialogue | meningococcal septicemia | Community engagement (religious leaders, elders etc), radio talk shows, house to house visits, radio ads (jingles) | hand-washing, avoiding gatherings, risk perceptions | Increased trust and adherence. Contact tracing successful and epidemic contained, trust increased and adherence to messages increased (e.g. reporting sick people to health facility). |
| **Teasdale (2011)** | UK | Focus groups | H1N1 pandemic | Government + author-developed messages | Government advice was stay at home if you have flu-like symptoms and continue to go to work if you do not (behaviour not tested in this study though) | Participants were sceptical about the feasibility and appropriateness of self-diagnosis of pandemic flu. Participants actively evaluated recommended actions according to their own beliefs and reasoning about flu and their perceptions of the costs of carrying out the recommended actions. For most people the experience of the H1N1 pandemic was relatively mild compared to previous influenza pandemics of the 20th century. Consequently, the recommendation to stay at home was perceived as inappropriate. Written government communications aiming to improve and maintain population health may also need to address the recipients' perspective. Attempts should be made to elicit and address common doubts and concerns, to reduce perceived barriers to recommended behaviours, emphasize benefits, to find ways to support people to adopt them and consider the likely contextual factors that may affect perceptions of the advice. |
| **Tully (2019)** | West Africa | Content analysis | Ebola | Hyperlinks - embedded content in social media, e.g.Twitter used for information dissemination. | Risk perceptions | Increased credibility and awareness of messages. Organizations produce proportionally fewer tweets than news outlets - maintaining an active Twitter presence allows response organisations to promote their work and position themselves centrally to the unfolding crisis and conversation. Results suggested that 'owned' and 'earned' media served as informational resources and promotional pieces for the organizations. Repeatedly linking to sites controlled by the organization and/or produced by news media and other reliable sources served as a low-cost way for organizations to keep pace with the rapid flow of information on social media. |



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**Table 2 – Characteristics of Quantitative Design Studies**

| | Country | Study Type | Health Threat | Intervention type | Target behaviour | Main findings |
|---|---|---|---|---|---|---|
| **Aburto (2009)** | Mexico | cross sectional representative household survey | H1N1 pandemic 2009 | | N/A - reporting on a range of preventative behaviours | Risk mediated behaviours in some cases e.g. if they perceived the threat of the virus to be high, they were in turn more likely to avoid crowds. People who did comply, were more likely to if they had knowledge about the virus. |
| **Bekalu (2017)** | USA | Experimental | Pandemic influenza | Video clips | washing hands frequently, staying home when sick and avoiding unnecessary travel | A non-narrative message format may be more effective than its narrative counterpart to communicate basic prevention information during public health emergencies. Compared with the narrative and/or fictional version, the more didactic and factual format was found to be more effective in changing knowledge and perceived response efficacy related to prevention of pandemic influenza. |
| **Chang (2012)** | Taiwan | Experiment (2 by 3 factorial design) | H1N1 pandemic | Newspapers | Risk perceptions | The amount of exposure to news stories increases its impact on the public. When a health issue is ambiguous, greater exposures to news coverage with high alarm frames, as opposed to low alarm frames, evoke more fear and increase participants' perceptions of the severity of the issue, as well as their vulnerability to it. Exposures to news do not increase prevention or treatment efficacy. Repetition exacerbates the effects of media exposure on perceived severity and vulnerability but not fear. The amount of exposure increases its impact. |
| **Daellenbach (2018)** | Australia, New Zealand | Quantitative online survey | Crisis | | Risk perceptions | Adaptable and personalised to be more effective. Segmentation based on Theory of Planned Behaviour variables. The study highlighting that vulnerability may be identified based on such factors as critical awareness, perceived barriers, and ultimately preparation undertaken, including community-related factors. |



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| **Davis, (2013)** | USA | Experimental design with 3 conditions | Influenza A; Influenza B; HINT | Posters | Hand washing | Prompts alone were not effective at increasing hand washing.<br><br>In the education building female bathroom soap usage significantly increased during the first poster prompt (following baseline), then fell below baseline during the second prompt (i.e., warning of influenza transmission). In the business building bathroom women's soap usage did not change significantly from baseline to first poster condition but decreased slightly on the second condition. Women in the education building used significantly more soap overall than women in the business building; there were no significant differences between men by building type. |
| **Hickey (2014)** | Quatar | Cross-sectional survey | H1N1 pandemic 2009 | Non-pharmaceutical interventions (e.g. using a thermometer, wearing a face-covering, hand washing, and household disinfection practices) | using a thermometer, wearing a face-covering, hand washing, and household disinfection practices | Attitudes towards recommended NPIs were generally negative or uncertain, limited experience applying these interventions (e.g., using a thermometer, wearing a face-covering) and inadequate hand washing and household disinfection practices |
| **Hoda (2016)** | Saudi Arabia | Survey | MERS | Promoting public awareness of MERS in the Saudi population | Risk awareness | The internet was the most commonly used source of information (39.5%) and the most endorsed channel for a MERS awareness campaign. Physicians were the preferred source of information (45.6%), followed by other health care providers (31.3%). In univariate multinomial logistic regression models, males and individuals aged ≤27 years were more likely to seek information from the internet than from physicians. Residents of southern and western Riyadh preferred physicians as a credible source of information over the Ministry of Health. |
| **Idoiaga (2016)** | Spain | Experimental | Flu epidemic | Mass media | Risk perceptions | The human interest framing increased the perception of risk, especially when the proximity of the epidemic was high; this effect was explained by people's emotional response. |



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| | | | | | | Furthermore, youth projected the risk towards 'the other' in order to protect their invulnerability identity. |
| **Jhummon-Mahadnac (2012)** | Australia | Cross-sectional survey | H1N1 pandemic 2009 | Public Education Campaigns | Working from home, postponing social gatherings, wearing facemasks, oseltamivir treatment if sick | 68% (175) reported behaviour change because of the pandemic. Gaps in knowledge included failure to identify certain high risk groups. Recall of government campaigns was significantly associated with a higher knowledge score. 60% (151) thought that authorities and media had exaggerated the threat; only 40% (101) would comply with recommended measures in a future pandemic. |
| **Johnson & Slovic (2015)** | USA | Experimental - 2 studies | Ebola | Mock story | Risk perceptions | Informing Americans about the small likelihood of post-21-days Ebola symptoms would not increase perceived risk and distrust and might diminish negative reactions to the media reporting a case who developed symptoms of Ebola after 21 days. Thus, public health officials wanting appropriate public responses to potential or actual epidemics may benefit from early communication of unpleasant infectious-disease facts before events reveal them, and signal officials' lack of preparedness for public reactions. |
| **Kavanagh (2011)** | Australia | cross sectional survey | H1N1 pandemic 2009 | official sources (health dept, schools) and unofficial sources (media, family and friends, health care providers) | quarantine | Compliance was higher when they understood the messages. make messages easy to understand for everyone. Official sources are trusted/legitimised more than unofficial sources |
| **Miczo (2013)** | USA | Cross-sectional survey | H1N1 pandemic | Campaigns on campus (flyers, posters, emails). Mass media (various, including television & radio) other sources (e.g. medical) | Various - Self-isolation, hand hygiene, respiratory etiquette, sanitizing frequently | No significant results found for the impact of the health messages on health behaviour performance. The most frequently mentioned messages students remembered were: to wash hands (56.9%), avoidance (23.5%) (e.g. staying away from others when feeling ill) and getting a flu shot/ vaccination (22.1%). Positive health self-concept was positively related to the argument message dimension and |



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| | | | | | touched surfaces, not attending public events if sick, vaccination, early intervention for high-risk students and staff members, and cancelling travel plans | flu-prevention behaviours; negative health self-concept was negatively related to the argument message dimension and hand washing.<br>Campus most frequently cited source and useful from campus health viewpoint. There was limited success in attempting to assess memorable messages using a dimensional approach. |
| **Prati (2011)** | Italy | Cross-sectional survey | H1N1 | Educational campaign | Various – all preventative behaviours | Recommended behaviours such as cleaning objects, social distancing and washing hands were related to all the psychosocial factors except for trust in the institutional response to the outbreak and trust in medical science. Trust in medical science was significantly associated with cleaning objects. To use tissues when sneezing was predicted by all the psychosocial factors, except for trust in medical science and control. These results showed that media trust, trust in the Ministry of Health (regardless of what was done to reduce this risk), worry and perceived severity of illness predicted all the recommended behaviours. Exposure to the Topo Gigio campaign did have a limited influence on complying with health related recommendations but only with regard the use of tissues when sneezing. |
| **Rim (2014)**<br><br>***(also reported in table 1)*** | USA | content analysis + correlating with survey results | H1N1 | News media | Various | The more info from health authorities, the more news coverage. The same correlation for severity of content. Increased information subsidies with salience of the severity attribute was linked with increased H1N1 salience in media coverage. But no relationship between the severity attribute use in media coverage, and the public perception of risk. |



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| | | | | | | Framing that effectively empowers the public to engage in desired behaviour should be further studied for the success of a public health campaign. |
| **Roess (2017)** | DR Congo | Pre- post interventional study | Ebola | Four films | Risk perceptions and awareness | a culturally and linguistically appropriate video-centred intervention was effective in improving knowledge, attitudes and behaviours related to EVD in Congo. The results also demonstrate retention of the knowledge one year after exposure to the intervention. |
| **Updegraff (2011)** | USA | Experimental design with 3 conditions | H1N1 | Posters/ signs placed besides hand sanitizer units around university campus | Use of hand sanitizer dispensers | All signs resulted in significantly greater usage than no sign, but they were not equally effective. Dispensers with the gain-framed signs had the greatest usage of all, with 66.4% more use than dispensers with no signs ($p < .001$). Loss-framed signs were associated a 58.4% increase in use over no sign ($p < .001$). The social norms signs (44.3% increase) and the perceived susceptibility signs (40.6% increase) were associated with somewhat lower increases in usage compared to the gain-framed and loss-framed signs, but both led to significantly more usage than no sign at all (both $p$'s $< .01$). Gain-framed signs received 12.5% more usage than dispensers in the other sign conditions combined, a significant difference ($p = .029$). Usage of sanitizer dropped consistently over time, closely mirroring temporal trends in public interest in H1N1. This study showed that the relatively simple strategy of placing theoretically grounded cue-to-action reminder signs at the point-of-use significantly promoted usage. The worst-performing sign emphasized people's susceptibility to contamination. |
| **Van der Weerd (2011)** | Netherlands | repeated measure cross-sectional telephone survey | H1N1 | | information-seeking | During the course of the pandemic the majority of respondents wanted to receive information from health services/health care providers and media. Wanted information on how to prevent infection, what to do in the event of illness, symptom, risks , consequences and number of infected cases. Trust in governmental information depressed |



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| | | | | | | by the time of period 3. At first, mistrust was because information was incomplete, kept secret or withheld. Later, because they thought the situation was exaggerated, information unclear and information was contradicted itself. Fear/worry was sig associated with intention to adopt protective measures during all time periods. |
| **Winters (2018)** | Sierra Leone | Cross-sectional | Ebola | A variety of sources: mostly electronic media, followed by community sources, print and new media. About half received government communications and most were exposed to more than 2 types of info. | Not reported | increased protective behaviours. Strong dose-response between info exposure and protective behaviours, however exposure to all sources (except electronic and print media) was associated with misconceptions |
| **Yardley (2011)** | UK | parallel-group pragmatic exploratory trial design | H1N1 pandemic | Website | hand washing | Increased hand washing, targeted/changed attitudes |
| **Zikmund-Fischer (2017)** | Netherlands | experiment | hypothetical influenza | mock news article | cover mouth, wash hands | Participants are more influenced by average than extreme case information, presenting both is counterproductive. |

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**Table 3 – Characteristics of Reviews and Narrative Reports**

| Author (date) | Country | Study Type | Health Threat | Intervention type | Target behaviour | Main findings |
|---|---|---|---|---|---|---|
| **Barrelet (2013)** | | Systematic Review | H1N1, H5N1 | | Risk perception, vaccine perception, rumours | Risk perceptions vary across time, social groups, cultures, and countries. Social resistance to vaccination is an enduring phenomenon that should not be downplayed by public health officials. Competing narratives (official statements and presentations, rumours, conspiracy theories, alternative worldviews and explanations, urban legends, hoaxes, etc.) about public health crises always coexist in the public sphere. Trust building is a key aspect of risk perceptions. |
| **Crouse Quinn (2008)** | USA | Commentary /proposal of a model for crisis and emergency and risk communication in a pandemic | Pandemics (general) | | Not reported | Overall proposed model: pre-disaster phase (ongoing risk education, partnership formation, community engagement, opportunity for deliberation on difficult policies and procedures); during an emergency (more one way, emergency risk communication, more need for immediate action, community partners' engagement in response); increased capacity and resilience of communities at times of disasters and pandemics |
| **Goldberg (2015)** | USA | Report | Ebola and others | Checklist | Risk awareness and perceptions | 5 stages: agree on a common goal, coordinate the leadership, develop a coordination strategy, launch a communication operation, maximize communication effectiveness |
| **Lin (2014)** | | Systematic review | H1N1 | Non-specific variety of interventions, including 'Websites', ' Commercial television', 'Health department' | Preventive behaviours such as hygiene and social distancing practices in 70% of the studies (N = 64), risk perceptions (70%, N = 64), levels of knowledge and awareness about the pandemic (53%, N | To reduce communication inequalities during a large scale emergency, such as a pandemic, public health officials should focus their communication efforts on the young, the less educated and the indigent because there is evidence that these are the people at risk of not knowing about the threat, perceiving the threat to be of low risk and ultimately being less likely to follow recommended behaviours. An honest reporting of what the threat looks like, through a presentation of known and unknown factors, seems to have a better impact on people's knowledge, attitudes and beliefs, |



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| | | | | | = 49), and emotional responses such as fear and worry (47%, N = 43).<br>Factors influencing the H1N1 vaccination acceptance rate (26%, N = 24) were also frequently investigated. | including trust in the way the government is handling the emergency. Consequently, there is some evidence that better knowledge and trust are likely to be associated with the adoption of recommended behaviours (i.e. immunization practices). Social networks and ties to the community are also drivers of better knowledge and compliance with preventive measures; these results suggest that non-traditional channels of communication (i.e. partnership with community leaders or organizations) should be used to reach out to the most vulnerable and those in need of a better understanding of the risks and actions needed to be able to protect themselves. Public health communication messages are still delivered at a literacy level that does not meet the needs of the less educated. |
| **Menon (2005)** | Singapore | Commentary – narrative review | SARS | | Risk perceptions | 1. Providing more information is more effective than less information, 2. it always better to over-react than to under-react. 3. fear and ignorance of any disease is worse than the disease itself. It reduces all to impotence and defeat. Information is an all-powerful tool to fight fear. It empowers people and allows them to become socially responsible. |
| **Menon (2006)** | Singapore | Narrative review/ commentary | SARS | Mixed - TV, technology | Not reported | Technology (The Infrared Fever Screening System), a SARS-dedicated TV channel, transparency, leadership (highest levels of government came into the picture very early in the crisis and took the lead), communication tools (moderating public fear through dialogue with grassroots leaders and community, briefings for foreign business groups, diplomats, religious groups, trade associations, cartoons, hotlines, songs on TV), earning the trust and confidence of Singaporeans, rebuttal of negative reports in the foreign media claiming that foreign visitors had been infected with SARS whilst transiting Singapore, outbreak communications across cultures. |
| **Rogers (2009)** | Australia | Deliberative forum | Influenza | hypothetical | Risk awareness | The forum wanted full and frank information about the potential risk and international developments including numbers of cases and fatalities. Forum members understood that predictions about the pandemic influenza were uncertain, |



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| | | | | | | this did not lead to loss of confidence in the experts or the information they imparted. Third, the forum recommended releasing geographically localizing information about initial cases. The forum recommended increasing use of television and websites, including those targeting youth and rural and remote groups as the pandemic developed. The forum recommended immediate activities to educate and build awareness and swift action if and when Australia has its first cases of PI. |
| **Santibañez (2015)** | USA | Commentary | Ebola | | N/A | 10-step approach can be adapted for infectious disease response communications (in this case Ebola) allowing community and faith-based organisations to help spread effective health messages |
| **Schiavo (2014)** | | Systematic review | Influenza, SARS | Media | Handwashing, immunizations, infection control precautions | most studies found behaviour changed |
| **Toppenberg (2018)** | Various | Rapid Review of Grey Literature | Ebola | | Awareness only | Literature strongly underlines the central importance of local communities. A one-size-fits-all approach does not work. For maximum effectiveness, local communities need to be involved with and own emergency risk communication processes, preferably well before an emergency occurs. Social media can open new avenues for communication but is not a general panacea and should not be viewed as a replacement for traditional modes of communication. In general, the gray literature indicates movement toward greater recognition of emergency risk communication as a vitally important element of public health. |
| **Vaughan (2009)** | Not reported | Commentary – narrative review | Pandemic Influenza | | Message acceptance | Barriers to accepting messages include, environmental factors, social and cultural characteristics, language preferences (translation, culturally sensitive) difficulty of and attitudes towards public health interventions |



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**Table 4 – Characteristics of Preprint Studies**

| | Country | Study Type | Health Threat | Intervention type | Target behaviour | Main findings |
|---|---|---|---|---|---|---|
| **Abu-Akel, Spitz & West (2020)** | Switzerland | Pilot experimental survey study | COVID-19 | Social distancing | Social distancing | The government official was more effective at garnering support for social distancing than the celebrity, particular in older respondents. Support and compliance relating to social distancing was higher in older people, despite them having lower percieved risk of threat. Females were overall more in support of social distancing. Higher support and compliance was related to greater concern about the situation, concern for others health, greater belief in social distancing measures and feeling more constrained by them. Inverse relationship between city size and practise of social distancing. |
| **Bilancini, (2020)** | Italy | Four-condition, between-subjects experiment. | COVID-19 | Official coronavirus-related messages | Behaviours relating to the COVID-19 emergency | No statistically significant difference between the 4 conditions. Authors conclude that stronger 'nudging' interventions may be needed to influence behaviour. |
| **Blagov (2020)** | USA | Online survey | COVID-19 | Public health messages worded to target different personality traits (self-centered, responsible, compassionate, avoidant & sociable). Looks like she were developed by the researchers. | Staying at home, physically distancing and practising handwashing | Agreeableness and conscientiousness predicted endorsement of social distancing and hygiene. Meanness and disinhibition (and overall psychopathy) as well as Machiavellianism (less so) predicted lower intent for social distancing. Together with boldness, the psychopathy traits predicted endorsement of risky, venturesome behavior under the pretend scenario that one is a disease carrier. Meanness and disinhibition predicted endorsement of knowing and possibly deliberate behaviour that puts others at risk of infection. |



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| **Bradshaw, (2020)** | Australia | Randomized experimental online study | COVID-19 | Online vignettes about the contact tracing app | Downloading and using a contact tracing app | Autonomy-supportive and controlling message framing did not differentially affect intended uptake. However, there was a main effect of information safety. Those in high information safety conditions reported higher intentions to use the application and to recommend it to others than those in low information safety conditions, regardless of message framing. Australians appeared more willing to assent to authority regarding contact tracing insofar as their data safety can be assured. |
| **Dai (2020)** | China | Cross-sectional online survey | COVID-19 | Information on government emergency management measures relating to COVID-19 | Protective behaviours, including preventive (i.e. wearing masks, disinfectants), avoidant (i.e. stringent quarantine, avoiding public places), and management of disease behaviours (i.e. seeking professional protection or treatment information, paying for preventive and therapeutic drugs) | Governmental information transparency, positive propaganda, rumour refutation, and supplies positively predict the protective behaviours. Individual factors such as perceived control, positive emotions, and risk perception mediate role in predicting protective behaviours. |
| **Everett, (2020)** | USA | online experimental survey study | COVID-19 | Social media (Facebook post) either from a citizen or leader. | Behavioural intentions for: Hand washing, avoiding gatherings, isolation, cancelling holidays | Stronger intentions if messages were shared by 'leader' than citizen, modest effects of using deontological and virtue-based messaging (compared to utilitarian) |
| **Goldberg, (2020)** | USA | Online Survey | COVID-19 | N/A - measured behaviours as a result of seeing | mask buying and mask wearing | Once the CDC recommendation had been disseminated for at least one full day, there were large increases in reported mask wearing and mask buying. |



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| | | | | national guidance from the CDC | | |
| **Heffner (2020)** | USA | Within subject design, online survey | COVID-19 | 2 messages about staying at home, one focused on 'fear', the other on 'altruism' | Intentions to self-isolate, emotional response | Both messages were associated with increased willingness to self-isolate. The prosocial message was more effective at boosting willingness to self-isolate if it produced a strong, positive, and arousing emotional response. Fear messages created a stronger emotional response. |
| **Jordan (2020)** | USA | Three-part online experimental study | COVID-19 | PH messages/ fliers about COVID-19 emphasising personal, public or personal & public protective messages. | Hand washing, social distancing, staying at home, wearing a mask. | In the early stages of the pandemic, prosocial (i.e. public) messaging was more effective. However, as the pandemic progressed, all three types of framing (i.e. prosocial/ public, self-interested/ personal and both) were equally as effective. Self-interested framing was never more effective than prosocial framing and did not improve efficacy of prosocial framing. |
| **Merkley, (2020)** | Canada | 2-part online experimental survey study | COVID-19 | News stories relating to the pandemic - signalling information from experts or non-experts. | Information on hygiene (i.e. cleaning surfaces), social gatherings, testing and scientific discoveries. | Study 1: We find that our respondents were on average 3 points more likely to select news stories with headlines featuring experts (p~0.007). News stories with headlines featuring experts scored 2 points higher in credibility (p<0.002). In support of H1, we find that the effect of receiving headlines featuring experts is heterogeneous across levels of anti-intellectualism. The marginal effects on story selection are shown in the left panel of Figure 2. Respondents with the lowest levels of anti-intellectualism are six points more likely to select stories featuring experts (p<0.001), but this effect disappears once the mid-point of the scale is reached.<br>Study 2: Our respondents were 19 points more likely to select COVID-19 news (p<0.001). They also viewed such news as 6 points more credible (p<0.001) and 19 points more important (p<0.001). Once again, |



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| | | | | | | anti-intellectualism strongly conditions news preferences. Respondents with the highest levels of trust in experts were 25 points more likely to select COVID-19 news (p<0.001), and viewed it as 8 points more credible (p<0.001) and 24 points more important (p<0.001). In all three cases this effect weakens as anti-intellectualism rises, though it does not entirely vanish for the most part. |
| **Wirz, (2020)** | USA | Online Survey | COVID-19 | N/A - survey asking whether people were practising guidelines (e.g. social distancing) and questions about barriers and communication. | Social distancing | Perceived effectiveness of social distancing for decreasing the risk of infection and perceived social norms related to social distancing were strongly correlated with reported social distancing, suggesting that individuals engaged in social distancing if they thought it worked and if they thought their peers were doing it. If they learned that their social distancing behaviour would help others (e.g., family, older members of the community) and that there are serious health consequences for people like them, they would engage in more social distancing. Respondents in this group reported often getting news from national news networks and social media, followed by newspapers/news magazines. Respondents reported having the highest levels of trust in information from the CDC, public health experts, and the WHO. These groups were followed by university scientists and friends and family. The White House, state and local governments, and the news media were among the least trusted sources for information. |



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# Appendix 5- Data quality table using Mixed Methods appraisal tool and AMSTAR

| Studies | | | | Criteria from the mixed methods appraisal tool | | | | | | | | | | | | | | | | | | | | | | | | | | |
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| | | | | S1 | S2 | 1.1 | 1.2 | 1.3 | 1.4 | 1.5 | 2.1 | 2.2 | 2.3 | 2.4 | 2.5 | 3.1 | 3.2 | 3.3 | 3.4 | 3.5 | 4.1 | 4.2 | 4.3 | 4.4 | 4.5 | 5.1 | 5.2 | 5.3 | 5.4 | 5.5 |
| 1. Aburto (2009) | cross sectional representative household survey | 4 | **** | | | | | | | | | | | | | | | | | | 1 | 1 | 1 | 0 | 1 | | | | | |
| 2. Avery (2009) | Content analysis | 5 | **** | 1 | 1 | 1 | 1 | 1 | 1 | 1 | | | | | | | | | | | | | | | | | | | | |
| 3. Berry (2007) | Content analysis | 4 | **** | 1 | 1 | 1 | 1 | 1 | 1 | 0 | | | | | | | | | | | | | | | | | | | | |
| 4. Bekalu (2017) | Pre/post intervention study | 4 | **** | 1 | 1 | | | | | | 1 | 1 | 1 | 0 | 1 | | | | | | | | | | | | | | | |
| 5. Basnyat & Lee (2014) | Qualitative | 4 | **** | 1 | 1 | 1 | 0 | 1 | 1 | 1 | | | | | | | | | | | | | | | | | | | | |
| 6. Bonwitt (2018) | Experimental | 4 | **** | 1 | 1 | 1 | 0 | 1 | 1 | 1 | | | | | | | | | | | | | | | | | | | | |
| 7. Chang (2012) | Experiment (2 by 3 factorial design) | 5 | ***** | 1 | 1 | | | | | | | | | | | | | | | | | | | | | 1 | 1 | 1 | 1 | 1 |
| 8. Cole & Watkins (2015) | interviews | 2 | ** | 1 | 1 | 1 | 1 | 0 | 0 | 0 | | | | | | | | | | | | | | | | | | | | |
| 9. Crouse Quinn (2008) | Commentary | | n/a | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 10. Crosier (2015) | Qualitative Interviews | 4 | **** | 1 | 1 | 1 | 1 | 1 | 1 | 0 | | | | | | | | | | | | | | | | | | | | |
| 11. Daellenbach (2018) | Quantitative online survey | 3 | *** | 1 | 1 | | | | | | | | | | | | | | | | 1 | 0 | 1 | 0 | 1 | | | | | |
| 12. Davis, (2013) | Experimental design with 3 conditions | 5 | ***** | 1 | 1 | | | | | | | | | | | 1 | 1 | 1 | 1 | 1 | | | | | | | | | | |
| 13. Davis (2015) | Qualitative interviews and focus groups | 3 | *** | 1 | 1 | 1 | 1 | 0 | 1 | 0 | | | | | | | | | | | | | | | | | | | | |
| 14. Freiman (2011) | face to face interviews | 5 | ***** | 1 | 1 | | | | | | | | | | | | | | | | 1 | 1 | 1 | 1 | 1 | | | | | |
| 15. Goldberg (2015) | Report | | n/a | | | | | | | | | | | | | | | | | | | | | | | | | | | |

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| 16. Gray (2012) | Focus groups | 5 | ***** | 1 | 1 | 1 | 1 | 1 | 1 | 1 | | | | | | | | | | | | | | | | | | | | |
| 17. Hickey (2014) | Cross-sectional survey | 5 | ***** | 1 | 1 | | | | | | | | | | | | | | | | 1 | 1 | 1 | 1 | 1 | | | | | |
| 18. Hoda (2016) | Survey | 2 | ** | 1 | 1 | | | | | | | | | | | | | | | | 0 | 0 | 1 | 0 | 1 | | | | | |
| 19. Holmes (2009) | Semi-structured interviews | 5 | **** | 1 | 1 | 1 | 1 | 1 | 1 | 1 | | | | | | | | | | | | | | | | | | | | |
| 20. Idoiaga (2016) | experimental | 2 | ** | 1 | 1 | | | | | | 0 | 0 | 1 | 0 | 1 | | | | | | | | | | | | | | | |
| 21. Jhummon-Mahadnac (2012) | Cross-sectional survey | 5 | ***** | 1 | 1 | | | | | | | | | | | | | | | | 1 | 1 | 1 | 1 | 1 | | | | | |
| 22. Johnson (2015) | Experimental – 2 studies (both studies) | 3 | *** | 1 | 1 | | | | | | 1 | 1 | 1 | 0 | 0 | | | | | | | | | | | | | | | |
| 23. Jones (2010) | Focus groups | 2 | ** | 1 | 1 | 0 | 0 | 1 | 0 | 1 | | | | | | | | | | | | | | | | | | | | |
| 24. Kavanagh (2011) | cross sectional survey | 4 | **** | 1 | 1 | | | | | | | | | | | | | | | | 1 | 1 | 1 | 0 | 1 | | | | | |
| 25. Lapka (2008) | Focus Group and Interviews | 5 | **** | 1 | 1 | 1 | 1 | 1 | 1 | 1 | | | | | | | | | | | | | | | | | | | | |
| 26. Lohiniva (2020) | Qualitative | 4 | **** | 1 | 1 | 1 | 1 | 1 | 0 | 1 | | | | | | | | | | | | | | | | | | | | |
| 27. Li (2016) | Focus group | 5 | **** | 1 | 1 | 1 | 1 | 1 | 1 | 1 | | | | | | | | | | | | | | | | | | | | |
| 28. Liao, (2010) | telephone interviews (surveys) | 4 | **** | | | | | | | | | | | | | | | | | | 1 | 1 | 1 | 0 | 1 | | | | | |
| 29. Lyu (2012) | Qualitative Interviews | 3 | *** | 1 | 1 | 1 | 0 | 1 | 1 | 0 | | | | | | | | | | | | | | | | | | | | |
| 30. Menon (2005) | Narrative review | | n/a | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 31. Menon (2006) | Narrative review/commentary | | n/a | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 32. Miczo (2013) | Cross-sectional survey | 2 | ** | 1 | 1 | | | | | | | | | | | | | | | | 0 | 0 | 1 | 0 | 1 | | | | | |
| 33. Mitchell (2014) | Focus groups and interviews | 5 | **** | 1 | 1 | 1 | 1 | 1 | 1 | 1 | | | | | | | | | | | | | | | | | | | | |

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| 34. Person, (2004) | Focus groups and rapid situational assessment | 2 | ** | 1 | 1 | 1 | 1 | 0 | 0 | 0 | | | | | | | | | | | | | | | | | | | | |
| 35. Prati (2011) | Cross-sectional survey | 3 | *** | 1 | 1 | | | | | | | | | | | | | | | | 1 | 0 | 1 | 0 | 1 | | | | | |
| 36. Qian (2020) | Content analysis | 2 | ** | 1 | 1 | 0 | 0 | 1 | 1 | 0 | | | | | | | | | | | | | | | | | | | | |
| 37. Qiu (2018) | Focus groups and interviews | 5 | ***** | 1 | 1 | 1 | 1 | 1 | 1 | 1 | | | | | | | | | | | | | | | | | | | | |
| 38. Roess (2017) | Pre-post interventional study | 3 | *** | 1 | 1 | 0 | 0 | 1 | 1 | 1 | | | | | | | | | | | | | | | | | | | | |
| 39. Rogers (2009) | Deliberative forum | 3 | *** | 1 | 1 | 1 | 1 | 1 | 0 | 1 | | | | | | | | | | | | | | | | | | | | |
| 40. Rim (2014) | Content analysis +correlation with survey results | 1 | * | 1 | 1 | 0 | 0 | 1 | 0 | 0 | | | | | | | | | | | | | | | | | | | | |
| 41. Rimi (2016) | ethnography | 3 | *** | 1 | 1 | 1 | 0 | 1 | 1 | 0 | | | | | | | | | | | | | | | | | | | | |
| 42. Santibañez (2015) | Commentary | | n/a | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 43. Sell (2017) | Content analysis of news coverage | 5 | **** | 1 | 1 | 1 | 1 | 1 | 1 | 1 | | | | | | | | | | | | | | | | | | | | |
| 44. Seltzer (2015) | Content analysis | 3 | *** | 1 | 1 | 1 | 0 | 1 | 0 | 0 | | | | | | | | | | | | | | | | | | | | |
| 45. Sumo (2019) | Interpersonal communicati on' and dialogue | | n/a | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 46. Teasdale (2011) | Focus groups | 4 | **** | 1 | 1 | 1 | 1 | 1 | 1 | 0 | | | | | | | | | | | | | | | | | | | | |
| 47. Toppenberg (2018) | Rapid Review of Grey Literature | 5 | ***** | 1 | 1 | 1 | 1 | 1 | 1 | 1 | | | | | | | | | | | | | | | | | | | | |
| 48. Tully (2019) | Content analysis | 3 | *** | 1 | 1 | 1 | 1 | 0 | 1 | 0 | | | | | | | | | | | | | | | | | | | | |
| 49. Updegraff (2011) | Experimental design with 3 conditions | 4 | ***** | 1 | 1 | | | | | | 1 | 1 | 1 | 0 | 1 | | | | | | | | | | | | | | | |

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| 50. | Vaughan (2009) | Narrative review | | n/a | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 51. | Van der Weerd (2011) | Cross-sectional telephone survey | 5 | **** | 1 | 1 | | | | | | | | | | | | | | | | 1 | 1 | 1 | 1 | 1 | | | | | |
| 52. | Winters (2018) | Cross-sectional | 5 | ***** | 1 | 1 | | | | | | | | | | | | | | | | 1 | 1 | 1 | 1 | 1 | | | | | |
| 53. | Yardley (2011) | parallel-group pragmatic exploratory trial design (RCT) | 4 | **** | 1 | 1 | | | | | | 1 | 1 | 1 | 0 | 1 | | | | | | | | | | | | | | | |
| 54. | Zikmund-Fischer (2017) | experiment | 2 | ** | 1 | 1 | | | | | | 0 | 0 | 0 | 1 | 1 | | | | | | | | | | | | | | | |

Data Quality check of Systematic Reviews with AMSTAR

| Citation | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 |
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| Schiavo et al | 1 | 1 | 1 | 1 | 1 | 0 | 1 | 1 | 1 | 0 | 1 |
| Lin et al | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 0 | 1 |
| Barrelet et al | 1 | 1 | 0 | 1 | 1 | 0 | 0 | 0 | 1 | 0 | 1 |

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| Pre-print Quality Check | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
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| Studies | | | | Criteria from the mixed methods appraisal tool | | | | | | | | | | | | | | | | | | | | | | | | | |
| | | | | S1 | S2 | 1.1 | 1.2 | 1.3 | 1.4 | 1.5 | 2.1 | 2.2 | 2.3 | 2.4 | 2.5 | 3.1 | 3.2 | 3.3 | 3.4 | 3.5 | 4.1 | 4.2 | 4.3 | 4.4 | 4.5 | 5.1 | 5.2 | 5.3 | 5.4 | 5.5 |
| 1. Abu-Akel (2020) | Pilot survey-based study (randomised) | 4 | **** | 1 | 1 | | | | | | 1 | 1 | 1 | 0† | 1 | | | | | | | | | | | | | | | |
| 2. Bilancini (2020) | Randomised controlled trial | 3 | *** | 1 | 1 | | | | | | 1 | 0 | 1 | 0 | 1 | | | | | | | | | | | | | | | |
| 3. Blagov (2020) | Cross-sectional survey | 2 | ** | 1 | 1 | | | | | | | | | | | | | | | | 1 | 0 | 0 | 0 | 1 | | | | | |
| 4. Bradshaw (2020) | Randomised controlled trial | 4 | **** | 1 | | | | | | | 1 | 1 | 0 | 1 | 1 | | | | | | | | | | | | | | | |
| 5. Dai (2020) | Cross sectional design survey | 4 | **** | 1 | 1 | | | | | | | | | | | 1 | 1 | 0† | 1 | 1 | | | | | | | | | | |
| 6. Everett (2020) | Randomised cross-over study | 1 | * | 1 | 0 | | | | | | 0 | 0 | 0 | 0 | 1 | | | | | | | | | | | | | | | |
| 7. Goldberg (2020) | National survey | 4 | **** | 1 | 1 | | | | | | | | | | | | | | | | 1 | 1 | 1 | 0 | 1 | | | | | |
| 8. Heffner (2020) | Cross-sectional analytic study | 4 | **** | 1 | 1 | | | | | | | | | | | 2 | 1 | 1 | 1 | 1 | | | | | | | | | | |
| 9. Jordan (2020) | Randomised controlled trial | 2 | ** | 1 | 1 | | | | | | 0 | 0 | 1 | 0 | 1 | | | | | | | | | | | | | | | |
| 10. Merkley (2020) | Randomised controlled trial | 1 | * | 1 | 1 | | | | | | 0 | 0 | 0 | 0 | 1 | | | | | | | | | | | | | | | |
| 11. Wirz (2020) | Online Survey | 3 | *** | 1 | 1 | | | | | | | | | | | | | | | | 1 | 1 | 0† | 0† | 1 | | | | | |

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# Appendix 6: Influences on effective public-health messaging

Most of the studies were qualitative work and reported on determinants of intent to adhering to guidelines by reporting what influences people's responses to public health messaging. These are: preconceptions and understanding of the threat, perceived susceptibility and perceived risk severity (threat appraisal(Rogers, 1975)).

*Preconceptions and understanding of the threat*

Of the 68 studies included, 26 specified that public health messaging included information about the threat which can be effective in identifying symptoms and changing behaviours. Some of these, including the only randomised control trial, reported changes in handing washing (Yardley, Miller, Schlotz, & Little, 2011), changes in knowledge and higher compliance to quarantine (Kavanagh et al., 2011). However, this particular study found the messages were not effective in encouraging quarantine behaviour suggesting that increasing knowledge alone did not change behaviour (Kavanagh et al., 2011). More comprehensive reporting on what the threat is and looks like, including presentation of known and unknown factors has a positive impact on people's knowledge, attitudes and beliefs. Self-reports, in the qualitative studies and in surveys, suggest changes in hygiene and unspecified protective behaviours following messages of how a virus (e.g. H1N1, SARS, Ebola) is transmitted, and the ability to idenitfy symptoms, and how to prevent transmision (Chidgzey, Davis, Williams, & Reeve, 2015; Gray et al., 2012; Hickey, Gagnon, & Jitthai, 2014; Hoda, 2016; R. Li, Xie, Yang, & Frost, 2016; Rim, Ha, & Kiousis, 2014; Teasdale & Yardley, 2011; Yardley et al., 2011). In the systematic reivew, public health messaging campaigns using posters (e.g. "*Catch it, Bin it, Kill it"*) increased knowledge about tramsission, and successfully increased self-reported prevention behaviours (hygiene when using tissues and sneezing). However, compliance to these behaviours was also predicted by worrry (Lin, Savoia, Agboola, & Viswanath, 2014). The systmatic review argued that individuals with higher education were better informed about H1N1. The review concluded that messages and communication are still delivered at a high literacy level (Lin, Savoia, et al., 2014). Whilst most of these papers highlighted that people will have different preconceptions and concerns to address in the public health messages, three papers also highlighted that people may have limited experiences of engaging in the preventative behaviour e.g. using face masks or using a thermometer (Hickey et al., 2014). Such cases highlighted a need for training/skill development as part of a public health campaign.

People that were able to identify symptoms of H1N1 were more likely to avoid crowds and washed hands more (Aburto et al., 2010). Framing and choice of language were found to be influential in behaviour and how individuals understand the threat (Teasdale & Yardley, 2011). Positively framed messages (emphasising the benefits of prevention behaviours) were considered 'robust contingency measures' (Basnyat & Lee, 2015). Gain-framed signs were significantly more effective in influencing the use of hand sanitizer as cue-to-action reminder signs at the point-of use during the H1N1 pandemic (Updegraff, Emanuel, Gallagher, & Steinman, 2011). The worst-performing signs emphasised people's susceptibility to contamination (Updegraff et al., 2011). Communication strategies were characterized in three case studies from the H1N1 pandemic from Italy, Hungary and England as "top-down" expert-led responses by public health communications specialists because there was no audience research. With no audience research, there was a perceived lack of planning and consideration to the tone, targeting or channelling of public health messages. Lack of consideration of these aspects led to low levels of public compliance and engagement with



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behaviour (Crosier, McVey, & French, 2015). A non-narrative message format (i.e., factual) may be more effective than a narrative counterpart to communicate basic prevention information during public health emergencies. Compared with the narrative and/or fictional version, the more didactic and factual format was found to be more effective in changing knowledge and perceived response efficacy related to prevention of pandemic influenza however, behaviour or intent were not measured in the study (Bekalu, Bigman, McCloud, Lin, & Viswanath, 2018). When messages provided the public with clear consistent information whilst admitting that information is evolving, reported the risks, and focused on the practical actions that people can do to reduce their risk and emphasized the efficacy of those actions, people were more likely to clean objects, wash hands and use tissues when sneezing (Prati, Pietrantoni, & Zani, 2011). A pre-print manuscript found that for one to perceive and understand the extent of a threat, messages that are framed in relation to identity (e.g. "don't be a spreader" vs "don't spread") were more likely to predict preventative behaviour following the Japanese Ministry of Health guidelines (Yonemitsu, 2020).

When individuals do not understand the message or the preventative behaviour being communicated, participants did not outright reject public health messages. Instead, people adjusted the key underlining message to what they thought made sense, and engaged in behaviour consistent with the adjusted message (Bonwitt et al., 2018). As there is general uncertainty around a pandemic flu it is hard for people to differentiate this from seasonal flu especially when symptoms are framed as "flu-like symptoms" (Teasdale & Yardley, 2011). In a qualitative study, people reported being more likely to follow guidelines when there were fewer perceived barriers to perform the recommended behaviours, when there are emphasised benefits and when contextual factors are considered such as feeling guilty and anxious about missing work and not wanting to let people down (Teasdale & Yardley, 2011). High levels of knowledge was associated with exposure to messages and was a mediator between different sources and complying with recommended protective behaviour. However, engaging with social media was associated with misconceptions of public health information (Winters et al., 2018).

*Threat appraisal: perceived susceptibility/vulnerability*

Increase in exposure to any type of information was associated with increased knowledge, and as level of outbreak increased (that increased the level of threat appraisal), protective behaviours increased, suggesting a strong dose-response (Johnson & Slovic, 2015; Winters et al., 2018). As perceptions of higher risk increase so did self-reported behaviours such as handing washing, wearing a facemask and using hand gels (Aburto et al., 2010). However, salience (i.e. prominence) in media coverage did not have a direct relationship with increasing perceived risk. One study found that there was no relationship between salience in media coverage of H1N1, or the severity of content and the public perception of risk (Rim et al., 2014).

Being worried about H1N1 was associated with media attention and number of H1N1 cases, and overall an important predictor of compliance with recommended preventative behaviours, such as using tissues, hand gel and washing hands (Lin, Savoia, et al., 2014). Worry led to effective compliance and increased adaptive behaviour in one included study (Gutteling, Terpstra, & Kerstholt, 2018). Evoking fear increased participants' perceptions of the severity of the issue and their vulnerability to the risk but did not increase prevention behaviours or treatment efficacy (Chang, 2012). Moreover, panic could be counterproductive (Zikmund-Fisher, Scherer, Knaus, Das, & Fagerlin, 2017); providing a "worse-case" scenario (that there is a high risk and there is a pandemic) and increasing knowledge of the health threat simultaneously, has been found to induce panic in



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those receiving the message (Jhummon-Mahadnac, Knott, & Marshall, 2012). Believability of messages was reduced by scepticism towards the media and governments; in these instances the threat was perceived as exaggerated and able to cause unnecessary panic which could influence low adoption of recommended action (Teasdale & Yardley, 2011). In a preprint manuscript, messages arousing an emotional response (such as fear) was more effective in boosting willingness to self-isolate alongside prosocial messages that are producing strong and positive emotional response (Heffner, 2020).

Perceiving a threat would mean that population warning systems (method where local, regional or national authorities contact the public en masse) can work but social pressure in addition to increasing perceived risk is also required (Gutteling et al., 2018). Worry about self or family members led to increased perceptions of disease severity and a stronger belief of being susceptible to the infection, leading to a greater compliance with recommended behaviours (Lin, Savoia, et al., 2014). Perceived susceptibility in the messaging (e.g. "this can happen to someone like me") was more likely to lead to engagement in physical distancing than the geographical closeness of the disease (e.g. cases in the local area) thereby appraising the threat as high (Cole & Watkins, 2015). If risk is perceived to be low (e.g. H1N1 pandemic likened to influenza pandemics) recommendations such as "stay at home" were perceived as extreme and inappropriate (Teasdale & Yardley, 2011). This then has implications for increasing distrust in the messages and sources. Behaviours such as usage of a sanitizer dropped over time mapping the temporal trends in public interest in H1N1 (Updegraff et al., 2011).

Individuals were more likely to be concerned about the risk to themselves rather than the risk they would pose to others (M. D. Davis, Stephenson, Lohm, Waller, & Flowers, 2015). A recent systematic review concluded that beliefs about personal susceptibility to H1N1, and perceiving it as potentially severe (infection rate), is linked with high levels of worry about self or family members at risk (Lin, Savoia, et al., 2014). Efficient public health messaging (where individuals were more likely to comply with behaviour (e.g. avoiding crowds; (Aburto et al., 2010) tapped into worry about self or family members by framing messages with positive social responsibility (Basnyat & Lee, 2015). Concern for family led to reporting more information-seeking and appraising the threat as a higher risk to self and their family (Cole & Watkins, 2015). Prosocial framing and focus on activating social norms were found in the grey literature to be effective in increased engagement in physical distancing (Jordan, 2020; Wirz 2020). Tailoring public health messages with compassion may increase acceptance of the messages (Blagov, 2020). Such prosocial framing in the public health messaging personalised the risk and increasing the perceived susceptibility and vulnerability.

### Increasing trust

An important finding that 60% of the included studies identified the need to increase the public's trust in the messages during pandemics and epidemics (Aburto et al., 2010; Cole & Watkins, 2015; Gray et al., 2012; Hickey et al., 2014; Holmes, Henrich, Hancock, & Lestou, 2009; Jhummon-Mahadnac et al., 2012; Johnson & Slovic, 2015; Kavanagh et al., 2011; Liao, Cowling, Lam, Ng, & Fielding, 2010; Lohiniva, Sane, Sibenberg, Puumalainen, & Salminen, 2020; Lyu et al., 2013; Person et al., 2004; Prati et al., 2011; Qiu et al., 2018; Sumo et al., 2019). Participants cited trust as the main reason they complied, and observational studies found trust to be essential in complying with preventative behaviours (R. Li et al., 2016; Menon & Goh, 2005), both when communicating public health advice from public health agencies (Li et al., 2016) or from the government (Menon & Goh, 2005). Arguably, the need to increase trust highlights that acceptability of the message is reliant on



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the right message given in the right way (framing cross-reference), by an appropriate messenger (cross-reference credible source) at the right time.

One of the systematic reviews, concluded that competing narratives should be addressed to support trust-building, and this can be achieved through involving the public with two-way communications in the design of the messages (Barrelet, Bourrier, Burton-Jeangros, & Schindler, 2013). Whilst the other systematic review, found that an honest reporting of what the threat looks like, through a presentation of known and unknown factors, seems to have a better impact on people's knowledge, attitudes and beliefs, including trust in the way the government is handling the emergency (Lin, Savoia, et al., 2014).

In a repeated measures survey during H1N1 pandemic, trust levels in the government decreased over time, higher levels of government trust led to higher intention to adhere to the guidelines and change in trust was due to conflicting messages at the same time points and as they change over time (van der Weerd, Timmermans, Beaujean, Oudhoff, & van Steenbergen, 2011). Anxiety was mitigated by messages that were prompt and delivered at the right time for the point of the pandemic/epidemic, transparent, recommended preventative behaviours, and provided evolving outbreak trends (i.e. giving information as it happens) (R. Li et al., 2016). People are more likely to engage in behaviour inconsistent with the message if messages lack clarity or there is inconsistent messaging both over time and between sources. This eventually results in individuals distrusting the message. Therefore, one paper argues besides information and excessive exposure of such information there is also a need for behavioural modelling (to see the behaviour by others) (O. L. Davis, Fante, & Jacobi, 2013). Information in messages needs to be relevant and trustworthy to influence behaviour change and can include: catastrophic potential, probability of dying, and reasons for exposure. Providing explanations for perceived differences from other countries, not being too optimistic and not hide information can increase overall trust towards authorities (Lohiniva et al., 2020). Attempting to increase knowledge in the context of low levels of public trust in the source can make a message ineffective, and mistrust can increase if the perception is that the information is exaggerated (Jhummon-Mahadnac et al., 2012; Teasdale & Yardley, 2011). When an epidemic (e.g. Ebola) seemed contained, and contact tracing was successful, trust increased and, in response, so did the adherence to messages (e.g. reporting to health facility) (Sumo et al., 2019).

*Multiple and credible sources*

Individuals seek information from multiple sources to meet different information needs (Cole & Watkins, 2015). Whilst official sources of information (such as governmental sources, or public health bodies) are seen as legitimate, individuals do not always find them useful (Toppenberg-Pejcic et al., 2019; Tully, Dalrymple, & Young, 2019) thereby seek other sources. When official sources take a long time to provide information, individuals rely on informal sources such as rumours, word of mouth, social media (Cole & Watkins, 2015; van der Weerd et al., 2011); the value of information is at its highest when the outbreak is emerging. If mass media do not report the outbreak in time (before unofficial/informal sources start reporting the outbreak) it fails to become the leading indicator that people use for their information (J. Li, Xu, Cuomo, Purushothaman, & Mackey, 2020). Delay can impact the public's trust in official sources (e.g. government) and has led to beliefs that the threat is exaggerated by government or news and media (J. Li et al., 2020; Teasdale & Yardley, 2011) especially when information is conflicting. Furthermore, a delay in addressing rumours



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circulating among the public increases the chances of apathy and communication fatigue (Mitchell et al., 2014).

The grey literature reported that individuals seek information from sources that they consider trustworthy these include public health experts, WHO and Centers for Disease Control and Prevention (CDC) and state and local governments; news media were the least trusted sources (Wirz 2020). Another preprint found that there was greater compliance and acceptance of messages when the messages came from government officials and credible sources (Abu-Akel, 2020). The effect on behaviour change (self-reported mask wearing and mask buying) from messaging was greater amongst individuals who trusted the Centers for Disease Control and Prevention (CDC), government agencies and scientists (M. H. Goldberg, Gustafson, A., Maibach, E.W., Ballew, M.T., Bergquist, P., Kotcher, J.E., Marlon, J.R., Rosenthal, S.A., Leiserowitz, A.,, 2020) and these messages were better received (Merkley, 2020). Explicit information, data safety and transparency were important (Bradshaw 2020) especially when describing protective behaviours (Dai, 2020). Whilst individuals generally preferred expert sources for the initial information-seeking, they eventually relied on other sources (and perceptions such as "anti-intellectualism" increased); eventually leading to discontinuing use of official sources for information during a pandemic (Merkley, 2020).

### Community engagement and social networks

Social networks and close ties to the community are drivers of better knowledge and compliance with preventive measures. A systematic review suggests that non-traditional channels of communication (i.e. partnership with community leaders or organizations) should be used to reach out to the most vulnerable (can include those who have a disability e.g. hearing/vision/intellectual) and those who are least literate as well as non-Native speakers. Public health communication messages are still delivered at a literacy level that does not meet the needs of the less educated. To reduce communication inequalities, those who are less likely to identify risk associated with the threat and so less likely to adhere to the behaviours should be targeted. These include the young, least educated and hard to reach communities (Lin, Savoia, et al., 2014). The public pay more attention if the community is engaged in the intervention (Schiavo, May Leung, & Brown, 2014) as communication is a dynamic process and to address the concerns, values, interests and priorities the public need to be partners in communication (Barrelet et al., 2013).

The inclusion of various communities increases the chances of developing materials with which all individuals can identify (Crosier et al., 2015). This in turn increases the chances of perceiving the message is relevant to their self and important people in their lives thus heightening risk perception, and changing behaviours (Crouse Quinn, 2008; Gray et al., 2012; Person et al., 2004). Individuals want information that fits with their experiences (Freiman et al., 2011; Teasdale & Yardley, 2011). Adaptable and personalised information, that is context-driven, is more effective in changing determinants of behaviour (e.g. especially in vulnerable groups) (Daellenbach, Parkinson, & Krisjanous, 2018; Qiu et al., 2018). Core messages should be consistent to ensure trust but also considerations need to be made to ensure the involvement of communities to address their needs (e.g. including community leaders, translations, access to resources). Message delivery should be appropriate for the targeted population; for example, social media can open new avenues but should not to other types of delivery (Toppenberg-Pejcic et al., 2019). There is a use for social media, as it can be effective in communicating messages (Lwin, Lu, Sheldenkar, & Schulz, 2018), fostering trust and providing opportunities for dialogue (Tully et al., 2019). Social media can be especially



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useful for reaching young people who trust social media channels (R. Li et al., 2016). Social-economic status (SES) is related to information access and understanding; those of lower SES were less likely to use a website, and more likely to find public health messages from TV and Radio to be confusing and contradictory than those of higher SES (Aburto et al., 2010).

The grey literature highlighted that developing prosocial messages and identity, increased willingness to self-isolate especially when producing a strong, positive emotional response (Heffner, 2020). Another preprint found that messages should not just target those who are at risk but those who do not perceive themselves at risk from COVID-19 (e.g. young people and hard to reach groups). Messages must also address barriers to physical distancing and outline benefits of carrying them out, emphasising altruism through messaging from credible sources using social media (Wirz 2020). Another preprint found that prosocial messages were the most effective type of framing, tapping into people's sense of morality especially at the beginning of a pandemic. This changed over time and self-interest messaging were just as effective (Jordan, 2020).

## Messages for sub-populations

### Young (Student populations) and older adult populations

Three studies examined perceptions and experiences of H1N1 messages with student populations (Idoiaga, De Montes, & Valencia, 2016; Miczo, Danhour, Lester, & Bryant, 2013; Mitchell et al., 2014). An experimental study on framing found that younger students (mean age = 21.05) perceived themselves as less vulnerable than older students (over the age of 60) in falling ill with flu when reading a news report that described a flu epidemic (Idoiaga et al., 2016). The authors suggest this may be due to the 'social invulnerability identity' of younger people. This is consistent with social cognitive approaches to health risk perception, which suggests people may be "unrealistically optimistic" about a particular health threat (Weinstein, 1987).

Campus and mass media were reported, by students, as the most common places for getting information about the H1N1. Students tend to perceive information from the university as more credible than the media (Mitchell et al., 2014).

Focus groups exploring reactions to two different campaign ideas (for a potential avian influenza epidemic in Australia) found that there was some confusion amongst older adults whereby they presumed the 'WIPE' part of the WASH, WIPE, and WEAR' message was referring to handkerchiefs, rather than disposable tissues. Older adults (and separately mothers) predominantly preferred messages which emphasised protecting others (e.g. that they should wear a mask if they were sick in order to protect others) (Jones, Waters, Holland, Bevins, & Iverson, 2010). On the other hand, regional and frequent travellers' groups thought it was equally important to protect themselves from other people's germs, and also protect others from their germs (Jones et al., 2010).

### Vulnerable groups

The importance of both protecting themselves and others was evident in findings from focus groups conducted with vulnerable groups in New Zealand during the influenza H1N1 2009 pandemic (Gray et al., 2012). People wanted messages about specific actions that they could take to protect themselves and their families during the pandemic.



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During the influenza H1N1 2009 pandemic, a cross-sectional survey was conducted with migrant workers (i.e., anyone who did not have Thai citizenship) to explore their perceptions of and ability to implement non-pharmaceutical interventions (NPIs; e.g., using face masks when sick) proposed by the WHO (Hickey et al., 2014). Certain groups of migrant workers were considered as a vulnerable group due to traditions in raising poultry and swine, poor personal hygiene and sanitation, low health literacy, limited access to healthcare, and high frequency of cross-border communication. Attitudes towards recommended NPIs were generally negative or ambivalent. Barriers included never having worn a face mask and not knowing the correct way to wear one. This highlights the need to educate vulnerable groups about performing NPIs during pandemics as migrants are at risk of propagating the spread of a pandemic virus. It is also important that public education campaigns can reach migrants living in remote and difficult to access areas, and to accommodate for the diverse cultural and linguistic needs.

The importance of ensuring that public health messages can reach different linguistic groups was also echoed by the Pacific Peoples' participants in the focus groups conducted by Gray et al. (Gray et al., 2012). Older people in the group are unlikely to understand English so messages should be communicated in the original languages. However, as noted by Person et al., (Person et al., 2004) this can be very difficult when an in-house translation service does not exist, and the rapidly evolving scientific evidence will challenge the turnaround time for developing, translating, and disseminating information. To help address this, the team prioritised key SARS documents for translation and back-translations conducted by professional translation services ensured accuracy. The translated information should be disseminated through culturally appropriate channels such as community visits, town hall meetings, and health and education and communication channels to complement mass media messages. This is similar to reflections of Singapore's experience in dealing with the SARS outbreak (Menon, 2006) where there was ongoing transparent dialogue with grassroots leaders and communities, and religious groups.

Social Economic Status (SES)

A cross-sectional survey with 1010 adult Italians during the influenza H1N1 2009 pandemic found that respondents who were women, and were facing economic hardship, were more likely to clean objects, wash their hands, and use tissues when sneezing (Prati et al., 2011).

A Mexican study, which explored the public's ability and willingness to adopt community mitigation efforts during the influenza H1N1 2009 pandemic, found that respondents from lower SES groups were more likely to report costs of masks being a barrier to their use (Aburto et al., 2010). Respondents from lower SES tertials were also more likely to report that mitigation recommendations were contradictory or confusing. The authors highlight that it is important for future campaigns to tailor the messaging to people of lower SES who are more likely to have lower literacy levels and reduced ability to interpret messages. Messages should be coordinated, consistent, and simplified for these groups.

*Evaluation of messages at salient points in an epidemic/pandemic*

This presents the papers that had specific aims for the time of the study (either beginning, during, or post a pandemic/epidemic) and these are mapped to recommendations developed in this review. Evaluations collecting data at different time points over the course of a pandemic can tell us about any changes in patterns of public behaviour or perceptions as the situation evolves.



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Table 1: Papers organised according to timepoints of epidemic/pandemic

| | Study | Aim of messages | Mapped to Recommendations | |
|---|---|---|---|---|
| **Beginning of epidemic/pandemic** | Yardley et al 2011°<br>Chang et al, 2012°<br>Teasdale et al, 2011°<br>Mitchell et al, 2014°<br>Kavanagh et al, 2011°<br>Freiman et al, 2001° | Increase knowledge and awareness | • Early announcements<br>• Consistency of messages<br>• Acknowledge uncertainties<br>• Timing of recommendations / announcements / alerts | **Addressing uncertainty and increasing trust** |
| | Aburto et al, 2010°<br>Hickey et al, 2014†<br>Roess et al, 2017†<br>Bonwitt et al, 2018†<br>Winters et al, 2018†<br>Lohiniva et al, 2020 ¥ | Empower use of behaviours to prevent spread of virus within the first 6 months | • Integrate community /community leaders in risk comms and into planning<br>• Social responsibility<br>• Tailor to increase accessibility | **Community engagement** |
| | | | • Different sources of information – ensure consistency<br>• Clear instructions<br>• Language to explain severity | **Message framing and unified messaging** |
| **During epidemic/pandemic** | Jhummon-Mahadnac et al, 2010°<br>Gray et al, 2012°<br>Prati et al, 2011°<br>Updegraff et al, 2011° | Assess efficacy of messages/ campaigns | • Responding to concerns questions<br>• Skilled / credible spokespeople<br>• Admitting mistakes<br>• Acknowledge uncertainties<br>• Acknowledge unknowns and knowns<br>• Labelling recommendations as interim | **Addressing uncertainty and increasing trust** |
| | Rim et al, 2014°<br>Johnson & Slovic, 2015†<br>Toppenberg et al, 2018†<br>Tully et al, 2019†<br>Davis et al, 2013‡ | Update and possibly introduce new behaviours based on new information<br>Continue with raising awareness and maintain behaviours | • Tailor information to users needs (accessibility)<br>• Use of local spokespeople and influencers<br>• Translate and use different sources with consistency in messaging | **Community engagement** |
| | Bekalu et al, 2017‡<br>Rimi et al 2016+<br>Avery et al, 2009+ | Manage misinformation | • Increase understanding of health threat<br>• Promote sense of personal control<br>• Social responsibility | **Message framing and unified messaging** |
| **Post** | Davis et al, 2015°<br>Basnyat & lee, 2014°<br>Miczo et al, 2013°<br>Crosier et al 2015°<br>Barrelet et al 2013°<br>Person et al, 2004•<br>Qui et al, 2017+•<br>Lyu et al, 2012• | Assess efficacy of messages/ campaigns<br>Manage misinformation | • Responding to concerns questions<br>• Skilled / credible spokespeople<br>• Admitting mistakes<br>• Acknowledge uncertainties<br>• Acknowledge unknowns and knowns | **Addressing uncertainty and increasing trust** |

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| | Study | Aim of messages | Mapped to Recommendations | |
|---|---|---|---|---|
| | Li et al 2016[+]<br><br>Sell et al 2017†<br>Sumo et al 2019†<br>Cole & Watkins, 2015† | Evaluation for lessons learnt and Preparation training | • Public release of evaluations and reviews | |
| | | | • Translate and use different sources with consistency in messaging<br>• Equity considerations | **Community engagement** |
| | | | • identify inconsistencies<br>• Core messages consistent<br>• understanding of health threat | **Message framing and unified messaging** |

Note a: °Study at beginning, during or post H1N1; ¥ Study during COVID-19; †Study at the beginning, during or post Ebola; ‡Study at beginning, during or post influenza season; [+]Study at beginning, during or post avian flu; • Study at beginning, during or post SARS

## Appendix 7: Rapid review findings mapped to the BPS COVID-19 Behavioural Science and Disease Prevention Psychological Guidance

| Recommendation in review | BPS Behavioural Science guidance |
|---|---|
| Address uncertainty | 2. Deliver messages from a credible source in relatable terms to the target audience |
| Engaging communities | 1. Minimise the 'I' and emphasise the 'we'<br><br>4. Identify what influences each preventative behaviour and ensure policies, messaging and interventions target all relevant drivers<br><br>6. Avoid unintended negative consequences |
| Unified messaging | 5. Clearly specify behaviours and their effectiveness<br><br>7. Create clear channels of access for health literacy<br><br>8. Use behavioural scientists and the psychological evidence base to support the COVID-19 response<br><br>9. Make a pledge to work together, through a multi-disciplinary approach. #CombatCovid19Together |
| Message framing | 3. Create worry but not fear |



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## AUTHORS

Professor Madelynne A. Arden is an HCPC registered Health Psychologist and a Professor of Health Psychology at Sheffield Hallam University, specialising in the development and evaluation of theory and evidence based behaviour change interventions.

Professor Christopher J. Armitage is an HCPC registered Health Psychologist, a Fellow of the Academy of Social Sciences and Professor of Health Psychology at the University of Manchester. He specialises in using psychological theory to develop tools to facilitate behaviour change among diverse populations.

Professor Lucie Byrne-Davis is an HCPC registered Health Psychologist, Principal Fellow of the Higher Education Academy, Fellow of the European Health Psychology Society and a Professor of Health Psychology at the University of Manchester, specialising in health professional behaviour change and global health.

Natalie Carr is a health psychology MSc student at Manchester Metropolitan University.

Professor Angel Chater is an HCPC registered Health Psychologist, a Professor in Health Psychology and Behaviour Change at the University of Bedfordshire and Leads the British Psychological Society's Division of Health Psychology and COVID-19 Behavioural Science and Disease Prevention Taskforce.

Dr Tracy Epton is a lecturer and researcher, at the University of Manchester, specialising in behaviour change.

Dr Daniela Ghio is a lecturer at the University of Salford, specialising in intervention development and health behaviour change.

Ella Guest is a Research Associate at the Centre for Appearance Research in Bristol and Stage 2 Trainee in final year at UWE, Bristol.

Professor Jo Hart is an HCPC registered Health Psychologist and a professor of health professional education at University of Manchester, specialising in health professional education and behaviour change.

Dr Sarah Hotham is a Senior Research Fellow and BPS Chartered Psychologist at the Centre for Health Services Studies, University of Kent specialising in applied heath research and training of healthcare professionals.

Dr Neil Howlett is a Senior Research Fellow in the Department of Psychology and Sport Sciences, University of Hertfordshire, specialising in physical activity and wider behaviour change methodology.

Dr Elizabeth Jenkinson is a Senior Lecturer in Health Psychology and an HCPC registered Health Psychologist at the University of the West of England, Bristol.

Dr Angelos P. Kassianos is a Senior Research Fellow, a Fellow of Higher Education Academy and an HCPC registered Health Psychologist, at UCL (Department of Applied Health Research) specialising in



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disease prevention, medical education as well as developing and evaluating evidence-based behaviour change digital interventions.

Dr Chris Keyworth is a Research Fellow, at The University of Manchester, specialising in health communication and behaviour change.

Dr Sadie Lawes- Wickwar is a HCPC registered Health Psychologist. Associate Fellow of BPS & Member of DHP. Visiting Lecturer in Health Psychology at City, University of London.

Daniel Scanlan has a MSc in Health Psychology and works as a Research Assistant at Education Support.

Sabina Stanescu is a Health Psychology Trainee and a PhD Candidate within Psychology at the University of Southampton.

Dr Natalia Stanulewicz is a Lecturer in Psychology at De Montfort University. Her research focuses on health behaviours, wellbeing as well as emotional and individual characteristics predicting prosocial behaviours.

Lisa Sutherland - Behavioural Insight Consultant. MSc in Health Psychology 20 years working in digital, social marketing and behavioural change intervention design.

Dr Rachael Thorneloe is a Senior Research Fellow in the Centre for Behavioural Science and Applied Psychology, Sheffield Hallam University, specialising in behaviour change.

Dr Mei Yee Tang is a Research Associate at the Behavioural Science Policy Research Unit at Newcastle University.

Daniella Watson is a Trainee Health Psychologist & Global Health PhD student (University of Southampton).

Juliette Westbrook is a Research Assistant at the University of Reading and studying MSc Health Psychology at the University of Bath.

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# PRISMA 2009 Checklist

| Section/topic | # | Checklist item | Reported on page # |
|---|---|---|---|
| **TITLE** | | | |
| Title | 1 | Identify the report as a systematic review, meta-analysis, or both. | 1 |
| **ABSTRACT** | | | |
| Structured summary | 2 | Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number. | 2 |
| **INTRODUCTION** | | | |
| Rationale | 3 | Describe the rationale for the review in the context of what is already known. | 3 |
| Objectives | 4 | Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS). | 3 |
| **METHODS** | | | |
| Protocol and registration | 5 | Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number. | 3 |
| Eligibility criteria | 6 | Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale. | 4 |
| Information sources | 7 | Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched. | 4 |
| Search | 8 | Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated. | S2-S3 |
| Study selection | 9 | State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis). | 4 |
| Data collection process | 10 | Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators. | 4 |
| Data items | 11 | List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made. | 4, S5 |
| Risk of bias in individual studies | 12 | Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis. | 4 |
| Summary measures | 13 | State the principal summary measures (e.g., risk ratio, difference in means). | N/A |
| Synthesis of results | 14 | Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., $I^2$) for each meta-analysis. | N/A |

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# PRISMA 2009 Checklist



| Section/topic | # | Checklist item | Reported on page # |
|---|---|---|---|
| Risk of bias across studies | 15 | Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies). | N/A |
| Additional analyses | 16 | Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified. | N/A |
| **RESULTS** | | | |
| Study selection | 17 | Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram. | 5 Figure 1 |
| Study characteristics | 18 | For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations. | S5 |
| Risk of bias within studies | 19 | Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12). | S41 |
| Results of individual studies | 20 | For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot. | N/A |
| Synthesis of results | 21 | Present results of each meta-analysis done, including confidence intervals and measures of consistency. | 5 |
| Risk of bias across studies | 22 | Present results of any assessment of risk of bias across studies (see Item 15). | 5 |
| Additional analysis | 23 | Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]). | 5 |
| **DISCUSSION** | | | |
| Summary of evidence | 24 | Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers). | 12 |
| Limitations | 25 | Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias). | 13 |
| Conclusions | 26 | Provide a general interpretation of the results in the context of other evidence, and implications for future research. | 13 |
| **FUNDING** | | | |
| Funding | 27 | Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review. | 4 |

*From:* Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(7): e1000097. doi:10.1371/journal.pmed1000097

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# What influences people's responses to public health messages for managing risks and preventing infectious diseases? A rapid systematic review of the evidence and recommendations



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| | |
|---|---|
| | Division of Medical Education<br>Byrne-Davis, Lucie; The University of Manchester School of Medical Sciences, Division of Medical Education<br>Keyworth, Christopher; University of Leeds Faculty of Medicine and Health, School of Psychology |
| Primary Subject Heading: | Public health |
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| Keywords: | Public health < INFECTIOUS DISEASES, COVID-19, Infection control < INFECTIOUS DISEASES |
| | |

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# What influences people's responses to public health messages for managing risks and preventing infectious diseases? A rapid systematic review of the evidence and recommendations

Daniela Ghio,[1,4] Sadie Lawes- Wickwar,[2] Mei Yee Tang,[3] Tracy Epton, [4] Neil Howlett, [5] Elizabeth Jenkinson,[6] Sabina Stanescu, [7] Juliette Westbrook, [8] Angelos P. Kassianos [9] Daniella Watson, [10] Lisa Sutherland, [11] Natalia Stanulewicz, [12] Ella Guest, [13] Daniel Scanlan, [14] Natalie Carr, [15] Angel Chater, [16] Sarah Hotham, [17] Rachael Thorneloe, [18] Christopher J Armitage [4,19] Madelynne A Arden, [18] Jo Hart, [4] Lucie Byrne-Davis, [4] Chris Keyworth [20]

As part of the Health Psychology Exchange [healthpsychologyexchange@outlook.com]

Correspondence to [tracy.epton@manchester.ac.uk]

[1] Department of Psychology, Faculty of Health and Society, University of Salford, Manchester, United Kingdom

[2] Department of Primary Care and Population Health, University College London, London, UK

[3] Behavioural Science Policy Research Unit, Population Health Sciences, Newcastle University, UK

[4] Manchester Centre for Health Psychology, University of Manchester, Manchester, UK

[5] Department of Psychology, Sports, and Geography, School of Life and Medical Sciences, University of Hertfordshire, Hertfordshire, UK

[6] Department of Health and Social Sciences, University of West England, Bristol, UK

[7] School of Psychology, University of Southampton, Southampton, United Kingdom

[8] Department of Psychology, University of Bath, Bath, UK

[9] Department of Applied Health Research, University College London, London, UK

[10] Human Development and Health, Faculty of Medicine, University of Southampton. UK

[11] Behavioural Insight, United Kingdom

[12] Faculty of Health and Life Sciences, School of Applied Social Sciences, De Montfort University, Leicester, United Kingdom

[13] Centre for Appearance Research, Department of Health and Social Sciences, Faculty of Health and Applied Sciences, University of the West of England, Bristol, United Kingdom

[14] Department of Communication, Policy, and Research, Education Support, London, UK

[15] Faculty of Health, Psychology, and Social Care, Manchester Metropolitan University, Manchester, UK

[16] Centre for Health, Wellbeing and Behaviour Change, University of Bedfordshire, Bedfordshire, UK

[17] Centre for Health Services Studies, University of Kent, Canterbury, United Kingdom

[18] Centre for Behavioural Science and Applied Psychology, Sheffield Hallam University, Sheffield, UK

[19]Manchester University NHS Foundation Trust, Manchester Academic Health Science Centre, Manchester; and NIHR Greater Manchester Patient Safety Translational Research Centre, University of Manchester, Manchester, UK

[20]School of Psychology, University of Leeds , Leeds, UK

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**Abstract**

***Background***

Individual behaviour changes, such as hand hygiene and physical distancing, are required on a population scale to reduce transmission of infectious diseases such as COVID-19. However, little is known about effective methods of communicating risk reducing information, and how populations might respond.

***Objective***

To synthesise evidence relating to what: a) characterises effective public health messages for managing risk and preventing infectious disease, b) influences people's responses to messages.

***Design***

A rapid systematic review was conducted. Protocol is published on Prospero CRD42020188704.

***Data sources***

Electronic databases were searched: Ovid Medline, Ovid PsycINFO and Healthevidence.org, and grey literature (PsyarXiv, OSF Preprints) up to May 2020.

***Study selection***

All study designs were included that: (a) evaluated public health messaging interventions targeted at adults, (b) concerned a communicable disease spread via primary route of transmission of respiratory and/or touch. Outcomes included preventative behaviours, perceptions/awareness and intentions. Non-English language papers were excluded.

***Synthesis***

Due to high heterogeneity studies were synthesised narratively focusing on determinants of intentions in the absence of measured adherence/preventative behaviours. Themes were developed independently by two researchers and discussed within team to reach consensus. Recommendations were translated from narrative synthesis to provide evidence-based methods in providing effective messaging.

***Results***

Sixty-eight eligible papers were identified. Characteristics of effective messaging include delivery by credible sources, community engagement, increasing awareness/knowledge, mapping to stage of epidemic/pandemic. To influence intent effectively, public health messages need to be acceptable, increase understanding/perceptions of health threat and perceived susceptibility.

***Discussion***

There are four key recommendations: (1) engage communities in development of messaging, (2) address uncertainty immediately and with transparency, (3) focus on



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unifying messages from sources, and (4) frame messages aimed at increasing understanding, social responsibility and personal control. Embedding principles of behavioural science into public health messaging is an important step towards more effective health-risk communication during epidemics/pandemics.

**Strengths and Limitations**

- Whilst we conducted a rapid review, we ensured that we completed it in a systematic manner with a broad initial search (e.g. no restriction on study design) to develop recommendations from lessons in risk communication that we can translate to the current pandemic.
- The rapid review included all study designs with high heterogeneity, some of which were low quality, so findings should be interpreted tentatively.
- The focus of most the studies included in the review was on determinants of intention and not behaviour, therefore we acknowledge that the recommendations may not lead to successful enactment of target behaviours (e.g. handwashing) even though they may be helpful in increasing intentions.
- We were unable to conduct backward and forward citation searching on the included studies, this may have resulted in relevant literature not being captured.
- We had strong stakeholder engagement as part of the team with different expertise of behaviour science and public health that provided feedback from initial design through to development of recommendations to be used by public health practitioners.

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**Background**

The outbreak of novel infectious diseases, including COVID-19, require rapid changes to existing (e.g. related to physical distance) and new (e.g. use of face coverings) behaviours in the context of uncertainties and often rapidly-evolving new knowledge. Public health messaging is one component of effective risk communication strategies to ensure sustained population level behaviour change. However, little is known about what characterises effective public health messages for dealing with infectious diseases and what factors influence the public's response.

Drawing from the Behaviour Change Wheel (BCW) [1] system for intervention development, the "COM-B" model of behaviour change proposes that public health messages should be designed as multi-component strategies to support people's capability (the knowledge/skills), opportunity (societal norms/physical resources) and motivation (the desire/habit) to act; factors at the heart of the behaviour change wheel.[1] This theory has been used in current guidance to highlight core considerations for reducing the spread of COVID-19 [2], namely, increasing knowledge of personal protective behaviours (capability), signposting and making available resources (opportunity), and explaining why behaviour change is important, while mitigating emotional reactions (motivation).

When building public health campaigns or interventions, it is important to consider past evidence to build a behavioural diagnosis using COM-B; accepted modes of delivery; and an evaluation of outcomes. It is crucial to understand public health messaging in rapidly changing epidemics/pandemics and for this, a review of the evidence base is needed.

***Review aims***

To conduct a rapid systematic review[3] to identify and synthesise evidence in order to provide evidence-based recommendations for designing and delivering public health messages for health authorities and social care organisations dealing with infectious disease outbreaks. This review aims to identify:

a) What influences people's responses to public health messages about health-risk communication

b) What characterises effective public health messages for managing risk and preventing disease during epidemics/pandemics

## Methods

Rapid systematic review methods[3] the searches started in May 2020 and analysis was completed in July 2020. Public health messages were characterised as: messages delivered by mobile-technology, news broadcasts/newspapers, posters, leaflets and press conferences.

***Information sources***

The following databases were searched: Ovid PsycINFO, Ovid MEDLINE, PsyArXiv, OSF Preprints, and healthevidence.org (search strategies are presented in Appendix 1). The



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protocol for this rapid systematic review is published on Prospero CRD42020188704. The SPICE criteria (Setting, Perspective, Phenomena of Interest, Comparison, Evaluation, Time Scope), was used to guide data extraction.[4] Our search strategy was piloted with a scoping review to ensure that the terms were capturing all relevant literature and to also choose which databases to search. These terms were then shared within the team and with public health practitioners and behaviour science experts for feedback using an iterative process to finalise our search terms.

***Patient and Public Involvement***

This work was a rapid response to a request by Public Health Practitioners to the Health Psychology Exchange consortium. The Patient Public Involvement and Engagement group within the Health Psychology Exchange group were consulted when developing the protocol.

***Ethical Approval***

No ethical approval was sought to undertake this rapid systematic review.

***Eligibility criteria***

All study designs were considered for inclusion (e.g. systematic reviews, empirical studies) and grey literature (e.g. guidelines, frameworks, and policy documents) with no date restrictions until 20 May 2020.

Papers were included if they:

- evaluated a public health messaging intervention targeted at adults aged 18 years and above (no limitations on population or region),
- concerned a communicable disease spread via primary route of transmission of respiratory and/or touch (human to human contact)
- were written in English.

Papers on HIV were excluded as they involved different preventative behaviours and therefore deemed to be out of scope of the review. Papers that focused exclusively on public health messaging for vaccination uptake (intentions and uptake) during epidemic/pandemics were noted and the findings synthesised in a separate review.[5]

To ensure a broad range of literature relating to epidemics/pandemics/ health crisis communication could be captured studies were not excluded based on outcome. However, outcomes of interest included preventative behaviours (e.g. handwashing, quarantining), perceptions (e.g. risk), intent, and awareness.

***Study selection***

Titles/abstracts (80% double screened) and full texts were screened by 15 authors (Figure 1; for further breakdown of the included studies see Appendix 2 in supplementary materials). Conflicts over inclusion (2.3% had disagreements) were resolved through discussions with 4 authors (Appendix 3 provides detail on each author's roles in screening, extraction and synthesis).

***Data extraction***



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Characteristics of the papers (e.g. type of message, quality of study), the type of health risk and results were extracted (Appendix 4). Four authors (JW, SS, NC, DS) screened and completed a data quality check using Mixed Methods Appraisal Tool[6] for the 54 individual papers, the 11 pre-prints and AMSTAR[7] for the systematic reviews (Appendix 5). Overall, there was a moderate agreement level between the reviewers with 61% level of agreement. Disagreements were resolved through discussion with moderators.

***Synthesis of results***

A narrative synthesis was conducted[8] to identify key themes with respect to: (a) what influenced people's responses to public health messages in general and for subpopulations in particular during salient time points (further details presented in Appendix 6) and (b) interpreted recommendations for effective public health messaging for managing risk and preventing disease during epidemics/pandemics, which are presented below. The synthesis involved combining the results of reviews and individual studies reporting (a) qualitative studies, (b) quantitative studies, and (c) both qualitative and quantitative studies, in order to describe the recommendations for effective delivery of public health messages. These were exported into NVivo (Version 12) to data manage the combined results of different papers. To establish trustworthiness in data analysis, discussions among several members of the study team were held at fortnightly intervals to develop the coding framework, and to discuss, refine, and group the emerging codes into overall explanatory themes. All study authors were involved in establishing the conceptual framework.

[insert figure 1 PRISMA Diagram]

## Results

A total of 68 papers rated as high to moderate-quality (about 50% of them scoring as high-quality and 32% of moderate-quality) were included: 3 systematic reviews, 54 individual peer-reviewed papers and 11 preprints. The papers focused mainly on Influenza A virus subtype H1N1 (n=20), Covid-19 (n=15) and Ebola (n=11), other diseases (n= 12) which have emerged at different timepoints in the last 50 years. The timelines from initial outbreaks are highlighted in Figure 2. The included studies were conducted at various time points (e.g., beginning, during or post) during these pandemics and eleven did not report the timing (Appendix 6). Key variables included: (a) behaviours (e.g. hand-washing, quarantine, using tissues, physical-distancing), (b) cognitive factors (e.g. increase in awareness, perceived risk), and (c) emotions (worry, anxiety). (See Appendix 5 for full characteristics of papers).

[insert figure 2 here Timelines of pandemics/epidemics]

A narrative analysis of the papers was conducted on what was mostly qualitative work that reported on determinants of intent to adhere to guidelines. These were organised according to preconceptions and understanding of the threat, perceived susceptibility, and perceived risk severity (threat appraisal). This narrative analysis is presented in



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Appendix 6. Across the different themes and sub-themes developed about community engagement, messages for sub-populations, increasing trust, perceptions and understanding of threat and threat appraisal we developed four areas of recommendations to provide evidence-based steps to be taken to provide effective public health messaging during pandemics/epidemics. These recommendations are cross-referenced to the narrative synthesis in Table 1 and the recommendations are reported below with evidence summarised.

| Recommendation | Cross-reference to narrative synthesis in appendix 6 |
|---|---|
| **1. Engaging with key stakeholders and communities** | |
| 1.a. Involve community leaders and others perceived as credible sources within the community | Community engagement<br><br>Messages for sub-populations<br><br>Increase trust |
| 1.b. Tailoring helps to make the key messages applicable to an individual's situation | |
| 1.c. Consider any difficulties accessing information and levels of literacy | |
| 1.d. Use different media for delivery and match delivery to the population's needs and perceptions | |
| **2. Addressing uncertainty immediately with transparency** | |
| 2.a. Address uncertainty and changing information that may exist during an ongoing public health crisis | Increase trust<br><br>Preconceptions and understanding threat<br><br>Timing – beginning of health-crisis |
| 2.b. Consistency and co-ordination between different sources of information | |
| 2.c. Be transparent: admit errors and unknowns whenever appropriate | |
| 2.d. Be transparent: identify sources of information | |
| **3. Unified messages** | |
| 3.a. Make core messages consistent | Increase trust<br><br>Threat appraisal<br><br>Preconceptions and understanding threat |
| 3.b. Identify inconsistencies across sources | |
| 3.c. Increase awareness of the risks of the virus to their own health and the health of others | |
| **4. Message framing** | |
| 4.a. Increase understanding of health threat | Preconceptions and understanding threat<br><br>Threat appraisal |
| 4.b. Consider social responsibility | |
| 4.c. Language choice to explain severity | |



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| | |
|---|---|
| 4.d. Promote sense of personal control | Community engagement |

Table 1: Recommendations mapped and cross-referenced to Narrative Synthesis

## *Recommendations*

Four recommendations were derived from the evidence (See Appendix 6 for a comprehensive report of the evidence on influences on effective public health messaging, messages for sub-populations and communication at salient points in an epidemic/pandemic). Results were synthesised in Figure 3, including the recommendations and influences on behaviours. The recommendations are (1) engaging with different communities, (2) addressing uncertainty immediately, with transparency, (3) unified messaging, and (4) message framing.

[insert figure 3 Diagram of synthesis of recommendations]

### *1. Engaging with key stakeholders and communities*

**1a. Involve community leaders and others perceived as credible sources within the community**

One high-quality study indicated that messages designed without input from the target population may lead to low levels of public adherence to behaviour change messages. [9] A moderate-quality review found that this is especially important when dealing with inconsistencies and changing information. [10]

A high-quality systematic review found that the public pay more attention to messages if the community is engaged in its development [11] as this will ensure that the information is relatable,[9] and addresses the concerns, values, interests and priorities of the community.[10] Consequently, this may lead to heightened perceptions of personal risk. Two studies of low to moderate-quality, suggested that one way to do so is to include community leaders, to find people that are trusted[12] and allow faith-based organisations to help.[13] One moderate-quality study found that culturally and linguistically appropriate messages (e.g. in videos) can also help retain long-term knowledge of preventative behaviours.[14]

A preprint study stated that over time preferred expert sources (e.g. government websites) are displaced by unofficial sources (e.g. social media) for information regarding epidemic/pandemics;[15] therefore, developing ties within the community (e.g. trusted spokesperson) can be helpful to provide accurate information. A high-quality study found that students tend to perceive information from their university (from their own communities) as more credible than the media. [16] One low-quality study found that



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community engagement is also important for quickly disseminating messages which are translated into different languages.[17, 18]

**1b. Tailoring helps to make the key messages applicable to an individual's situation**

A high-quality systematic review found that those who are less likely to accept they are at personal risk of the threat (e.g. the young, least educated and hard to engage communities) are also less likely to adhere to the recommended behaviours.[19] This has important implications for reducing inequalities through tailored public health messaging. Social networks and having close ties to the community are drivers of better knowledge and behaviour change. Partnership with community leaders and/or community organizations should be used to reach out to the most vulnerable (this may include those who have a disability e.g. hearing/vision) and those who are least literate as well as non-Native speakers.[19]

Individuals desire information that fits with their experiences.[20-23] Studies of moderate-quality have shown that adaptable and personalised information, that is context-driven, is more effective in changing determinants of behaviour, especially in vulnerable groups.[24, 25] One moderate-quality study and two high-quality studies showed that messages should be tailored to take into account: (a) different levels of perceived susceptibility (e.g., younger adults see themselves as less vulnerable than older adults[26]), (b) likelihood of misunderstanding instructions (e.g., older people thought a campaign was referring to handkerchiefs rather than disposable tissues)[27] and (c) skills needed to enact the behaviour (e.g., migrant workers didn't know how to wear a face-covering).[28] One low-quality study found that there may be differences in message preferences (e.g., older adults and mothers preferred messages that emphasised the protection of others).[27]

**1c. Consider any difficulties accessing information and levels of literacy**

Messages are typically delivered at a high literacy level.[19] When individuals do not understand the message, they may engage in behaviours that reflect their understanding.[29] Some target groups, such as those with low literacy levels, could particularly benefit from a simplified, clear, and consistent message. [30] Pilot-testing messages can help identify phrasing that can be confusing or unfamiliar.[31]

It is also important to ensure public health messages reach non-native speakers by using translated materials as shown in a high-quality study.[32] Older people in non-native groups in particular may not understand English. Thus, messages should be communicated in the original languages and through generations. Translated information should be disseminated through culturally appropriate channels such as community visits, town hall meetings, and health and education and communication channels to complement mass



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media messages. This reinforces the need for community leaders (recommendation 1a) in aiding with translations and identifying the appropriate channels.

Some people have limited experiences of engaging in recommended behaviours (e.g. using face-coverings or a thermometer).[33] These limited experiences highlight a need for training/skill development to be included as part of a public health campaign. This will improve health literacy and self-efficacy, especially when it refers to 'new' behaviours. Including training/skill development fits in with taking a COM-B model approach in developing public health messaging as it increases an individual's physical/psychological capabilities. Benefits to taking this approach could be enhanced with equally improving motivation (reflective and automatic) by considering other recommendations (e.g. recommendation 4: Message framing) as well as considering opportunity for behaviour (social and physical) which can identify potential barriers (e.g. social norms).

**1d. Use different media for delivery and match delivery to the population's needs and perceptions**

Message delivery should be appropriate for the targeted population. For example, two studies of moderate to high-quality suggested that social media can reach younger people.[34, 35] A study of moderate-quality showed that social media can be effective in communicating messages fostering trust and providing opportunities for dialogue.[36] However, as highlighted in two studies of high-quality, there are disadvantages as (a) social media is also associated with misconceptions, such as what causes the disease[37] and (b) those without internet access (or a device) would be excluded.[38] A high-quality study found that those of lower socioeconomic status SES were less likely to use a website, and more likely to find public health messages from TV and Radio to be confusing and contradictory than those of higher SES.[30] In specific situations for example, messages through the use of posters in bathrooms to increase handwashing need to not just have prompts for the behaviour but also messages about transmission as a high-quality study found that prompts alone do not increase handwashing.[39] This is consistent with the Health Belief Model where cues of action can trigger behaviour but requires cognitive representations of perceived susceptibility and perceived barriers/costs to action.

## *2. Addressing uncertainty immediately and with transparency*

**2a. Address uncertainty and changing information that may exist during an ongoing public health crisis**



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Public health messages with emerging epidemics/pandemics are likely to involve much uncertainty about the virus and the appropriate preventative behaviours. A high-quality systematic review found that honest reporting about the threat, through a presentation of known and unknown factors, increases people's knowledge and makes attitudes and beliefs more positive; it also increases trust in the way the government is handling the emergency.[19]

As epidemics/pandemics develop the information and recommended behaviours may change. Changing information decreases trust in the government.[40] However, three studies of moderate to high-quality reported that prompt (i.e. giving information as it happens), stage appropriate (i.e., not seen as too lenient or too extreme for the risk level), and transparent messaging (i.e. that includes recommended preventative behaviours) reduces anxiety around the reported health risk.[41]

**2b. Consistency and co-ordination between different sources of information**

Different sources of information may give information and behaviour change instructions that are inconsistent with each other, especially as knowledge rapidly changes over time. Differences in the information and behaviour change instructions between countries should be explained. Information should be transparent, not be hidden and predictions should not be too optimistic – this will increase overall trust towards authorities.[42, 43]

**2c. Be transparent: admit errors and unknowns whenever appropriate**

A high-quality study showed that if uncertainties are not acknowledged or transparency is not provided, the consequences of the errors are addressed and this can reduce trust in agencies, thus reducing information acceptance and compliance with recommendations.[44]

Two studies of moderate to high-quality found that information needs to be released as early as possible at the start of the outbreak, whenever there is conflicting information and even if there are unknowns.[45, 46] Honest, open[25] and explicit information, transparency and assurances of personal data safety are important[47] especially when promoting preventative behaviours.[48] A low-quality study found that if official sources report the outbreak before unofficial/informal sources, then they become the leading indicator that people use for their information.[17] Delay can impact the public's trust in official sources that leads to beliefs that the threat is exaggerated by government or news media[22] and increases the chance of apathy and communication fatigue.[16]

**2d. Be transparent: identify sources of information**



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Two high-quality studies reported that attempting to increase knowledge in the context of low levels of public trust in the source can make a message ineffective; and mistrust can increase if the perception is that the information is exaggerated[22, 49] or if the outbreak is perceived as uncontrolled.[50] Three preprint studies reported that trust is key to the acceptance of messaging that can lead to behaviour change (e.g. using masks, physical distance), thus there is greater adherence when the messages come from trusted sources. [15, 51, 52] Sources that potentially can be perceived as credible by the general population can include public health experts, organisations (e.g. Centers for Disease Control and Prevention) and state and local governments.[53, 54] A study found that believability of messages was reduced by scepticism towards the media and governments; in these instances, the threat was perceived as exaggerated and able to cause unnecessary panic which could influence low adoption of recommended behaviours.[22] Therefore, it is important to ensure transparency highlighting that information sources are credible and legitimate.[55]

### *3. Unified messages*

**3a. Deliver consistent, clear, core messages about risk and preventative behaviour across sources within the same time points**

Where possible, messages should be unified across sources; as shown by two high-quality studies, this is especially important for those groups who have lower literacy (e.g., lower SES) as they are more likely to perceive messages as confusing and inconsistent.[30, 38] A moderate-quality review found that unifying messages result in a greater understanding of the health message, greater perceived risk and clear guidelines for behaviour;[10, 30] these all contribute to increased trust. A high-quality study found that a repeated measures survey during the H1N1 pandemic, showed trust levels in the government decreased over time and this decrease was due to conflicting messages at concurrent time-points.[40] In two moderate-quality studies it was found that if messages from different sources provide clear, consistent instructions these are more likely to be recalled and adhered to.[56] When messages provided the public with clear consistent information, whilst admitting that information is evolving, reported the risks, and focused on risk-reducing actions, people were more likely to perform protective behaviours (e.g., clean objects, wash hands and use tissues when sneezing).[57]

**3b. Identify inconsistencies in messages from uncontrolled sources, especially when addressing key preventative behaviours**

Whilst official sources of information (such as governmental sources, or public health bodies) are perceived as legitimate, individuals do not always find them useful as



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highlighted in one high-quality study and one moderate-quality study,[35, 36] and consequently seek other sources (e.g. unofficial sources such as family or social media). In a moderate-quality study, it was found that individuals seek information from multiple sources to meet different information needs.[45] Different unofficial sources of information may give information and behaviour change instructions that are inconsistent with each other and official sources. Additionally, as found in a high-quality study, unofficial sources may detract from understanding of preventative behaviours.[58] It is important to acknowledge and address the inconsistencies in unofficial sources.

When addressing inconsistency, a high-quality study found that it is also important to recognise that some messages may appear inconsistent due to varying levels of personal risk of different groups (conditional messages). To improve clarity these conditional messages should be identified and where possible explained or dispelled.[32]

**3c. Increase the public's awareness of the risks of the virus to their own health and the health of others**

Public health messaging that included information about the threat can be effective in identifying symptoms and changing behaviours. From six high-quality studies, it was found that the key aim of public health messages early in an epidemic/pandemic is typically to increase knowledge and awareness of the health-risk (see Appendix 6 for papers and priorities mapped onto timepoints).[16, 20, 22, 55, 59, 60] However, knowledge about the virus alone is not sufficient to change behaviour;[59] acknowledgement of unknown factors, how to identify symptoms, and how to prevent contracting and spreading the virus is also necessary as highlighted by five high-quality studies and a high-quality systematic review, two moderate-quality and one low-quality study. [19, 22, 32, 33, 41, 55, 61, 62] A low-quality study found that when perceived risk is low, ignoring recommended behaviours is rationalised.[63]

## *4. Message framing*

**4a. Increase factual knowledge of all aspects of a virus (e.g. symptoms) and benefits of preventative behaviour using an appropriate message frame**

Six high-quality studies found that framing and choice of language are influential in how individuals understand the threat and in turn behave. [22, 64-68] Positively framed messages (emphasising the benefits of prevention behaviours) may be effective.[64] For example, gain-framed signs (e.g., "stay healthy this season. Sanitize your hands") are more effective in influencing the use of hand sanitizer than signs that emphasised people's susceptibility to contamination.[65]



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Non-narrative messages (i.e., factual) are more effective than narrative messages (e.g. storytelling such as a movie *Contagion*) in changing knowledge and perceived response efficacy related to prevention of influenza.[68] Factual and scientific knowledge in messages in the media can positively influence risk perceptions.[66] Whilst formal information (presented from credible sources) increased understanding.[69]

**4b. Consider framing messages around social responsibility and norms**

Cohesive social networks and having close ties to the community are drivers of better knowledge and compliance with preventive measures. There is some moderate-quality evidence to suggest that framing messages about others' risk, in addition to your own risk, is effective in increasing information seeking.[45] In a high-quality systematic review, it was found that being worried (about self or family members at risk) was an important predictor of compliance with recommended preventative behaviours, such as using tissues, hand gel and washing hands.[19] The effects of worry about others at risk on compliance with preventative behaviours can potentially be amplified when combined with messages about being socially responsible. A high-quality study found framing messages with positive social responsibility to be useful for the public.[64]

The grey literature highlighted that developing prosocial messages and promoting positive emotional appeals, increased willingness to self-isolate especially when producing a strong, positive emotional response (such as fear).[70] Prosocial framing is effective in changing behaviour; this may be due to inducing compassion, activating social norms, altruism or moral duties.[54, 71]

**4c. Choice of language needs to be clear and appropriate to understanding the magnitude of risk**

A high-quality systematic review and a moderate-quality study, found that messages should sufficiently increase worry and perceived severity of risk to self and others[19, 45] so as highlighted in a high-quality study, recommendations are seen as proportionate[22] and behaviour change is more likely to occur.[19] However, in two high-quality studies, it was found that inducing too much fear has mixed results on behaviour: it can be counterproductive if this leads to panic.[49, 72]

There can also be confusion about the use of technical terms. A high-quality study found that it is difficult to differentiate between pandemic flu and seasonal flu especially when symptoms are framed as "flu-like symptoms." [22]

**4d. Frame the message to emphasise positive beliefs about one's own health and that preventative behaviour is within their control**



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A study of moderate-quality found that when messages provided the public with clear consistent information (whilst admitting that information and evidence is evolving), focused on the practical actions that people can take to reduce their risk and emphasized the efficacy of those actions, people were more likely to perform preventative behaviours.[57] This may also further empower them and help them become socially responsible as shown in a low-quality study.[73] People are more likely to follow guidelines when there are fewer perceived barriers to perform recommended behaviours, when benefits are emphasised, and when contextual factors (e.g., anxiety about missed work) are addressed.[22]

Recommendation of increasing self-efficacy through messaging is well supported in theoretical frameworks about behaviour change and risk communication for example, the Theory of Planned Behaviour[74], Health Belief Model[75], Protection Motivation Theory[76], the Health Action Process Approach[77], COM-B model[1] and Social Cognitive Theory[78, 79]. Increasing self-efficacy has positive implications on planning, intention possibly then behaviour. This review includes limited but high quality evidence that people want messages about specific actions that they could take to protect themselves and their families during the epidemic/pandemic[32] with messages that emphasise the benefits of prevention behaviours (positively framed) potentially effective in increasing uptake and compliance.[64]. informing the individual regarding preventative actions to stop the spread of the disease is particularly important in the early stages (see Appendix 6 for breakdown of timepoints). In a moderate-quality study, it was suggested that including non-health benefits may improve adherence to recommendations. [63]

## Discussion

The aim of this systematic review was to synthesis evidence-based recommendations for designing and delivering public health messaging. We found that inconsistent messages across different sources can be detrimental to any public health messaging campaigns,[80] which is particularly true in times of public health-crisis and uncertainty.[81] Miscommunication, or contradictions can result in lower compliance of preventative behaviours. Literacy is fundamental to the understanding and acceptance of any public health messaging campaign. Therefore, public health messaging should focus on clarity, simplicity, transparency and unified messaging even if tailoring to different groups regardless of level of literacy. Even when the message is understood, different populations have different barriers to address in order to change behaviour. A COM-B behavioural diagnosis[1] can help to inform public health strategies (e.g., lower SES groups find costs of face-coverings a barrier)[30] as suggested in recent British Psychological Society guidance (see appendix 7 to see how current recommendations map onto the guidance).[2] Our review suggests that unified messaging can increase trust[45] build community resilience[32], and increase perceived risk and knowledge of threat[49]. Furthermore, consistent messaging through different time points during an evolving



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pandemic are important to consider for context-specific recommendations; these were rarely considered or reported in the studies included in the review.

We found evidence of mode of delivery as an important consideration of public health messaging. Specifically, there should be careful consideration of how to communicate public health messages, which should target specific populations. For example, making use of social media platforms to target younger populations,[16, 34] or the radio as a delivery method for older populations.[20] Translations for different cultures, and the inclusion of community leaders as part of public health message campaigns, should also be prioritised. This would ensure that all populations are reached (and not stigmatized), and the knowledge, concerns, cultural values, interests, and priorities of the targeted populations are all considered.[82] However, as noted in one of the studies this can be very difficult when an in-house translation service does not exist, and the rapidly evolving scientific evidence will challenge the turnaround time for developing, translating, and disseminating information.[83]

***Limitations***

We acknowledge the present rapid systematic review has limitations. Although we searched multiple databases systematically it is possible that relevant research was excluded from this review since we did not have the resources to translate non-English language papers in such a short space of time or conduct backward and forward citation searching. The inclusion of unpublished literature in the review means some findings may change once these papers have been published in peer-reviewed journals. However, our inclusive approach means a range of types of messages and a variety of factors related to what influences public perceptions of messages have been identified and used to inform recommendations for messaging during current and future epidemic/pandemics.

The aim of this rapid review was to synthesise lessons learnt from previous epidemics/pandemics to provide evidence-based recommendations about what characteristics create effective messaging. The focus of most studies was on determinants of intent and not behaviour, which may have implications on successful enactment of target behaviours. As highlighted in theories (such as health action process approach[77]) intention formation is part of the process and key to planning and more work is needed to understand the translation into action. Inclusion of different infectious disease (although the messaging would be of similar behaviours) may have included different contextual influences that we could not account for when synthesising the data (e.g. different countries and different social norms or political influences). Furthermore, more work is needed to understand the moderating effects of individual differences on message acceptance.

Much of our evidence is consistent with components of relevant health behaviour models, (e.g., Protection Motivation Theory,[76] Health Action Process Approach,[77] and Health Belief Model[75]) with some studies reporting the use of the models in their design. These models state that people perform protective behaviours when they perceive (a)



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the health threat as sufficiently threatening (i.e., they are vulnerable to a severe risk), (b) the recommended behaviours to be effective, and (c) they have self-efficacy to perform the behaviour. We found evidence that such cognitive appraisals were important considerations for developing public health messages, but further research is needed to examine effects on behaviour change.

***Conclusions***

Increasing knowledge and awareness of health risks alone is unlikely to be sufficient to increase understanding of risk and subsequent risk reduction behaviours, and adherence to recommendations. Rather, taking a multifaceted approach to public health messaging which considers all relevant drivers of behaviour (social, psychological and environmental factors), is recommended. Our four key recommendations should be considered when designing and delivering public health messages: engage communities in the development of public health messaging, using credible and legitimate sources, address uncertainty immediately and with transparency, focus on unified messages from all sources, and develop messages aimed at increasing understanding, inducing social responsibility and empowering personal control. These are being translated into practical guidelines for agencies. Embedding these principles of behavioural science into public health messaging is an important step towards more effective health-risk communication.

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**Figure Captions:**

Figure 1: PRISMA Diagram

Figure 2: Timelines of pandemics/epidemics included in review

Note:1Centre for Disease Control & Prevention (CDC). Years of Ebola Virus Disease Outbreak. 2019. Available from: https://www.cdc.gov/vhf/ebola/history/chronology.html; 2.WHO. H5N1 avian influenza: Timeline of major events. 2012. Available from: https://www.who.int/influenza/human_animal_interface/H5N1_avian_influenza_update.pdf; 3CDC. CDC SARS Response Timeline. 2013. Available from: https://www.cdc.gov/about/history/sars/timeline.htm; 4CDC. 2009 H1N1 Pandemic Timeline. Available from: https://www.cdc.gov/flu/pandemic-resources/2009-pandemic-timeline.html; 5WHO. Rolling updates on coronavirus disease (COVID-19). 2020. Available from: https://www.who.int/emergencies/diseases/novel-coronavirus-2019/events-as-they-happen

Figure 3: Diagram of synthesis of recommendations and influences on behaviour



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## References


1. Michie S, van Stralen MM, West R. The behaviour change wheel: A new method for characterising and designing behaviour change interventions. Implement Sci. 2011;6(1):42.
2. Chater AA, M.A.; Armitage C.J., Byre-Davis, L.; Chadwick, P.; Drury, J.; Hart, J.; Lewis, L.; McBride, E.; Perriard-Abdoh, S.; Thompson, S.; Whittaker, E.; O'Connor, D.;. Behavioural science and disease prevention: psychological guidance: British Psychological Society; 2020 [
3. Dobbins M. Rapid review guidebook. National Collaborating Centre for Methods and Tools, Hamilton, ON. 2017.
4. Booth A. Clear and present questions: formulating questions for evidence based practice. Library Hi Tech. 2006;24(3):355-68.
5. Lawes-Wickwar SG, Daniela; Tang, Mei Yee; Keyworth, Chris; Stanescu, Sabina; Westbrook, Juliette; Jenkinson Elizabeth; Kassianos, Angelos, P.; Scanlan, Daniel; Garnett Natalie; Laidlaw, Lynn; Byrne-Davis, Lucie; Chater, Angel; Epton Tracy; and members of the Health Psychology Exchange (HPX) Public Health MEssaging Review Group A rapid review of responses to public health messages encouraging vaccination to prevent the spread of a pandemic or epidemic virus. in submission. 2020.
6. Hong QN, Gonzalez-Reyes A, Pluye P. Improving the usefulness of a tool for appraising the quality of qualitative, quantitative and mixed methods studies, the Mixed Methods Appraisal Tool (MMAT). Journal of evaluation in clinical practice. 2018;24(3):459-67.
7. Shea BJ, Grimshaw JM, Wells GA, Boers M, Andersson N, Hamel C, et al. Development of AMSTAR: a measurement tool to assess the methodological quality of systematic reviews. BMC Med Res Methodol. 2007;7(1):10.
8. Popay J, Roberts H, Sowden A, Petticrew M, Arai L, Rodgers M, et al. Guidance on the conduct of narrative synthesis in systematic reviews. A product from the ESRC methods programme Version. 2006;1:b92.
9. Crosier A, McVey D, French J. 'By failing to prepare you are preparing to fail': Lessons from the 2009 H1N1 'swine flu' pandemic. [References]: European Journal of Public Health. Vol.25(1), 2015, pp. 135-139.; 2015.
10. Barrelet C, Bourrier M, Burton-Jeangros C, Schindler M. Unresolved issues in risk communication research: the case of the H1N1 pandemic (2009-2011). Influenza other respi. 2013;7 Suppl 2:114-9.
11. Schiavo R, May Leung M, Brown M. Communicating risk and promoting disease mitigation measures in epidemics and emerging disease settings. Pathog Glob Health. 2014;108(2):76-94.
12. Lyu SY, Chen RY, Wang SF, Weng YL, Peng EY, Lee MB. Perception of spokespersons' performance and characteristics in crisis communication: experience of the 2003 severe acute respiratory syndrome outbreak in Taiwan. J Formos Med Assoc. 2013;112(10):600-7.
13. Santibanez S, Siegel V, O'Sullivan M, Lacson R, Jorstad C. Health communications and community mobilization during an Ebola response: partnerships with community and faith-based organizations. Public Health Rep. 2015;130(2):128-33.
14. Roess AA, Di Peppi R, Kinzoni EA, Molouania M, Kennedy E, Ibata SR, et al. Knowledge Gained and Retained from a Video-Centered, Community-Based Intervention for Ebola Prevention, Congo. J Health Commun. 2017;22(11):913-22.
15. Merkley E, Loewen PJ, Owen T, Ruths D. Anti-intellectualism and Information Preferences during the COVID-19 Pandemic. 2020.




For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

16. Mitchell T, Massoudi M, Swerdlow DL, Dee DL, Gould LH, Kutty PK, et al. Swine flu in college: early campus response to outbreak control measures. Am J Health Behav. 2014;38(3):448-64.
17. Menon KU. SARS revisited: managing "outbreaks" with "communications". Ann Acad Med Singapore. 2006;35(5):361-7.
18. Vaughan E, Tinker T. Effective health risk communication about pandemic influenza for vulnerable populations. Am J Public Health. 2009;99 Suppl 2:S324-32.
19. Lin L, Savoia E, Agboola F, Viswanath K. What have we learned about communication inequalities during the H1N1 pandemic: a systematic review of the literature. BMC Public Health. 2014;14:484.
20. Freiman AJ, Montgomery JP, Green JJ, Thomas DL, Kleiner AM, Boulton ML. Did H1N1 influenza prevention messages reach the vulnerable population along the Mississippi Gulf Coast? J Public Health Manag Pract. 2011;17(1):52-8.
21. Qian X, Ren R, Wang Y, Guo Y, Fang J, Wu ZD, et al. Fighting against the common enemy of COVID-19: a practice of building a community with a shared future for mankind. Infect. 2020;9(1):34.
22. Teasdale E, Yardley L. Understanding responses to government health recommendations: public perceptions of government advice for managing the H1N1 (swine flu) influenza pandemic. Patient Educ Couns. 2011;85(3):413-8.
23. Davis MD, Stephenson N, Lohm D, Waller E, Flowers P. Beyond resistance: social factors in the general public response to pandemic influenza. BMC Public Health. 2015;15:436.
24. Daellenbach K, Parkinson J, Krisjanous J. Just how prepared are you? An application of marketing segmentation and theory of planned behavior for disaster preparation. [References]: Journal of Nonprofit & Public Sector Marketing. Vol.30(4), 2018, pp. 413-443.; 2018.
25. Qiu W, Chu C, Hou X, Rutherford S, Zhu B, Tong Z, et al. A Comparison of China's Risk Communication in Response to SARS and H7N9 Using Principles Drawn From International Practice. Disaster med. 2018;12(5):587-98.
26. Idoiaga N, De Montes LG, Valencia J-F. Communication and representation of risk in health crises: The influence of framing and group identity. [References]: Revista de Psicologia Social. Vol.31(1), 2016, pp. 59-74.; 2016.
27. Jones SC, Waters L, Holland O, Bevins J, Iverson D. Developing pandemic communication strategies: Preparation without panic. [References]: Journal of Business Research. Vol.63(2), 2010, pp. 126-132.; 2010.
28. Hickey JE, Gagnon AJ, Jitthai N. Knowledge about pandemic influenza preparedness among vulnerable migrants in Thailand. Health Promot Internation. 2016;31(1):124-32.
29. Bonwitt J, Dawson M, Kandeh M, Ansumana R, Sahr F, Brown H, et al. Unintended consequences of the 'bushmeat ban' in West Africa during the 2013-2016 Ebola virus disease epidemic. Soc Sci Med. 2018;200:166-73.
30. Aburto NJ, Pevzner E, Lopez-Ridaura R, Rojas R, Lopez-Gatell H, Lazcano E, et al. Knowledge and adoption of community mitigation efforts in Mexico during the 2009 H1N1 pandemic. Am J Prev Med. 2010;39(5):395-402.
31. Lapka C, Jupka K, Wray RJ, Jacobsen H. Applying cognitive response testing in message development and pre-testing. [References]: Health Education Research. Vol.23(3), 2008, pp. 467-476.; 2008.
32. Gray L, MacDonald C, Mackie B, Paton D, Johnston D, Baker MG. Community responses to communication campaigns for influenza A (H1N1): a focus group study. BMC Public Health. 2012;12:205.
33. Hickey J, Gagnon AJ, Jitthai N. Pandemic preparedness: perceptions of vulnerable migrants in Thailand towards WHO-recommended non-pharmaceutical interventions: a cross-sectional study. BMC Public Health. 2014;14:665.



For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

34. Seltzer EK, Jean NS, Kramer-Golinkoff E, Asch DA, Merchant RM. The content of social media's shared images about Ebola: A retrospective study. [References]: Public Health. Vol.129(9), 2015, pp. 1273-1277.; 2015.
35. Toppenberg-Pejcic D, Noyes J, Allen T, Alexander N, Vanderford M, Gamhewage G. Emergency risk communication: Lessons learned from a rapid review of recent gray literature on Ebola, Zika, and yellow fever. [References]: Health Communication. Vol.34(4), 2019, pp. 437-455.; 2019.
36. Tully M, Dalrymple KE, Young R. Contextualizing nonprofits' use of links on Twitter during the West African Ebola virus epidemic. [References]: Communication Studies. Vol.70(3), 2019, pp. 313-331.; 2019.
37. Winters M, Jalloh MF, Sengeh P, Jalloh MB, Conteh L, Bunnell R, et al. Risk Communication and Ebola-Specific Knowledge and Behavior during 2014-2015 Outbreak, Sierra Leone. Emerg Infect Dis. 2018;24(2):336-44.
38. Avery EJ, Kim S. Anticipating or precipitating crisis? Health agencies may not be heeding best practice advice in avian flu press releases. [References]: Journal of Public Relations Research. Vol.21(2), 2009, pp. 187-197.; 2009.
39. Davis OL, Fante RM, Jacobi LL. The effectiveness of sign prompts to increase hand washing behaviors in restrooms. [References]: North American Journal of Psychology. Vol.15(3), 2013, pp. 565-576.; 2013.
40. van der Weerd W, Timmermans DR, Beaujean DJ, Oudhoff J, van Steenbergen JE. Monitoring the level of government trust, risk perception and intention of the general public to adopt protective measures during the influenza A (H1N1) pandemic in The Netherlands. BMC Public Health. 2011;11:575.
41. Li R, Xie R, Yang C, Frost M. Perceptions on the risk communication strategy during the 2013 avian influenza A/H7N9 outbreak in humans in China: a focus group study. Western pac. 2016;7(3):21-8.
42. Lohiniva AL, Sane J, Sibenberg K, Puumalainen T, Salminen M. Understanding coronavirus disease (COVID-19) risk perceptions among the public to enhance risk communication efforts: a practical approach for outbreaks, Finland, February 2020. Euro Surveill. 2020;25(13):04.
43. Rogers WA, Street JM, Braunack-Mayer AJ, Hiller JE, FluViews T. Pandemic influenza communication: views from a deliberative forum. Health Expect. 2009;12(3):331-42.
44. Holmes BJ, Henrich N, Hancock S, Lestou V. Communicating with the public during health crises: Experts' experiences and opinions. [References]: Journal of Risk Research. Vol.12(6), 2009, pp. 793-807.; 2009.
45. Cole J, Watkins C. International employees' concerns during serious disease outbreaks and the potential impact on business continuity: Lessons identified from the 2014-15 West African Ebola outbreak. J Bus Contin Emer Plan. 2015;9(2):149-62.
46. Johnson BB, Slovic P. Fearing or fearsome Ebola communication? Keeping the public in the dark about possible post-21-day symptoms and infectiousness could backfire. [References]: Health, Risk & Society. Vol.17(5-6), 2015, pp. 458-471.; 2015.
47. Bradshaw EL, Ryan R, Noetel M, Saeri AK, Slattery P, Grundy E, et al. Information safety assurances affect intentions to use COVID-19 contact tracing applications, regardless of autonomy-supportive or controlling message framing. 2020.
48. Dai B, Fu D, Meng G, Liu B, Li Q, Liu X. The effects of governmental and individual predictors on COVID-19 protective behaviors in China: a path analysis model. Public Administration Review. 2020.
49. Jhummon-Mahadnac ND, Knott J, Marshall C. A cross-sectional study of pandemic influenza health literacy and the effect of a public health campaign. BMC Res Notes. 2012;5:377.
50. Sumo J, George G, Weah V, Skrip L, Rude JM, Clement P, et al. Risk communication during disease outbreak response in post-Ebola Liberia: experiences in Sinoe and Grand Kru counties. Pan Afr Med J. 2019;33(Suppl 2):4.



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51. Abu-Akel A, Spitz A, West R. Who is listening? Spokesperson Effect on Communicating Social and Physical Distancing Measures During the COVID-19 Pandemic. 2020.
52. Goldberg M, Gustafson A, Maibach E, Ballew MT, Bergquist P, Kotcher J, et al. Mask-wearing increases after a government recommendation: A natural experiment in the US during the COVID-19 pandemic. 2020.
53. Berry TR, Wharf-Higgins J, Naylor PJ. SARS wars: an examination of the quantity and construction of health information in the news media. Health Commun. 2007;21(1):35-44.
54. Wirz CD, Schwakopf JM, Brossard D, Brown LD, Brauer M. Self-reported compliance and attitudes about social distancing during the COVID-19 outbreak Preprint.
55. Yardley L, Miller S, Schlotz W, Little P. Evaluation of a Web-based intervention to promote hand hygiene: exploratory randomized controlled trial. J Med Internet Res. 2011;13(4):e107.
56. Miczo N, Danhour E, Lester KE, Bryant J. Memorable messages and the H1N1 flu virus. [References]: Western Journal of Communication. Vol.77(5), 2013, pp. 625-644.; 2013.
57. Prati G, Pietrantoni L, Zani B. Compliance with recommendations for pandemic influenza H1N1 2009: the role of trust and personal beliefs. Health Educ Res. 2011;26(5):761-9.
58. McVernon J, Mason K, Petrony S, Nathan P, LaMontagne AD, Bentley R, et al. Recommendations for and compliance with social restrictions during implementation of school closures in the early phase of the influenza A (H1N1) 2009 outbreak in Melbourne, Australia. BMC Infect Dis. 2011;11:257.
59. Kavanagh AM, Bentley RJ, Mason KE, McVernon J, Petrony S, Fielding J, et al. Sources, perceived usefulness and understanding of information disseminated to families who entered home quarantine during the H1N1 pandemic in Victoria, Australia: a cross-sectional study. BMC Infect Dis. 2011;11:2.
60. Chang C. News coverage of health-related issues and its impacts on perceptions: Taiwan as an example. [References]: Health Communication. Vol.27(2), 2012, pp. 111-123.; 2012.
61. Hoda J. Identification of information types and sources by the public for promoting awareness of Middle East respiratory syndrome coronavirus in Saudi Arabia. Health Educ Res. 2016;31(1):12-23.
62. Rim H, Ha JH, Kiousis S. The evidence of compelling arguments in agenda building: Relationships among public information subsidies, media coverage, and risk perceptions during a pandemic outbreak. [References]: Journal of Communication Management. Vol.18(1), 2014, pp. 101-116.; 2014.
63. Rimi NA, Sultana R, Ishtiak-Ahmed K, Rahman MZ, Hasin M, Islam MS, et al. Understanding the failure of a behavior change intervention to reduce risk behaviors for avian influenza transmission among backyard poultry raisers in rural Bangladesh: a focused ethnography. BMC Public Health. 2016;16(1):858.
64. Basnyat I, Lee ST. Framing of Influenza A (H1N1) pandemic in a Singaporean newspaper. Health Promot Internation. 2015;30(4):942-53.
65. Updegraff JA, Emanuel AS, Gallagher KM, Steinman CT. Framing flu prevention-An experimental field test of signs promoting hand hygiene during the 2009-2010 H1N1 pandemic. [References]: Health Psychology. Vol.30(3), 2011, pp. 295-299.; 2011.
66. Sell TK, Boddie C, McGinty EE, Pollack K, Smith KC, Burke TA, et al. Media Messages and Perception of Risk for Ebola Virus Infection, United States. Emerg Infect Dis. 2017;23(1):108-11.
67. Liao Q, Cowling B, Lam WT, Ng MW, Fielding R. Situational awareness and health protective responses to pandemic influenza A (H1N1) in Hong Kong: a cross-sectional study. PLoS ONE. 2010;5(10):e13350.
68. Bekalu MA, Bigman CA, McCloud RF, Lin LK, Viswanath K. The relative persuasiveness of narrative versus non-narrative health messages in public health emergency communication: Evidence from a field experiment. Prev Med. 2018;111:284-90.



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69. Liao Q, Cowling B, Lam WT, Ng MW, Fielding R. Situational awareness and health protective responses to pandemic influenza a (H1N1) in Hong Kong: A cross-sectional study. [References]: PLoS ONE. Vol.5 2010, ArtID e13350.; 2010.
70. Heffner J, Vives ML, FeldmanHall O. Emotional responses to prosocial messages increase willingness to self-isolate during the COVID-19 pandemic. 2020.
71. Jordan J, Yoeli E, Rand D. Don't get it or don't spread it? Comparing self-interested versus prosocially framed COVID-19 prevention messaging. 2020.
72. Zikmund-Fisher BJ, Scherer AM, Knaus M, Das E, Fagerlin A. Discussion of Average versus Extreme Case Severity in Pandemic Risk Communications. Emerg Infect Dis. 2017;23(4):706-8.
73. Menon KU, Goh KT. Transparency and trust: Risk communications and the Singapore experience in managing SARS. [References]: Journal of Communication Management. Vol.9(4), 2005, pp. 375-383.; 2005.
74. Ajzen I. The theory of planned behavior. Organizational behavior and human decision processes. 1991;50(2):179-211.
75. Becker MH. The health belief model and personal health behavior. Health education monographs. 1974;2:324-473.
76. Rogers RW. A protection motivation theory of fear appeals and attitude change. The journal of psychology. 1975;91(1):93-114.
77. Schwartzer R. Self-efficacy in the adoption and maintenance of health behaviours: Theoretical approaches and a new model. Self-efficacy: Thought control of action Washington, DC: Hemisphere. 1992:217-43.
78. Bandura A. Social Cognitive Theory: An Agentic Perspective. Annual Review of Psychology. 2001;52(1):1-26.
79. Bandura A. Social cognitive theory of mass communication. Media psychology. 2001;3(3):265-99.
80. Hernandez-Garcia I, Gimenez-Julvez T. Assessment of Health Information About COVID-19 Prevention on the Internet: Infodemiological Study. JMIR Public Health Surveill. 2020;6(2):e18717.
81. Hooker C, Capon A, Leask J. Communicating about risk: strategies for situations where public concern is high but the risk is low. Public health res. 2017;27(1):15.
82. Lin L, Jung M, McCloud RF, Viswanath K. Media use and communication inequalities in a public health emergency: a case study of 2009-2010 pandemic influenza A virus subtype H1N1. Public Health Rep. 2014;129 Suppl 4:49-60.
83. Person B, Sy F, Holton K, Govert B, Liang A, National Center for Inectious Diseases SCOT. Fear and stigma: the epidemic within the SARS outbreak. Emerg Infect Dis. 2004;10(2):358-63.

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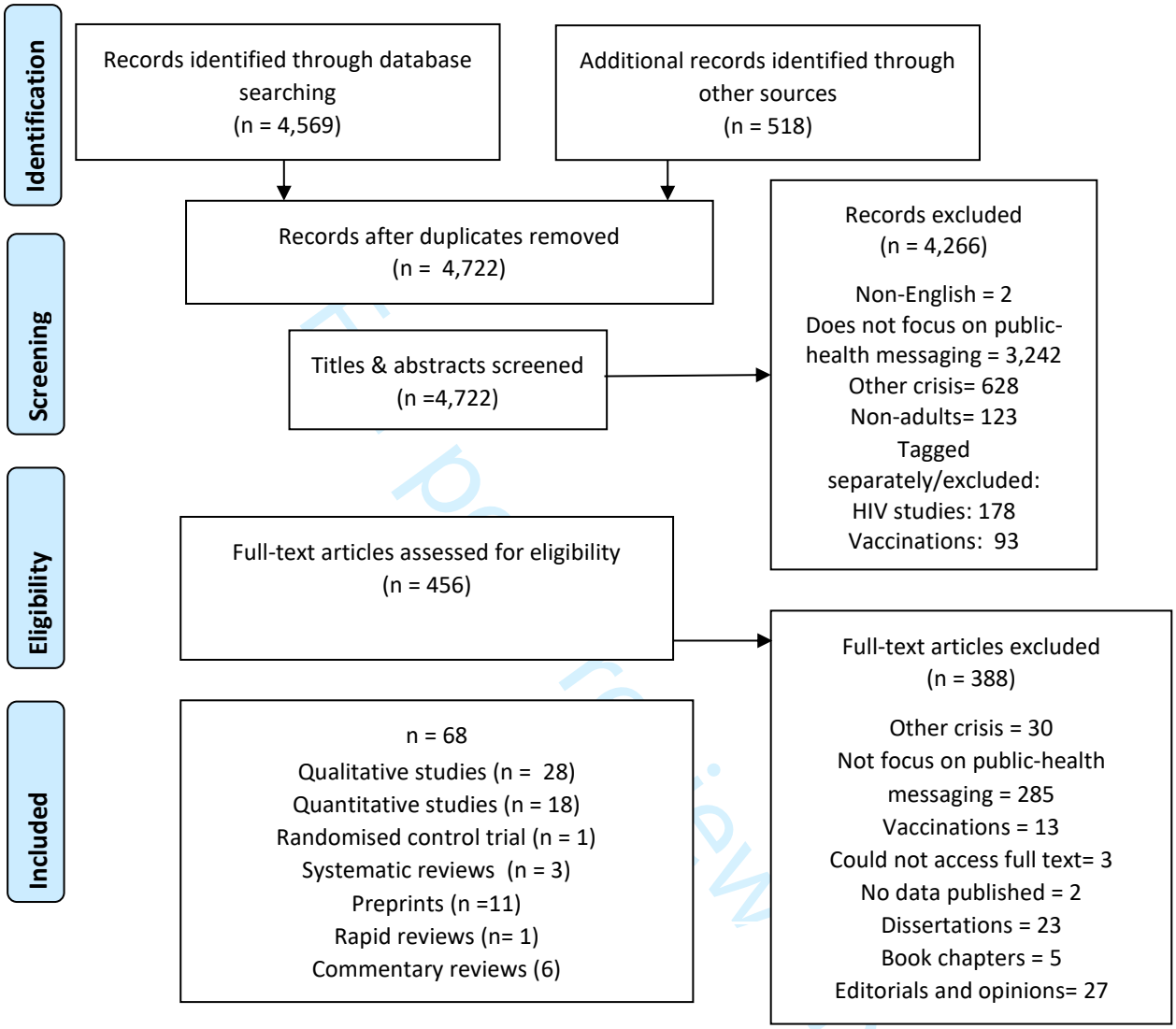



Figure 1: Prisma Diagram

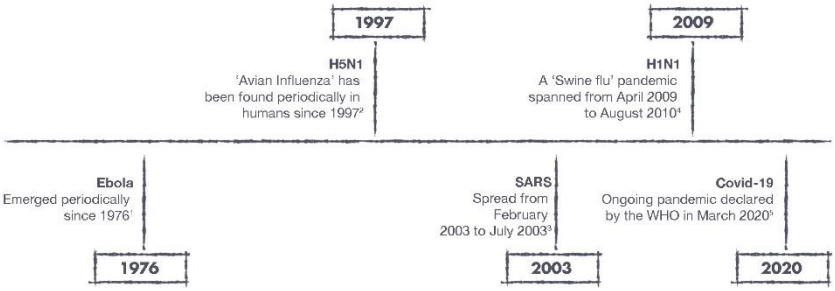



Figure 2: Timelines of pandemics/epidemics included in review

Note:[1]Centre for Disease Control & Prevention (CDC). Years of Ebola Virus Disease Outbreak. 2019. Available from: https://www.cdc.gov/vhf/ebola/history/chronology.html; [2]WHO. H5N1 avian influenza: Timeline of major events. 2012. Available from: https://www.who.int/influenza/human_animal_interface/H5N1_avian_influenza_update.pdf; [3]CDC. CDC SARS Response Timeline. 2013. Available from: https://www.cdc.gov/about/history/sars/timeline.htm; [4]CDC. 2009 H1N1 Pandemic Timeline. Available from: https://www.cdc.gov/flu/pandemic-resources/2009-pandemic-timeline.html; [5]WHO. Rolling updates on coronavirus disease (COVID-19). 2020. Available from: https://www.who.int/emergencies/diseases/novel-coronavirus-2019/events-as-they-happen

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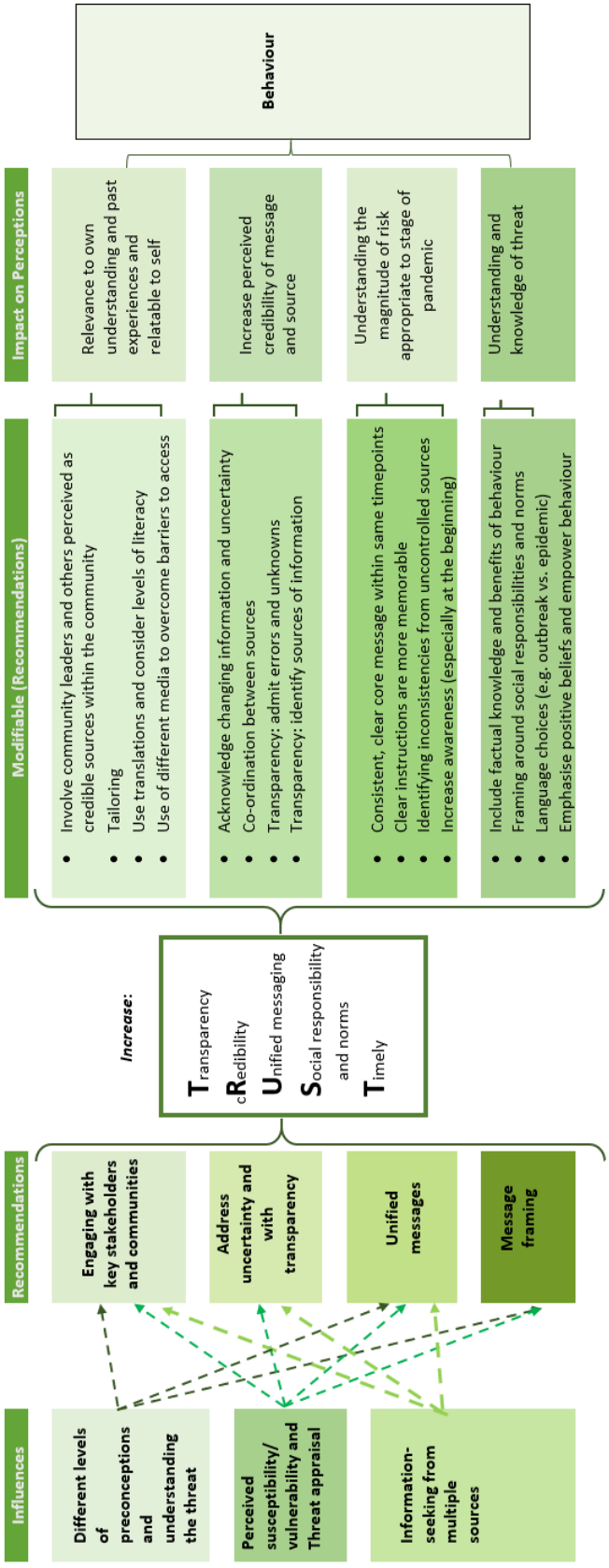



Figure 3: Diagram of synthesis of recommendations and influences on behaviour

# Supplementary Appendix

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# Appendix 1: Search Terms

*Ovid PsycINFO & Ovid MEDLINE*

| # | Search Terms | Results | Notes (e.g. where search terms were adapted from) |
|---|---|---|---|
| ***Health and risk communication terms*** | | | |
| 1 | public service announcements/ | 370 | |
| 2 | communication/ or messages/ or persuasive communication/ or scientific communication/ or social communication/ | 65306 | Imploded and selected a few narrower terms as the others were irrelevant |
| 3 | ((public health or health or risk or emergenc* or crisis* or catastroph* or disaster* or outbreak*) adj3 (communication* or campaign* or information* or plan* or message* or uncertaint* or alert* or awareness or recommendation* or guideline* or guidance or measure*)).tw. | 54841 | |
| 4 | 1 or 2 or 3 | 116708 | |
| ***Pandemic and epidemic terms*** | | | |
| 5 | exp Pandemics/ | 473 | |
| 6 | exp Epidemics/ | 3399 | |
| 7 | exp Influenza/ | 1396 | Includes Swine Flu, no MeSH terms for SARs, MERs, and Ebola so these searched as keywords only |
| 8 | exp Emergency Preparedness/ | 1192 | |
| 9 | ("SARS" or coronavirus or severe acute respiratory syndrome).tw. | 587 | |
| 10 | (ebola or ebolavirus or ebola virus).tw. | 411 | |
| 11 | ("MERS-CoV" or "MERS or Middle East Respiratory Syndrome coronavirus").tw. | 11 | |
| 12 | (influenza or swine flu or H1N1 or pH1N1* or pdmH1N1* or nH1N1*).tw. | 2490 | |
| 13 | (pandemic* or epidemic*).tw. | 14247 | |
| 14 | 5 or 6 or 7 or 8 or 9 or 10 or 11 or 12 or 13 | 17926 | |
| 15 | 4 and 14 | 1525 | |

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*Health Evidence (Reviews)*

[(Public health messag*) OR ("Risk communication") AND Pandemic OR Epidemic] (224 results)

*Grey Literature Searches*

OSF Preprints + PsyArXiv Preprints

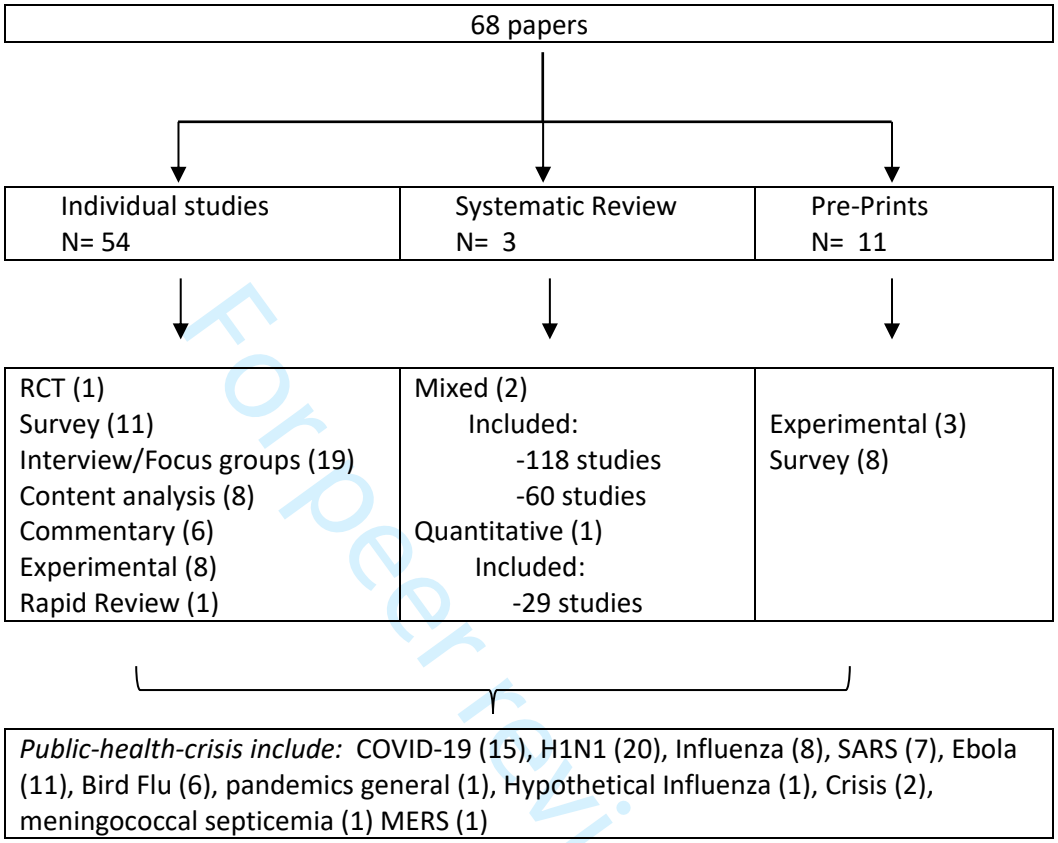
| **Search category** | **Search terms** |
|---|---|
| Intervention | (Public health messag* OR communication OR plan OR alert OR health campaign OR health recommendation* OR public information OR medical information OR public awareness OR community engagement OR risk reduction OR health promotion) |
| Phenomena of interest | (Pandemic OR catastrophe OR cris?s OR outbreak OR emergency OR Coronavirus OR Covid* OR SARS* OR ebola* OR MERS OR epidemic) |
| Combination | (Public health messag* OR communication OR plan OR alert OR health campaign OR health recommendation* OR public information OR medical information OR public awareness OR community engagement OR risk reduction OR health promotion) AND (Pandemic OR catastrophe OR cris?s OR outbreak OR emergency OR Coronavirus OR Covid* OR SARS* OR ebola* OR MERS OR epidemic) |
| Limitations | Arts & Humanities; Life Sciences; Medicine and Health Sciences; Physical Sciences and Mathematics; Social and Behavioral Sciences |
| Results | 518 |

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## Appendix 2: Breakdown of studies




## Appendix 3: screening, data extraction and synthesis details

| *Task* | *Authors* |
|---|---|
| *Screening of titles and abstracts (1st and 2nd)* | NH, EJ, SS, JW, APK, DW, LS, NS, EG, DS, NC, DG, MYT, SLW, CK |
| *Conflicts resolved* | DG, MYT, SLW, CK |
| *Development of guidance document* | DG, MYT, SLW, CK, TE, AC, SH, LBD |
| *Pilot data extraction* | AG, SLW |
| *Data extraction* | NH, EJ, SS, JW, APK, DW, LS, NS, EG, DS, NC, DG, MYT, SLW, CK |
| *Quality check* | JW, SS, DG, MYT, NC, DS |
| *Analysis* | DG, MYT, SLW, CK |
| *Iterations and discussions* | DG, MYT, SLW, CK, TE, NH, EJ, AC, SH, RT, CJA, MAA, JH, LBD |
| *Agreed on final categories* | DG, MYT, SLW, CK, TE, NH, EJ, AC, SH, RT, CJA, MAA, JH, LBD |



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Appendix 4: Data and study characteristics tables 1-4

**Table 1: Characteristics of Qualitative Studies**

| | Country | Study Type | Health Threat | Intervention type | Target behaviour | Main findings |
|---|---|---|---|---|---|---|
| **Avery (2009)** | USA | Content Analysis of avian flu press releases | Avian flu | Press releases issued by federal and global health agencies | Not reported | The threat was not localized; outlets for publics to contact for more information were not always present, and there was inconsistency in how avian flu was referenced. There was a steady increase in the volume of releases since 2004, and public health information officers generally seemed to recognize the importance of partnerships and timeliness in pandemic preparedness. Public health communicators must establish a strong sense of efficacy in an informed public to insure that audiences will adhere to protocol rather than succumb to hysteria in the face of a wide-scale crisis such as pandemic flu. They must also provide information accessible to all publics, including those who do not have Internet access. |
| **Basnyat & Lee (2014)** | Singapore | Qualitative | H1N1 pandemic | Singapore Ministry of Health press release in newspapers | N/A | The media did convey useful information for managing the pandemic from the Ministry of Health press release (i.e. social responsibility, benefits of lockdown) and these were often in the tone/ semantics of colonial-era ideologies that are present within the culture in Singapore and reflect that of the government. Beneficial for the government to provide more than basic information, such as about coping with anxiety, and consider how other types of media (e.g. social media, blogs) can be used too. |
| **Berry (2007)** | Canada | Content analysis of the construction of news reports on health topics | SARS/other infectious diseases/general health | Media reports - newspaper, radio, television, and internet news on health-related topics | Not reported | The results of the content analysis showed that in 2003, there was far more information available in the news media on SARS and West Nile Virus than on other health topics with greater population prevalence, such as obesity or heart disease. The number of articles about SARS in 1 year was greater than for any other individual topic across all 5 years, with the exception of smoking (which in 2003 had only 36 articles, compared to SARS, which had 164) and “other” topics. Expert sources were cited far more often than nonexperts. |
| **Bonwitt (2018)** | Sierra Leone & Guinea | Longitudinal qualitative/ | Ebola | posters | Various preventative behaviours | All study participants, irrespective of age or gender, were aware of wild mammals acting as a source of transmission for Ebola virus disease (EVD); respondents gave a variety of answers when asked |



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| | | | | | | |
|---|---|---|---|---|---|---|
| | | ethnographic study | | | | what species of animal could transmit Ebola - this confusion may have rested in the content of public health messages, which were inconsistent in the species shown to be potentially hazardous; People simply refused to believe that wild meat could pose any health risk; Those that believed or partially believed in the link between EVD and wild meat adopted various strategies to mitigate the perceived risk of infection. Most admitted to only refraining from eating those animals that they understood as posing a risk for EVD; The widespread suspicion over the risks posed by wild animals prompted discussions during village meetings to discuss the veracity of public health messages.<br>Rather than outright rejection of these messages, people elaborated situated hypotheses to make sense of the conflicting and incomplete information they had received. This process of contextualisation helped to bridge the disjuncture between fear and the highly routine nature of eating meat. |
| **Cole & Watkins (2015)** | Liberia, Sierra Leone or Guinea | interviews | Ebola | WHO, CDC, professional media (international, local) informal information e.g. rumours | Information-seeking | Their recall can be grouped into three broad source categories (official, international and local media, and informal) Throughout later information-seeking behaviour, interviewees maintained a distinction between the three categories. This affected how they trusted and processed information from each, although generally they tended to use all three simultaneously. Multiple source information gathering, especially following up from informal information (personal experiences) because they felt that was more rumour and wanted confirmation from official sources but felt official sources were slow with the information. Consistency provided reassurance and helped to be trusted. perception of risk, triggered by hearing that someone they knew personally, or felt an affinity to, had contracted the virus. know who employees see as their community and 'people like me'. Behavioural changes seemed to kick in at the border between the near and real at-risk rather than at the border between far at-risk and near at-risk. Those in the far at-risk group generally showed more concern for people they knew in higher risk categories than for themselves. |
| **Crosier (2015)** | UK. Italy, Hungary | Qualitative interviews | H1N1 | Government poster | Public compliance and engagement | The study found a lack of planning and a low value attached to the skills required to produce effective communications. There was a dearth of good quality audience research to inform the development of communications. Little thought had been given to |



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| | | | | | | the tone, targeting or channelling of messages. Instead, communications were characterized by a 'one size fits all' and a 'top down', expert-led response. There was also little effort to evaluate the impact of communications, but where this was done, very low levels of public compliance and engagement with key behavioural messages were found. |
| **Davis (2015)** | Australia & Scotland | Qualitative interviews and focus groups | H1N1 pandemic 2009 | Broadcast and electronic media | Hygiene (e.g. covering mouth when coughing or sneezing, washing hands, cleaning surfaces), social isolation, vaccination | Participants wanted information but to be able to interpret advice based on own situation. Information on hygiene and social isolation endorsed but seen as short-term solution as being social is a central part of life. Participants had an individualistic approach to risk, decide whether to adopt PH behaviours based on risk to themselves, gender norms play a part in understanding and engagement with hygiene and social isolation. |
| **Freiman (2011)** | USA | face to face interviews | H1N1 pandemic 2009 | TV, newspaper, internet, family/friends, workplace, radio, other | Reporting on a range of preventative behaviours | Variable - did not always have the desired effect, or an effect at all. relied on the internet - however a lot of the target population did not have access to this - and thus were missed. Even those with internet access were unlikely to visit the CDC's website to learn about H1N1 |
| **Gray (2012)** | New Zealand | Focus groups | H1N1 pandemic 2009 | Ministry of Health campaigns | General protective behaviours | Four major themes: personal and community risk, building community strategies, responsibility and information sources. People wanted messages about specific actions that they could take to protect themselves and their families and to mitigate any consequences. They wanted transparent and factual communication where both good and bad news is conveyed by people who they could trust. |
| **Holmes (2009)** | Canada | Semi-structured interviews | crisis | Hypothetical | Risk perceptions | The notion of uncertainty – and the difficulties associated with communicating in uncertain situations – arose frequently. Recommendations: is to set and get agreement on a goal or goals (there may be different goals set for different audiences or specific situations). View all other aspects of communications planning and implementation through this ethical lens. Media representatives should be engaged immediately in discussions about a potential emerging infectious disease outbreak, including the role media should play and how the public health community can help them fulfil that role. Building trust with the public. |



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| **Jones (2010)** | Australia | Focus groups | Bird Flu | Campaigns prepared by advertising agency | Hand washing and Hygiene and self-protection measures (e.g. | In the early stages of a pandemic (i.e., prior to the identification of cases within a given country), communications should focus on increasing awareness of the disease and communicating important, but simple, protective behaviours to reduce the risk of transmission. Social marketing campaigns will also need to have an associated strategy for their social marketing campaigns' targeting intermediaries—such as general practitioners and other medical personnel, schools, business owners, and commercial and public organisations which could be utilized to disseminate information and resources. |
| **Lapka, (2008)** | USA | Focus groups and interviews | Influenza | | Various | Cognitive Response Testing (CRT) is easily applied to various subject matters and effectively highlights problematic terminology and phrases, allowing for beneficial and efficient message revisions. CRT is a straightforward, appropriate and useful method in health promotion message development and pre-testing. |
| **Li (2016)** | China | focus group | H7N9 | News from various sources | Risk perception | Most people learnt about virus from tv, info was felt to be released in timely and transparent manner, most useful was info about preventative behaviours, and evolving outbreak trends. Gov info perceived as trustworthy, but young people trusted social media more and wanted gov to publish news in such way. Health recommendations seen as easy to follow and clear. Anxiety affected by disease severity, distance from pandemic, high level of news coverage, trust in gov ability to fight the issue. Health communication needs to be specific and practical |
| **Liao, (2010)** | Hong Kong | telephone interviews | H1N1 pandemic 2009 | | hand hygiene and social distancing | There was a relationship between trust in formal information and understanding (positive), trust in information info and susceptibility (negative) |
| **Lohiniva (2020)** | Finland | Qualitative | COVID-19 | e-mail and social media | Risk perceptions | Risk perception domain: catastrophic - potential strong wording describing the epidemic, catastrophe, worldwide threat. Concept - Emotional response. Recommendation - Avoid downplaying strong feelings. Concept - Localised epidemic; stigmatising attitude towards foreign nationals. Recommendation - Humanise infected people by telling stories. Location of transmission: public transportation linked with airports and foreign passengers. Concept - Crowded places; stigmatising attitudes towards foreign nationals. Recommendation - Emphasise handwashing and cough etiquette as effective ways to prevent COVID-19. Concept - Unreliable |



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| | | | | | | information Recommendation - Repeat information and provide an explanation (reason). |
| **Lyu (2012)** | Taiwan | Qualitative Interviews | SARS | News reports/ public announcements | Perception of spokesperson | The essential/favourable characteristics for spokespersons should include: having professional capability, having better media interactions, involvement in policy making, having trustworthiness, and having favourable (or positive) personality and traits. |
| **Mitchell (2014)** | USA | Focus groups | Influenza A H1N1 | University mobile message alerts | Various preventative behaviours | Reported concern and commitment to recommendations decreased rapidly. Initial university messaging and response was critical in shaping participants' later perceptions. |
| **Person, (2004)** | USA | Focus groups and rapid situational assessment | SARS | Community outreach strategies | N/A Community resilience | SARS-related stigmatization was occurring more frequently within the Asian community than from outsiders directed toward the Asian community. The team also found that those persons with SARS-like symptoms who used traditional herbal physicians and pharmacies were less likely to be referred to, or seek out, public health officials, suggesting that further research into strategies to reach this population is needed. Conducting community visits also showed that CDC was responding to the needs of the community at risk for SARS-related fear, stigmatization, and discrimination and was modelling positive behaviours to the public. |
| **Qian (2020)** | China | Content analysis | COVID-19 | Newspapers | N/A | The Chinese mass media news lagged when reporting the major developments of the viral spread. Prevention and control procedures, medical treatment, and research are major themes of the press but mainly focus on the whole society, while instructions on personal and individual prevention, clinic and medicine choices, and detection need to be further enhanced. |
| **Qiu (2018)** | China | qualitative comparative case study | SARS, H7N9 | Media | Risk perceptions and awareness | 8 key principles of risk communication<br>Trust is the basis by maintaining an open and honest attitude and engage key stakeholders.<br>Delaying in reporting cases caused distrust, need accurate, timely, honest reporting, transparency helped subdue rumours, maintain social stability and helped maintain trust, and ensured more cooperation to reduce the spread of the disease. Being honest and open, planning well, being empathetic and caring, Accepting and involving the public as a partner regular feedback from the public and interaction between government and community. |



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| **Rim (2014)** | USA | content analysis + correlating with survey results | H1N1 | News media | Various | The more info from health authorities, the more news coverage. The same correlation for severity of content. Increased information subsidies with salience of the severity attribute was linked with increased H1N1 salience in media coverage. But no relationship between the severity attribute use in media coverage, and the public perception of risk. Framing that effectively empowers the public to engage in desired behaviour should be further studied for the success of a public health campaign. |
| **Rimi (2016)** | Bangladesh | ethnography | avian influenza | Posters | Prevention (handling poultry) | Incorporating non-health benefits might improve acting on recommendations.<br>More people heard about the bird flu after the intervention, and had more awareness around the disease, as well as performed preventative behaviours. Not all behaviours changed in the desired direction though (e.g., selling sick poultry, separating it). Reported behaviours were not consistent with the observations done by the researchers though. People rationalised ignoring recommended behaviours - low perception of risk - the flu not present among their birds, also finance issues and not worrying about getting ill. Also inconvenience was mentioned, as well as social pressure not to follow the recommendations, and scepticism about the necessity of the intervention - some conspiracy theories. |
| **Sell (2017)** | USA | Content analysis of news coverage | Ebola | News coverage | risk perception | more scientific knowledge to be used in media coverage. Certain risk messages about Ebola were used more frequently than others by US news media, which may have affected risk perception during the outbreak. Some messages increase, some decrease risk perception. |
| **Seltzer (2015)** | USA | Content analysis | Ebola | Instagram/ Flickr pictures relating to Ebola | Risk awareness | Images of health care workers and professionals [308 (25%)], West Africa [75 (6%)], the Ebola virus [59 (5%)], and artistic renderings of Ebola [64 (5%)]. Also identified were images with accompanying embedded text related to Ebola and associated: facts [68 (6%)], fears [40 (3%)], politics [46 (4%)], and jokes [284 (23%). Image sharing platforms are being used for information exchange about public health crises, like Ebola. Use differs by platform and discerning these differences can help inform future uses for health care professionals and researchers seeking to assess public fears and misinformation or provide targeted education/awareness interventions. |



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| **Sumo (2019)** | Liberia | Interpersonal communication' and dialogue | meningococcal septicemia | Community engagement (religious leaders, elders etc), radio talk shows, house to house visits, radio ads (jingles) | hand-washing, avoiding gatherings, risk perceptions | Increased trust and adherence. Contact tracing successful and epidemic contained, trust increased and adherence to messages increased (e.g. reporting sick people to health facility). |
| **Teasdale (2011)** | UK | Focus groups | H1N1 pandemic | Government + author-developed messages | Government advice was stay at home if you have flu-like symptoms and continue to go to work if you do not (behaviour not tested in this study though) | Participants were sceptical about the feasibility and appropriateness of self-diagnosis of pandemic flu. Participants actively evaluated recommended actions according to their own beliefs and reasoning about flu and their perceptions of the costs of carrying out the recommended actions. For most people the experience of the H1N1 pandemic was relatively mild compared to previous influenza pandemics of the 20th century. Consequently, the recommendation to stay at home was perceived as inappropriate. Written government communications aiming to improve and maintain population health may also need to address the recipients' perspective. Attempts should be made to elicit and address common doubts and concerns, to reduce perceived barriers to recommended behaviours, emphasize benefits, to find ways to support people to adopt them and consider the likely contextual factors that may affect perceptions of the advice. |
| **Tully (2019)** | West Africa | Content analysis | Ebola | Hyperlinks - embedded content in social media, e.g.Twitter used for information dissemination. | Risk perceptions | Increased credibility and awareness of messages. Organizations produce proportionally fewer tweets than news outlets - maintaining an active Twitter presence allows response organisations to promote their work and position themselves centrally to the unfolding crisis and conversation. Results suggested that 'owned' and 'earned' media served as informational resources and promotional pieces for the organizations. Repeatedly linking to sites controlled by the organization and/or produced by news media and other reliable sources served as a low-cost way for organizations to keep pace with the rapid flow of information on social media. |



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**Table 2 – Characteristics of Quantitative Design Studies**

| | Country | Study Type | Health Threat | Intervention type | Target behaviour | Main findings |
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| **Aburto (2009)** | Mexico | cross sectional representative household survey | H1N1 pandemic 2009 | | N/A - reporting on a range of preventative behaviours | Risk mediated behaviours in some cases e.g. if they perceived the threat of the virus to be high, they were in turn more likely to avoid crowds. People who did comply, were more likely to if they had knowledge about the virus. |
| **Bekalu (2017)** | USA | Experimental | Pandemic influenza | Video clips | washing hands frequently, staying home when sick and avoiding unnecessary travel | A non-narrative message format may be more effective than its narrative counterpart to communicate basic prevention information during public health emergencies. Compared with the narrative and/or fictional version, the more didactic and factual format was found to be more effective in changing knowledge and perceived response efficacy related to prevention of pandemic influenza. |
| **Chang (2012)** | Taiwan | Experiment (2 by 3 factorial design) | H1N1 pandemic | Newspapers | Risk perceptions | The amount of exposure to news stories increases its impact on the public. When a health issue is ambiguous, greater exposures to news coverage with high alarm frames, as opposed to low alarm frames, evoke more fear and increase participants' perceptions of the severity of the issue, as well as their vulnerability to it. Exposures to news do not increase prevention or treatment efficacy. Repetition exacerbates the effects of media exposure on perceived severity and vulnerability but not fear. The amount of exposure increases its impact. |
| **Daellenbach (2018)** | Australia, New Zealand | Quantitative online survey | Crisis | | Risk perceptions | Adaptable and personalised to be more effective. Segmentation based on Theory of Planned Behaviour variables. The study highlighting that vulnerability may be identified based on such factors as critical awareness, perceived barriers, and ultimately preparation undertaken, including community-related factors. |



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| **Davis, (2013)** | USA | Experimental design with 3 conditions | Influenza A; Influenza B; HINT | Posters | Hand washing | Prompts alone were not effective at increasing hand washing.<br><br>In the education building female bathroom soap usage significantly increased during the first poster prompt (following baseline), then fell below baseline during the second prompt (i.e., warning of influenza transmission). In the business building bathroom women's soap usage did not change significantly from baseline to first poster condition but decreased slightly on the second condition. Women in the education building used significantly more soap overall than women in the business building; there were no significant differences between men by building type. |
| **Hickey (2014)** | Quatar | Cross-sectional survey | H1N1 pandemic 2009 | Non-pharmaceutical interventions (e.g. using a thermometer, wearing a face-covering, hand washing, and household disinfection practices) | using a thermometer, wearing a face-covering, hand washing, and household disinfection practices | Attitudes towards recommended NPIs were generally negative or uncertain, limited experience applying these interventions (e.g., using a thermometer, wearing a face-covering) and inadequate hand washing and household disinfection practices |
| **Hoda (2016)** | Saudi Arabia | Survey | MERS | Promoting public awareness of MERS in the Saudi population | Risk awareness | The internet was the most commonly used source of information (39.5%) and the most endorsed channel for a MERS awareness campaign. Physicians were the preferred source of information (45.6%), followed by other health care providers (31.3%). In univariate multinomial logistic regression models, males and individuals aged ≤27 years were more likely to seek information from the internet than from physicians. Residents of southern and western Riyadh preferred physicians as a credible source of information over the Ministry of Health. |
| **Idoiaga (2016)** | Spain | Experimental | Flu epidemic | Mass media | Risk perceptions | The human interest framing increased the perception of risk, especially when the proximity of the epidemic was high; this effect was explained by people's emotional response. |



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| | | | | | | Furthermore, youth projected the risk towards 'the other' in order to protect their invulnerability identity. |
| **Jhummon-Mahadnac (2012)** | Australia | Cross-sectional survey | H1N1 pandemic 2009 | Public Education Campaigns | Working from home, postponing social gatherings, wearing facemasks, oseltamivir treatment if sick | 68% (175) reported behaviour change because of the pandemic. Gaps in knowledge included failure to identify certain high risk groups. Recall of government campaigns was significantly associated with a higher knowledge score. 60% (151) thought that authorities and media had exaggerated the threat; only 40% (101) would comply with recommended measures in a future pandemic. |
| **Johnson & Slovic (2015)** | USA | Experimental - 2 studies | Ebola | Mock story | Risk perceptions | Informing Americans about the small likelihood of post-21-days Ebola symptoms would not increase perceived risk and distrust and might diminish negative reactions to the media reporting a case who developed symptoms of Ebola after 21 days. Thus, public health officials wanting appropriate public responses to potential or actual epidemics may benefit from early communication of unpleasant infectious-disease facts before events reveal them, and signal officials' lack of preparedness for public reactions. |
| **Kavanagh (2011)** | Australia | cross sectional survey | H1N1 pandemic 2009 | official sources (health dept, schools) and unofficial sources (media, family and friends, health care providers) | quarantine | Compliance was higher when they understood the messages. make messages easy to understand for everyone. Official sources are trusted/legitimised more than unofficial sources |
| **Miczo (2013)** | USA | Cross-sectional survey | H1N1 pandemic | Campaigns on campus (flyers, posters, emails). Mass media (various, including television & radio) other sources (e.g. medical) | Various - Self-isolation, hand hygiene, respiratory etiquette, sanitizing frequently | No significant results found for the impact of the health messages on health behaviour performance. The most frequently mentioned messages students remembered were: to wash hands (56.9%), avoidance (23.5%) (e.g. staying away from others when feeling ill) and getting a flu shot/ vaccination (22.1%). Positive health self-concept was positively related to the argument message dimension and |



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| | | | | | touched surfaces, not attending public events if sick, vaccination, early intervention for high-risk students and staff members, and cancelling travel plans | flu-prevention behaviours; negative health self-concept was negatively related to the argument message dimension and hand washing.<br>Campus most frequently cited source and useful from campus health viewpoint. There was limited success in attempting to assess memorable messages using a dimensional approach. |
| **Prati (2011)** | Italy | Cross-sectional survey | H1N1 | Educational campaign | Various – all preventative behaviours | Recommended behaviours such as cleaning objects, social distancing and washing hands were related to all the psychosocial factors except for trust in the institutional response to the outbreak and trust in medical science. Trust in medical science was significantly associated with cleaning objects. To use tissues when sneezing was predicted by all the psychosocial factors, except for trust in medical science and control. These results showed that media trust, trust in the Ministry of Health (regardless of what was done to reduce this risk), worry and perceived severity of illness predicted all the recommended behaviours. Exposure to the Topo Gigio campaign did have a limited influence on complying with health related recommendations but only with regard the use of tissues when sneezing. |
| **Rim (2014)**<br><br>***(also reported in table 1)*** | USA | content analysis + correlating with survey results | H1N1 | News media | Various | The more info from health authorities, the more news coverage. The same correlation for severity of content. Increased information subsidies with salience of the severity attribute was linked with increased H1N1 salience in media coverage. But no relationship between the severity attribute use in media coverage, and the public perception of risk. |



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| | | | | | | Framing that effectively empowers the public to engage in desired behaviour should be further studied for the success of a public health campaign. |
| **Roess (2017)** | DR Congo | Pre- post interventional study | Ebola | Four films | Risk perceptions and awareness | a culturally and linguistically appropriate video-centred intervention was effective in improving knowledge, attitudes and behaviours related to EVD in Congo. The results also demonstrate retention of the knowledge one year after exposure to the intervention. |
| **Updegraff (2011)** | USA | Experimental design with 3 conditions | H1N1 | Posters/ signs placed besides hand sanitizer units around university campus | Use of hand sanitizer dispensers | All signs resulted in significantly greater usage than no sign, but they were not equally effective. Dispensers with the gain-framed signs had the greatest usage of all, with 66.4% more use than dispensers with no signs ($p < .001$). Loss-framed signs were associated a 58.4% increase in use over no sign ($p < .001$). The social norms signs (44.3% increase) and the perceived susceptibility signs (40.6% increase) were associated with somewhat lower increases in usage compared to the gain-framed and loss-framed signs, but both led to significantly more usage than no sign at all (both $p$'s $< .01$). Gain-framed signs received 12.5% more usage than dispensers in the other sign conditions combined, a significant difference ($p = .029$). Usage of sanitizer dropped consistently over time, closely mirroring temporal trends in public interest in H1N1. This study showed that the relatively simple strategy of placing theoretically grounded cue-to-action reminder signs at the point-of-use significantly promoted usage. The worst-performing sign emphasized people's susceptibility to contamination. |
| **Van der Weerd (2011)** | Netherlands | repeated measure cross-sectional telephone survey | H1N1 | | information-seeking | During the course of the pandemic the majority of respondents wanted to receive information from health services/health care providers and media. Wanted information on how to prevent infection, what to do in the event of illness, symptom, risks , consequences and number of infected cases. Trust in governmental information depressed |



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| | | | | | | by the time of period 3. At first, mistrust was because information was incomplete, kept secret or withheld. Later, because they thought the situation was exaggerated, information unclear and information was contradicted itself. Fear/worry was sig associated with intention to adopt protective measures during all time periods. |
| **Winters (2018)** | Sierra Leone | Cross-sectional | Ebola | A variety of sources: mostly electronic media, followed by community sources, print and new media. About half received government communications and most were exposed to more than 2 types of info. | Not reported | increased protective behaviours. Strong dose-response between info exposure and protective behaviours, however exposure to all sources (except electronic and print media) was associated with misconceptions |
| **Yardley (2011)** | UK | parallel-group pragmatic exploratory trial design | H1N1 pandemic | Website | hand washing | Increased hand washing, targeted/changed attitudes |
| **Zikmund-Fischer (2017)** | Netherlands | experiment | hypothetical influenza | mock news article | cover mouth, wash hands | Participants are more influenced by average than extreme case information, presenting both is counterproductive. |

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**Table 3 – Characteristics of Reviews and Narrative Reports**

| Author (date) | Country | Study Type | Health Threat | Intervention type | Target behaviour | Main findings |
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| **Barrelet (2013)** | | Systematic Review | H1N1, H5N1 | | Risk perception, vaccine perception, rumours | Risk perceptions vary across time, social groups, cultures, and countries. Social resistance to vaccination is an enduring phenomenon that should not be downplayed by public health officials. Competing narratives (official statements and presentations, rumours, conspiracy theories, alternative worldviews and explanations, urban legends, hoaxes, etc.) about public health crises always coexist in the public sphere. Trust building is a key aspect of risk perceptions. |
| **Crouse Quinn (2008)** | USA | Commentary /proposal of a model for crisis and emergency and risk communication in a pandemic | Pandemics (general) | | Not reported | Overall proposed model: pre-disaster phase (ongoing risk education, partnership formation, community engagement, opportunity for deliberation on difficult policies and procedures); during an emergency (more one way, emergency risk communication, more need for immediate action, community partners' engagement in response); increased capacity and resilience of communities at times of disasters and pandemics |
| **Goldberg (2015)** | USA | Report | Ebola and others | Checklist | Risk awareness and perceptions | 5 stages: agree on a common goal, coordinate the leadership, develop a coordination strategy, launch a communication operation, maximize communication effectiveness |
| **Lin (2014)** | | Systematic review | H1N1 | Non-specific variety of interventions, including 'Websites', ' Commercial television', 'Health department' | Preventive behaviours such as hygiene and social distancing practices in 70% of the studies (N = 64), risk perceptions (70%, N = 64), levels of knowledge and awareness about the pandemic (53%, N | To reduce communication inequalities during a large scale emergency, such as a pandemic, public health officials should focus their communication efforts on the young, the less educated and the indigent because there is evidence that these are the people at risk of not knowing about the threat, perceiving the threat to be of low risk and ultimately being less likely to follow recommended behaviours. An honest reporting of what the threat looks like, through a presentation of known and unknown factors, seems to have a better impact on people's knowledge, attitudes and beliefs, |



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| | | | | | = 49), and emotional responses such as fear and worry (47%, N = 43).<br>Factors influencing the H1N1 vaccination acceptance rate (26%, N = 24) were also frequently investigated. | including trust in the way the government is handling the emergency. Consequently, there is some evidence that better knowledge and trust are likely to be associated with the adoption of recommended behaviours (i.e. immunization practices). Social networks and ties to the community are also drivers of better knowledge and compliance with preventive measures; these results suggest that non-traditional channels of communication (i.e. partnership with community leaders or organizations) should be used to reach out to the most vulnerable and those in need of a better understanding of the risks and actions needed to be able to protect themselves. Public health communication messages are still delivered at a literacy level that does not meet the needs of the less educated. |
| **Menon (2005)** | Singapore | Commentary – narrative review | SARS | | Risk perceptions | 1. Providing more information is more effective than less information, 2. it always better to over-react than to under-react. 3. fear and ignorance of any disease is worse than the disease itself. It reduces all to impotence and defeat. Information is an all-powerful tool to fight fear. It empowers people and allows them to become socially responsible. |
| **Menon (2006)** | Singapore | Narrative review/ commentary | SARS | Mixed - TV, technology | Not reported | Technology (The Infrared Fever Screening System), a SARS-dedicated TV channel, transparency, leadership (highest levels of government came into the picture very early in the crisis and took the lead), communication tools (moderating public fear through dialogue with grassroots leaders and community, briefings for foreign business groups, diplomats, religious groups, trade associations, cartoons, hotlines, songs on TV), earning the trust and confidence of Singaporeans, rebuttal of negative reports in the foreign media claiming that foreign visitors had been infected with SARS whilst transiting Singapore, outbreak communications across cultures. |
| **Rogers (2009)** | Australia | Deliberative forum | Influenza | hypothetical | Risk awareness | The forum wanted full and frank information about the potential risk and international developments including numbers of cases and fatalities. Forum members understood that predictions about the pandemic influenza were uncertain, |



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| | | | | | | this did not lead to loss of confidence in the experts or the information they imparted. Third, the forum recommended releasing geographically localizing information about initial cases. The forum recommended increasing use of television and websites, including those targeting youth and rural and remote groups as the pandemic developed. The forum recommended immediate activities to educate and build awareness and swift action if and when Australia has its first cases of PI. |
| **Santibañez (2015)** | USA | Commentary | Ebola | | N/A | 10-step approach can be adapted for infectious disease response communications (in this case Ebola) allowing community and faith-based organisations to help spread effective health messages |
| **Schiavo (2014)** | | Systematic review | Influenza, SARS | Media | Handwashing, immunizations, infection control precautions | most studies found behaviour changed |
| **Toppenberg (2018)** | Various | Rapid Review of Grey Literature | Ebola | | Awareness only | Literature strongly underlines the central importance of local communities. A one-size-fits-all approach does not work. For maximum effectiveness, local communities need to be involved with and own emergency risk communication processes, preferably well before an emergency occurs. Social media can open new avenues for communication but is not a general panacea and should not be viewed as a replacement for traditional modes of communication. In general, the gray literature indicates movement toward greater recognition of emergency risk communication as a vitally important element of public health. |
| **Vaughan (2009)** | Not reported | Commentary – narrative review | Pandemic Influenza | | Message acceptance | Barriers to accepting messages include, environmental factors, social and cultural characteristics, language preferences (translation, culturally sensitive) difficulty of and attitudes towards public health interventions |



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**Table 4 – Characteristics of Preprint Studies**

| | Country | Study Type | Health Threat | Intervention type | Target behaviour | Main findings |
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| **Abu-Akel, Spitz & West (2020)** | Switzerland | Pilot experimental survey study | COVID-19 | Social distancing | Social distancing | The government official was more effective at garnering support for social distancing than the celebrity, particular in older respondents. Support and compliance relating to social distancing was higher in older people, despite them having lower percieved risk of threat. Females were overall more in support of social distancing. Higher support and compliance was related to greater concern about the situation, concern for others health, greater belief in social distancing measures and feeling more constrained by them. Inverse relationship between city size and practise of social distancing. |
| **Bilancini, (2020)** | Italy | Four-condition, between-subjects experiment. | COVID-19 | Official coronavirus-related messages | Behaviours relating to the COVID-19 emergency | No statistically significant difference between the 4 conditions. Authors conclude that stronger 'nudging' interventions may be needed to influence behaviour. |
| **Blagov (2020)** | USA | Online survey | COVID-19 | Public health messages worded to target different personality traits (self-centered, responsible, compassionate, avoidant & sociable). Looks like she were developed by the researchers. | Staying at home, physically distancing and practising handwashing | Agreeableness and conscientiousness predicted endorsement of social distancing and hygiene. Meanness and disinhibition (and overall psychopathy) as well as Machiavellianism (less so) predicted lower intent for social distancing. Together with boldness, the psychopathy traits predicted endorsement of risky, venturesome behavior under the pretend scenario that one is a disease carrier. Meanness and disinhibition predicted endorsement of knowing and possibly deliberate behaviour that puts others at risk of infection. |



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| **Bradshaw, (2020)** | Australia | Randomized experimental online study | COVID-19 | Online vignettes about the contact tracing app | Downloading and using a contact tracing app | Autonomy-supportive and controlling message framing did not differentially affect intended uptake. However, there was a main effect of information safety. Those in high information safety conditions reported higher intentions to use the application and to recommend it to others than those in low information safety conditions, regardless of message framing. Australians appeared more willing to assent to authority regarding contact tracing insofar as their data safety can be assured. |
| **Dai (2020)** | China | Cross-sectional online survey | COVID-19 | Information on government emergency management measures relating to COVID-19 | Protective behaviours, including preventive (i.e. wearing masks, disinfectants), avoidant (i.e. stringent quarantine, avoiding public places), and management of disease behaviours (i.e. seeking professional protection or treatment information, paying for preventive and therapeutic drugs) | Governmental information transparency, positive propaganda, rumour refutation, and supplies positively predict the protective behaviours. Individual factors such as perceived control, positive emotions, and risk perception mediate role in predicting protective behaviours. |
| **Everett, (2020)** | USA | online experimental survey study | COVID-19 | Social media (Facebook post) either from a citizen or leader. | Behavioural intentions for: Hand washing, avoiding gatherings, isolation, cancelling holidays | Stronger intentions if messages were shared by 'leader' than citizen, modest effects of using deontological and virtue-based messaging (compared to utilitarian) |
| **Goldberg, (2020)** | USA | Online Survey | COVID-19 | N/A - measured behaviours as a result of seeing | mask buying and mask wearing | Once the CDC recommendation had been disseminated for at least one full day, there were large increases in reported mask wearing and mask buying. |



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| | | | | national guidance from the CDC | | |
| **Heffner (2020)** | USA | Within subject design, online survey | COVID-19 | 2 messages about staying at home, one focused on 'fear', the other on 'altruism' | Intentions to self-isolate, emotional response | Both messages were associated with increased willingness to self-isolate. The prosocial message was more effective at boosting willingness to self-isolate if it produced a strong, positive, and arousing emotional response. Fear messages created a stronger emotional response. |
| **Jordan (2020)** | USA | Three-part online experimental study | COVID-19 | PH messages/ fliers about COVID-19 emphasising personal, public or personal & public protective messages. | Hand washing, social distancing, staying at home, wearing a mask. | In the early stages of the pandemic, prosocial (i.e. public) messaging was more effective. However, as the pandemic progressed, all three types of framing (i.e. prosocial/ public, self-interested/ personal and both) were equally as effective. Self-interested framing was never more effective than prosocial framing and did not improve efficacy of prosocial framing. |
| **Merkley, (2020)** | Canada | 2-part online experimental survey study | COVID-19 | News stories relating to the pandemic - signalling information from experts or non-experts. | Information on hygiene (i.e. cleaning surfaces), social gatherings, testing and scientific discoveries. | Study 1: We find that our respondents were on average 3 points more likely to select news stories with headlines featuring experts (p~0.007). News stories with headlines featuring experts scored 2 points higher in credibility (p<0.002). In support of H1, we find that the effect of receiving headlines featuring experts is heterogeneous across levels of anti-intellectualism. The marginal effects on story selection are shown in the left panel of Figure 2. Respondents with the lowest levels of anti-intellectualism are six points more likely to select stories featuring experts (p<0.001), but this effect disappears once the mid-point of the scale is reached.<br>Study 2: Our respondents were 19 points more likely to select COVID-19 news (p<0.001). They also viewed such news as 6 points more credible (p<0.001) and 19 points more important (p<0.001). Once again, |



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| | | | | | | |
|---|---|---|---|---|---|---|
| | | | | | | anti-intellectualism strongly conditions news preferences. Respondents with the highest levels of trust in experts were 25 points more likely to select COVID-19 news (p<0.001), and viewed it as 8 points more credible (p<0.001) and 24 points more important (p<0.001). In all three cases this effect weakens as anti-intellectualism rises, though it does not entirely vanish for the most part. |
| **Wirz, (2020)** | USA | Online Survey | COVID-19 | N/A - survey asking whether people were practising guidelines (e.g. social distancing) and questions about barriers and communication. | Social distancing | Perceived effectiveness of social distancing for decreasing the risk of infection and perceived social norms related to social distancing were strongly correlated with reported social distancing, suggesting that individuals engaged in social distancing if they thought it worked and if they thought their peers were doing it. If they learned that their social distancing behaviour would help others (e.g., family, older members of the community) and that there are serious health consequences for people like them, they would engage in more social distancing. Respondents in this group reported often getting news from national news networks and social media, followed by newspapers/news magazines. Respondents reported having the highest levels of trust in information from the CDC, public health experts, and the WHO. These groups were followed by university scientists and friends and family. The White House, state and local governments, and the news media were among the least trusted sources for information. |



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# Appendix 5- Data quality table using Mixed Methods appraisal tool and AMSTAR

| Studies | | | | Criteria from the mixed methods appraisal tool | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | S1 | S2 | 1.1 | 1.2 | 1.3 | 1.4 | 1.5 | 2.1 | 2.2 | 2.3 | 2.4 | 2.5 | 3.1 | 3.2 | 3.3 | 3.4 | 3.5 | 4.1 | 4.2 | 4.3 | 4.4 | 4.5 | 5.1 | 5.2 | 5.3 | 5.4 | 5.5 |
| 1. Aburto (2009) | cross sectional representative household survey | 4 | **** | | | | | | | | | | | | | | | | | | 1 | 1 | 1 | 0 | 1 | | | | | |
| 2. Avery (2009) | Content analysis | 5 | **** | 1 | 1 | 1 | 1 | 1 | 1 | 1 | | | | | | | | | | | | | | | | | | | | |
| 3. Berry (2007) | Content analysis | 4 | **** | 1 | 1 | 1 | 1 | 1 | 1 | 0 | | | | | | | | | | | | | | | | | | | | |
| 4. Bekalu (2017) | Pre/post intervention study | 4 | **** | 1 | 1 | | | | | | 1 | 1 | 1 | 0 | 1 | | | | | | | | | | | | | | | |
| 5. Basnyat & Lee (2014) | Qualitative | 4 | **** | 1 | 1 | 1 | 0 | 1 | 1 | 1 | | | | | | | | | | | | | | | | | | | | |
| 6. Bonwitt (2018) | Experimental | 4 | **** | 1 | 1 | 1 | 0 | 1 | 1 | 1 | | | | | | | | | | | | | | | | | | | | |
| 7. Chang (2012) | Experiment (2 by 3 factorial design) | 5 | ***** | 1 | 1 | | | | | | | | | | | | | | | | | | | | | 1 | 1 | 1 | 1 | 1 |
| 8. Cole & Watkins (2015) | interviews | 2 | ** | 1 | 1 | 1 | 1 | 0 | 0 | 0 | | | | | | | | | | | | | | | | | | | | |
| 9. Crouse Quinn (2008) | Commentary | | n/a | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 10. Crosier (2015) | Qualitative Interviews | 4 | **** | 1 | 1 | 1 | 1 | 1 | 1 | 0 | | | | | | | | | | | | | | | | | | | | |
| 11. Daellenbach (2018) | Quantitative online survey | 3 | *** | 1 | 1 | | | | | | | | | | | | | | | | 1 | 0 | 1 | 0 | 1 | | | | | |
| 12. Davis, (2013) | Experimental design with 3 conditions | 5 | ***** | 1 | 1 | | | | | | | | | | | 1 | 1 | 1 | 1 | 1 | | | | | | | | | | |
| 13. Davis (2015) | Qualitative interviews and focus groups | 3 | *** | 1 | 1 | 1 | 1 | 0 | 1 | 0 | | | | | | | | | | | | | | | | | | | | |
| 14. Freiman (2011) | face to face interviews | 5 | ***** | 1 | 1 | | | | | | | | | | | | | | | | 1 | 1 | 1 | 1 | 1 | | | | | |
| 15. Goldberg (2015) | Report | | n/a | | | | | | | | | | | | | | | | | | | | | | | | | | | |

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|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 16. Gray (2012) | Focus groups | 5 | ***** | 1 | 1 | 1 | 1 | 1 | 1 | 1 | | | | | | | | | | | | | | | | | | | | |
| 17. Hickey (2014) | Cross-sectional survey | 5 | ***** | 1 | 1 | | | | | | | | | | | | | | | | 1 | 1 | 1 | 1 | 1 | | | | | |
| 18. Hoda (2016) | Survey | 2 | ** | 1 | 1 | | | | | | | | | | | | | | | | 0 | 0 | 1 | 0 | 1 | | | | | |
| 19. Holmes (2009) | Semi-structured interviews | 5 | **** | 1 | 1 | 1 | 1 | 1 | 1 | 1 | | | | | | | | | | | | | | | | | | | | |
| 20. Idoiaga (2016) | experimental | 2 | ** | 1 | 1 | | | | | | 0 | 0 | 1 | 0 | 1 | | | | | | | | | | | | | | | |
| 21. Jhummon-Mahadnac (2012) | Cross-sectional survey | 5 | ***** | 1 | 1 | | | | | | | | | | | | | | | | 1 | 1 | 1 | 1 | 1 | | | | | |
| 22. Johnson (2015) | Experimental – 2 studies (both studies) | 3 | *** | 1 | 1 | | | | | | 1 | 1 | 1 | 0 | 0 | | | | | | | | | | | | | | | |
| 23. Jones (2010) | Focus groups | 2 | ** | 1 | 1 | 0 | 0 | 1 | 0 | 1 | | | | | | | | | | | | | | | | | | | | |
| 24. Kavanagh (2011) | cross sectional survey | 4 | **** | 1 | 1 | | | | | | | | | | | | | | | | 1 | 1 | 1 | 0 | 1 | | | | | |
| 25. Lapka (2008) | Focus Group and Interviews | 5 | **** | 1 | 1 | 1 | 1 | 1 | 1 | 1 | | | | | | | | | | | | | | | | | | | | |
| 26. Lohiniva (2020) | Qualitative | 4 | **** | 1 | 1 | 1 | 1 | 1 | 0 | 1 | | | | | | | | | | | | | | | | | | | | |
| 27. Li (2016) | Focus group | 5 | **** | 1 | 1 | 1 | 1 | 1 | 1 | 1 | | | | | | | | | | | | | | | | | | | | |
| 28. Liao, (2010) | telephone interviews (surveys) | 4 | **** | | | | | | | | | | | | | | | | | | 1 | 1 | 1 | 0 | 1 | | | | | |
| 29. Lyu (2012) | Qualitative Interviews | 3 | *** | 1 | 1 | 1 | 0 | 1 | 1 | 0 | | | | | | | | | | | | | | | | | | | | |
| 30. Menon (2005) | Narrative review | | n/a | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 31. Menon (2006) | Narrative review/commentary | | n/a | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 32. Miczo (2013) | Cross-sectional survey | 2 | ** | 1 | 1 | | | | | | | | | | | | | | | | 0 | 0 | 1 | 0 | 1 | | | | | |
| 33. Mitchell (2014) | Focus groups and interviews | 5 | **** | 1 | 1 | 1 | 1 | 1 | 1 | 1 | | | | | | | | | | | | | | | | | | | | |

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|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 34. Person, (2004) | Focus groups and rapid situational assessment | 2 | ** | 1 | 1 | 1 | 1 | 0 | 0 | 0 | | | | | | | | | | | | | | | | | | | | |
| 35. Prati (2011) | Cross-sectional survey | 3 | *** | 1 | 1 | | | | | | | | | | | | | | | | 1 | 0 | 1 | 0 | 1 | | | | | |
| 36. Qian (2020) | Content analysis | 2 | ** | 1 | 1 | 0 | 0 | 1 | 1 | 0 | | | | | | | | | | | | | | | | | | | | |
| 37. Qiu (2018) | Focus groups and interviews | 5 | ***** | 1 | 1 | 1 | 1 | 1 | 1 | 1 | | | | | | | | | | | | | | | | | | | | |
| 38. Roess (2017) | Pre-post interventional study | 3 | *** | 1 | 1 | 0 | 0 | 1 | 1 | 1 | | | | | | | | | | | | | | | | | | | | |
| 39. Rogers (2009) | Deliberative forum | 3 | *** | 1 | 1 | 1 | 1 | 1 | 0 | 1 | | | | | | | | | | | | | | | | | | | | |
| 40. Rim (2014) | Content analysis +correlation with survey results | 1 | * | 1 | 1 | 0 | 0 | 1 | 0 | 0 | | | | | | | | | | | | | | | | | | | | |
| 41. Rimi (2016) | ethnography | 3 | *** | 1 | 1 | 1 | 0 | 1 | 1 | 0 | | | | | | | | | | | | | | | | | | | | |
| 42. Santibañez (2015) | Commentary | | n/a | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 43. Sell (2017) | Content analysis of news coverage | 5 | **** | 1 | 1 | 1 | 1 | 1 | 1 | 1 | | | | | | | | | | | | | | | | | | | | |
| 44. Seltzer (2015) | Content analysis | 3 | *** | 1 | 1 | 1 | 0 | 1 | 0 | 0 | | | | | | | | | | | | | | | | | | | | |
| 45. Sumo (2019) | Interpersonal communicati on' and dialogue | | n/a | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 46. Teasdale (2011) | Focus groups | 4 | **** | 1 | 1 | 1 | 1 | 1 | 1 | 0 | | | | | | | | | | | | | | | | | | | | |
| 47. Toppenberg (2018) | Rapid Review of Grey Literature | 5 | ***** | 1 | 1 | 1 | 1 | 1 | 1 | 1 | | | | | | | | | | | | | | | | | | | | |
| 48. Tully (2019) | Content analysis | 3 | *** | 1 | 1 | 1 | 1 | 0 | 1 | 0 | | | | | | | | | | | | | | | | | | | | |
| 49. Updegraff (2011) | Experimental design with 3 conditions | 4 | ***** | 1 | 1 | | | | | | 1 | 1 | 1 | 0 | 1 | | | | | | | | | | | | | | | |

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|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 50. | Vaughan (2009) | Narrative review | | n/a | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 51. | Van der Weerd (2011) | Cross-sectional telephone survey | 5 | **** | 1 | 1 | | | | | | | | | | | | | | | | 1 | 1 | 1 | 1 | 1 | | | | | |
| 52. | Winters (2018) | Cross-sectional | 5 | ***** | 1 | 1 | | | | | | | | | | | | | | | | 1 | 1 | 1 | 1 | 1 | | | | | |
| 53. | Yardley (2011) | parallel-group pragmatic exploratory trial design (RCT) | 4 | **** | 1 | 1 | | | | | | 1 | 1 | 1 | 0 | 1 | | | | | | | | | | | | | | | |
| 54. | Zikmund-Fischer (2017) | experiment | 2 | ** | 1 | 1 | | | | | | 0 | 0 | 0 | 1 | 1 | | | | | | | | | | | | | | | |

Data Quality check of Systematic Reviews with AMSTAR

| Citation | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Schiavo et al | 1 | 1 | 1 | 1 | 1 | 0 | 1 | 1 | 1 | 0 | 1 |
| Lin et al | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 0 | 1 |
| Barrelet et al | 1 | 1 | 0 | 1 | 1 | 0 | 0 | 0 | 1 | 0 | 1 |

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| Pre-print Quality Check | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Studies | | | | Criteria from the mixed methods appraisal tool | | | | | | | | | | | | | | | | | | | | | | | | | |
| | | | | S1 | S2 | 1.1 | 1.2 | 1.3 | 1.4 | 1.5 | 2.1 | 2.2 | 2.3 | 2.4 | 2.5 | 3.1 | 3.2 | 3.3 | 3.4 | 3.5 | 4.1 | 4.2 | 4.3 | 4.4 | 4.5 | 5.1 | 5.2 | 5.3 | 5.4 | 5.5 |
| 1. Abu-Akel (2020) | Pilot survey-based study (randomised) | 4 | **** | 1 | 1 | | | | | | 1 | 1 | 1 | 0† | 1 | | | | | | | | | | | | | | | |
| 2. Bilancini (2020) | Randomised controlled trial | 3 | *** | 1 | 1 | | | | | | 1 | 0 | 1 | 0 | 1 | | | | | | | | | | | | | | | |
| 3. Blagov (2020) | Cross-sectional survey | 2 | ** | 1 | 1 | | | | | | | | | | | | | | | | 1 | 0 | 0 | 0 | 1 | | | | | |
| 4. Bradshaw (2020) | Randomised controlled trial | 4 | **** | 1 | | | | | | | 1 | 1 | 0 | 1 | 1 | | | | | | | | | | | | | | | |
| 5. Dai (2020) | Cross sectional design survey | 4 | **** | 1 | 1 | | | | | | | | | | | 1 | 1 | 0† | 1 | 1 | | | | | | | | | | |
| 6. Everett (2020) | Randomised cross-over study | 1 | * | 1 | 0 | | | | | | 0 | 0 | 0 | 0 | 1 | | | | | | | | | | | | | | | |
| 7. Goldberg (2020) | National survey | 4 | **** | 1 | 1 | | | | | | | | | | | | | | | | 1 | 1 | 1 | 0 | 1 | | | | | |
| 8. Heffner (2020) | Cross-sectional analytic study | 4 | **** | 1 | 1 | | | | | | | | | | | 2 | 1 | 1 | 1 | 1 | | | | | | | | | | |
| 9. Jordan (2020) | Randomised controlled trial | 2 | ** | 1 | 1 | | | | | | 0 | 0 | 1 | 0 | 1 | | | | | | | | | | | | | | | |
| 10. Merkley (2020) | Randomised controlled trial | 1 | * | 1 | 1 | | | | | | 0 | 0 | 0 | 0 | 1 | | | | | | | | | | | | | | | |
| 11. Wirz (2020) | Online Survey | 3 | *** | 1 | 1 | | | | | | | | | | | | | | | | 1 | 1 | 0† | 0† | 1 | | | | | |

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# Appendix 6: Influences on effective public-health messaging

Most of the studies were qualitative work and reported on determinants of intent to adhering to guidelines by reporting what influences people's responses to public health messaging. These are: preconceptions and understanding of the threat, perceived susceptibility and perceived risk severity (threat appraisal(Rogers, 1975)).

*Preconceptions and understanding of the threat*

Of the 68 studies included, 26 specified that public health messaging included information about the threat which can be effective in identifying symptoms and changing behaviours. Some of these, including the only randomised control trial, reported changes in handing washing (Yardley, Miller, Schlotz, & Little, 2011), changes in knowledge and higher compliance to quarantine (Kavanagh et al., 2011). However, this particular study found the messages were not effective in encouraging quarantine behaviour suggesting that increasing knowledge alone did not change behaviour (Kavanagh et al., 2011). More comprehensive reporting on what the threat is and looks like, including presentation of known and unknown factors has a positive impact on people's knowledge, attitudes and beliefs. Self-reports, in the qualitative studies and in surveys, suggest changes in hygiene and unspecified protective behaviours following messages of how a virus (e.g. H1N1, SARS, Ebola) is transmitted, and the ability to idenitfy symptoms, and how to prevent transmision (Chidgzey, Davis, Williams, & Reeve, 2015; Gray et al., 2012; Hickey, Gagnon, & Jitthai, 2014; Hoda, 2016; R. Li, Xie, Yang, & Frost, 2016; Rim, Ha, & Kiousis, 2014; Teasdale & Yardley, 2011; Yardley et al., 2011). In the systematic reivew, public health messaging campaigns using posters (e.g. "*Catch it, Bin it, Kill it"*) increased knowledge about tramsission, and successfully increased self-reported prevention behaviours (hygiene when using tissues and sneezing). However, compliance to these behaviours was also predicted by worrry (Lin, Savoia, Agboola, & Viswanath, 2014). The systmatic review argued that individuals with higher education were better informed about H1N1. The review concluded that messages and communication are still delivered at a high literacy level (Lin, Savoia, et al., 2014). Whilst most of these papers highlighted that people will have different preconceptions and concerns to address in the public health messages, three papers also highlighted that people may have limited experiences of engaging in the preventative behaviour e.g. using face masks or using a thermometer (Hickey et al., 2014). Such cases highlighted a need for training/skill development as part of a public health campaign.

People that were able to identify symptoms of H1N1 were more likely to avoid crowds and washed hands more (Aburto et al., 2010). Framing and choice of language were found to be influential in behaviour and how individuals understand the threat (Teasdale & Yardley, 2011). Positively framed messages (emphasising the benefits of prevention behaviours) were considered 'robust contingency measures' (Basnyat & Lee, 2015). Gain-framed signs were significantly more effective in influencing the use of hand sanitizer as cue-to-action reminder signs at the point-of use during the H1N1 pandemic (Updegraff, Emanuel, Gallagher, & Steinman, 2011). The worst-performing signs emphasised people's susceptibility to contamination (Updegraff et al., 2011). Communication strategies were characterized in three case studies from the H1N1 pandemic from Italy, Hungary and England as "top-down" expert-led responses by public health communications specialists because there was no audience research. With no audience research, there was a perceived lack of planning and consideration to the tone, targeting or channelling of public health messages. Lack of consideration of these aspects led to low levels of public compliance and engagement with



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behaviour (Crosier, McVey, & French, 2015). A non-narrative message format (i.e., factual) may be more effective than a narrative counterpart to communicate basic prevention information during public health emergencies. Compared with the narrative and/or fictional version, the more didactic and factual format was found to be more effective in changing knowledge and perceived response efficacy related to prevention of pandemic influenza however, behaviour or intent were not measured in the study (Bekalu, Bigman, McCloud, Lin, & Viswanath, 2018). When messages provided the public with clear consistent information whilst admitting that information is evolving, reported the risks, and focused on the practical actions that people can do to reduce their risk and emphasized the efficacy of those actions, people were more likely to clean objects, wash hands and use tissues when sneezing (Prati, Pietrantoni, & Zani, 2011). A pre-print manuscript found that for one to perceive and understand the extent of a threat, messages that are framed in relation to identity (e.g. "don't be a spreader" vs "don't spread") were more likely to predict preventative behaviour following the Japanese Ministry of Health guidelines (Yonemitsu, 2020).

When individuals do not understand the message or the preventative behaviour being communicated, participants did not outright reject public health messages. Instead, people adjusted the key underlining message to what they thought made sense, and engaged in behaviour consistent with the adjusted message (Bonwitt et al., 2018). As there is general uncertainty around a pandemic flu it is hard for people to differentiate this from seasonal flu especially when symptoms are framed as "flu-like symptoms" (Teasdale & Yardley, 2011). In a qualitative study, people reported being more likely to follow guidelines when there were fewer perceived barriers to perform the recommended behaviours, when there are emphasised benefits and when contextual factors are considered such as feeling guilty and anxious about missing work and not wanting to let people down (Teasdale & Yardley, 2011). High levels of knowledge was associated with exposure to messages and was a mediator between different sources and complying with recommended protective behaviour. However, engaging with social media was associated with misconceptions of public health information (Winters et al., 2018).

*Threat appraisal: perceived susceptibility/vulnerability*

Increase in exposure to any type of information was associated with increased knowledge, and as level of outbreak increased (that increased the level of threat appraisal), protective behaviours increased, suggesting a strong dose-response (Johnson & Slovic, 2015; Winters et al., 2018). As perceptions of higher risk increase so did self-reported behaviours such as handing washing, wearing a facemask and using hand gels (Aburto et al., 2010). However, salience (i.e. prominence) in media coverage did not have a direct relationship with increasing perceived risk. One study found that there was no relationship between salience in media coverage of H1N1, or the severity of content and the public perception of risk (Rim et al., 2014).

Being worried about H1N1 was associated with media attention and number of H1N1 cases, and overall an important predictor of compliance with recommended preventative behaviours, such as using tissues, hand gel and washing hands (Lin, Savoia, et al., 2014). Worry led to effective compliance and increased adaptive behaviour in one included study (Gutteling, Terpstra, & Kerstholt, 2018). Evoking fear increased participants' perceptions of the severity of the issue and their vulnerability to the risk but did not increase prevention behaviours or treatment efficacy (Chang, 2012). Moreover, panic could be counterproductive (Zikmund-Fisher, Scherer, Knaus, Das, & Fagerlin, 2017); providing a "worse-case" scenario (that there is a high risk and there is a pandemic) and increasing knowledge of the health threat simultaneously, has been found to induce panic in



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those receiving the message (Jhummon-Mahadnac, Knott, & Marshall, 2012). Believability of messages was reduced by scepticism towards the media and governments; in these instances the threat was perceived as exaggerated and able to cause unnecessary panic which could influence low adoption of recommended action (Teasdale & Yardley, 2011). In a preprint manuscript, messages arousing an emotional response (such as fear) was more effective in boosting willingness to self-isolate alongside prosocial messages that are producing strong and positive emotional response (Heffner, 2020).

Perceiving a threat would mean that population warning systems (method where local, regional or national authorities contact the public en masse) can work but social pressure in addition to increasing perceived risk is also required (Gutteling et al., 2018). Worry about self or family members led to increased perceptions of disease severity and a stronger belief of being susceptible to the infection, leading to a greater compliance with recommended behaviours (Lin, Savoia, et al., 2014). Perceived susceptibility in the messaging (e.g. "this can happen to someone like me") was more likely to lead to engagement in physical distancing than the geographical closeness of the disease (e.g. cases in the local area) thereby appraising the threat as high (Cole & Watkins, 2015). If risk is perceived to be low (e.g. H1N1 pandemic likened to influenza pandemics) recommendations such as "stay at home" were perceived as extreme and inappropriate (Teasdale & Yardley, 2011). This then has implications for increasing distrust in the messages and sources. Behaviours such as usage of a sanitizer dropped over time mapping the temporal trends in public interest in H1N1 (Updegraff et al., 2011).

Individuals were more likely to be concerned about the risk to themselves rather than the risk they would pose to others (M. D. Davis, Stephenson, Lohm, Waller, & Flowers, 2015). A recent systematic review concluded that beliefs about personal susceptibility to H1N1, and perceiving it as potentially severe (infection rate), is linked with high levels of worry about self or family members at risk (Lin, Savoia, et al., 2014). Efficient public health messaging (where individuals were more likely to comply with behaviour (e.g. avoiding crowds; (Aburto et al., 2010) tapped into worry about self or family members by framing messages with positive social responsibility (Basnyat & Lee, 2015). Concern for family led to reporting more information-seeking and appraising the threat as a higher risk to self and their family (Cole & Watkins, 2015). Prosocial framing and focus on activating social norms were found in the grey literature to be effective in increased engagement in physical distancing (Jordan, 2020; Wirz 2020). Tailoring public health messages with compassion may increase acceptance of the messages (Blagov, 2020). Such prosocial framing in the public health messaging personalised the risk and increasing the perceived susceptibility and vulnerability.

### Increasing trust

An important finding that 60% of the included studies identified the need to increase the public's trust in the messages during pandemics and epidemics (Aburto et al., 2010; Cole & Watkins, 2015; Gray et al., 2012; Hickey et al., 2014; Holmes, Henrich, Hancock, & Lestou, 2009; Jhummon-Mahadnac et al., 2012; Johnson & Slovic, 2015; Kavanagh et al., 2011; Liao, Cowling, Lam, Ng, & Fielding, 2010; Lohiniva, Sane, Sibenberg, Puumalainen, & Salminen, 2020; Lyu et al., 2013; Person et al., 2004; Prati et al., 2011; Qiu et al., 2018; Sumo et al., 2019). Participants cited trust as the main reason they complied, and observational studies found trust to be essential in complying with preventative behaviours (R. Li et al., 2016; Menon & Goh, 2005), both when communicating public health advice from public health agencies (Li et al., 2016) or from the government (Menon & Goh, 2005). Arguably, the need to increase trust highlights that acceptability of the message is reliant on



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the right message given in the right way (framing cross-reference), by an appropriate messenger (cross-reference credible source) at the right time.

One of the systematic reviews, concluded that competing narratives should be addressed to support trust-building, and this can be achieved through involving the public with two-way communications in the design of the messages (Barrelet, Bourrier, Burton-Jeangros, & Schindler, 2013). Whilst the other systematic review, found that an honest reporting of what the threat looks like, through a presentation of known and unknown factors, seems to have a better impact on people's knowledge, attitudes and beliefs, including trust in the way the government is handling the emergency (Lin, Savoia, et al., 2014).

In a repeated measures survey during H1N1 pandemic, trust levels in the government decreased over time, higher levels of government trust led to higher intention to adhere to the guidelines and change in trust was due to conflicting messages at the same time points and as they change over time (van der Weerd, Timmermans, Beaujean, Oudhoff, & van Steenbergen, 2011). Anxiety was mitigated by messages that were prompt and delivered at the right time for the point of the pandemic/epidemic, transparent, recommended preventative behaviours, and provided evolving outbreak trends (i.e. giving information as it happens) (R. Li et al., 2016). People are more likely to engage in behaviour inconsistent with the message if messages lack clarity or there is inconsistent messaging both over time and between sources. This eventually results in individuals distrusting the message. Therefore, one paper argues besides information and excessive exposure of such information there is also a need for behavioural modelling (to see the behaviour by others) (O. L. Davis, Fante, & Jacobi, 2013). Information in messages needs to be relevant and trustworthy to influence behaviour change and can include: catastrophic potential, probability of dying, and reasons for exposure. Providing explanations for perceived differences from other countries, not being too optimistic and not hide information can increase overall trust towards authorities (Lohiniva et al., 2020). Attempting to increase knowledge in the context of low levels of public trust in the source can make a message ineffective, and mistrust can increase if the perception is that the information is exaggerated (Jhummon-Mahadnac et al., 2012; Teasdale & Yardley, 2011). When an epidemic (e.g. Ebola) seemed contained, and contact tracing was successful, trust increased and, in response, so did the adherence to messages (e.g. reporting to health facility) (Sumo et al., 2019).

*Multiple and credible sources*

Individuals seek information from multiple sources to meet different information needs (Cole & Watkins, 2015). Whilst official sources of information (such as governmental sources, or public health bodies) are seen as legitimate, individuals do not always find them useful (Toppenberg-Pejcic et al., 2019; Tully, Dalrymple, & Young, 2019) thereby seek other sources. When official sources take a long time to provide information, individuals rely on informal sources such as rumours, word of mouth, social media (Cole & Watkins, 2015; van der Weerd et al., 2011); the value of information is at its highest when the outbreak is emerging. If mass media do not report the outbreak in time (before unofficial/informal sources start reporting the outbreak) it fails to become the leading indicator that people use for their information (J. Li, Xu, Cuomo, Purushothaman, & Mackey, 2020). Delay can impact the public's trust in official sources (e.g. government) and has led to beliefs that the threat is exaggerated by government or news and media (J. Li et al., 2020; Teasdale & Yardley, 2011) especially when information is conflicting. Furthermore, a delay in addressing rumours



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circulating among the public increases the chances of apathy and communication fatigue (Mitchell et al., 2014).

The grey literature reported that individuals seek information from sources that they consider trustworthy these include public health experts, WHO and Centers for Disease Control and Prevention (CDC) and state and local governments; news media were the least trusted sources (Wirz 2020). Another preprint found that there was greater compliance and acceptance of messages when the messages came from government officials and credible sources (Abu-Akel, 2020). The effect on behaviour change (self-reported mask wearing and mask buying) from messaging was greater amongst individuals who trusted the Centers for Disease Control and Prevention (CDC), government agencies and scientists (M. H. Goldberg, Gustafson, A., Maibach, E.W., Ballew, M.T., Bergquist, P., Kotcher, J.E., Marlon, J.R., Rosenthal, S.A., Leiserowitz, A.,, 2020) and these messages were better received (Merkley, 2020). Explicit information, data safety and transparency were important (Bradshaw 2020) especially when describing protective behaviours (Dai, 2020). Whilst individuals generally preferred expert sources for the initial information-seeking, they eventually relied on other sources (and perceptions such as "anti-intellectualism" increased); eventually leading to discontinuing use of official sources for information during a pandemic (Merkley, 2020).

### Community engagement and social networks

Social networks and close ties to the community are drivers of better knowledge and compliance with preventive measures. A systematic review suggests that non-traditional channels of communication (i.e. partnership with community leaders or organizations) should be used to reach out to the most vulnerable (can include those who have a disability e.g. hearing/vision/intellectual) and those who are least literate as well as non-Native speakers. Public health communication messages are still delivered at a literacy level that does not meet the needs of the less educated. To reduce communication inequalities, those who are less likely to identify risk associated with the threat and so less likely to adhere to the behaviours should be targeted. These include the young, least educated and hard to reach communities (Lin, Savoia, et al., 2014). The public pay more attention if the community is engaged in the intervention (Schiavo, May Leung, & Brown, 2014) as communication is a dynamic process and to address the concerns, values, interests and priorities the public need to be partners in communication (Barrelet et al., 2013).

The inclusion of various communities increases the chances of developing materials with which all individuals can identify (Crosier et al., 2015). This in turn increases the chances of perceiving the message is relevant to their self and important people in their lives thus heightening risk perception, and changing behaviours (Crouse Quinn, 2008; Gray et al., 2012; Person et al., 2004). Individuals want information that fits with their experiences (Freiman et al., 2011; Teasdale & Yardley, 2011). Adaptable and personalised information, that is context-driven, is more effective in changing determinants of behaviour (e.g. especially in vulnerable groups) (Daellenbach, Parkinson, & Krisjanous, 2018; Qiu et al., 2018). Core messages should be consistent to ensure trust but also considerations need to be made to ensure the involvement of communities to address their needs (e.g. including community leaders, translations, access to resources). Message delivery should be appropriate for the targeted population; for example, social media can open new avenues but should not to other types of delivery (Toppenberg-Pejcic et al., 2019). There is a use for social media, as it can be effective in communicating messages (Lwin, Lu, Sheldenkar, & Schulz, 2018), fostering trust and providing opportunities for dialogue (Tully et al., 2019). Social media can be especially



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useful for reaching young people who trust social media channels (R. Li et al., 2016). Social-economic status (SES) is related to information access and understanding; those of lower SES were less likely to use a website, and more likely to find public health messages from TV and Radio to be confusing and contradictory than those of higher SES (Aburto et al., 2010).

The grey literature highlighted that developing prosocial messages and identity, increased willingness to self-isolate especially when producing a strong, positive emotional response (Heffner, 2020). Another preprint found that messages should not just target those who are at risk but those who do not perceive themselves at risk from COVID-19 (e.g. young people and hard to reach groups). Messages must also address barriers to physical distancing and outline benefits of carrying them out, emphasising altruism through messaging from credible sources using social media (Wirz 2020). Another preprint found that prosocial messages were the most effective type of framing, tapping into people's sense of morality especially at the beginning of a pandemic. This changed over time and self-interest messaging were just as effective (Jordan, 2020).

## Messages for sub-populations

### Young (Student populations) and older adult populations

Three studies examined perceptions and experiences of H1N1 messages with student populations (Idoiaga, De Montes, & Valencia, 2016; Miczo, Danhour, Lester, & Bryant, 2013; Mitchell et al., 2014). An experimental study on framing found that younger students (mean age = 21.05) perceived themselves as less vulnerable than older students (over the age of 60) in falling ill with flu when reading a news report that described a flu epidemic (Idoiaga et al., 2016). The authors suggest this may be due to the 'social invulnerability identity' of younger people. This is consistent with social cognitive approaches to health risk perception, which suggests people may be "unrealistically optimistic" about a particular health threat (Weinstein, 1987).

Campus and mass media were reported, by students, as the most common places for getting information about the H1N1. Students tend to perceive information from the university as more credible than the media (Mitchell et al., 2014).

Focus groups exploring reactions to two different campaign ideas (for a potential avian influenza epidemic in Australia) found that there was some confusion amongst older adults whereby they presumed the 'WIPE' part of the WASH, WIPE, and WEAR' message was referring to handkerchiefs, rather than disposable tissues. Older adults (and separately mothers) predominantly preferred messages which emphasised protecting others (e.g. that they should wear a mask if they were sick in order to protect others) (Jones, Waters, Holland, Bevins, & Iverson, 2010). On the other hand, regional and frequent travellers' groups thought it was equally important to protect themselves from other people's germs, and also protect others from their germs (Jones et al., 2010).

### Vulnerable groups

The importance of both protecting themselves and others was evident in findings from focus groups conducted with vulnerable groups in New Zealand during the influenza H1N1 2009 pandemic (Gray et al., 2012). People wanted messages about specific actions that they could take to protect themselves and their families during the pandemic.



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During the influenza H1N1 2009 pandemic, a cross-sectional survey was conducted with migrant workers (i.e., anyone who did not have Thai citizenship) to explore their perceptions of and ability to implement non-pharmaceutical interventions (NPIs; e.g., using face masks when sick) proposed by the WHO (Hickey et al., 2014). Certain groups of migrant workers were considered as a vulnerable group due to traditions in raising poultry and swine, poor personal hygiene and sanitation, low health literacy, limited access to healthcare, and high frequency of cross-border communication. Attitudes towards recommended NPIs were generally negative or ambivalent. Barriers included never having worn a face mask and not knowing the correct way to wear one. This highlights the need to educate vulnerable groups about performing NPIs during pandemics as migrants are at risk of propagating the spread of a pandemic virus. It is also important that public education campaigns can reach migrants living in remote and difficult to access areas, and to accommodate for the diverse cultural and linguistic needs.

The importance of ensuring that public health messages can reach different linguistic groups was also echoed by the Pacific Peoples' participants in the focus groups conducted by Gray et al. (Gray et al., 2012). Older people in the group are unlikely to understand English so messages should be communicated in the original languages. However, as noted by Person et al., (Person et al., 2004) this can be very difficult when an in-house translation service does not exist, and the rapidly evolving scientific evidence will challenge the turnaround time for developing, translating, and disseminating information. To help address this, the team prioritised key SARS documents for translation and back-translations conducted by professional translation services ensured accuracy. The translated information should be disseminated through culturally appropriate channels such as community visits, town hall meetings, and health and education and communication channels to complement mass media messages. This is similar to reflections of Singapore's experience in dealing with the SARS outbreak (Menon, 2006) where there was ongoing transparent dialogue with grassroots leaders and communities, and religious groups.

Social Economic Status (SES)

A cross-sectional survey with 1010 adult Italians during the influenza H1N1 2009 pandemic found that respondents who were women, and were facing economic hardship, were more likely to clean objects, wash their hands, and use tissues when sneezing (Prati et al., 2011).

A Mexican study, which explored the public's ability and willingness to adopt community mitigation efforts during the influenza H1N1 2009 pandemic, found that respondents from lower SES groups were more likely to report costs of masks being a barrier to their use (Aburto et al., 2010). Respondents from lower SES tertials were also more likely to report that mitigation recommendations were contradictory or confusing. The authors highlight that it is important for future campaigns to tailor the messaging to people of lower SES who are more likely to have lower literacy levels and reduced ability to interpret messages. Messages should be coordinated, consistent, and simplified for these groups.

*Evaluation of messages at salient points in an epidemic/pandemic*

This presents the papers that had specific aims for the time of the study (either beginning, during, or post a pandemic/epidemic) and these are mapped to recommendations developed in this review. Evaluations collecting data at different time points over the course of a pandemic can tell us about any changes in patterns of public behaviour or perceptions as the situation evolves.



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Table 1: Papers organised according to timepoints of epidemic/pandemic

| | Study | Aim of messages | Mapped to Recommendations | |
|---|---|---|---|---|
| **Beginning of epidemic/pandemic** | Yardley et al 2011°<br>Chang et al, 2012°<br>Teasdale et al, 2011°<br>Mitchell et al, 2014°<br>Kavanagh et al, 2011°<br>Freiman et al, 2001° | Increase knowledge and awareness | • Early announcements<br>• Consistency of messages<br>• Acknowledge uncertainties<br>• Timing of recommendations / announcements / alerts | **Addressing uncertainty and increasing trust** |
| | Aburto et al, 2010°<br>Hickey et al, 2014†<br>Roess et al, 2017†<br>Bonwitt et al, 2018†<br>Winters et al, 2018†<br>Lohiniva et al, 2020 ¥ | Empower use of behaviours to prevent spread of virus within the first 6 months | • Integrate community /community leaders in risk comms and into planning<br>• Social responsibility<br>• Tailor to increase accessibility | **Community engagement** |
| | | | • Different sources of information – ensure consistency<br>• Clear instructions<br>• Language to explain severity | **Message framing and unified messaging** |
| **During epidemic/pandemic** | Jhummon-Mahadnac et al, 2010°<br>Gray et al, 2012°<br>Prati et al, 2011°<br>Updegraff et al, 2011° | Assess efficacy of messages/ campaigns | • Responding to concerns questions<br>• Skilled / credible spokespeople<br>• Admitting mistakes<br>• Acknowledge uncertainties<br>• Acknowledge unknowns and knowns<br>• Labelling recommendations as interim | **Addressing uncertainty and increasing trust** |
| | Rim et al, 2014°<br>Johnson & Slovic, 2015†<br>Toppenberg et al, 2018†<br>Tully et al, 2019†<br>Davis et al, 2013‡ | Update and possibly introduce new behaviours based on new information<br>Continue with raising awareness and maintain behaviours | • Tailor information to users needs (accessibility)<br>• Use of local spokespeople and influencers<br>• Translate and use different sources with consistency in messaging | **Community engagement** |
| | Bekalu et al, 2017‡<br>Rimi et al 2016+<br>Avery et al, 2009+ | Manage misinformation | • Increase understanding of health threat<br>• Promote sense of personal control<br>• Social responsibility | **Message framing and unified messaging** |
| **Post** | Davis et al, 2015°<br>Basnyat & lee, 2014°<br>Miczo et al, 2013°<br>Crosier et al 2015°<br>Barrelet et al 2013°<br>Person et al, 2004•<br>Qui et al, 2017+•<br>Lyu et al, 2012• | Assess efficacy of messages/ campaigns<br>Manage misinformation | • Responding to concerns questions<br>• Skilled / credible spokespeople<br>• Admitting mistakes<br>• Acknowledge uncertainties<br>• Acknowledge unknowns and knowns | **Addressing uncertainty and increasing trust** |

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| | Study | Aim of messages | Mapped to Recommendations | |
|---|---|---|---|---|
| | Li et al 2016[+]<br><br>Sell et al 2017†<br>Sumo et al 2019†<br>Cole & Watkins, 2015† | Evaluation for lessons learnt and Preparation training | • Public release of evaluations and reviews | |
| | | | • Translate and use different sources with consistency in messaging<br>• Equity considerations | **Community engagement** |
| | | | • identify inconsistencies<br>• Core messages consistent<br>• understanding of health threat | **Message framing and unified messaging** |

Note a: °Study at beginning, during or post H1N1; ¥ Study during COVID-19; †Study at the beginning, during or post Ebola; ‡Study at beginning, during or post influenza season; [+]Study at beginning, during or post avian flu; • Study at beginning, during or post SARS

## Appendix 7: Rapid review findings mapped to the BPS COVID-19 Behavioural Science and Disease Prevention Psychological Guidance

| Recommendation in review | BPS Behavioural Science guidance |
|---|---|
| Address uncertainty | 2. Deliver messages from a credible source in relatable terms to the target audience |
| Engaging communities | 1. Minimise the 'I' and emphasise the 'we'<br><br>4. Identify what influences each preventative behaviour and ensure policies, messaging and interventions target all relevant drivers<br><br>6. Avoid unintended negative consequences |
| Unified messaging | 5. Clearly specify behaviours and their effectiveness<br><br>7. Create clear channels of access for health literacy<br><br>8. Use behavioural scientists and the psychological evidence base to support the COVID-19 response<br><br>9. Make a pledge to work together, through a multi-disciplinary approach. #CombatCovid19Together |
| Message framing | 3. Create worry but not fear |



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## AUTHORS

Professor Madelynne A. Arden is an HCPC registered Health Psychologist and a Professor of Health Psychology at Sheffield Hallam University, specialising in the development and evaluation of theory and evidence based behaviour change interventions.

Professor Christopher J. Armitage is an HCPC registered Health Psychologist, a Fellow of the Academy of Social Sciences and Professor of Health Psychology at the University of Manchester. He specialises in using psychological theory to develop tools to facilitate behaviour change among diverse populations.

Professor Lucie Byrne-Davis is an HCPC registered Health Psychologist, Principal Fellow of the Higher Education Academy, Fellow of the European Health Psychology Society and a Professor of Health Psychology at the University of Manchester, specialising in health professional behaviour change and global health.

Natalie Carr is a health psychology MSc student at Manchester Metropolitan University.

Professor Angel Chater is an HCPC registered Health Psychologist, a Professor in Health Psychology and Behaviour Change at the University of Bedfordshire and Leads the British Psychological Society's Division of Health Psychology and COVID-19 Behavioural Science and Disease Prevention Taskforce.

Dr Tracy Epton is a lecturer and researcher, at the University of Manchester, specialising in behaviour change.

Dr Daniela Ghio is a lecturer at the University of Salford, specialising in intervention development and health behaviour change.

Ella Guest is a Research Associate at the Centre for Appearance Research in Bristol and Stage 2 Trainee in final year at UWE, Bristol.

Professor Jo Hart is an HCPC registered Health Psychologist and a professor of health professional education at University of Manchester, specialising in health professional education and behaviour change.

Dr Sarah Hotham is a Senior Research Fellow and BPS Chartered Psychologist at the Centre for Health Services Studies, University of Kent specialising in applied heath research and training of healthcare professionals.

Dr Neil Howlett is a Senior Research Fellow in the Department of Psychology and Sport Sciences, University of Hertfordshire, specialising in physical activity and wider behaviour change methodology.

Dr Elizabeth Jenkinson is a Senior Lecturer in Health Psychology and an HCPC registered Health Psychologist at the University of the West of England, Bristol.

Dr Angelos P. Kassianos is a Senior Research Fellow, a Fellow of Higher Education Academy and an HCPC registered Health Psychologist, at UCL (Department of Applied Health Research) specialising in



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disease prevention, medical education as well as developing and evaluating evidence-based behaviour change digital interventions.

Dr Chris Keyworth is a Research Fellow, at The University of Manchester, specialising in health communication and behaviour change.

Dr Sadie Lawes- Wickwar is a HCPC registered Health Psychologist. Associate Fellow of BPS & Member of DHP. Visiting Lecturer in Health Psychology at City, University of London.

Daniel Scanlan has a MSc in Health Psychology and works as a Research Assistant at Education Support.

Sabina Stanescu is a Health Psychology Trainee and a PhD Candidate within Psychology at the University of Southampton.

Dr Natalia Stanulewicz is a Lecturer in Psychology at De Montfort University. Her research focuses on health behaviours, wellbeing as well as emotional and individual characteristics predicting prosocial behaviours.

Lisa Sutherland - Behavioural Insight Consultant. MSc in Health Psychology 20 years working in digital, social marketing and behavioural change intervention design.

Dr Rachael Thorneloe is a Senior Research Fellow in the Centre for Behavioural Science and Applied Psychology, Sheffield Hallam University, specialising in behaviour change.

Dr Mei Yee Tang is a Research Associate at the Behavioural Science Policy Research Unit at Newcastle University.

Daniella Watson is a Trainee Health Psychologist & Global Health PhD student (University of Southampton).

Juliette Westbrook is a Research Assistant at the University of Reading and studying MSc Health Psychology at the University of Bath.

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# PRISMA 2009 Checklist

| Section/topic | # | Checklist item | Reported on page # |
|---|---|---|---|
| **TITLE** | | | |
| Title | 1 | Identify the report as a systematic review, meta-analysis, or both. | 1 |
| **ABSTRACT** | | | |
| Structured summary | 2 | Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number. | 2 |
| **INTRODUCTION** | | | |
| Rationale | 3 | Describe the rationale for the review in the context of what is already known. | 3 |
| Objectives | 4 | Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS). | 3 |
| **METHODS** | | | |
| Protocol and registration | 5 | Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number. | 3 |
| Eligibility criteria | 6 | Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale. | 4 |
| Information sources | 7 | Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched. | 4 |
| Search | 8 | Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated. | S2-S3 |
| Study selection | 9 | State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis). | 4 |
| Data collection process | 10 | Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators. | 4 |
| Data items | 11 | List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made. | 4, S5 |
| Risk of bias in individual studies | 12 | Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis. | 4 |
| Summary measures | 13 | State the principal summary measures (e.g., risk ratio, difference in means). | N/A |
| Synthesis of results | 14 | Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., $I^2$) for each meta-analysis. | N/A |

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# PRISMA 2009 Checklist



| Section/topic | # | Checklist item | Reported on page # |
|---|---|---|---|
| Risk of bias across studies | 15 | Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies). | N/A |
| Additional analyses | 16 | Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified. | N/A |
| **RESULTS** | | | |
| Study selection | 17 | Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram. | 5 Figure 1 |
| Study characteristics | 18 | For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations. | S5 |
| Risk of bias within studies | 19 | Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12). | S41 |
| Results of individual studies | 20 | For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot. | N/A |
| Synthesis of results | 21 | Present results of each meta-analysis done, including confidence intervals and measures of consistency. | 5 |
| Risk of bias across studies | 22 | Present results of any assessment of risk of bias across studies (see Item 15). | 5 |
| Additional analysis | 23 | Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]). | 5 |
| **DISCUSSION** | | | |
| Summary of evidence | 24 | Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers). | 12 |
| Limitations | 25 | Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias). | 13 |
| Conclusions | 26 | Provide a general interpretation of the results in the context of other evidence, and implications for future research. | 13 |
| **FUNDING** | | | |
| Funding | 27 | Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review. | 4 |

*From:* Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(7): e1000097. doi:10.1371/journal.pmed1000097

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